



# Power Play: Investing in Energy Infrastructure for the AI Era

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The world is facing a significant energy deficit. This was true before AI and the war in Iran exacerbated the shortfall. The kinds of investments underway today will shape the future of the energy landscape and its ability to support an evolving tech-forward economy. Businesses are responding through legacy and innovative ways both to alleviate current constraints and to scale up for the long run.

To address immediate constraint issues, companies are sidestepping grid backlogs by offering readily dispatchable power sources and modular units. As the grid expands to meet growing demand, multi-industry and electrical equipment companies are enabling utilities to expand generation, transmission, and delivery. Renewables cannot be overlooked as a scaling solution as the AI era exponentially increases the demand for energy. Solar in particular is among the fastest-to-deploy sources of new generation. When paired with battery energy storage systems, these technologies can provide the kind of around-the-clock, predictable power that AI infrastructure requires. Commercial nuclear is another important component of the longer-term energy supply makeup, with the near-term focus on extending the lifetime of existing plants while looking ahead to longer-term expansion. All that said, even as the grid scales significantly and the mix of energy supply diversifies, natural gas is likely to remain a primary and backup source of power.

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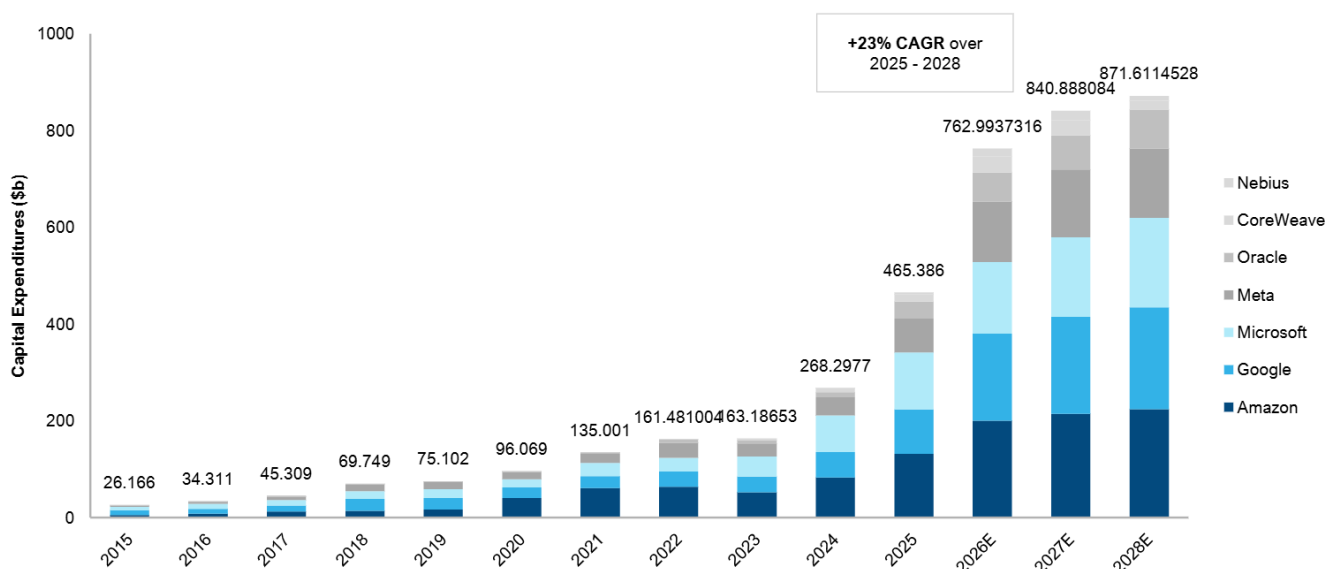
## The Macro and Demand Landscape

The grid is under significant pressure. US energy generation hit a record 4.43 terawatt hours in 2025, according to the US Energy Information Administration; generation will likely break that record again this and next year due primarily to large data centers. Energy demand driven by AI data centers globally is expected to quadruple over the next four years, according to the International Energy Agency. To get a sense of how great this demand is and will be in the coming years, we can look to cloud-related investments, growth in data centers and AI workloads, and energy prices.

One of the most telling data points has been cloud-related investments. Our Semiconductors Research Team has tracked a decade of data-center-related investments via capital expenditures (capex) at several major cloud service providers to finance the setup of physical data centers as well as the electronic equipment needed to fill them. The data shows that the current buildout is unlike any other. Capex has surged over the last three years, rising from \$161 billion in 2022 to \$465 billion in 2025. This reflects a compound annual growth rate of +42%, a jump up from comparable three-year rates of +29% (2019-2022) and +30% (2016-2019).

Spending is expected to step up even further to \$762 billion (+64% year-over-year) in 2026. If that happens, this would mark the third year in a row that cloud-related capital investments have risen more than 60% year over year. Prevailing estimates also expect growth to slow after this growth spurt, but as we saw throughout 2025, consensus tends to underestimate the amount of spending in latter years, so future spending could well exceed today's expectations.

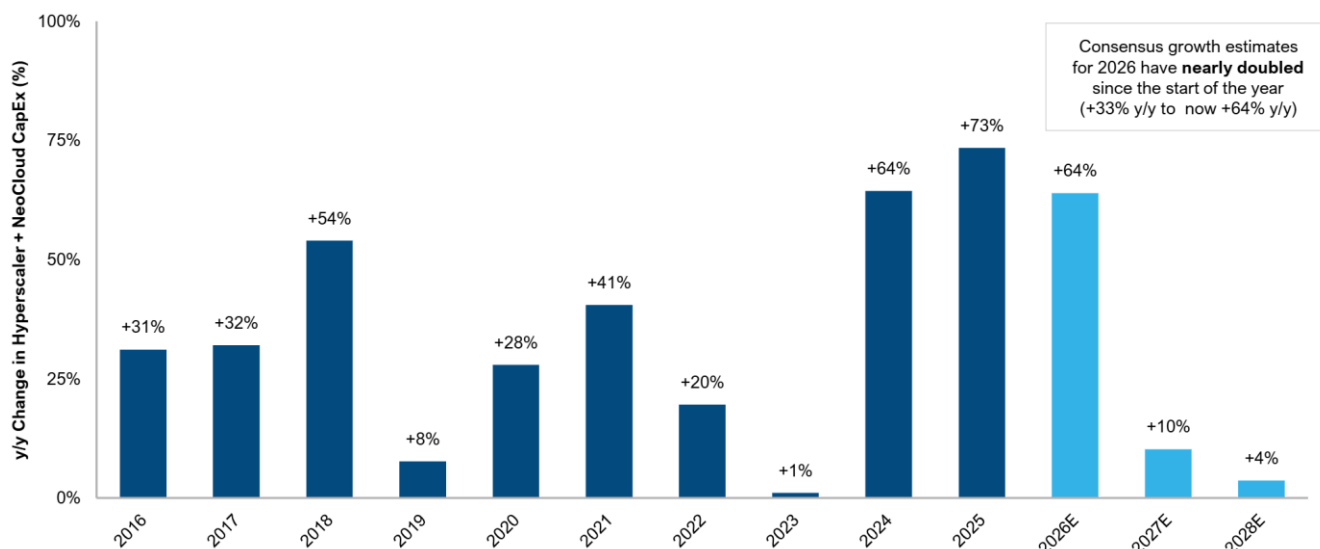
Figure 1: Aggregate Cloud Capex (2015 - 2028E)



Source: Deutsche Bank Research, Bloomberg Finance LP. Note: Dataset available upon request. Forecast data is based on median Bloomberg consensus estimates. Information is presented on an annual basis given the lack of quarterly detail by company and dispersion in consensus estimates going forward.



Figure 2: Annual Growth in Aggregate Cloud Capex (2016 - 2018E)



Source: Deutsche Bank Research, Bloomberg Finance LP. Note: Dataset available upon request. Forecast data above is based on median Bloomberg consensus estimates. Information is presented on an annual basis given the lack of quarterly detail by company and dispersion in consensus estimates going forward.

Data center AI workloads paint a similar AI-driven energy demand picture. Data centers represent about 5% of total US electricity usage today, according to estimates from BloombergNEF and the Department of Energy. This will likely expand with the adoption of AI, as well as with continued growth in the digital economy.

Based on midrange estimates from McKinsey, total global data center capacity demand is set to increase from 82 gigawatts in 2025 to 220 gigawatts in 2030, amounting to a 22% annual growth rate. This estimate sees generative AI workloads growing at 29% on average, plus 11% from non-AI workloads. Generative AI workloads could grow to 35% of total global data center activity by 2028, up from a mid-single-digit percentage as of 2023, according to estimates from Boston Consulting Group.

AI workloads require greater power density (essentially the amount of electricity a system needs per square foot) and more intensive liquid cooling, driving up energy consumption. Data center companies like Digital Realty (DLR) and Equinix (EQIX), for example, are already seeing these trends play out in their businesses. Digital Realty recently said it is rolling out higher-density deployments with pre-installed liquid cooling to meet AI demand. Equinix has noted that 60% of its larger deals in the last quarter of 2025 were driven by AI, up from 50% earlier last year. The power density for these deals, according to Equinix, is 33% higher than for non-AI deals.

These factors exacerbate concerns around US electricity prices and grid capacity, which were already mounting even before the war in Iran elicited a spike in global energy prices. After an extended period of stable growth in the US – over the last decade, electricity output grew at about a 0.7% annualized rate – data center electricity demand estimates are raising concerns about a knock-on



impact to household utility bills as well as grid capacity. The energy problem is just as acute for central bankers. Traditionally, Federal Reserve policymakers would tend to look through the temporary inflationary impact of an energy shock (see US Economic Perspectives: Impact of oil price shocks a "frack"-tion of what they once were). However, the backdrop to the latest energy shock is one in which inflation has persistently exceeded the Federal Reserve's 2% goalpost for the better part of the last five years. This raises the risk that inflation expectations could become unanchored. In short, the energy supply outlook carries implications across a breadth of economic issues, from interest rates to technology stocks.

These concerns could result in policy favorable to companies increasing energy supply, as energy prices will likely be top of mind for voters ahead of the midterm elections this November. As we recently outlined, historically, the relationship between gasoline prices and consumer energy spending suggests that a 1 cent rise in gasoline prices (on a sustained basis) increases spending on energy goods and services. This thus reduces income that can be spent on non-energy items – by \$1.15 billion, amounting to a sizable "energy tax." Mapping the impact of potential energy price shocks to consumers, we find that at a sustained \$100 per barrel oil price, the projected tax benefits to consumers from last year's tax legislation (the "One Big Beautiful Bill Act") would still exceed the drag from the energy tax increase. However, at \$150 per barrel, the implied energy tax increase would more than wipe out the tax cuts to consumers from that legislation and present a more serious threat to the outlook for consumer spending.

The sum total of this is that supply is already struggling to keep up. Wait times to connect a data center to the grid are exceeding four years (based on comments from Google representatives and other industry reports). Efforts are underway to increase supply, including expanding grid capacity, as well as AI players preparing to build their own sources of power that operate "behind-the-meter" (developed on the customer's side and not delivered through the public grid). Capex into investor-owned electric utilities has been steadily increasing and is projected to climb to nearly \$250 billion in 2029 (according to the Edison Electric Institute). The White House's introduction in early March of a Ratepayer Protection Pledge urging AI companies to build or buy additive energy further highlights the race to bolster energy supply to meet AI needs. While the exposure across regions will vary, the conflict in Iran will add another layer of constraint to energy supply, as 20% of global oil and 20% of LNG pass through the Strait of Hormuz. Companies are looking to fill the gaps in energy supply over a spectrum of problem sets solving for immediate constraints, for scale, and for consistency.





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## Solving for Constraint: Readily Deployable Power

The most direct response to AI-driven power constraints is coming from companies that enable data center operators to bring electricity online without waiting for grid expansion. These companies are delivering dispatchable on-site or near-site power. While there are existing manufacturers providing these kinds of products, there are an increasing number of companies adapting existing technology to enter this AI-adjacent space. FTAI Aviation (FTAI), which traditionally operates in the airline industry, and BorgWarner (BWA), an auto industry supplier, are examples of this kind of shift.

FTAI Aviation launched FTAI Power in December 2025 with the aim of converting the company's CFM56 engines into gas turbines that can deliver energy to data centers. The engine is prevalent, with enough in operation globally that one takes off every two seconds somewhere in the world. FTAI Power will offer its aeroderivative turbine "FTAI Mod-1," a mobile, trailer-mounted conversion unit that can be delivered and installed in under two weeks. As a 25-megawatt unit, it is smaller, which allows for flexibility in stacking and growing data centers. Deliveries will likely begin around December 2026, with approximately 100 units in production per year starting in 2027.

BorgWarner is taking a similar approach. The company is an auto industry supplier that designs and manufactures powertrain systems but will partner with Turbocell (a subsidiary of the AI-optimized data center infrastructure company Endeavour) to co-develop an integrated, modular power generation system. This system would provide primary and backup power for AI data centers and would serve as a "bridge" solution, meaning data center operators can install it quickly for power now but can keep it on to provide additional power even as longer-term energy infrastructure is built up.

Modular power generation systems like this one are designed with the high and often fluctuating power demands of AI computing in mind. Modular products also comprise standardized, interchangeable units that are more flexible in applications than the custom builds commonly found in legacy systems. These modular units, as well as the integrated battery systems and electronic components, can be scaled up or down based on specific data center needs.

Beyond quick-install energy solutions, companies further up the supply chain are adapting to the energy demanded by growth in AI. Woodward, for example, is an aerospace-industrial hybrid company whose power generation products have historically been focused on the industrial gas turbine market, but it is now broadening into supplying components for reciprocating engines and marine engines being repurposed as behind-the-meter power solutions. |



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## Solving for Scale: Grid Expansion and Infrastructure Development

Meeting AI-driven electricity demand forecasts will require system-wide capacity and infrastructure expansion. Investments are happening simultaneously but across pathways that range from near-term grid expansion and enablement platforms, near- and longer-term renewables, and long-duration commercial nuclear supply. In terms of grid expansion, efforts span both generation capacity and transmission and distribution (T&D) infrastructure to accommodate higher loads and load density, more stringent reliability requirements, increasingly complex grid architectures, and longer-duration power needs associated with AI workloads. GE Vernova (GEV), for example, manufactures combined-cycle and peaker gas turbines (currently the most common source of baseload power in the US) and also provides high-voltage T&D equipment including transformers, high-voltage direct current (HVDC) solutions, circuit breakers, switchgear, power-sending devices, and grid automation systems. Some of its products also supply the renewable and commercial nuclear sources of power.

As existing grid infrastructure faces a critical deficit, it is impossible to ignore renewables as a scaling solution. Although US onshore wind investment faces challenges due to unfavorable policy headwinds, solar, alongside other renewables, is among the fastest-to-deploy sources of new generation. When paired with battery energy storage systems (BESS), these technologies can meet the 24/7 power needs of AI-focused data centers. There are indications that as energy demand driven by AI increases, so too do investments in renewable energy sources. A recent study from the University of Zurich, for example, found that solar and BESS have been the primary drivers of increased energy investment in regions with pre-existing data center capacity (based on ChatGPT infrastructure in the US). Increased investments in gas play a role too, but these have been offset by declines in grid-connected gas plants. These developments are currently based on investment decisions, but policy drivers could add further impetus should the environmental toll stemming from AI usage take an increasingly prominent place in political discourse.

Solar energy is well suited to the energy demand landscape due to the technology's fast scalability, low to zero marginal costs, and favorable levelized cost of energy (LCOE). As a result, the sector is seeing an increasing number of power purchase agreements (PPAs) with large technology players, including Amazon (AMZN) and Google (GOOGL). These arrangements are also conducive to a buildout of development projects that are supporting equipment suppliers.

Clearway Energy (CWEN), for example, operates energy generation assets in the US, with a specific focus on clean tech. It is positioning itself to serve the growing demand for power in the US driven by hyperscalers and data centers. The company has announced 2 gigawatts of PPAs for these projects, with possibilities for additional volumes. Additionally, it has outlined 8-13 gigawatts of capacity via co-located data center complexes, utilizing a mix of renewables and storage in addition to gas.

The scale-up of renewables also depends on component and equipment suppliers. First Solar (FSLR), for example, is a US-based solar technology



company that designs and manufactures advanced thin-film modules (a kind of solar photovoltaic panel) for utility-scale projects. Nextpower (NXT), meanwhile, is an example of a clean tech equipment and systems provider, selling trackers (the mechanical systems that hold and rotate panels to follow the sun), software, electrical systems, and the foundations upon which solar technology is mounted. Solar electrical equipment also includes inverters, which convert solar-generated direct current electricity into distributable alternating current electricity delivered to both residential and commercial customers.

Commercial nuclear is another important component of longer-term energy supply, with small modular reactors (SMRs) a key part of the development pipeline. Businesses are focused on extending the lifespan of current reactors while also preparing for the possible buildout of new ones. In the near term, the priority is the maintenance, repair, and life extension of existing nuclear infrastructure. There are 54 nuclear power plants in the United States (amounting to 94 nuclear reactors), according to 2024 data from the US Energy Information Administration. These produce a net generating capacity of nearly 97 gigawatts and make up about 20% of US energy usage.

Maintenance of these plants requires the replacement and upgrading of critical reactor components, modernization of control systems, and other projects that allow current reactors to operate safely and consistently for longer periods. Suppliers such as BWX Technologies (BWXT) and Curtiss-Wright (CW) provide operating components and systems for nuclear plants, including reactor hardware, fuel-related technologies, pumps, valves, and control systems.

At the same time, the industry is preparing for potential new build activity, particularly around advanced reactor designs and SMRs. Development work spans reactor components, fuel technologies, and digital and mechanical systems designed for newer reactor platforms. BWX Technologies' work on advanced fuel and reactor components and Curtiss-Wright's involvement in systems and controls applicable to newer reactor designs are examples of how suppliers are positioning for a more modular and scalable nuclear build-out over time.



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## Solving for Consistency

At the core of scaling AI infrastructure is the need for stable, reliable, and low-cost power. While wind and solar power generation will undoubtedly be part of the mix, natural gas can be expected to continue to play a sizable role as both a primary source of power and a backup source of power to intermittent power sources such as wind. Even with significant renewable penetration, the grid will still require backup generation capacity. Natural gas, in our view, will remain the likely fuel selected for this purpose, due to its global availability (via pipeline or liquefied natural gas imports), energy density, price, and ability to be transported and stored on site for behind-the-grid power. Natural gas also avoids many of the political hurdles that constrain other energy sources. It does not face the regulatory and political hurdles of traditional nuclear and SMRs, nor is it falling out of favor in Western markets to the same degree as coal. As such, it is likely that natural gas will continue to be in demand for both baseload and backup power generation.

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## The Bottom Line

Projected energy demand tied to the growing AI ecosystem and the broader digital economy is rising faster than grid capacity and interconnection timelines can adjust. Businesses are responding to address immediate constraint issues, questions of scale, and the need for consistent primary and backup generation. This will shape not only the trajectory of the AI buildout, but have reverberating impacts across the broader economy, its competitiveness, and critical infrastructure.



## Appendix 1

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# What Iran means for the dollar: a perfect storm for the petrodollar

March 24, 2026

**The long-term legacy of the Iran conflict for the dollar could be the way it tests the foundations of the petrodollar regime.** If fault lines are further exposed, there could be significant downstream effects to the dollar's use in global trade and savings, and the dollar's role as the world's reserve currency.

**The world saves in dollars in large part because it pays in dollars.** The dollar's dominance in cross-border trade is arguably built on the petrodollar: globally traded oil is priced and invoiced in USD. This arrangement can be traced to a deal struck in 1974 where Saudi Arabia agreed to price oil in USD and invest surpluses in USD assets, in exchange for US security guarantees. Because oil is a core input to global manufacturing and transport, there is a natural incentive for global value chains to dollarize, and global surpluses to accumulate in USD.

**The foundations of the petrodollar regime have been under pressure even before this conflict.** Most Middle East oil is now sold to Asia not the US; sanctioned oil from Russia and Iran has already been trading off dollar rails; Saudi Arabia has been localizing defence, and experimenting with forms of non-dollar payment infrastructure such as Project mBridge.

**The current conflict may expose further fault lines, by challenging the US security umbrella** for Gulf infrastructure and the maritime security for global trade in oil. Damage to Gulf economies could encourage an unwind in their foreign asset savings held largely in dollars. In this context, reports that the passage for ships through the Strait of Hormuz may be granted in exchange for oil payments in yuan should be closely followed. The conflict could be remembered as a key catalyst for erosion in petrodollar dominance, and the beginnings of the petroyuan.

**A bigger risk could come if the world begins to move away from globally traded oil and gas itself,** to more resilient sources of energy including domestically available fuels, renewable energy, and nuclear power. The energy choices of the Global South, Europe and North Asia will be key to track. A move away from oil could be as powerful as the pressure to price it in other currencies.

**A world that becomes more self-sufficient in defence and energy would also be a world that holds less USD reserves.** The huge strategic importance of the Middle East to the dollar's role as the world's reserve currency should not be underestimated. The current conflict may be the perfect storm for the petrodollar.

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## What Iran means for the dollar: a perfect storm for the petrodollar

**The dollar is the world's reserve currency for a very simple reason: the world pays for global goods and services in dollars and is willing to save resulting surpluses in dollar assets.** From 1945-1971, the dollar was "backed" by gold – namely, global central banks were able to exchange \$35 for 1 oz at the Fed. This was the foundation of the international monetary system known as Bretton Woods. In 1971, the US broke the dollar's link to gold. Since then, the dollar has been in a purely fiat regime – one that is backed by the sovereign credit worthiness of the US and willingness of the world to save in its debt.

**Enduring support for the fiat dollar arguably comes from the dominance of the dollar in the pricing of cross-border trade.** Globally traded goods and services are largely priced in USD, with payments exchanged over US controlled payment rails. Global surpluses are thus built in USD and mostly invested back into US assets. Corporates are incentivized to save and borrow in the currency of their payables and receivables, banking systems are dollarized, and central banks save in dollars to act as effective lenders of last resort. This drives demand for USD reserves. See more on this link [here](#).

**A crucial anchor to this system is the petrodollar:** the fact that most globally traded oil is priced and invoiced in dollars. Because oil is so central to global manufacturing processes – from petrochemicals, fertilizers and transport, to running factories and offices – companies are incentivized to price end products in dollars as a natural currency hedge to a key cost. This is a big reason global goods and services are priced and traded in US dollars.

**The reason oil is priced in dollars can be traced back to the 1974 petrodollar arrangement between the US and Saudi Arabia.** In simple terms, Saudi Arabia agreed to price its oil exports in USD and invest oil surpluses into US Treasuries. In exchange, the US provided security guarantees and military protection. The rest of the GCC followed. Structural foreign demand for US debt lowered the funding costs of the US government, acting as an indirect means of payment for the security umbrella extended to the region. To the extent that this arrangement supported broader global invoicing and saving in USD, it has been crucial to the USD's role as the world's reserve currency.

**The current conflict has arguably shaken some core foundations of the petrodollar regime: the security-for-oil-pricing arrangement.** US military assets and bases in the Gulf have come under attack in the war. Oil infrastructure in the Gulf has also been hit. And the US ability to provide the maritime security to ensure the global flow of oil has been challenged with the closure of Hormuz. The US security umbrella has been fundamentally tested.

**The legacy of this conflict for the dollar could be the ways in which it tests the foundations of the petrodollar regime.** In the long-run, if the world uses less oil, the Gulf draws more deeply on existing dollar savings, if the Gulf moves closer to Asia in its trade and investment relationships, and eventually prices less oil in dollars – there could be significant downstream effects to the dollar's usage in global trade and savings.

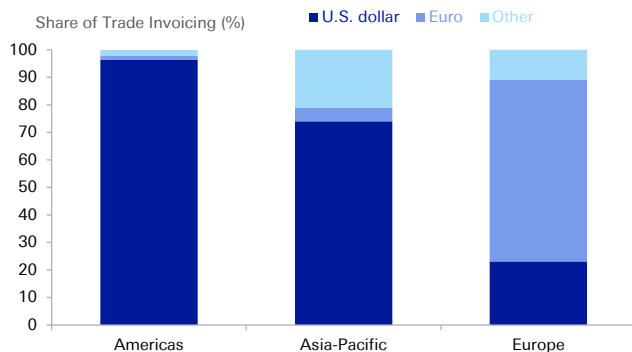


## The system was already under threat: many changes have been afoot

A lot had already begun to change before the US-Israel-Iran war broke out. We highlight a few key shifts that were underway:

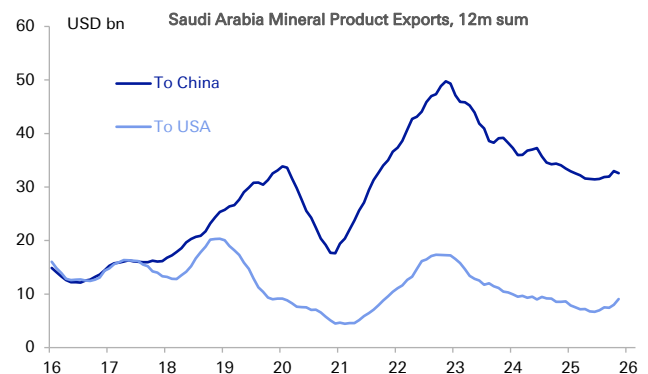
1. **The US was no longer the biggest buyer of Middle East oil.** With the shale revolution making the US energy independent, Saudi Arabia had been selling more than four times as much oil to China as to US. 85% of Middle East crude oil goes to Asia. This already introduced a fundamental instability with reports that China was keen for more oil to be invoiced in yuan.
2. **Saudi Arabia was already looking to localize more of its own defence.** Under Vision 2030, Saudi Arabia had been targeting an increase in the domestic content of military spending to 50%, looking to reduce dependence on foreign imported arms.
3. **Saudi Arabia had joined Project mBridge and signed FX swap lines with China.** Project mBridge is an initiative involving the PBOC, HKMA, Bank of Thailand, and the central banks of the UAE and Saudi Arabia. It uses blockchain technology to facilitate payments in the central bank digital currencies of each country. Crucially it does not depend on USD correspondent banking or SWIFT, and is at the minimum viable stage. The rails to transact outside of the dollar have already been built.
4. **Sanctions on Russia and Iran meant significant oil trade was already taking place outside of dollar rails.** Sales of Russian and Iranian oil have been priced and transacted in a range of local currencies from ruble, yuan, rupee with non-dollar payment infrastructure having been used.

Figure 1: The dominance of the dollar in cross-border trade has a lot to do with the petrodollar



Source : Deutsche Bank, Federal Reserve

Figure 2: There were already signs of instability to petrodollar foundations: Saudi Arabia sells four times as much oil to China now as to the US



Source : Deutsche Bank, CEIC



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## The conflict has introduced new sources of instability for the petrodollar

**First, the US security umbrella is being tested with Gulf economies facing attacks on US bases, oil fields and infrastructure in a war that began with US military action in the region.** Aside from the Gulf, pain has arguably been disproportionately felt by traditional US allies: Europe, Japan and Korea that are large energy importers exposed to the Straits.

**Second, direct diplomacy rather than US-led maritime security has been the conduit for oil to pass through the Strait of Hormuz.** Select tankers heading for China, India and Japan have been allowed passage. Bilateral relations have served countries.

**Third, and crucially there have been reports that Iran is negotiating with eight countries to allow passage of ships through the Strait of Hormuz if oil payments are paid in yuan** ([The Telegraph](#), [CNN](#), [Asia Times](#), [Chosun](#)). This adds further credence to the enormous power of oil pricing. The conflict could be the catalyst for erosion in petrodollar dominance and the beginnings of the petroyuan.

**To be sure, there are numerous mitigants to consider.** While the US is no longer the biggest global buyer of oil, it may well be on track to be the world's biggest supplier. As we argued in our [earlier piece](#), if the US controlled the entire Western Hemisphere's oil – either directly or by proxy – it would have more reserves than OPEC, putting it in a better position to set the terms of global oil trade.

**In a best-case scenario, the dollar could remain the dominant currency for oil trade if the US controls a majority of the world's supply,** although further development of reserves would be required to produce surpluses sufficient to be the world's biggest *tradeable* supplier.

**In a worst case scenario, one may see a fracturing of global oil pricing along trade routes and corridors.** One might imagine Middle East oil passing through the Straits of Hormuz to Asia being priced in yuan, while oil from the Western Hemisphere sold across the Atlantic and Pacific to historical US allies may be priced in USD.

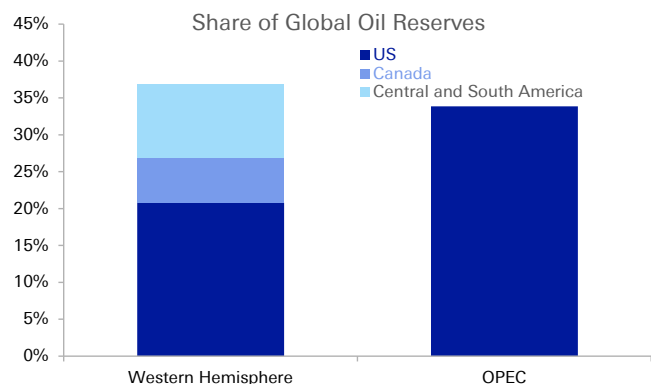
**Another important mitigant is that Gulf economies are very invested in the USD.** The GCC runs USD pegs that are backed by enormous USD savings. To maintain pegs, it makes sense for them to add weight with USD receivables. Any sign of moving away from USD revenues could invite self-fulfilling attacks on pegs.

**However, the Gulf may need to unwind dollar savings due to this crisis itself.** The war has already led to damage of oil and civil infrastructure in Gulf economies. Escalation or a prolonged conflict could worsen this (see [here](#) for what's at stake for Gulf economies and [here](#) for our take on escalation risks). Hydrocarbon economies could suffer both from near-term hits to physical production and reduced long-term demand for fossil fuels (more below). Moreover, services sectors such as finance, tourism, aviation may face scarring if there remains an overhang of security risk. All of this could mean Gulf economies need to redeploy more of their externally held dollar savings domestically to support local economies and potentially reorient diversification programs to a different regional climate. To be sure, reserves are plentiful: the MENA region has roughly USD2tn in central bank managed reserves, and around USD6tn in sovereign wealth funds, according to Global SWF. But if USD



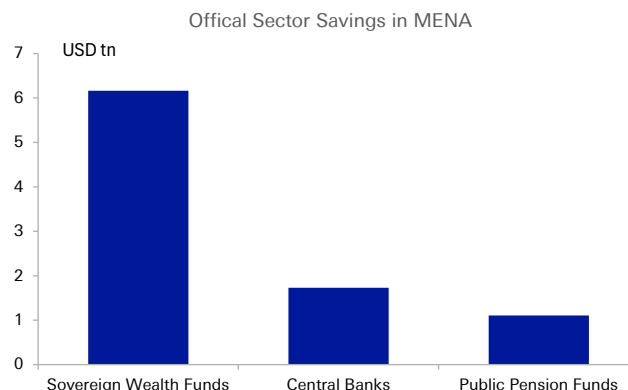
savings are being drawn down for domestic investments, this could reduce the hurdle to other currency changes.

Figure 3: If the US controlled Western Hemispheric oil, it would have more reserves than all of OPEC



Source : Deutsche Bank, Energy Institute

Figure 4: A majority of the Gulf region's savings are held in sovereign wealth funds



Source : Deutsche Bank, Global SWF

Perhaps the biggest long-term risk comes not just from the Gulf moving away from dollar pricing, but if the world begins to move away from oil itself to which we turn next.

## Bigger energy shifts could be in the offing: the world may diversify away from traded fossil fuels to domestic energy

**There are many parallels between now and the 1970s.** Most obviously: we have now seen a second major oil and gas shock this decade. In the 1970s, the 1973 oil embargo was followed by the 1979 Iranian Revolution, while this time, Russia's invasion of Ukraine (2022) has been followed by the ongoing US-Israel-Iran war. Global prices may remain structurally higher beyond the end of the current conflict if production facilities in the Gulf are materially damaged. And the historic weaponization of the Straits could introduce more risk premium into energy that moves on ships. Increasing self-sufficiency and domestic resilience would make sense even if oil prices came down.

**The 1973 oil embargo by Arab nations on Western economies motivated big energy shifts in efficiency, diversification, and reserve building.** It accelerated the development of alternative oil sources in Canada, Gulf of Mexico, Alaska, and the North Sea that helped the OECD diversify away from Middle East oil. It also led to the creation of Strategic Petroleum Reserves, and created the political will for initial investments in renewable and nuclear power.

**The US may see the least change.** Unlike the 1970s, the US is now energy independent in oil & gas. US industry will not be hurt immediately in the way that economies not receiving physical oil will be, and US oil producers will benefit. While there could be a consumption shock from higher pump prices, the US arguably retains the option of an export ban to manage domestic prices. WTI-Brent-Oman spreads already reflect the very different prices being paid around the world. The US growth impact is thus the most ambiguous and likely the least negative (see [here](#) and [here](#)). This should mean the US remains relatively more invested in fossil fuels.



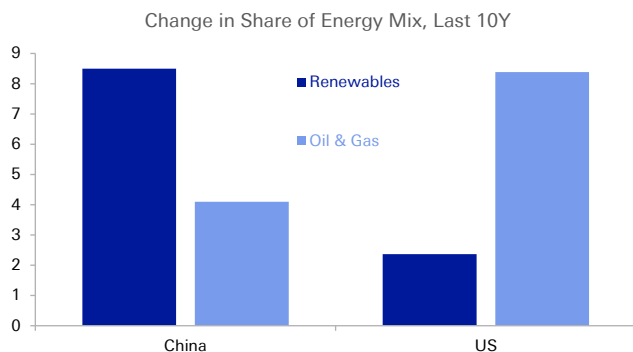
**But energy-dependent regions like Europe, Asia and many parts of the Global South – that are facing multiple threats to global trade in fossil fuels – arguably have three key options:**

- **The first option is to source more fossil fuels domestically.** Regions that have the ability to explore and develop more of their own oil and gas may be more motivated to do so, from the UK to Brazil, while Europe and parts of Asia may also turn back to domestic sources of coal. Even if the world does not move away from fossil fuels, global trade in it could well decline, and this is what matters for FX.
- **The second option is to double down on renewable energy.** Renewables are far more appealing on relative cost today than they were in the 1970s, due to the enormous industrial build out in China. China produces 80% of world's solar panels, 70% of the wind turbines, and 70% of lithium batteries ([Science](#)). China's overcapacity may well have been strategic, with China now extremely well placed to supply into new demand. Many Global South economies could accelerate this shift. But ultimately, this second option also comes with new dependencies on a mostly single industrial actor.
- **Genuine energy independence may only be created by a third option: nuclear power.** One could perhaps expect a significant focus on this in Europe, Japan (see [here](#)), and South Korea – traditional US allies. Just like their defence buildouts, nuclear power will have very long multi-year lead times but could have very significant FX implications.

**To be sure, there will be limits to how much the world can fully move away from oil.** Many industries rely on the fractionalization of oil for manufacturing. And as we have argued in our [earlier work](#), oil is crucial to militaries: while passenger cars can run on electricity - military jets, tanks and navy ships are likely to be powered by oil for a long time. Nevertheless, there is scope for greater parts of the energy mix to transition to more reliable, secure, and potentially cheaper sources of energy. If the world shifts away from globally traded oil and gas, to more domestic sources of fuel, renewable or nuclear power, the most obvious long-term impact would be reduced oil and gas deficits in Europe and North Asia, and reduced energy surpluses in the Middle East. Reduced global oil trade would also create more room for pricing of goods and services to shift away from the USD.

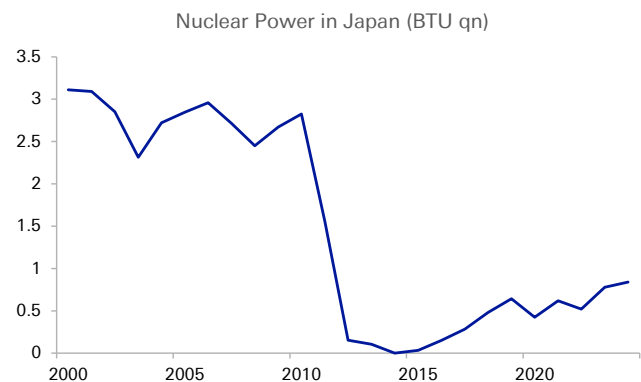
**The move away from oil could be as powerful as the pressure to price it in other currencies. The petrodollar system could be at risk from both the 'petro' and the 'dollar' legs.**

Figure 5: China has been leaning more towards renewable energy, while the US has been doubling down on fossil fuels



Source : Deutsche Bank, CEIC, IEA

Figure 6: Japan and Europe could ramp back up on nuclear power to build more energy independence



Source : Deutsche Bank, CEIC



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## Conclusion: What does this all mean for the dollar?

In the near term, we have [argued](#) that US energy independence might be worth some safe-haven premium on the dollar. The US is the only major economic region in the world that is both energy independent and is itself far from a battleground. However, there have been important offsetting factors from growing US fiscal risk around greater military spending, to the unwind of UST holdings in Asia and the Middle East to defend currencies (see pieces [here](#) and [here](#)) which mean the dollar has not strengthened much in this crisis even in the short term.

**The more interesting and lasting impact of this crisis on the dollar could be on the long-term foundations, namely the willingness of the world to price traded goods and services in dollars and to save resulting surpluses in dollar assets.**

**The foundations of the petrodollar regime were under pressure even before this conflict.** There were already signs of instability to the long-standing arrangement to price GCC oil in dollars in exchange for security: most Middle East oil is sold to Asia now not the US; sanctioned oil has already been trading off dollar rails; Saudi Arabia has been localizing defence and experimenting with non-dollar payment rails alongside other Global South central banks.

**The current conflict may have exposed further fault lines**, by challenging the US security umbrella for Gulf infrastructure, the maritime security for global trade in oil, and encouraging a potential unwind in Gulf dollar savings. In this context, reports that the passage of ships through the Strait of Hormuz may be granted in exchange for oil payments in yuan should be closely followed. The conflict could be the catalyst for erosion in petrodollar dominance and the beginnings of the petroyuan.

**A bigger risk perhaps comes if the world begins to move away from globally traded oil and gas itself**, to more resilient sources of energy from domestically available fuels, renewable energy, and nuclear power. The energy choices of the Global South, Europe and North Asia will be particularly crucial to track. The move away from oil could be as powerful as the pressure to price it in other currencies.

**A world that is looking to become more self-sufficient in defence and energy would also likely be a world that will hold less USD reserves.** If economic security is now a key part of national security, building greater energy resilience should be seen as complementary to the increased focus on defence spending. Reserves hitherto held in foreign assets will increasingly be needed to build domestic capacity that supports strategic autonomy (see [here](#) and [here](#)). Future surpluses could themselves reduce and be accumulated less in dollars. These are long-term dollar negatives.

**The huge strategic importance of the Middle East to the dollar's role as the world's reserve currency should not be underestimated. The current conflict may be the perfect storm for the petrodollar.**





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## Thematic FX Framework Pieces

This note is part of a broader series of thematic FX notes that explores the evolving nexus between payments, energy and defence as key pillars to the dollar's long-term status. Other FX pieces that can be read alongside this include:

[FX Special Report: What Venezuela has to do with the dollar](#)

[FX Special Report: What do stablecoins mean for dollar dominance?](#)

[FX Special Report: Who can come home?](#)

[FX Special Report: The geopolitics of reserves: shifting security alliances will matter for the USD](#)

[EM Special Publication: Geoeconomics 101: implications for the dollar, emerging markets and beyond](#)

[FX Blog: On defense spending, bond markets, home bias & the yen](#)

[EM Blog: Iran - a further impetus away from recycling savings to the US?](#)



## Appendix 1

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# The House View: Snapshot The Impact of the Middle East

Deutsche Bank

## Macro views

### World

- Global growth at 3.4% in 2026, 3.3% in 2027.
- Rationale:** Trade uncertainty is easing, a global structural shift towards fiscal impulse is boosting growth, and accelerating impact of AI continues to drive the broader macro outlook.
- Key risks:** Escalating tensions in the Gulf and oil market impact; AI sell-off sparks major market rotation; and tariff uncertainty is back after IEEPA tariffs are struck down.

### Europe

- Growth resilient pre Iran conflict. GDP to pick up from 1% in 2025 (Q4/Q4) to 1.5% in 2026. The German fiscal easing is becoming visible, with orders rising strongly.
- Rationale:** Now, higher energy costs and uncertainty could weigh. Much depends on breadth and duration of conflict.
- Key risks:** Conflict adds upside inflation risk, downside growth risk. The current level of energy costs could crowd out 0.25pp of GDP in 2026. Geopolitical uncertainty so far could cost another 0.25pp of GDP.

### China

- Government lowered 2026 growth target to 4.5 - 5% range for "risk prevention and structural reforms." Still, China's economy is setting course to reflation.
- Rationale:** Resilience to oil shock owing to lower reliance on oil for energy and increased stockpiles. Higher oil prices might also help end China's deflation. Renminbi's internationalization will accelerate, fostering the CNY's appreciation during 2026-27.
- Key risks:** Escalated geopolitical tension disrupting global supply chains. Deepening property sector downturn.

### United States

- 2026 growth is projected at 2.6%, supported by easy financial conditions, fiscal support and fading trade uncertainty.
- Rationale:** While labor market data shows volatility, we anticipate stabilization with unemployment remaining near recent levels (~4.4%) through year end. PCE inflation forecast has risen given recent outperformance and a lift from energy prices.
- Key risks:** A more substantial oil price shock could weaken real disposable income; financial conditions might tighten, perhaps driven by concerns around AI disruption or (private) credit risks.

### Germany

- 1.5% GDP in 2026 and 2027, driven by government and private consumption, as well as an increase in investment spending.
- Rationale:** Fiscal impulse key to German growth. Expansionary pick-up in defence and infrastructure public spending.
- Key risks:** Germany remains highly exposed to the energy cost shock and consequent deterioration of its current account. However, maintains solid NIIP so is better able to buffer against external shocks.

### India

- 7.5% yoy GDP for FY27 (FY26 at 7.6%).
- Rationale:** Less growth risks following finalized deals with the US and India-EU free trade agreement. Unless Brent breaches sustainably above \$90/bbl, FY27 fiscal deficit target of 4.3% of GDP looks likely to be achieved.
- Key risks:** Current account deficit could rise to \$65bn in FY27 (~1.5% of GDP) vs. our \$45.6 bn (1.0% of GDP) baseline, under Middle East conflict. Rupee tests its all-time lows; keep year-end target of USD/INR unchanged at 90.

## Key downside risks

- H Geopolitics.** Less than three months into 2026, geopolitics continues to drive volatility. The US policies toward Venezuela and Greenland, and the US/Israeli military strikes on Iran suggest markets need to be prepared for what comes next.
- M AI.** Questions over AI's impact on economic disruption, especially the labour market, continues to mount, although this discussion comes as much of the global economy sees a period of solid economic momentum.
- M Trade wars.** US 10%-15% retaliation against SCOTUS decision throws tariffs back into the spotlight, while US-China rivalry will continue to bifurcate supply chains, with critical minerals and semiconductors as ongoing flashpoints for escalation.



## Market views

|                                                                                    |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>Market Sentiment</b> | <ul style="list-style-type: none"> <li>Despite the rapidly developing Middle East conflict, its market impact remains contained so far.</li> <li>Global stocks excluding the US continue a seemingly unstoppable rally, even with choppy earnings, high valuations, and a difficult macro backdrop. Rising bond yields are due to the unwinding of expected rate cuts and an 'AI scare,' not higher oil prices.</li> <li>AI-driven boom-and-bust narratives will keep dominating market sentiment and shaping the broader macroeconomic outlook.</li> </ul> |
|    | <b>Equities</b>         | <ul style="list-style-type: none"> <li>S&amp;P 500 year-end 2026 target: 8000; STOXX 600 year-end 2026 target: 650.</li> <li>Recent declines suggest unwind of previous positioning build-up rather than changes with fundamental drivers.</li> <li>Overweight US and Europe; underweight Japan; neutral emerging markets.</li> </ul>                                                                                                                                                                                                                       |
| <b>%</b>                                                                           | <b>Rates</b>            | <ul style="list-style-type: none"> <li>10yr UST at 4.45% year-end 2026; long Dec-26 €STR as a hedge; long UST5Y spreads (target -21bp); EUR 10s30s TP (target 120bp, stop 85).</li> <li>Maintain higher equilibrium view in the US due to strong growth, stabilizing labour market, increased government spending and easy financial conditions.</li> </ul>                                                                                                                                                                                                 |
|    | <b>Credit</b>           | <ul style="list-style-type: none"> <li>Year-end 2026 targets: \$IG 105bp; \$HY 380bp. €IG 103bps; €HY 355bp.</li> <li>Moderate spread widening and volatility into 2026. Monitoring HY under geopolitical duress.</li> <li>Preference for €-credit over \$-credit due to healthier EU banks, lagged ECB easing &amp; FX-hedged yield advantage.</li> </ul>                                                                                                                                                                                                  |
|    | <b>FX</b>               | <ul style="list-style-type: none"> <li>Year-end 2026 targets: EUR/USD to 1.25; USD/JPY to 145.</li> <li>Dollar currently benefitting from global supply shock and higher energy prices. Longer-term forecasts point to weakness though contingent on geopolitical stabilisation.</li> </ul>                                                                                                                                                                                                                                                                 |
|  | <b>Oil &amp; Gold</b>   | <ul style="list-style-type: none"> <li>Scenario 1- Reopening Strait: Brent ~\$80/bbl, falls to \$70/bbl.</li> <li>Scenario 2- Ambiguous closure: Brent \$80-100/bbl despite OPEC adding supply in April.</li> <li>Scenario 3 - Full scale enforced closure: Brent \$130-150/bbl.</li> <li>Gold: Year-end 2026 target: \$6,000/oz; average \$5,500/oz through 2026 and 6,200 through 2027.</li> </ul>                                                                                                                                                        |
|  | <b>Monetary Policy</b>  | <ul style="list-style-type: none"> <li>Fed: 25bps cut in Sep, bringing Fed Funds rate into 3.5-3.75% range.</li> <li>ECB: 2.0% terminal rate reached; 25bps hike by mid-2027.</li> <li>BoJ: 25bps hike in Apr &amp; Oct, taking policy rate to 1.25% by YE.</li> <li>BoE: 25bps cuts in Jun &amp; Nov, taking Bank Rate to 3.25%.</li> <li>PBoC: Unchanged at 1.4% through 2026.</li> </ul>                                                                                                                                                                 |

### Key macro & markets forecasts

| GDP growth (%) |       |       | Central bank policy rate (%)    |         |         |         | Key market forecasts |         |         |         |
|----------------|-------|-------|---------------------------------|---------|---------|---------|----------------------|---------|---------|---------|
|                | 2026F | 2027F |                                 | Current | Mar 26F | Jun 26F |                      | Current | Jun-26F | Dec-26F |
| Global         | 3.4   | 3.3   | US: Federal Funds Rate          | 3.625   | 3.625   | 3.625   | US 10Y yield (%)     | 4.15    | 4.30    | 4.45    |
| US             | 2.6   | 2.1   | Eurozone: Deposit Facility Rate | 2.00    | 2.00    | 2.00    | EUR 10Y yield (%)    | 2.87    | 2.90    | 3.10    |
| Eurozone       | 1.1   | 1.3   | Japan: Policy Balance Rate      | 0.75    | 0.75    | 1.00    | S&P 500              | 6781    | 7450    | 8000    |
| Germany        | 1.5   | 1.5   | UK: Bank Rate                   | 3.75    | 3.75    | 3.50    | Gold                 | 5196    | 5300    | 6000    |
| Japan          | 0.8   | 1.0   | China: 7d OMO Rate              | 1.40    | 1.40    | 1.40    | EUR/USD              | 1.16    | 1.20    | 1.25    |
| UK             | 1.1   | 1.4   |                                 |         |         |         | USD/JPY              | 158     | -       | 145     |
| China          | 4.5   | 4.3   |                                 |         |         |         |                      |         |         |         |

### 2026 Macro events calendar

| March 2026 |    |                          | April 2026 |    |                          | May 2026 |    |                       |
|------------|----|--------------------------|------------|----|--------------------------|----------|----|-----------------------|
| 18         | CA | Bank of Canada Decision  | 2          | US | ECB Economic Bulletin    | 4        | EZ | ECB Decision          |
| 18         | US | Fed Decision             | 8          | US | FOMC Meeting Minutes     | 14       | EZ | ECB Economic Bulletin |
| 18-19      | JN | Bank of Japan Decision   | 27-28      | JN | Bank of Japan Decision   | 20       | US | FOMC Meeting Minutes  |
| 19         | UK | Bank of England Decision | 29         | US | Federal Reserve Decision |          |    |                       |
| 19         | EZ | ECB Decision             | 29         | CA | Bank of Canada Decision  |          |    |                       |
|            |    |                          | 30         | UK | Bank of England Decision |          |    |                       |
|            |    |                          | 30         | EZ | ECB Decision             |          |    |                       |



### Recent editions

- [The Impact of the Middle East](#), 10 March 2026
- [Anything but dull](#), 1 December 2025
- [A rocky equilibrium](#), 24 September 2025

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Deutsche Bank  
Research

# Tracking the effects of AI on the US labor market

March 10, 2026

Matthew Luzzetti | Brett Ryan | Justin Weidner | Amy Yang |  
Sourav Dasgupta | Suvir Ranjan

**Deutsche Bank Research Institute**

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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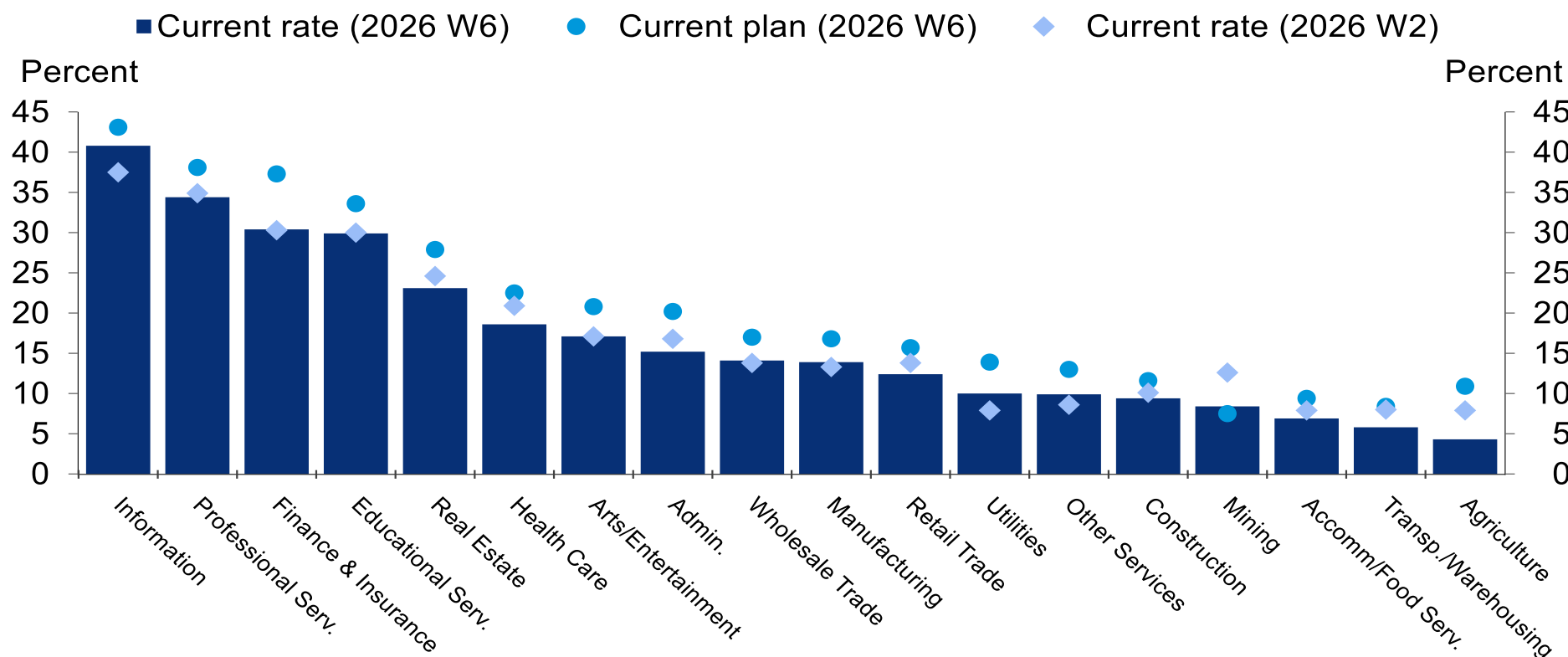


# 1. Firm AI adoption

# Most sectors have well below 50% AI adoption with little change over the past month, but adoption is expected to rise



## AI adoption rate

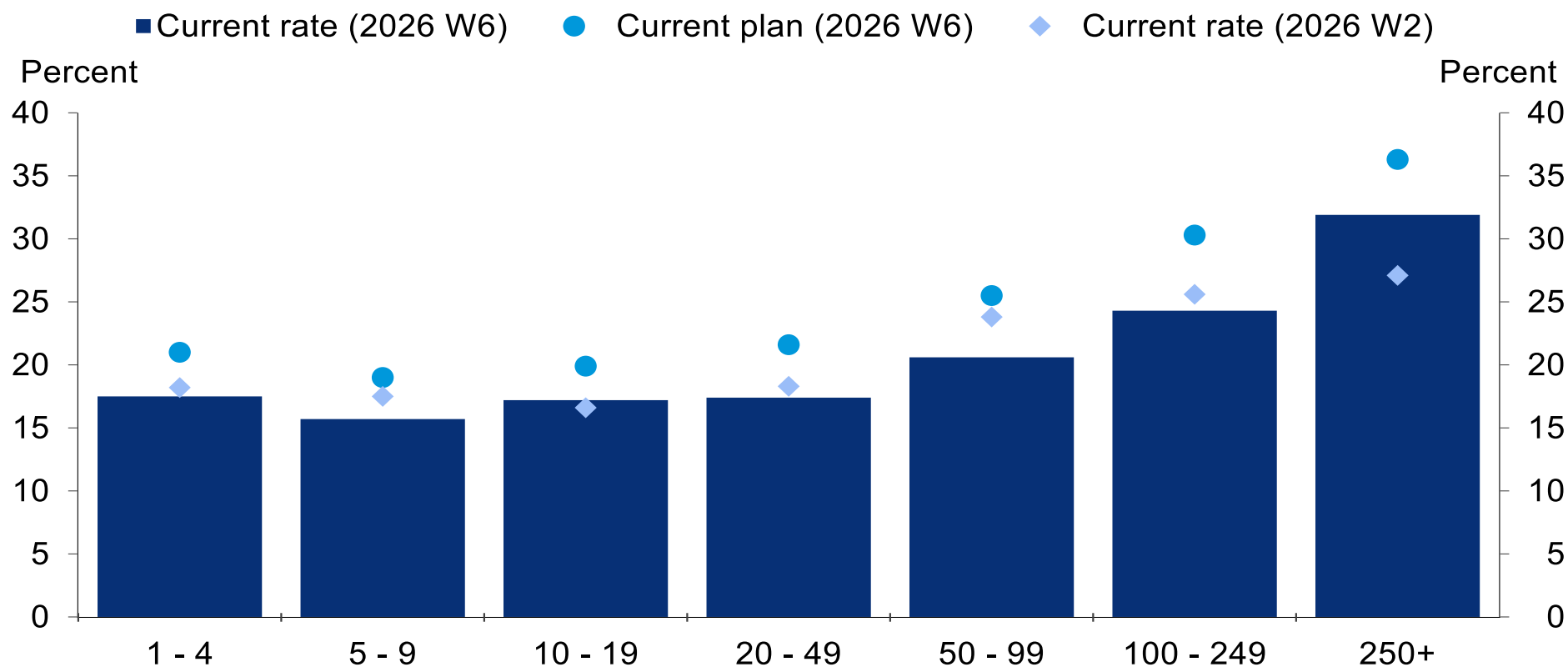


Source: USCB, Macrobond, Deutsche Bank

# Larger firms have higher rates of AI adoption; generally true across sectors as well



## AI adoption by firm size

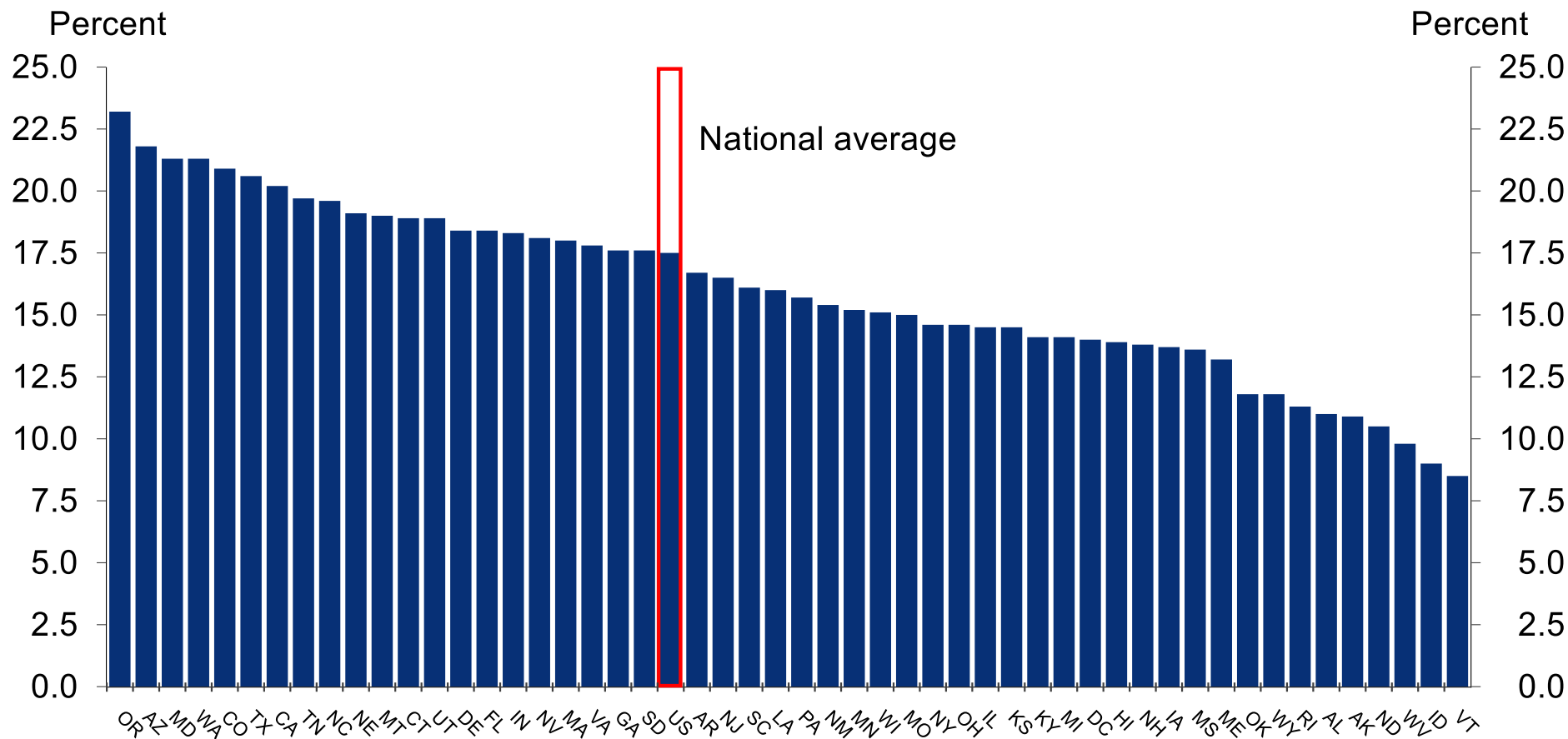


Source: USCB, Macrobond, Deutsche Bank

# Significant variation in cross-state AI adoption, with high adoption states nearly 3x low adoption; national avg. near 17%



## AI adoption by state (data as of 2026 W6)



Source: USCB, Macrobond, Deutsche Bank



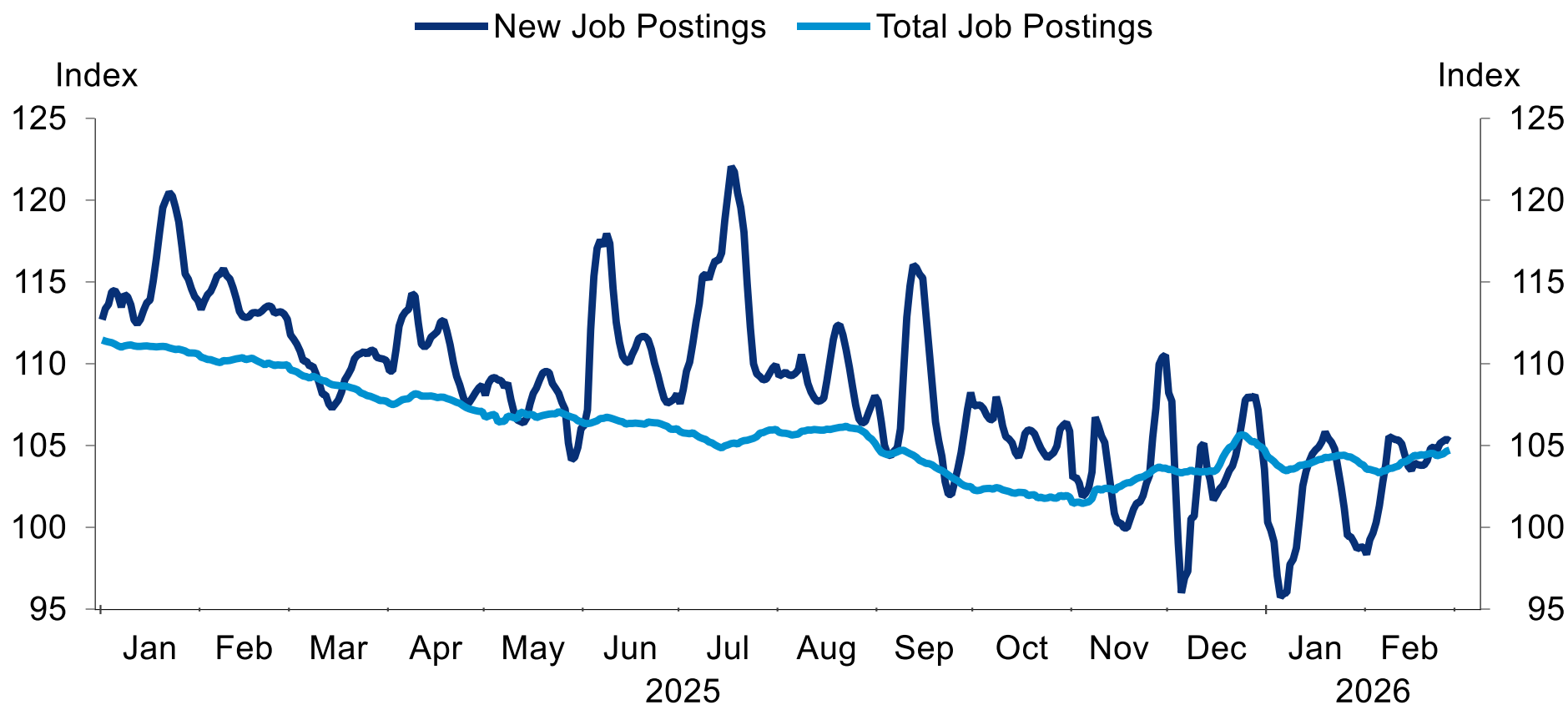
## **2. Developments in job openings**



Total job postings have picked up somewhat since late 2025



## Indeed Job Postings

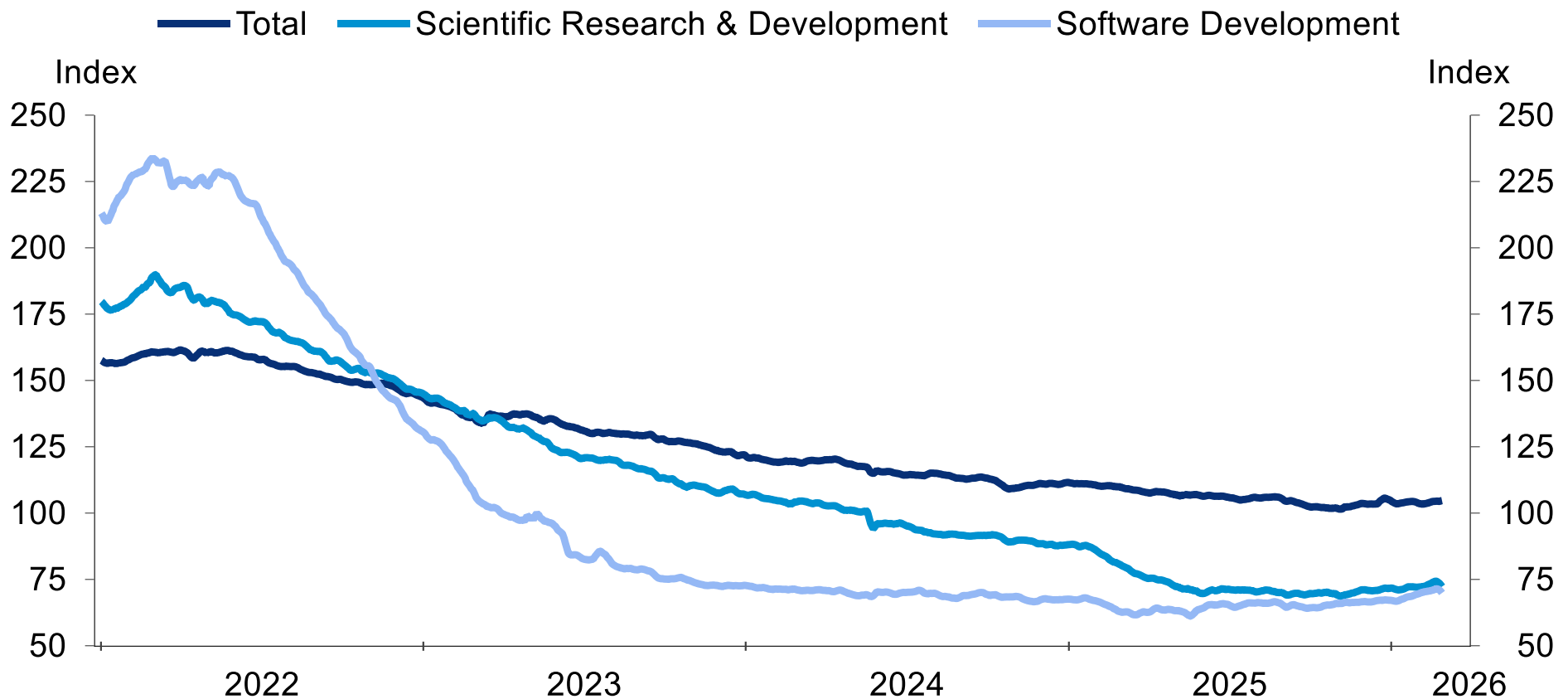


Source: Indeed Hiring Lab, Macrobond, Deutsche Bank



# Job openings in some AI-sensitive occupations have fallen faster than overall in past few years but have stabilized

## Indeed Job Postings

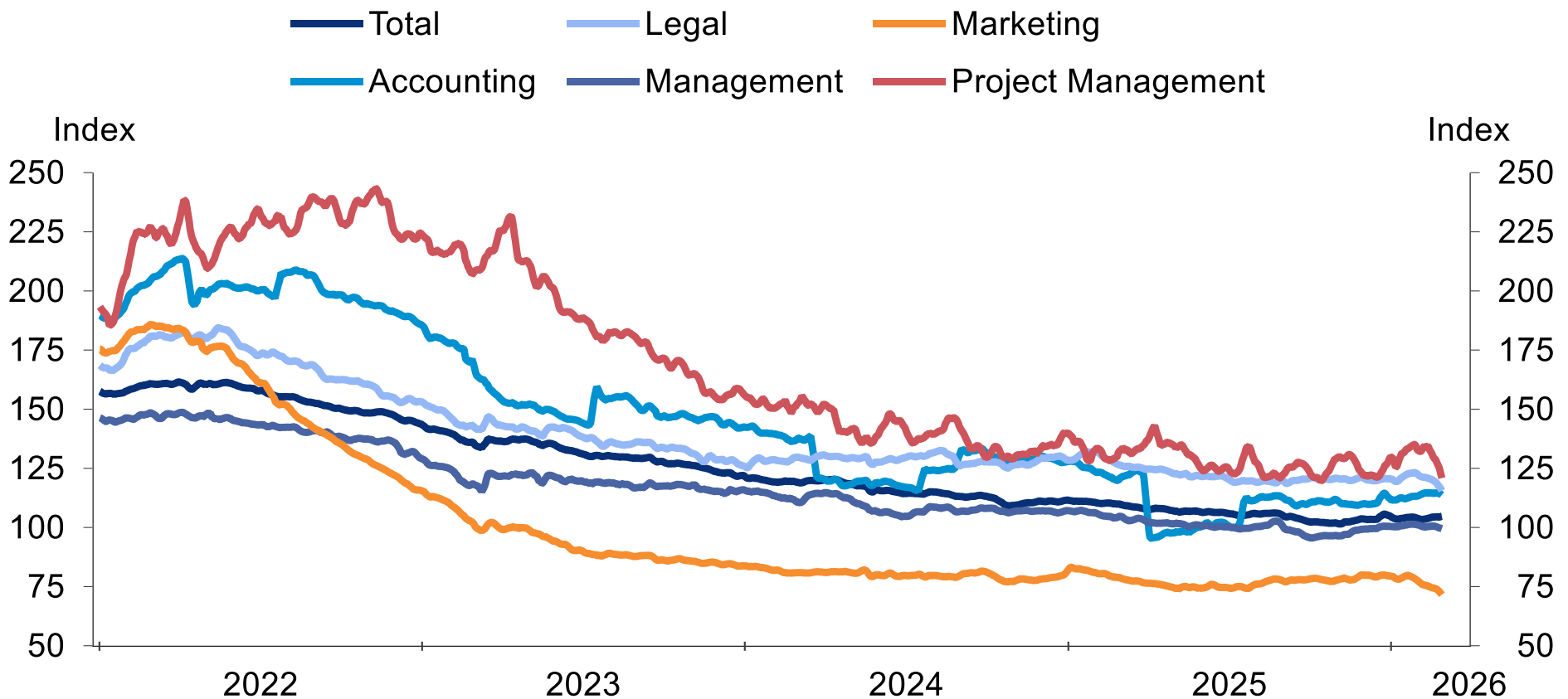


Source: Indeed Hiring Lab, Macrobond, Deutsche Bank



# No noticeable deterioration in job openings for occupations that have been a market worry recently

## Indeed Job Postings

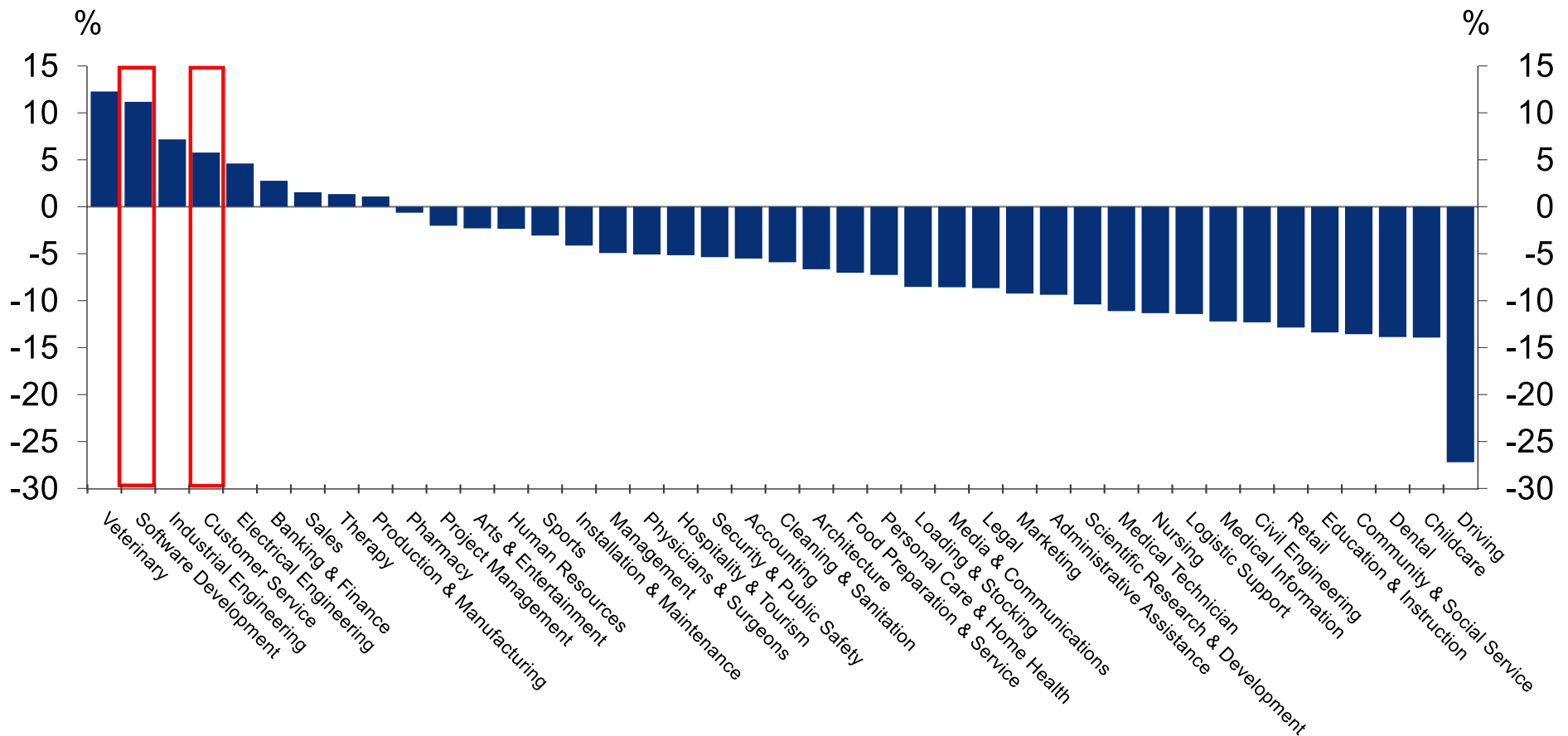


Source: Indeed Hiring Lab, Macrobond, Deutsche Bank



# No clear relationship between change in job postings over the next year and substitutability with AI

## Year-over-year change in job postings (data as of 2/27/2026)

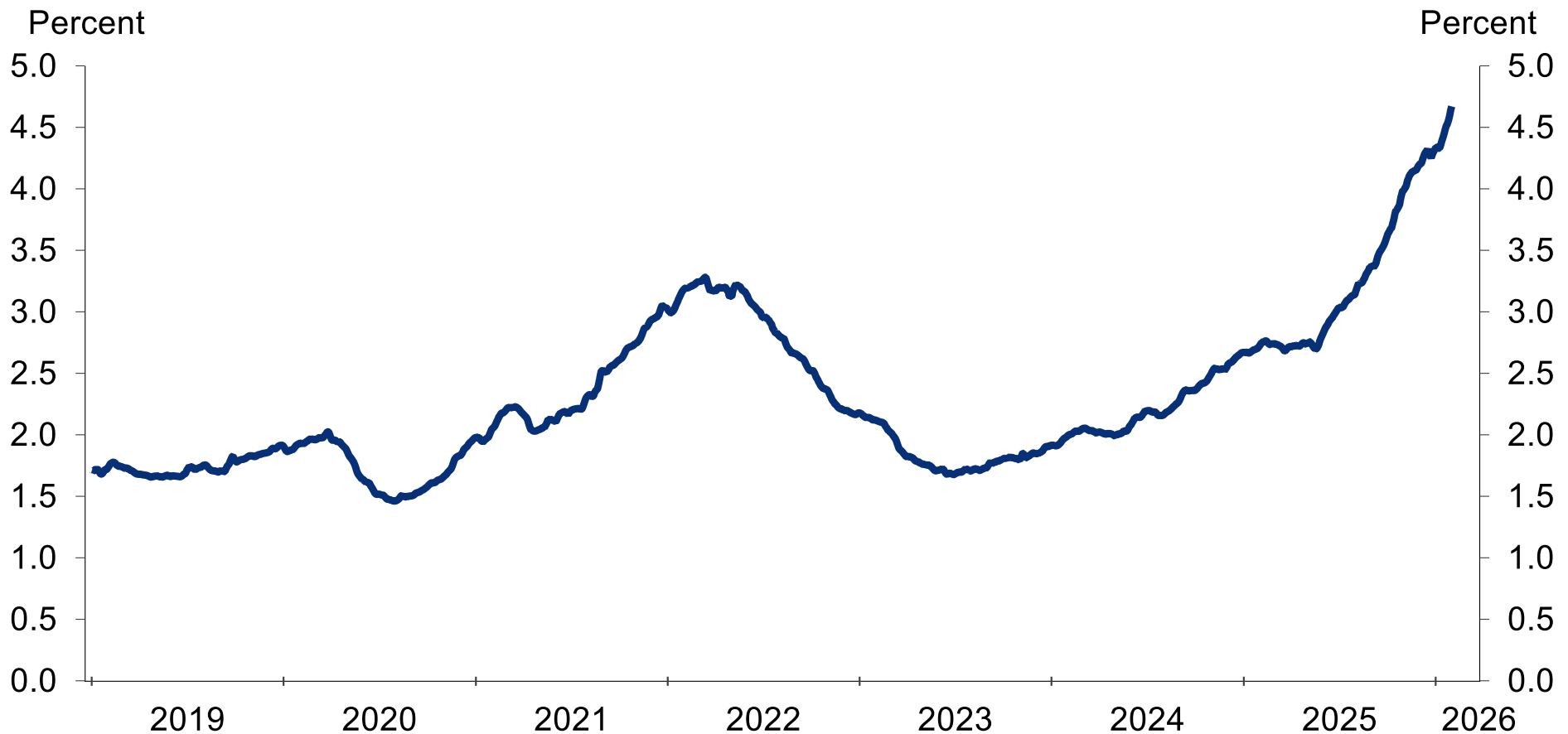


Source: Indeed Hiring Lab, Macrobond, Deutsche Bank



# Share of job postings containing AI terms is still low but rising sharply in recent months

## Indeed AI Tracker, US Share of Job Postings Mentioning AI Terms



Source: Indeed Hiring Lab, Macrobond, Deutsche Bank



# Over 2025, AI exposed jobs saw the largest share and change in job postings mentioning AI

## Nearly half of data and analytics roles (45%) include at least one AI mention in their postings

Top 15 occupational groups ranked by share of postings containing AI terms in Dec 2025

| Occupational group                      | Jan 2025 | Dec 2025 | Change (pp) |
|-----------------------------------------|----------|----------|-------------|
| Data & Analytics                        | 38.8%    | 44.9%    | +6.1        |
| Software Development                    | 28.5%    | 39.7%    | +11.2       |
| IT Systems & Solutions                  | 20.9%    | 24.7%    | +3.8        |
| Scientific Research & Development       | 16.0%    | 20.7%    | +4.7        |
| Industrial Engineering                  | 12.0%    | 18.7%    | +6.7        |
| Marketing                               | 8.4%     | 14.9%    | +6.5        |
| IT Infrastructure, Operations & Support | 8.7%     | 13.4%    | +4.6        |
| Electrical Engineering                  | 7.1%     | 12.1%    | +5          |
| Banking & Finance                       | 7.3%     | 10.1%    | +2.8        |
| Media & Communications                  | 7.0%     | 9.4%     | +2.4        |
| Human Resources                         | 4.4%     | 8.8%     | +4.4        |
| Legal                                   | 4.3%     | 8.3%     | +3.9        |
| Project Management                      | 4.3%     | 7.4%     | +3.1        |
| Accounting                              | 4.2%     | 6.0%     | +1.7        |
| Sales                                   | 2.9%     | 5.9%     | +3          |

Source: [Indeed](#), Deutsche Bank





# Jobs with AI mentions in knowledge work sectors

## Jobs with AI mentions are defying recent trends in many knowledge work sectors

Job postings with AI mentions compared to all postings in selected occupational group, Feb 2020 to Dec 2025



Source: Indeed. 3-month moving average indexed to Feb 1, 2020

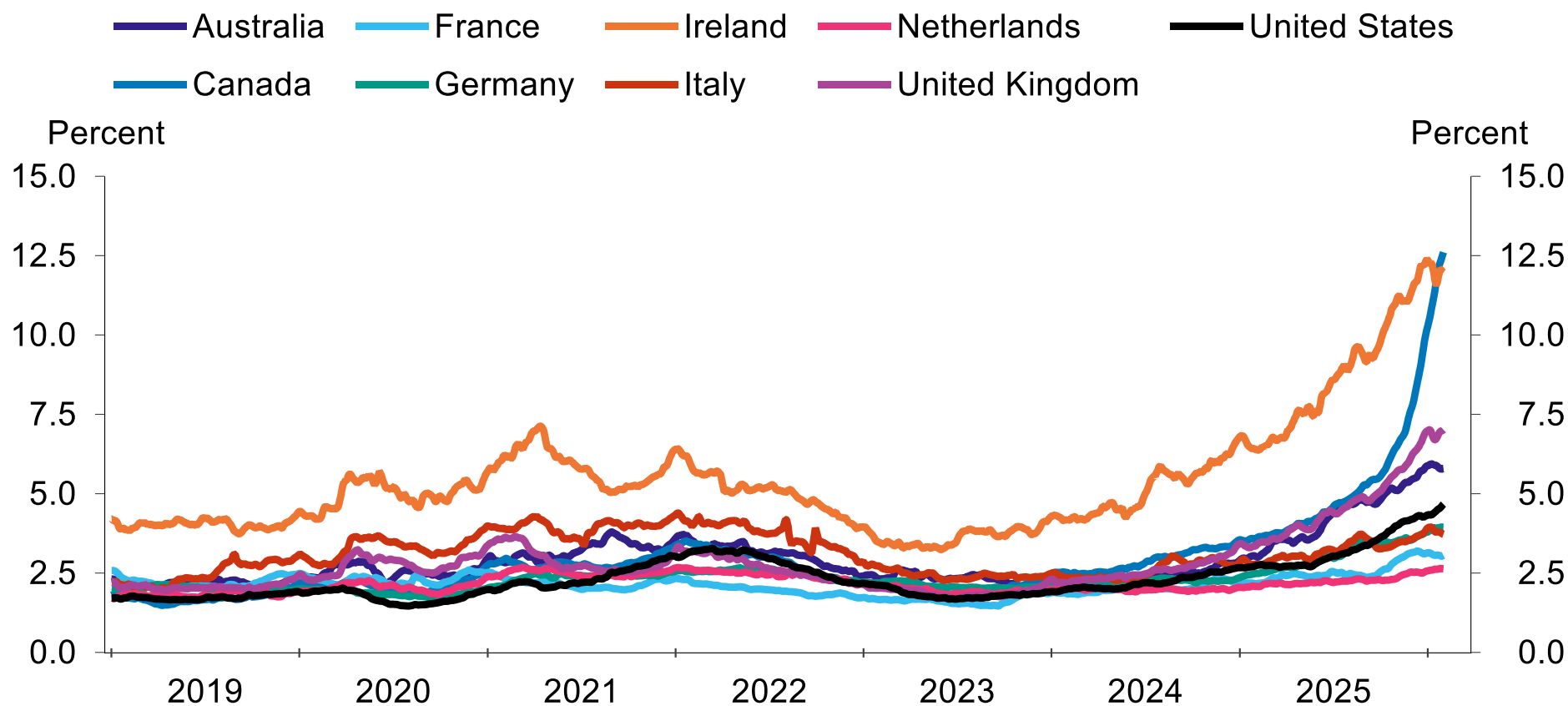


Source: [Indeed](#), Deutsche Bank

# Share of job postings referencing AI is rising across countries



## Indeed AI Tracker, Share of Job Postings Mentioning AI Terms



Source: Indeed Hiring Lab, Macrobond, Deutsche Bank

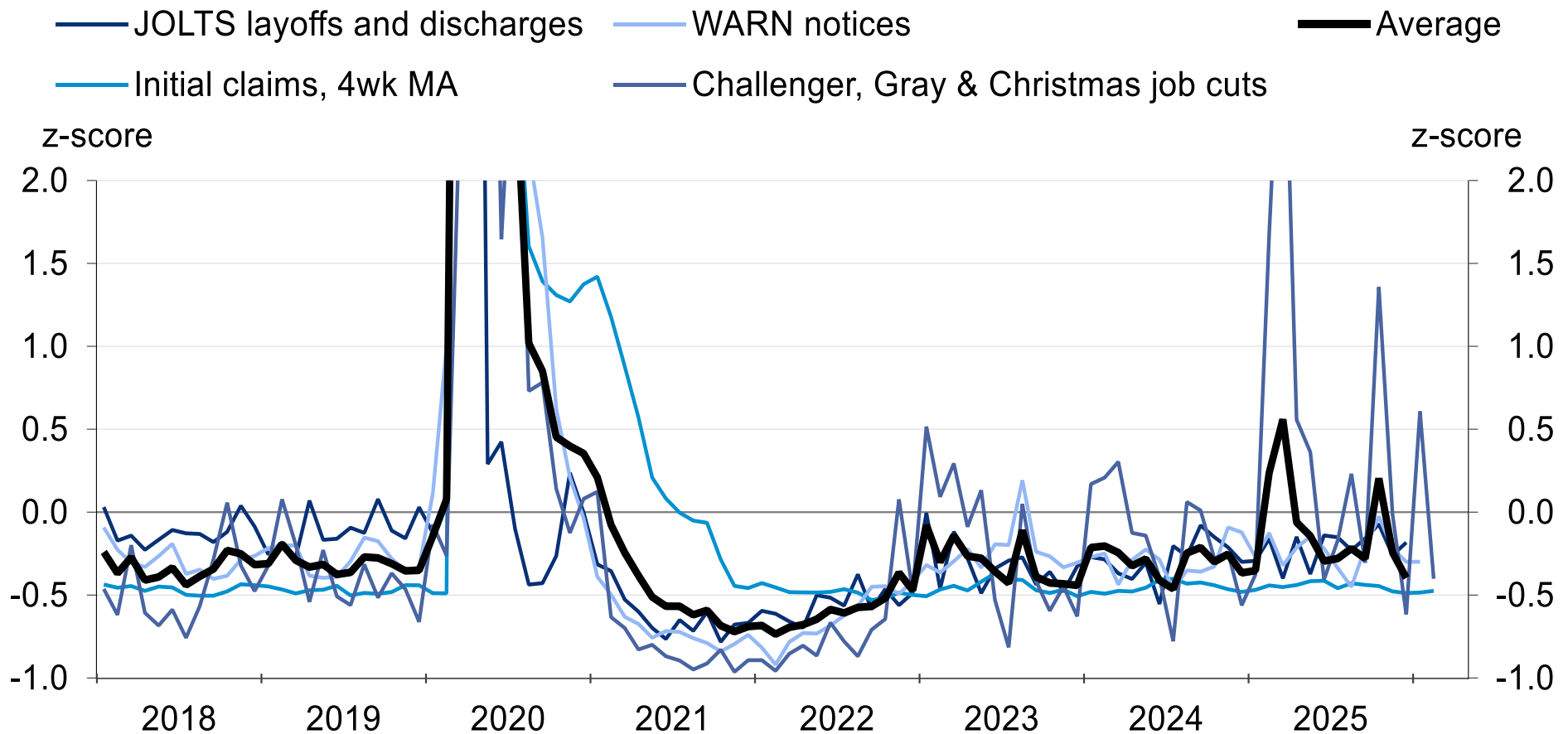


## 3. Layoffs



# Limited evidence of a change in the low-firing environment across key layoffs indicators

## Layoff indicators

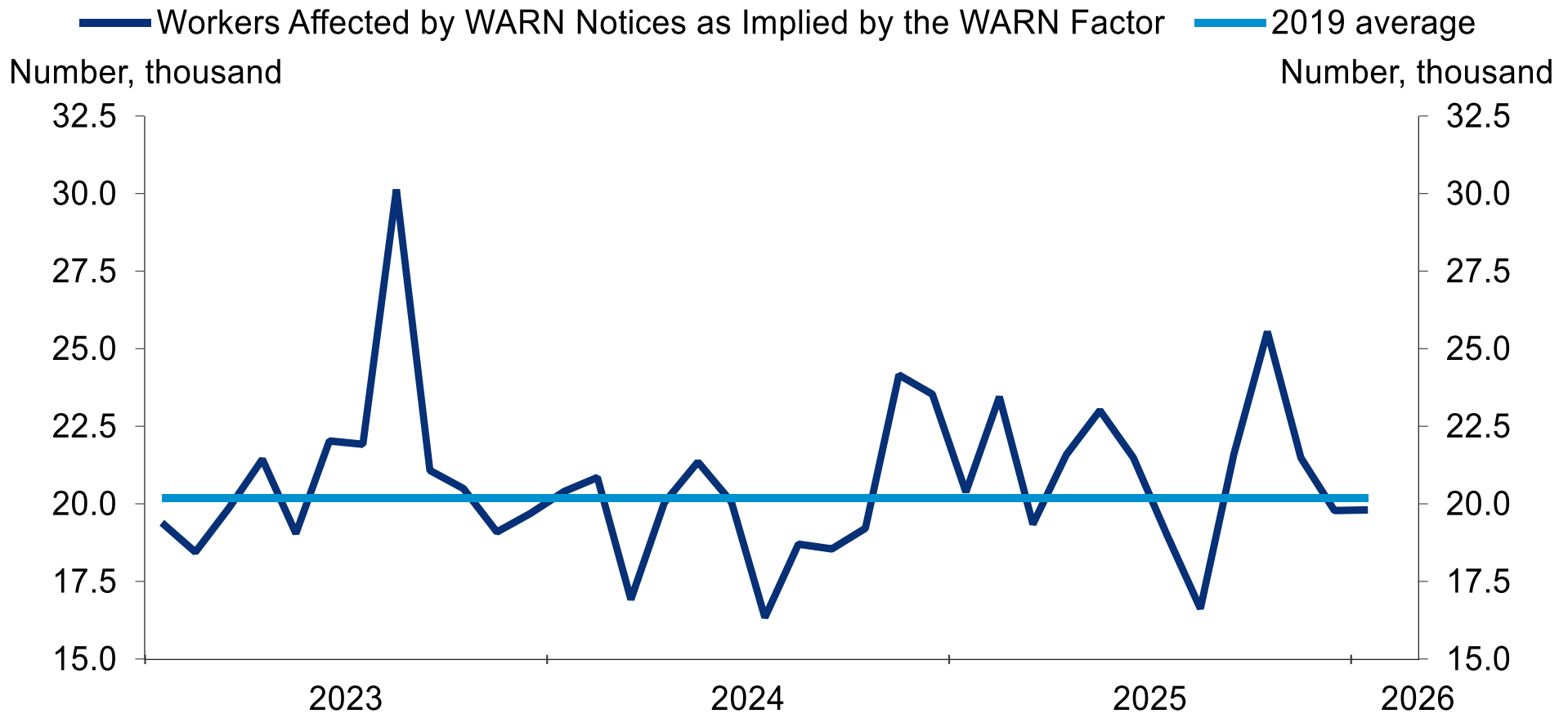


Source: DOL, BLS, ICPSR, Challenger, Gray & Christmas, Inc., Macrobond, Deutsche Bank



# WARN indicators show no broad-based impact of AI as layoffs remain low

## WARN Notices

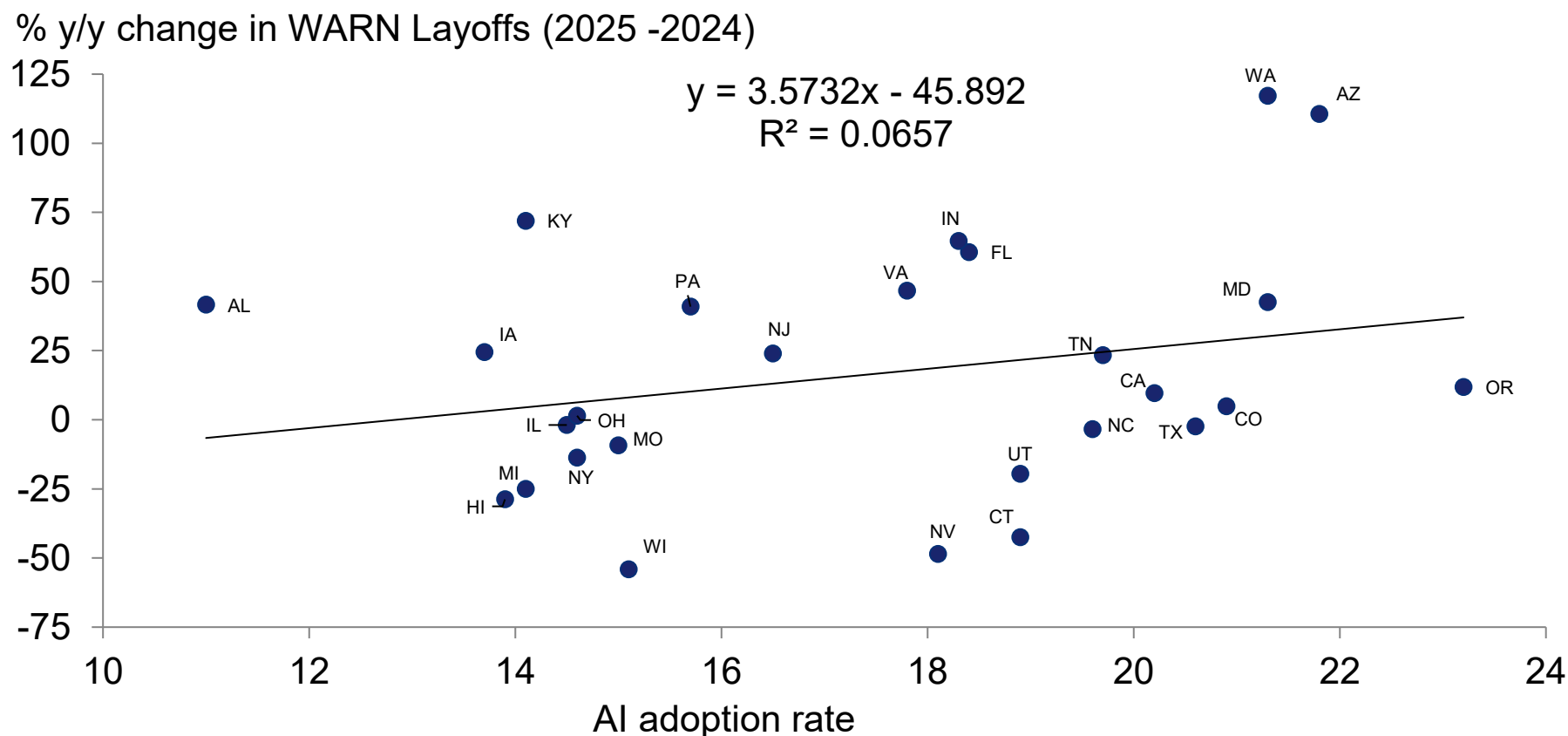


Source: ICPSR, Macrobond, Deutsche Bank

# Limited relationship between AI adoption rate and year-over-year change in WARN notices



AI adoption rate (x-axis) versus %yoy WARN (y-axis)

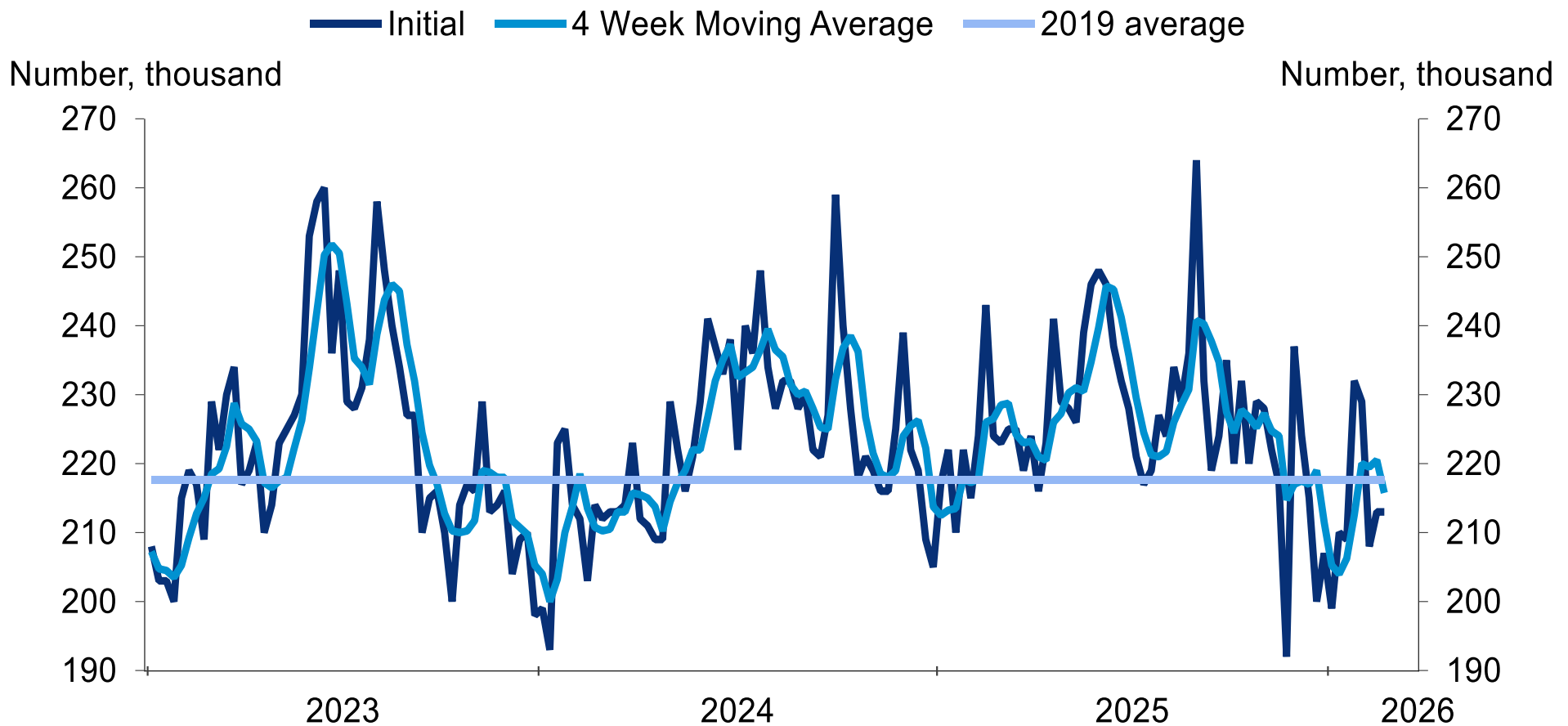


Source: Census, FRB Cleveland, Haver Analytics, Macrobond, Deutsche Bank



# Initial jobless claims also remain low

## Initial claims



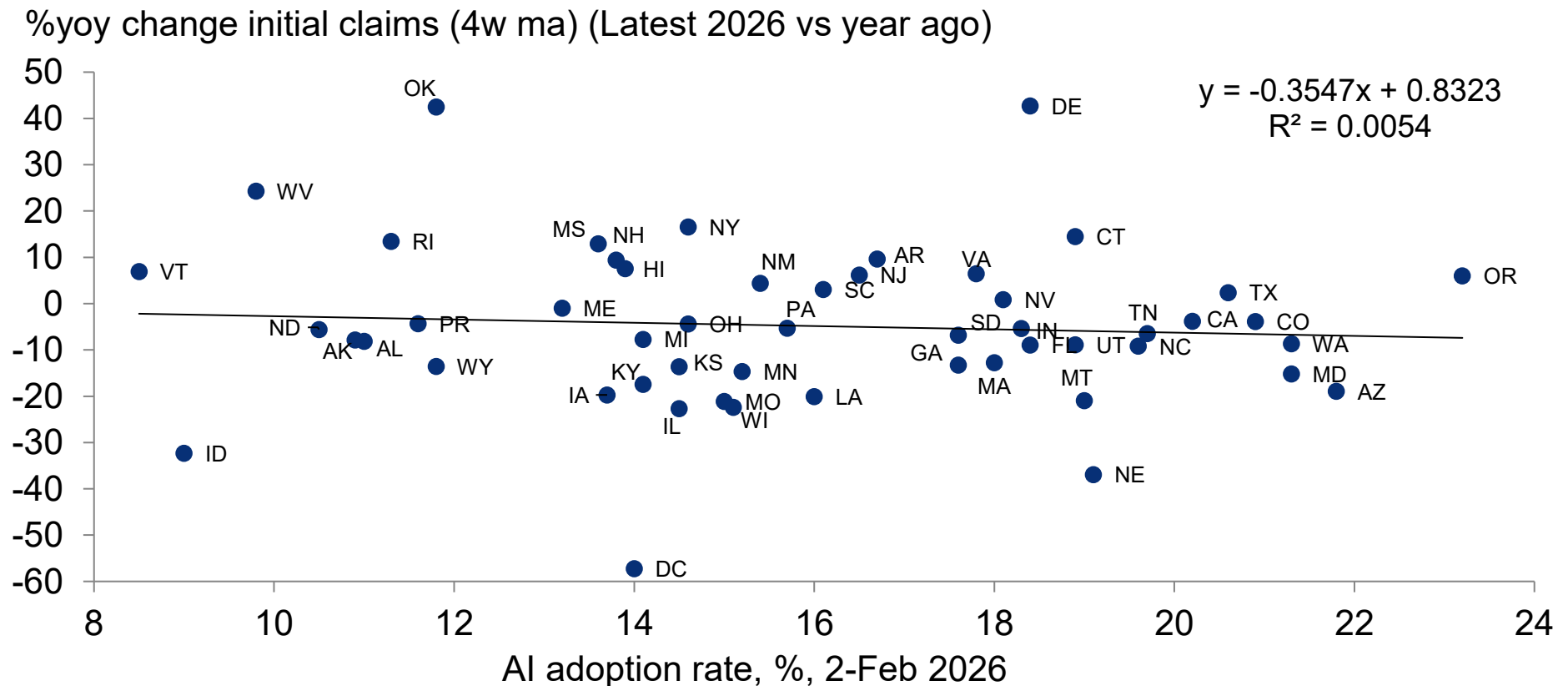
Source: DOL, Macrobond, Deutsche Bank





# Slight negative relationship between AI adoption rates across states and change in initial jobless claims

AI adoption rate (x-axis) vs. %yoy 4wk moving average of initial jobless claims

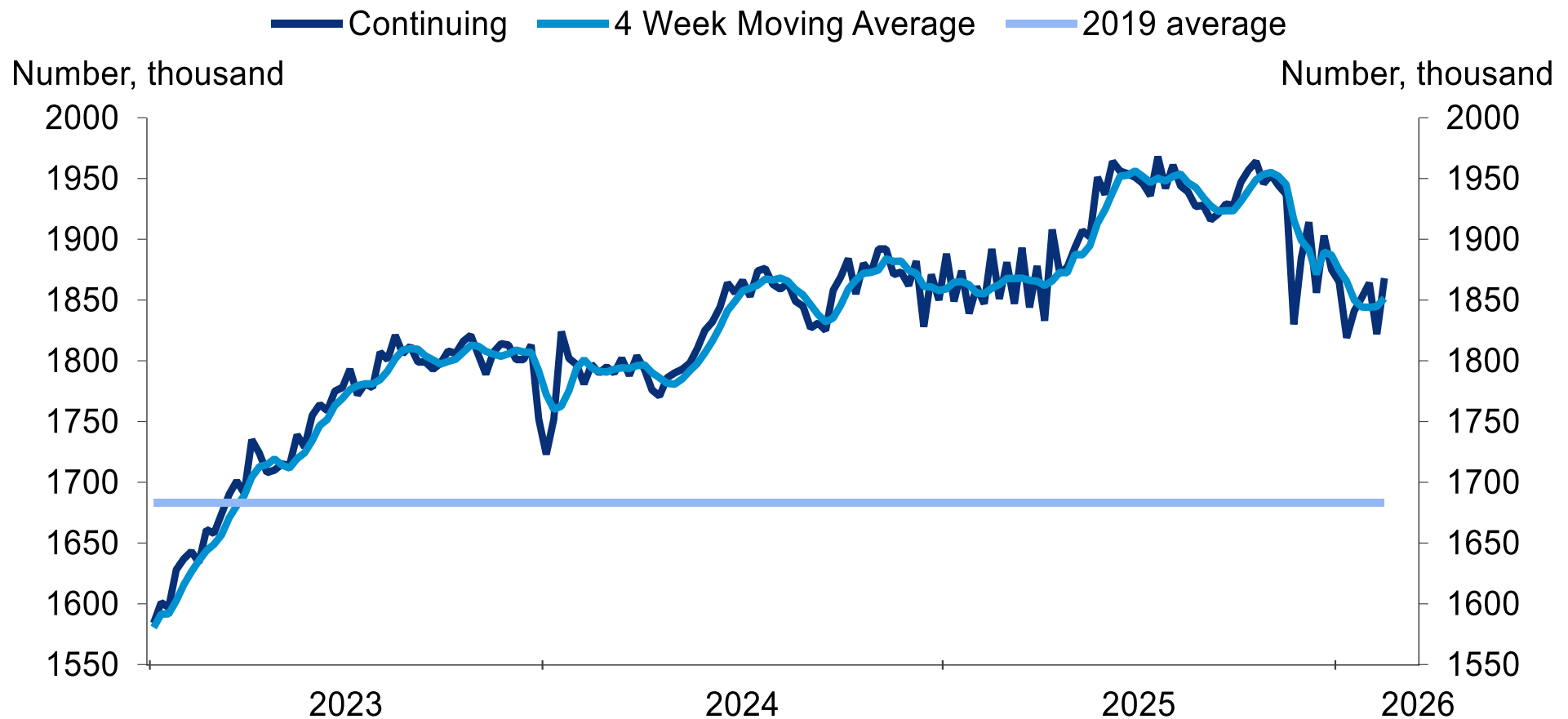


Source: Census, DoL, Haver Analytics, Macrobond, Deutsche Bank



Continuing claims have moderated and are consistent with readings over past few years, though they are higher than in '19

## Continuing claims

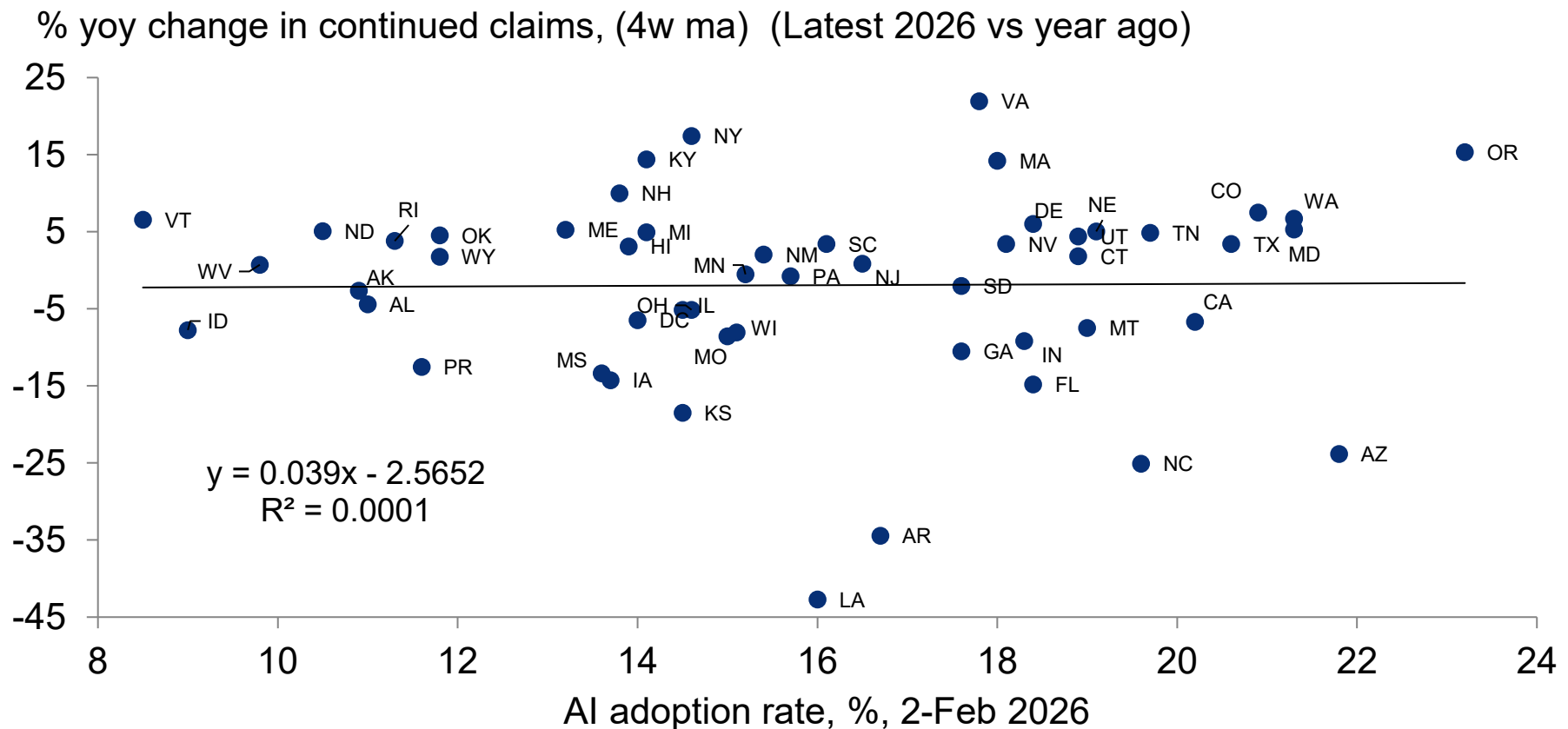


Source: DOL, Macrobond, Deutsche Bank



# Slight positive relationship between state-level AI adoption rate and yearly change in continuing claims

AI adoption rate (x-axis) versus %yoy change in 4wk moving average of continuing claims (y-axis)

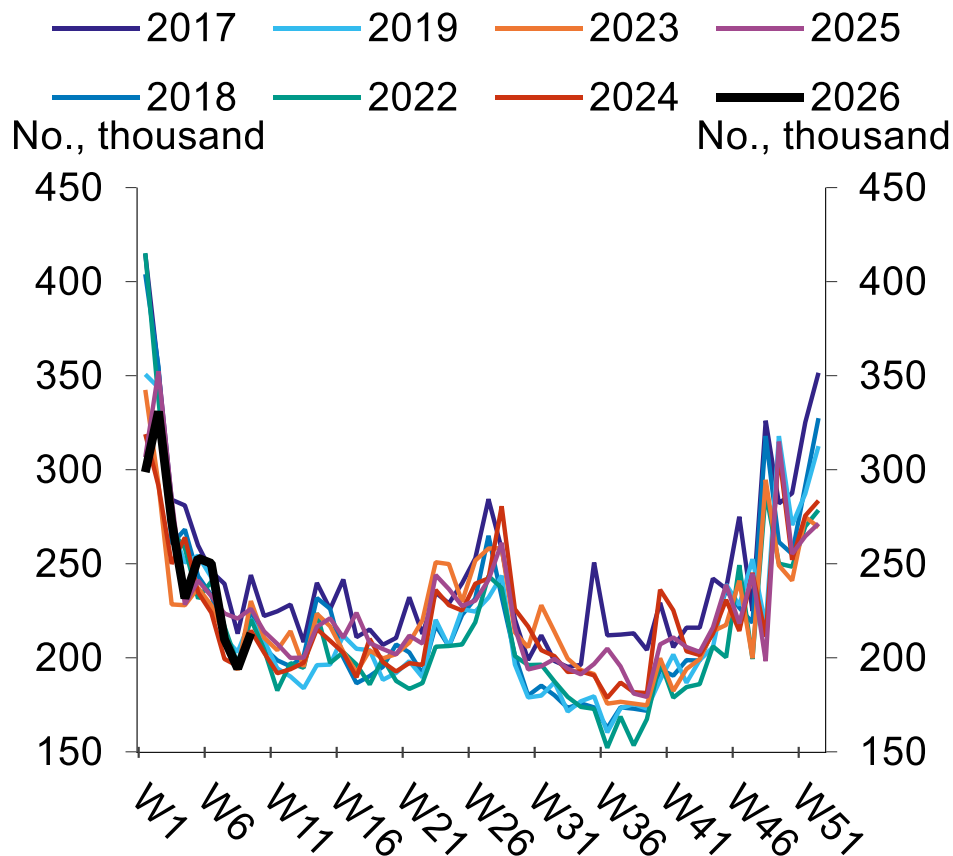


Source: Census, DoL, Haver Analytics, Macrobond, Deutsche Bank



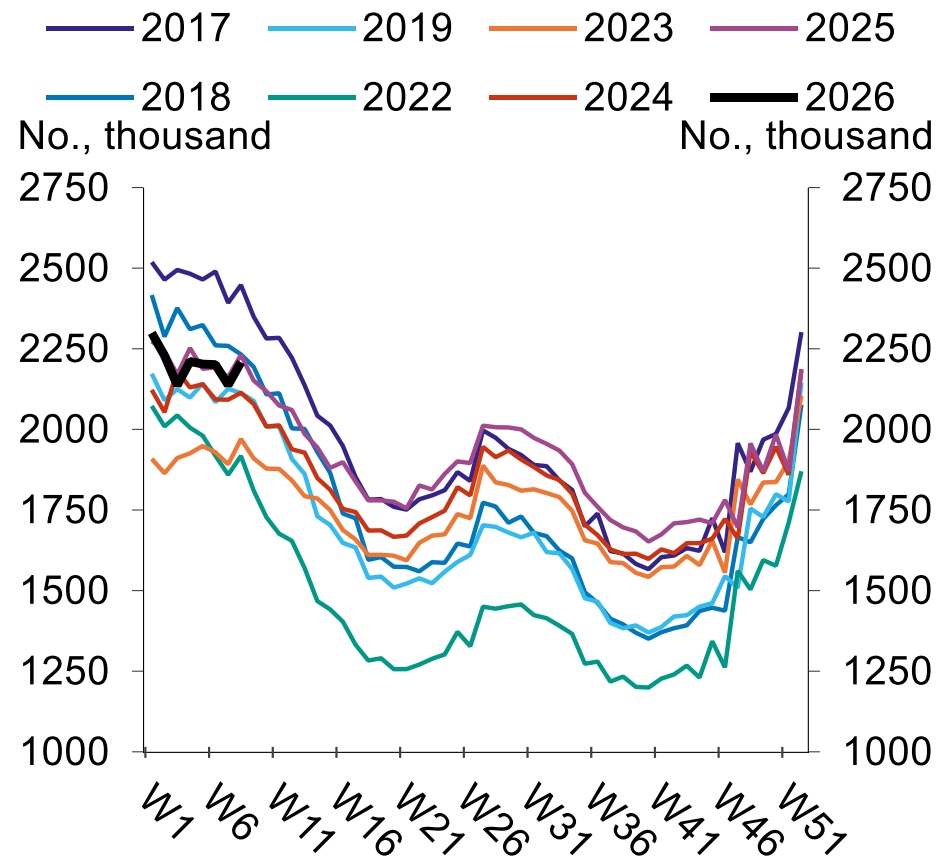
# Jobless claims data versus the past few years in NSA terms – initial claims low, continuing in the middle

## NSA initial claims



Source: DOL, Macrobond, Deutsche Bank

## NSA continuing claims



Source: DOL, Macrobond, Deutsche Bank

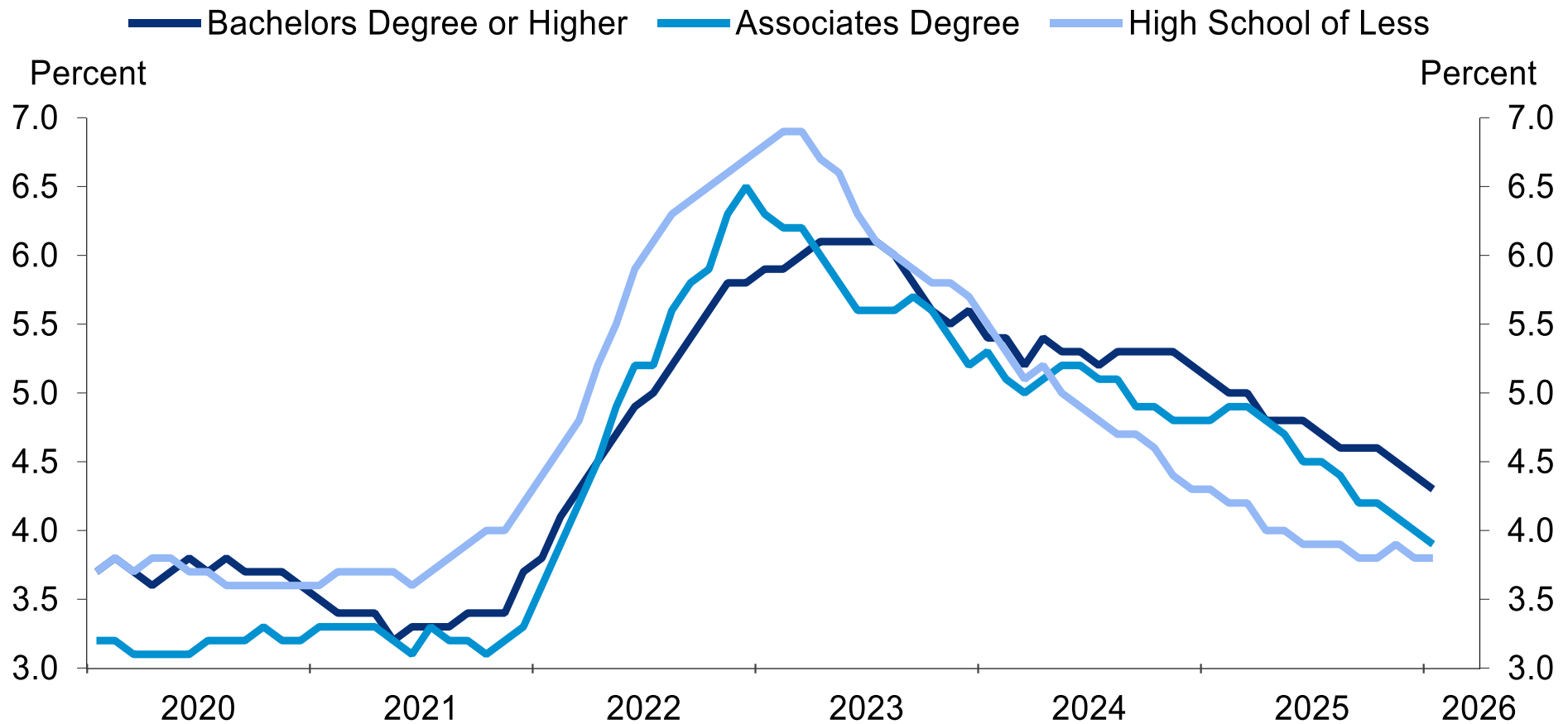


## 4. Wages



# Wage growth remains the highest for bachelor's degree or higher

## Atlanta Fed Wage Growth Tracker, 12 Month Moving Average

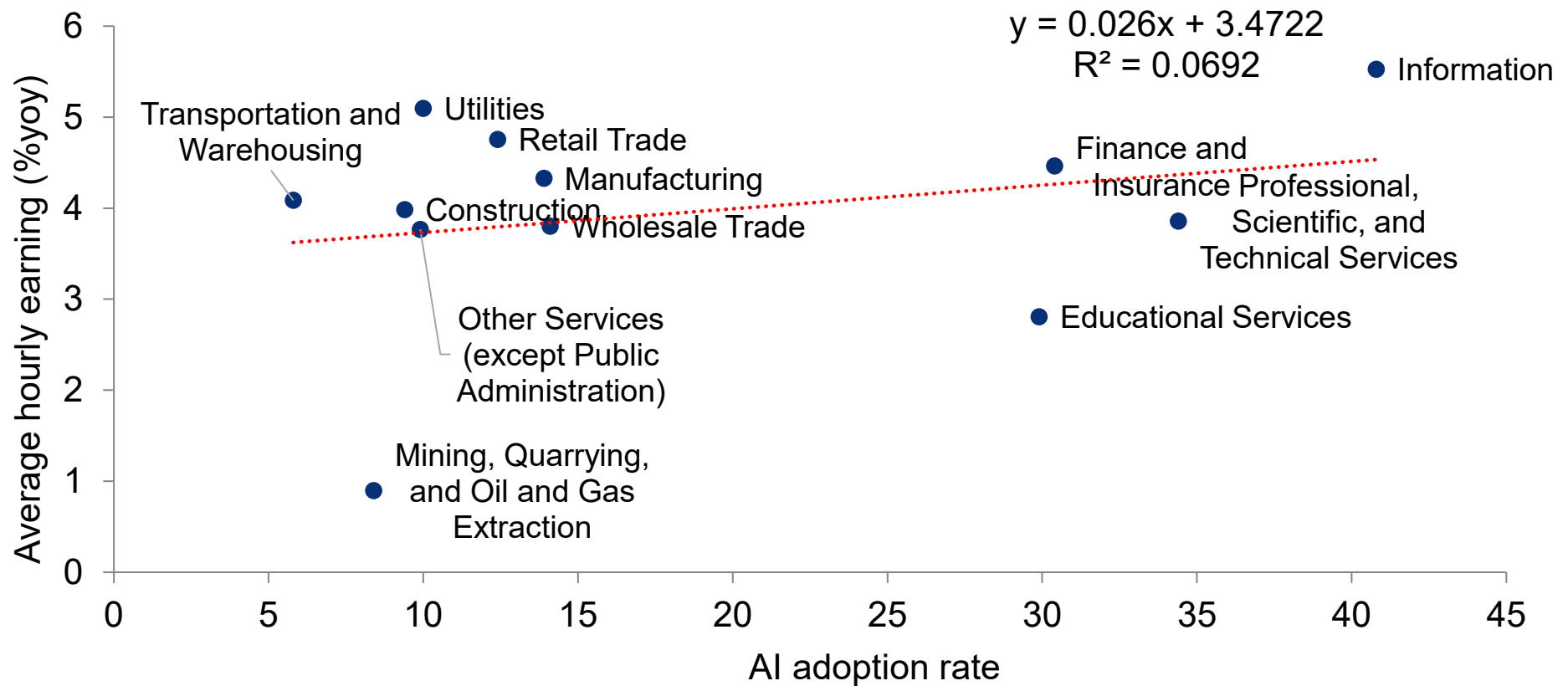


Source: Atlanta Fed, Macrobond, Deutsche Bank



# A positive relationship between wage growth and AI adoption according to average hourly earnings

AI adoption rate (x-axis) versus average hourly earnings (%yoy, y-axis)



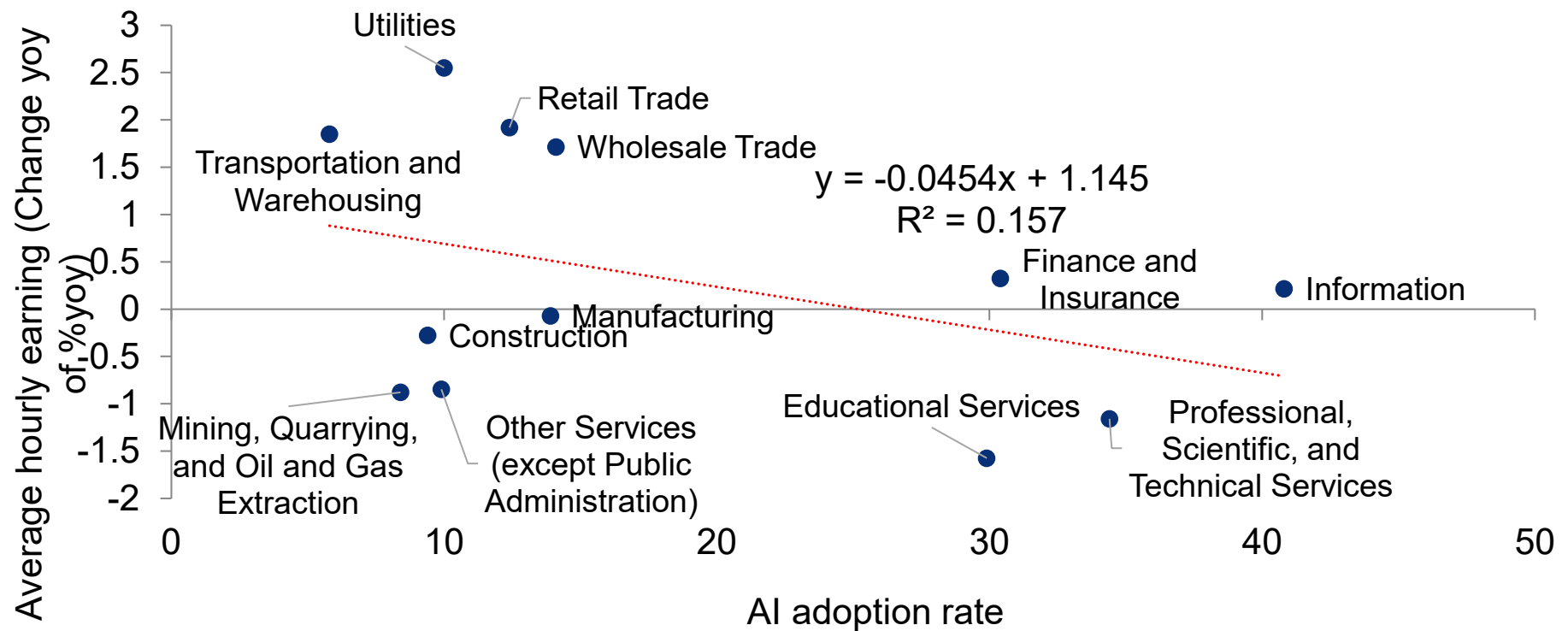
Source: BLS, Census, Haver Analytics, Macrobond, Deutsche Bank





# Sectors with higher AI adoption have shown more significant deceleration in wage growth on average

AI adoption rate (x-axis) versus average hourly earnings (Change yoy of %yoy, y-axis)

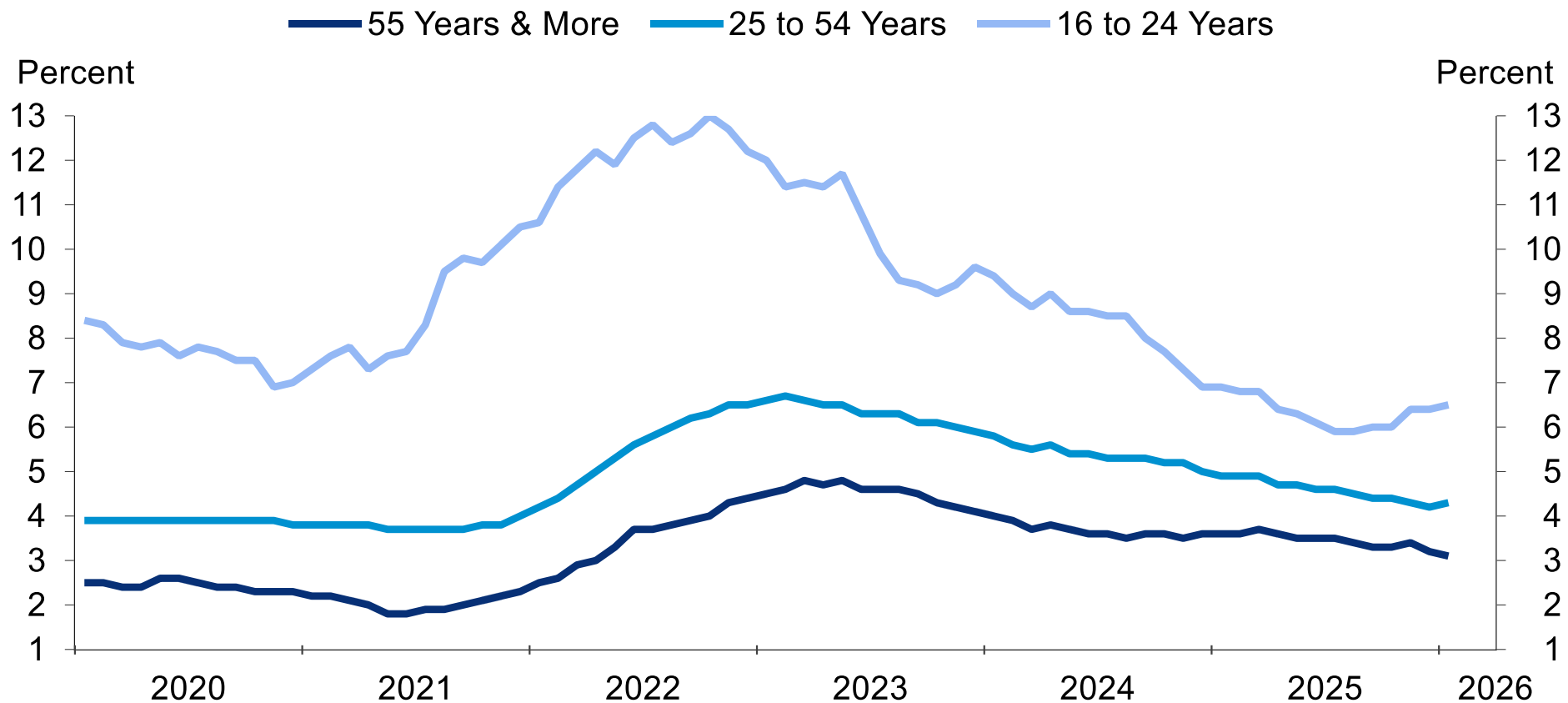


Source: BLS, Census, Haver Analytics, Macrobond, Deutsche Bank



Younger workers (16-24 ages) have seen the most noticeable uptrend in wage growth in recent months

## Atlanta Fed Wage Growth Tracker, 12 Month Moving Average

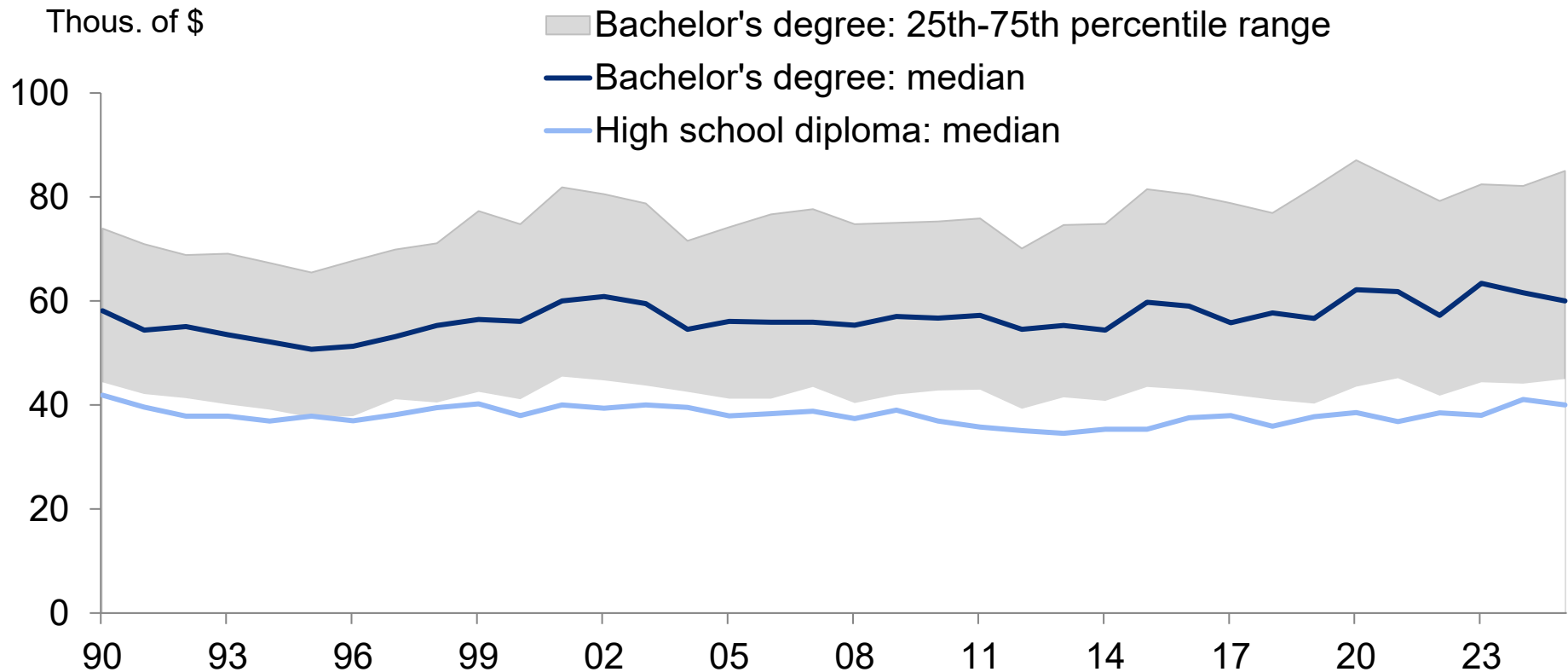


Source: Atlanta Fed, Macrobond, Deutsche Bank

# Wage premium persists for college degrees



## Distribution of Annual Wages for Recent College Graduates:



Source: NY Fed, Deutsche Bank

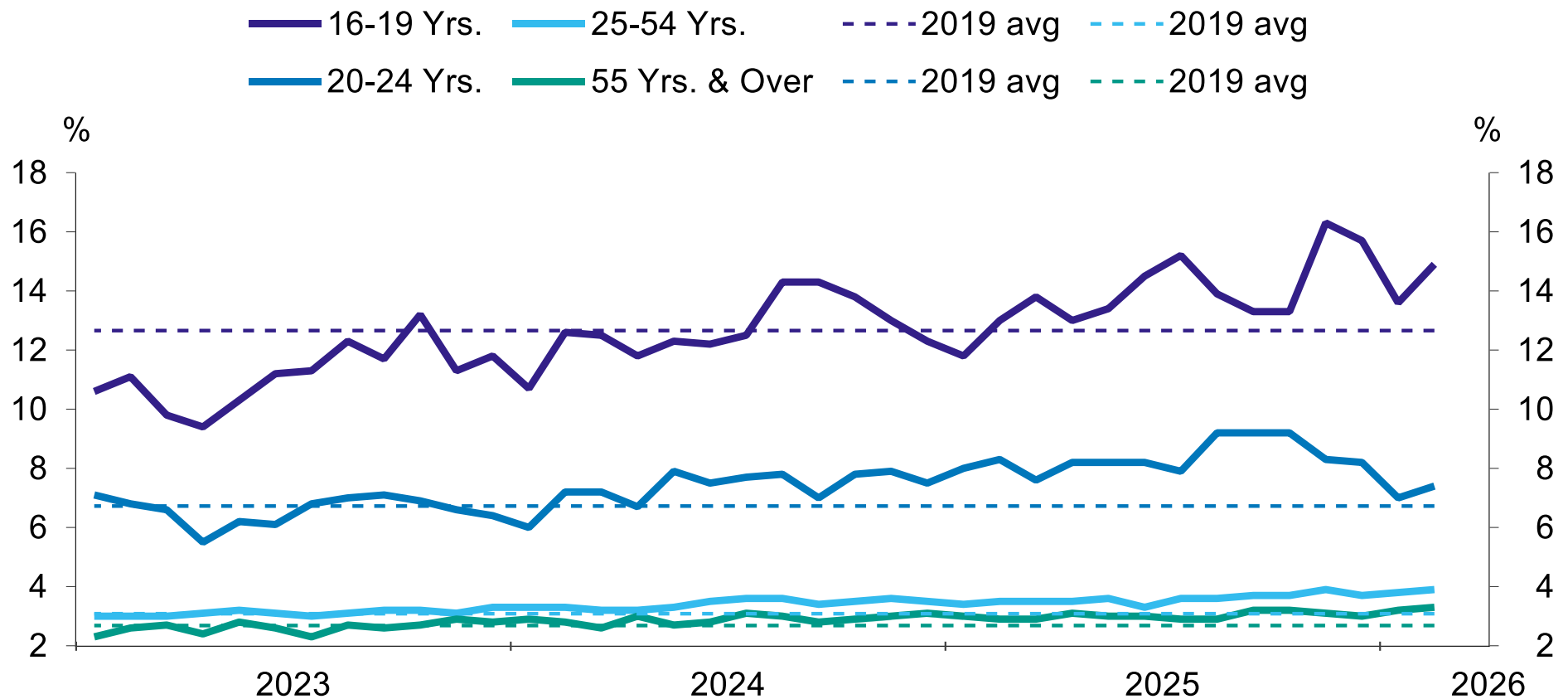


## **5. Labor developments for young workers**



# More noticeable rise in unemployment rates for younger age groups, though some improvement in recent months

## Unemployment rate by age

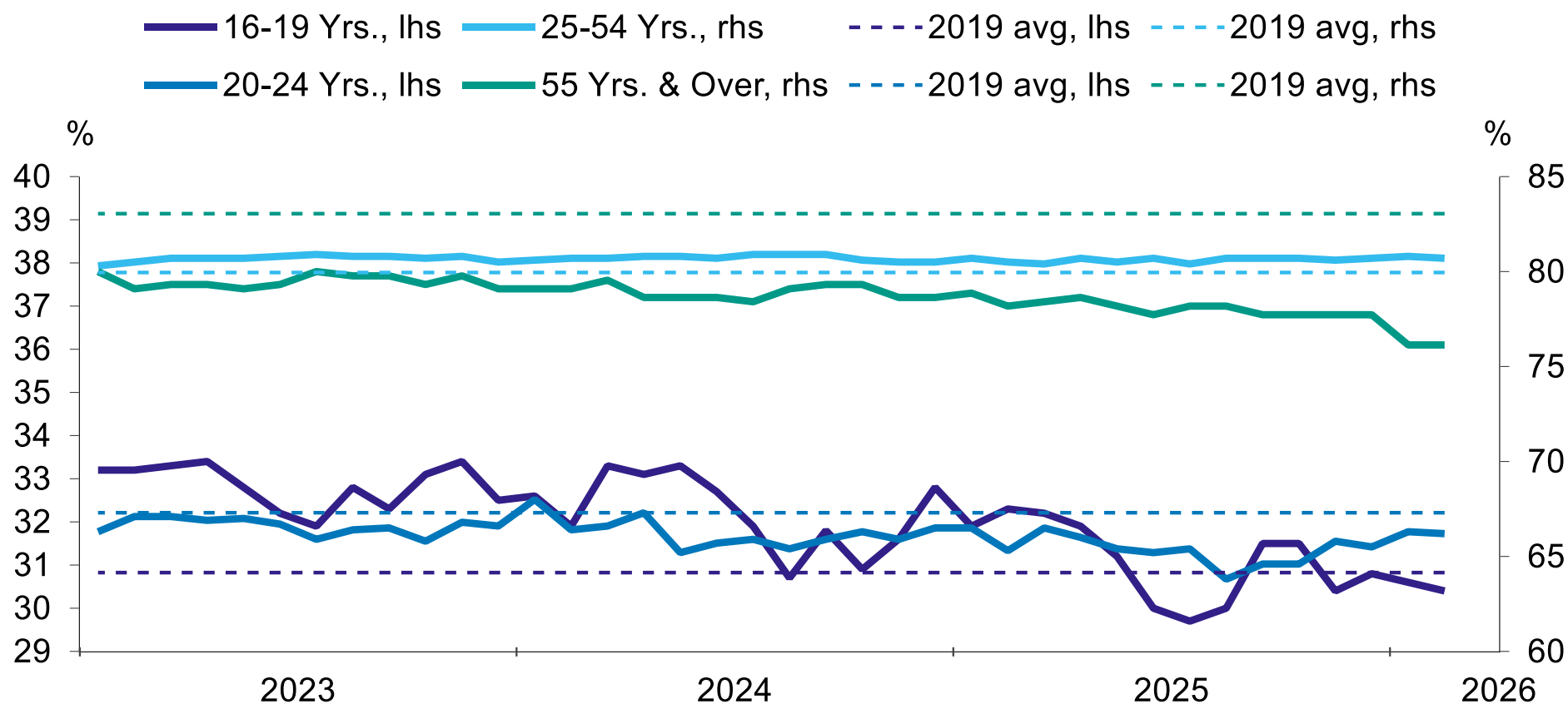


Source: BLS, Macrobond, Deutsche Bank

# But employment-to-population ratios for younger individuals is not far from 2019 levels



## Employment to population ratio by age

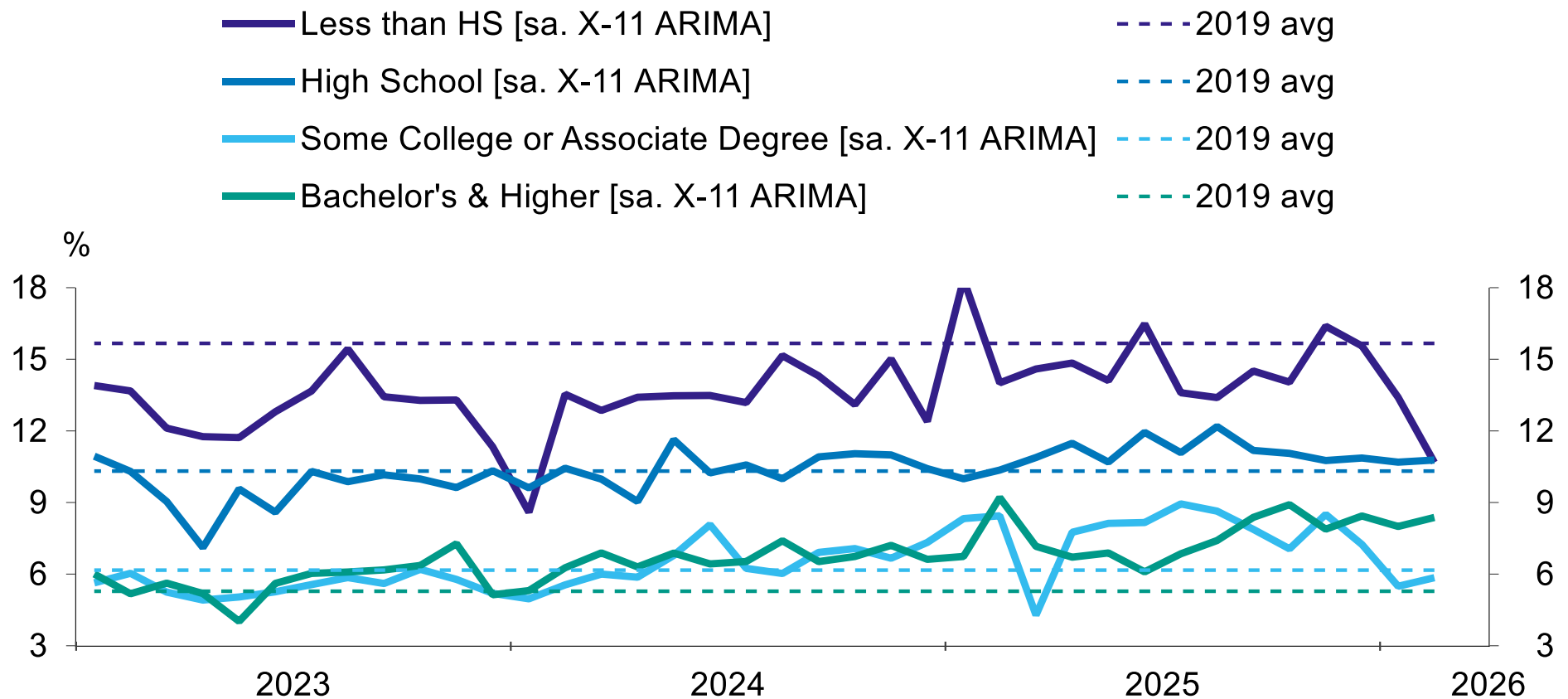


Source: BLS, Macrobond, Deutsche Bank



# Unemployment rate for young workers (16-24) w/ bachelor's degree is elevated, while other education groups are not

## Unemployment by education for age 16-24 (not enrolled in school)

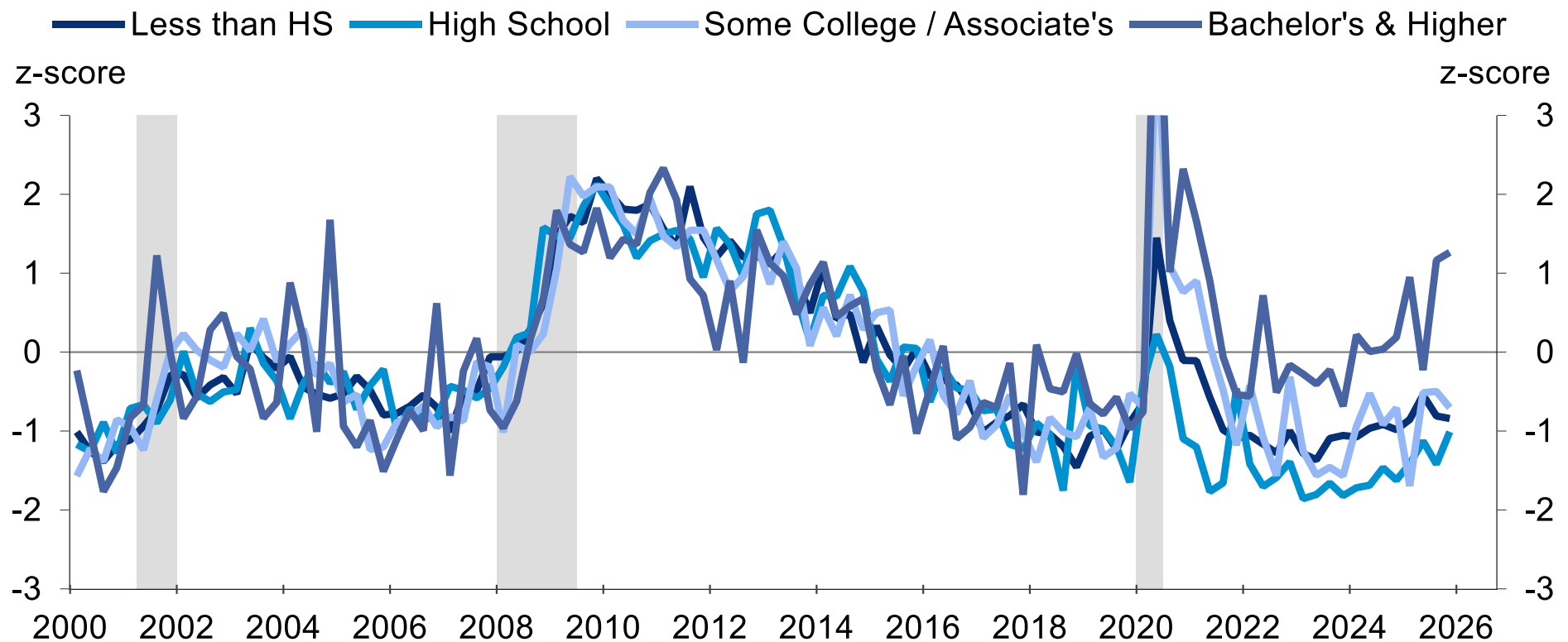


Source: BLS, Macrobond, Deutsche Bank



# Unemployment rate for younger workers with bachelor's degree looks elevated relative to 2000-2019 history

z-scores (2000-2019): unemployment rate by education age 16-24 (not enrolled in school)



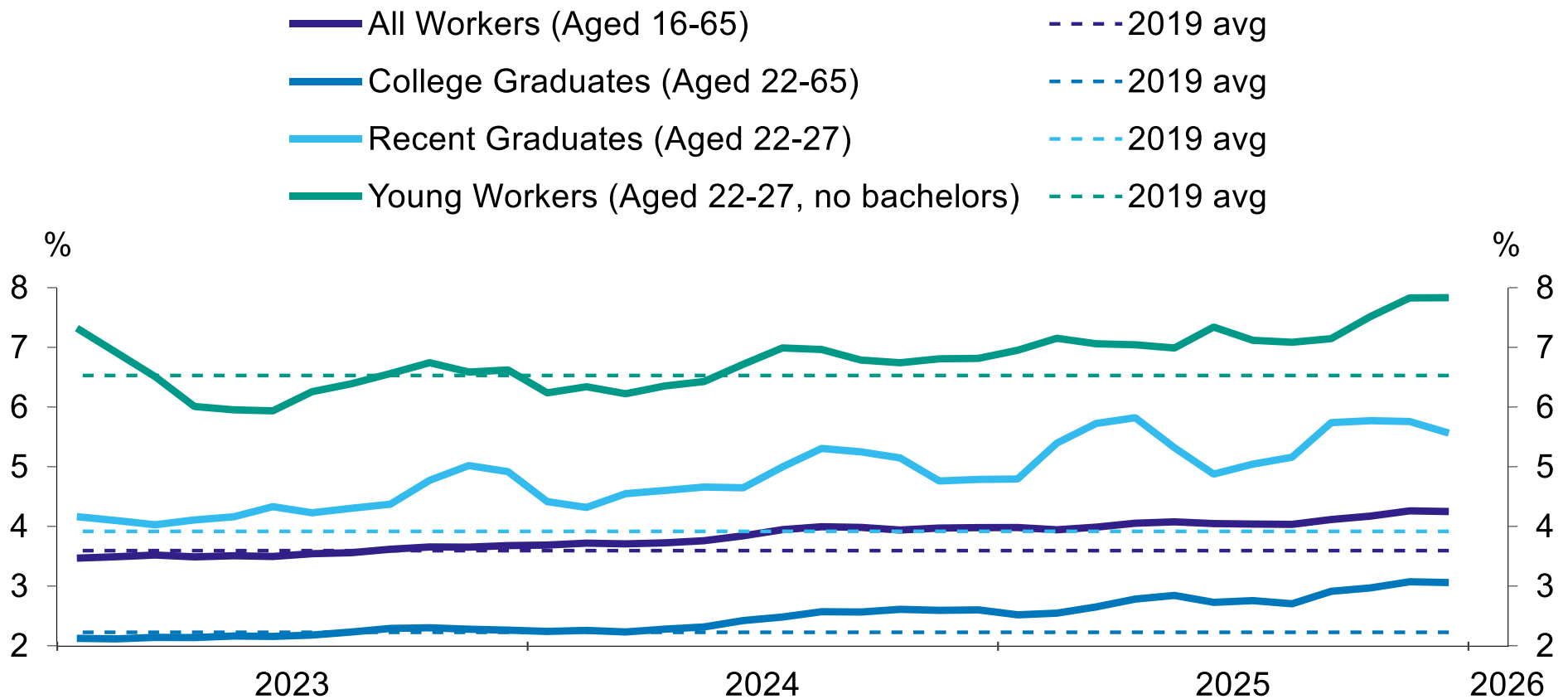
Source: BLS, Macrobond, Deutsche Bank





# NY Fed finds that unemployment rates are up for recent college graduates

## Unemployment for recent grads versus other groups

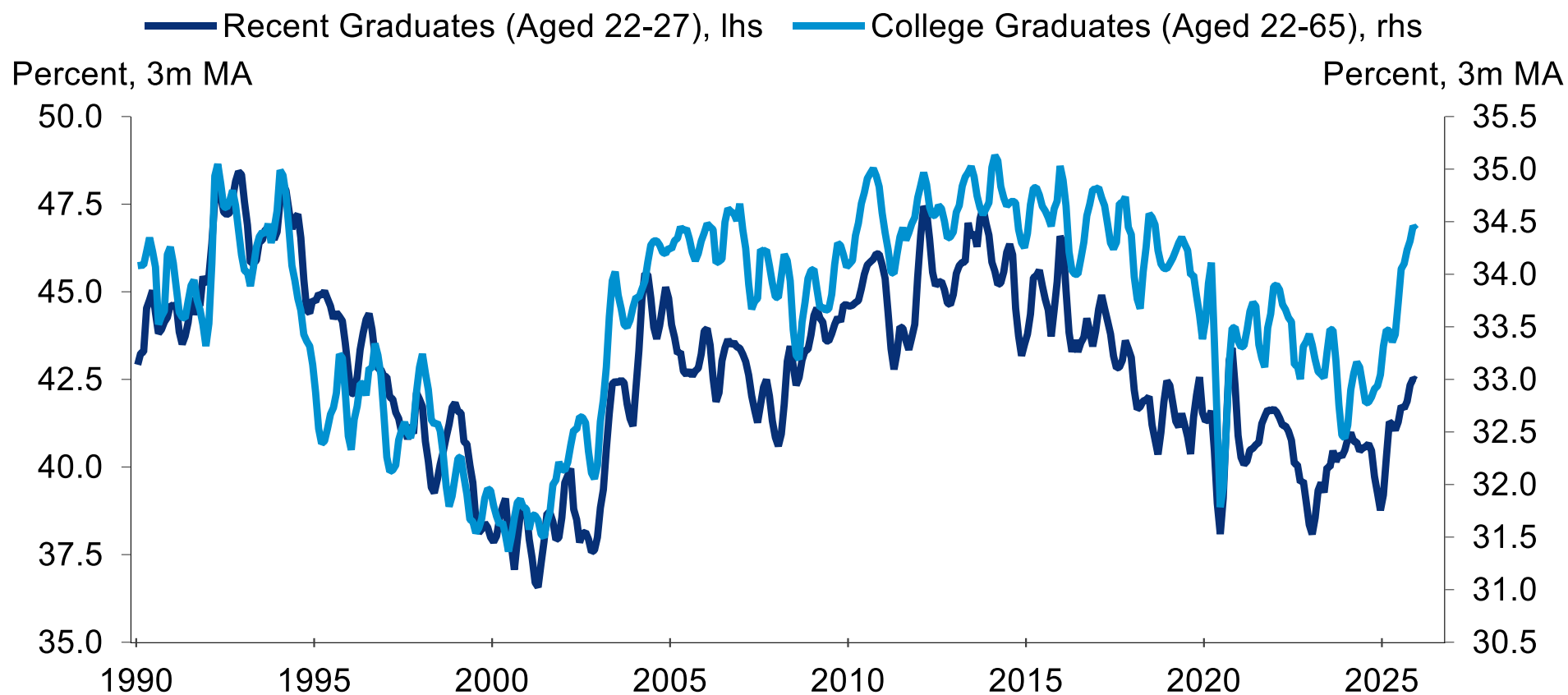


Source: New York Fed, Macrobond, Deutsche Bank

And underemployment rate near the highest in a few years



## Underemployment rates for college graduates



Source: New York Fed, Macrobond, Deutsche Bank

Source: NY Fed, Deutsche Bank



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Deutsche Bank  
Research

# Innovation, Jobs and Inflation:

## Lessons from 250 Years of Disruption

March 4, 2026

Jim Reid | Galina Pozdnyakova | Henry Allen | Rajsekhar Bhattacharyya

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank  
Research Institute

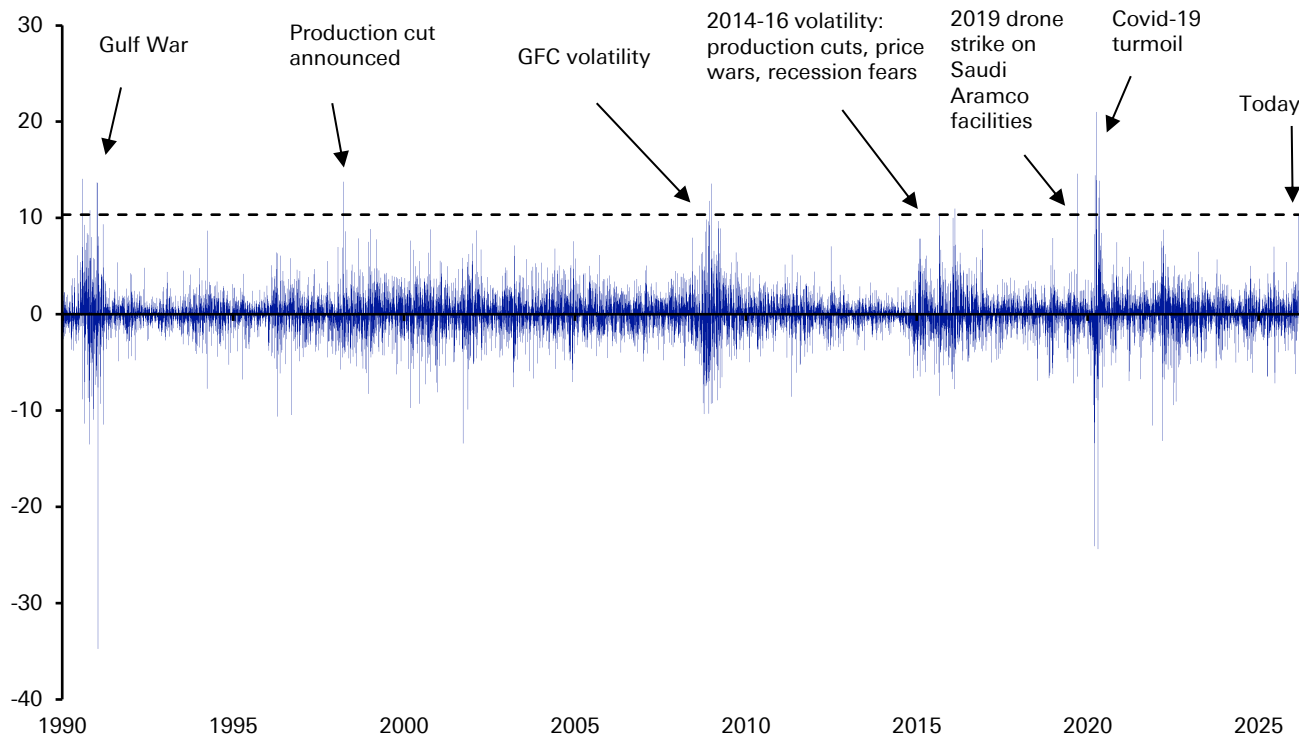


| Content                                                 | Page               |
|---------------------------------------------------------|--------------------|
| Putting the oil moves in historic context               | <a href="#">02</a> |
| What history tells us about impact of AI..              | <a href="#">03</a> |
| 2026 so far: Rotation, dispersion and value             | <a href="#">20</a> |
| Does the US really need a rate cut... unless tech melts | <a href="#">31</a> |

The weekend strikes on Iran have knocked AI concerns from the front page for now... Brent crude oil prices are up over +10% this morning at time of writing... which makes it the 15<sup>th</sup> biggest daily jump since 1990... to get into the top 5 it would need to be up around +14%...



## Daily Change in Brent crude oil price (%)



Source : Bloomberg Finance LP, Deutsche Bank.

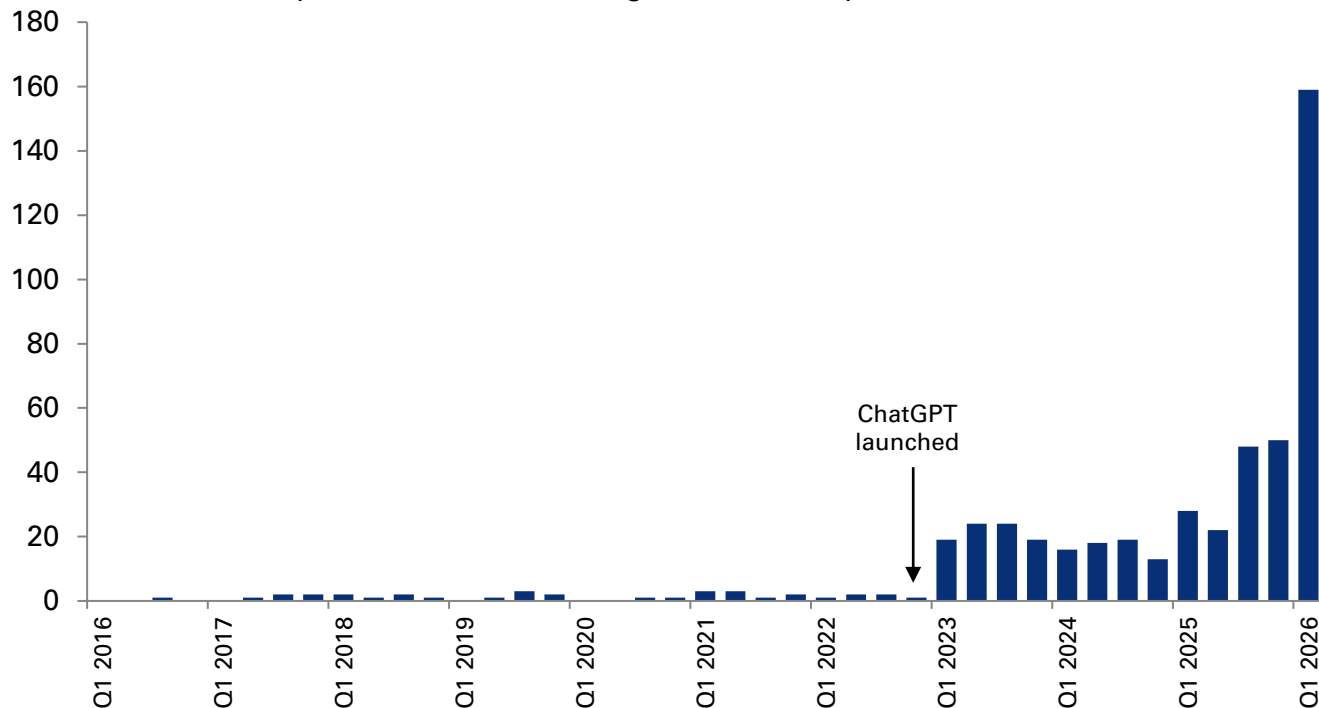


**What history tells us about impact of AI..**

AI disruption has been a steadily growing theme since ChatGPT was launched.... However in Q1 2026 mentions have tripled from Q4 2025 in US Earnings calls.... The world has gone AI-disruption crazy!



## Mentions of AI Disruptions in Global Earnings Call Transcripts



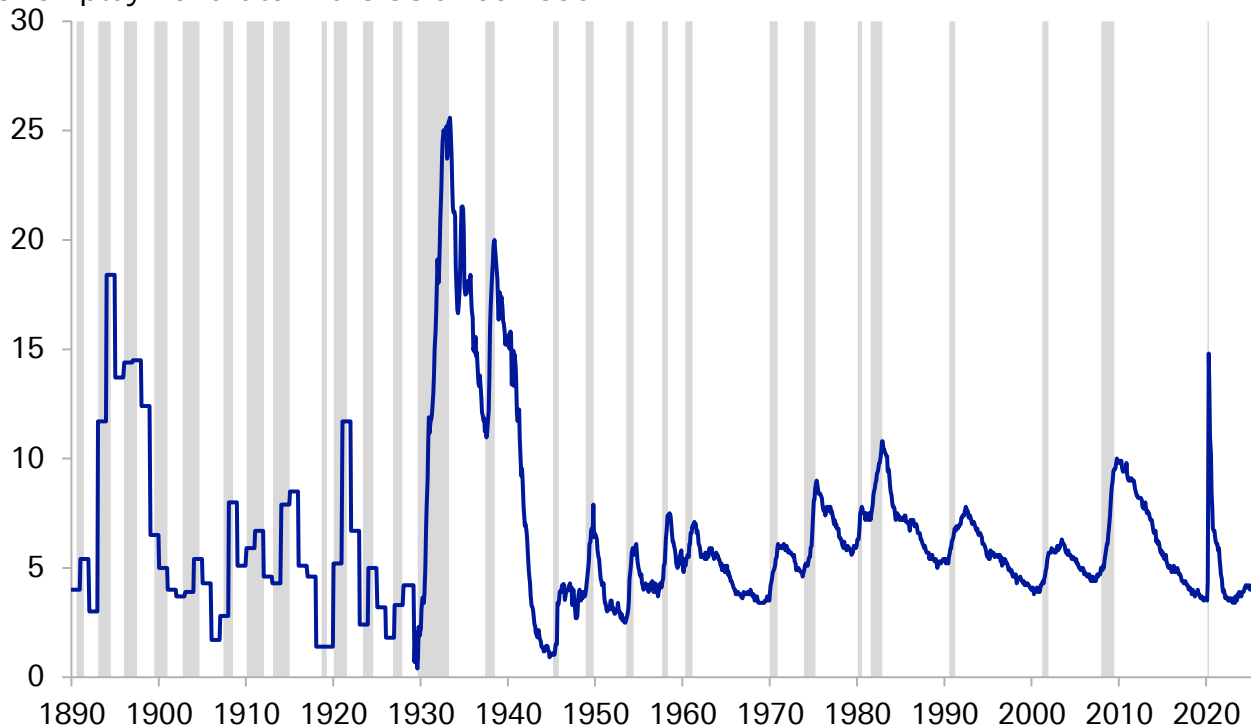
Source : Bloomberg Finance LP, Deutsche Bank., Last update : Feb 26



Unemployment rarely rises >1pp outside recessions. In this cycle, the low of 3.4% in H1 2023 was the joint lowest since 1953, possibly reflecting post-pandemic labour hoarding. So if AI were to “destroy” the labour market in aggregate, it would be the first major innovation in history to do so...



## Unemployment rate in the US since 1890

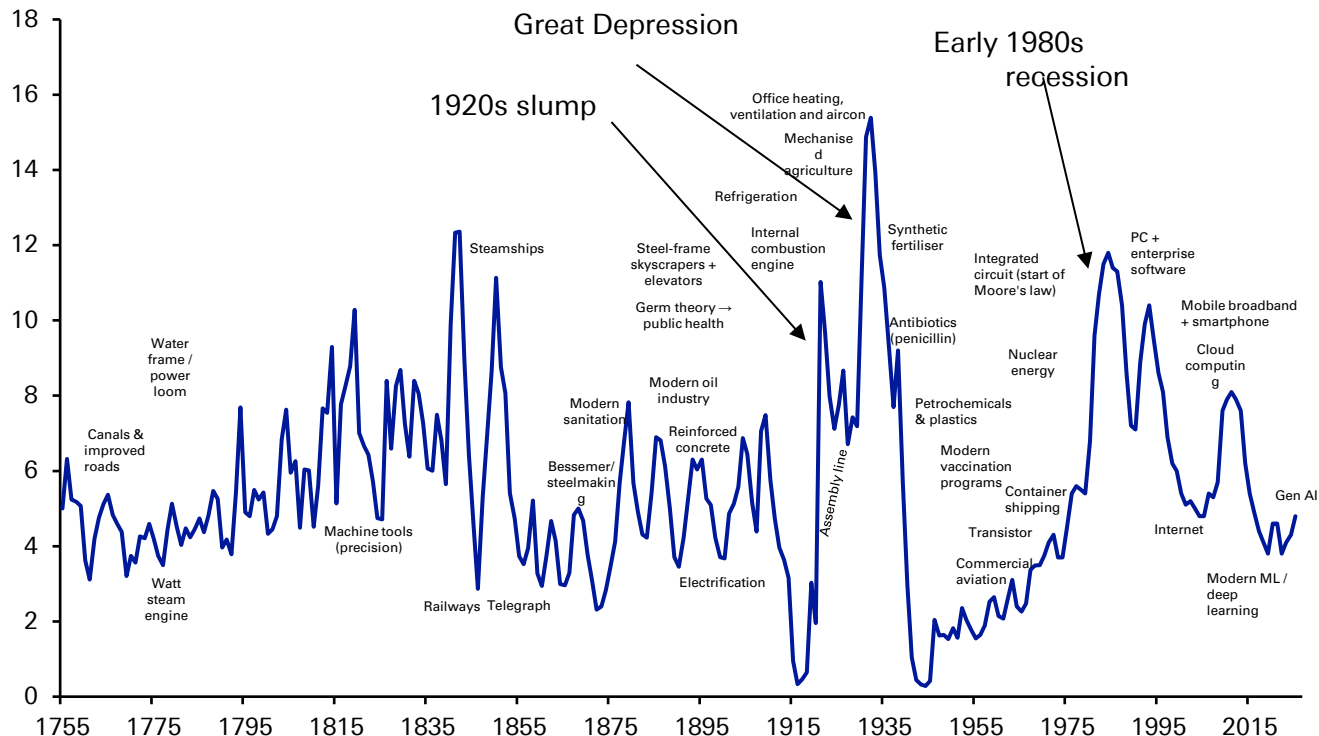


Source: Finaeon, BLS, Deutsche Bank

Since the start of the first Industrial Revolution, humans have been extraordinarily innovative with unemployment driven by macro factors like recessions, not new technologies... This time would have to be very different if AI destroys labour



## Unemployment rate in UK since 1755 and major innovations since the first Industrial Revolution

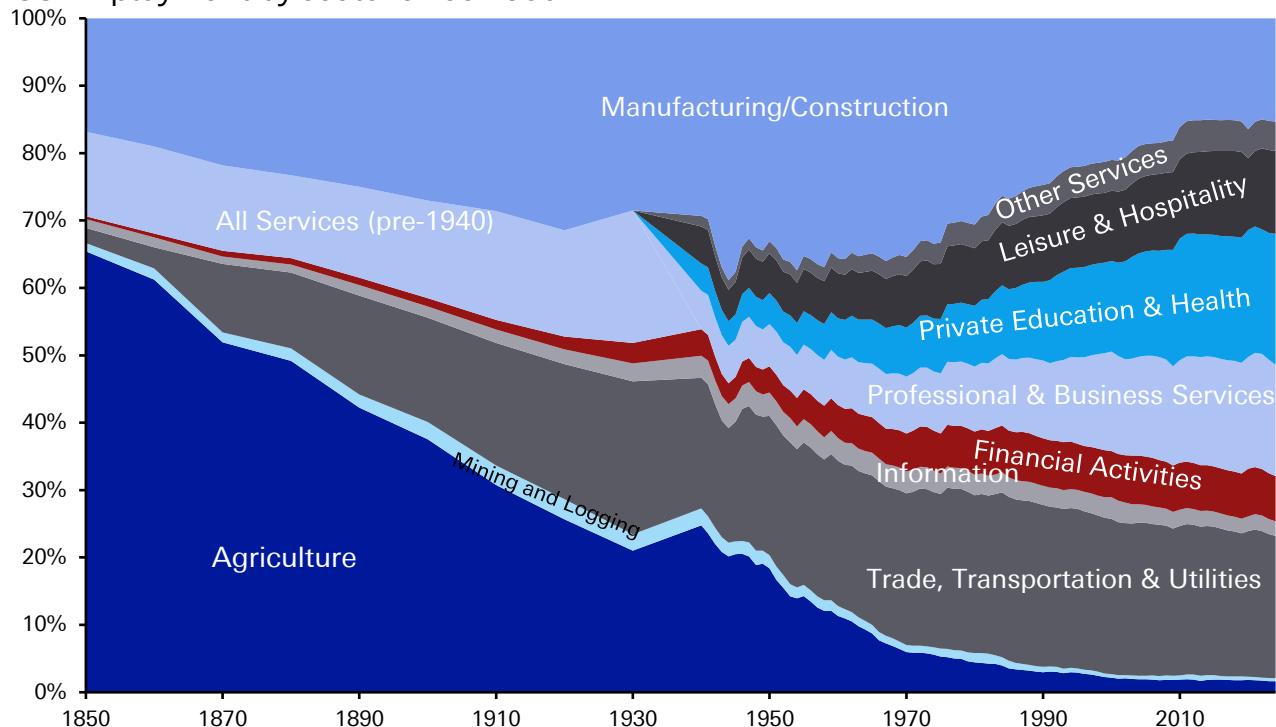


Source: Bank of England: Millennium of Macroeconomic Data, ONS, Deutsche Bank

All through history technology has shifted jobs around, and AI will shift it around again, but the most likely scenario is not huge job losses but adaptation and future jobs we didn't know existed or were needed.



## US Employment by sector since 1850

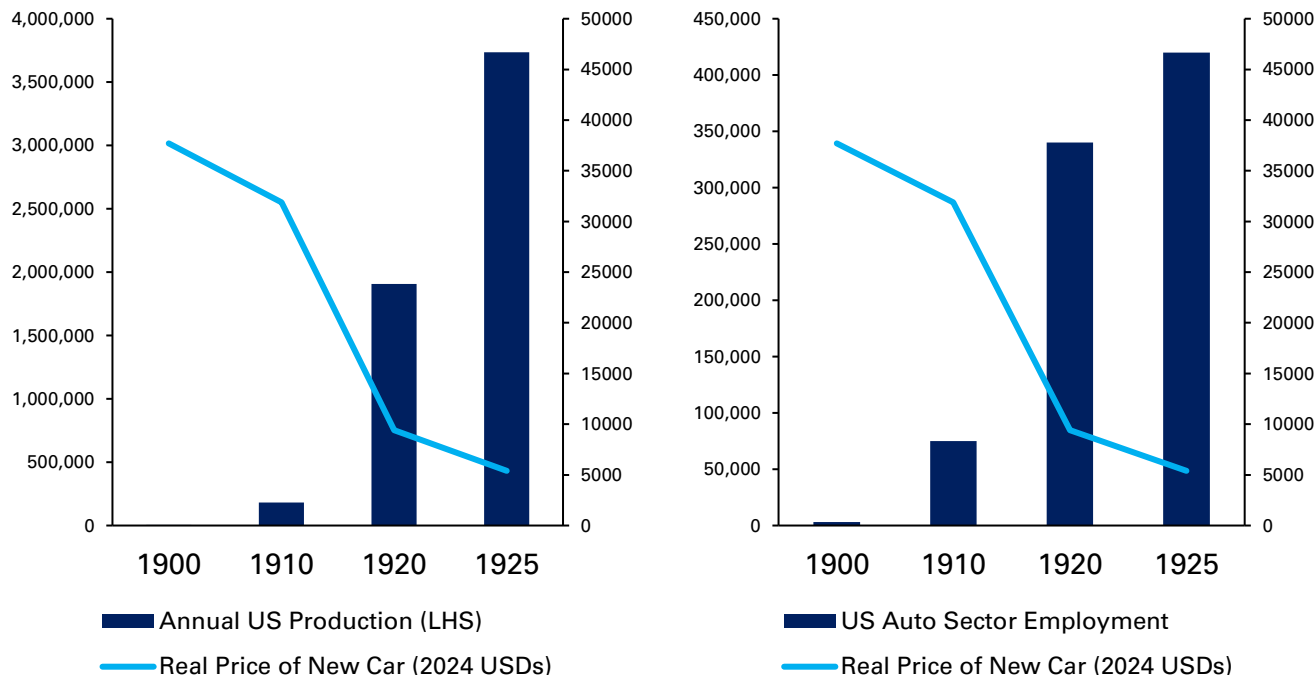


Source: Haver Analytics, US Census, Deutsche Bank

In the early days of the auto industry, Ford's move to the assembly line for the Model T chassis (c.1913) ushered in a new era of automated production. It also sparked widespread fears of deskilling and job losses. Yet mass production dramatically reduced the cost of a car, unleashing a surge in demand and ultimately creating far more jobs—not only in auto manufacturing, but across the broader economy, from suburban homebuilding to tourism and a wide range of related commerce.

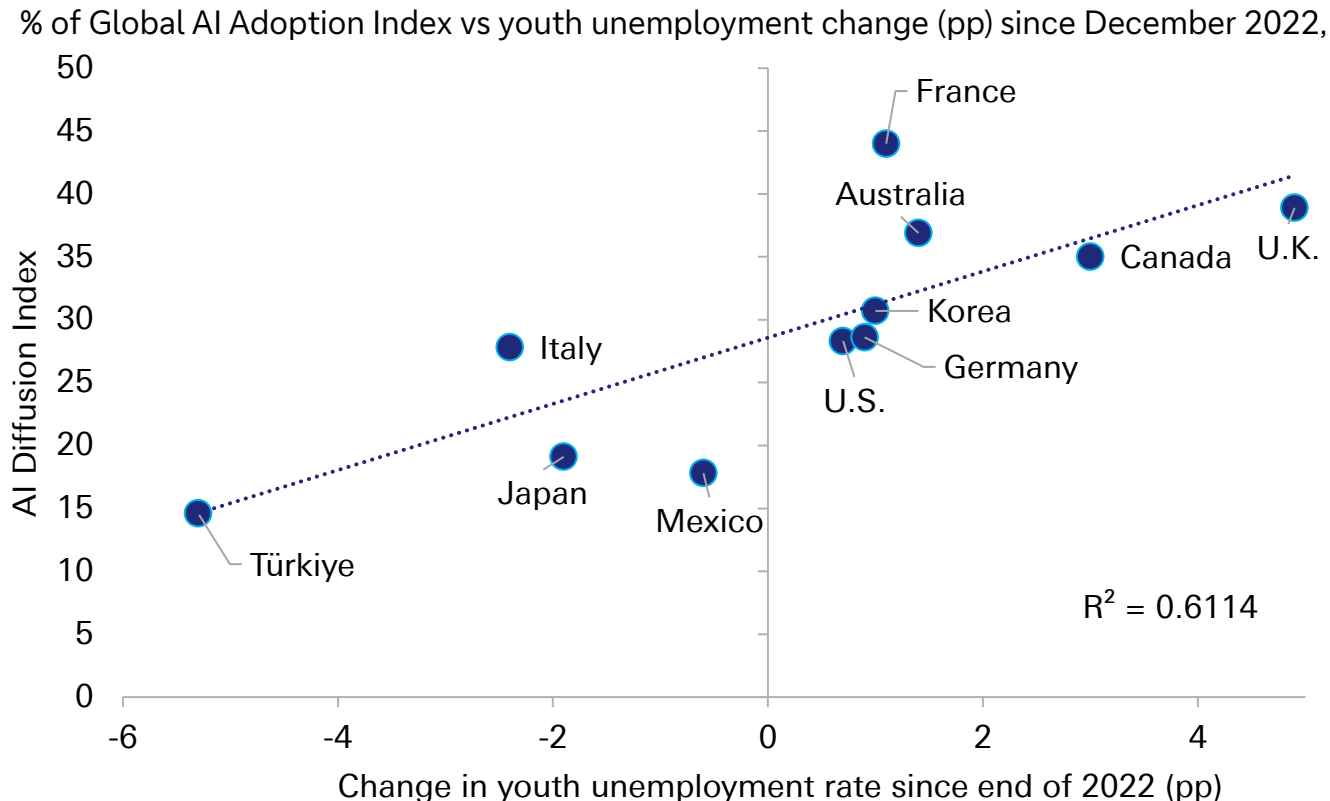


## Real adjusted price of a new US car and Auto production and employment numbers.



Source: dbLumina, Deutsche Bank

However, since ChatGPT was launched there is some evidence to suggest those economies with the highest adoption rates are seeing increases in youth unemployment and vice-versa.

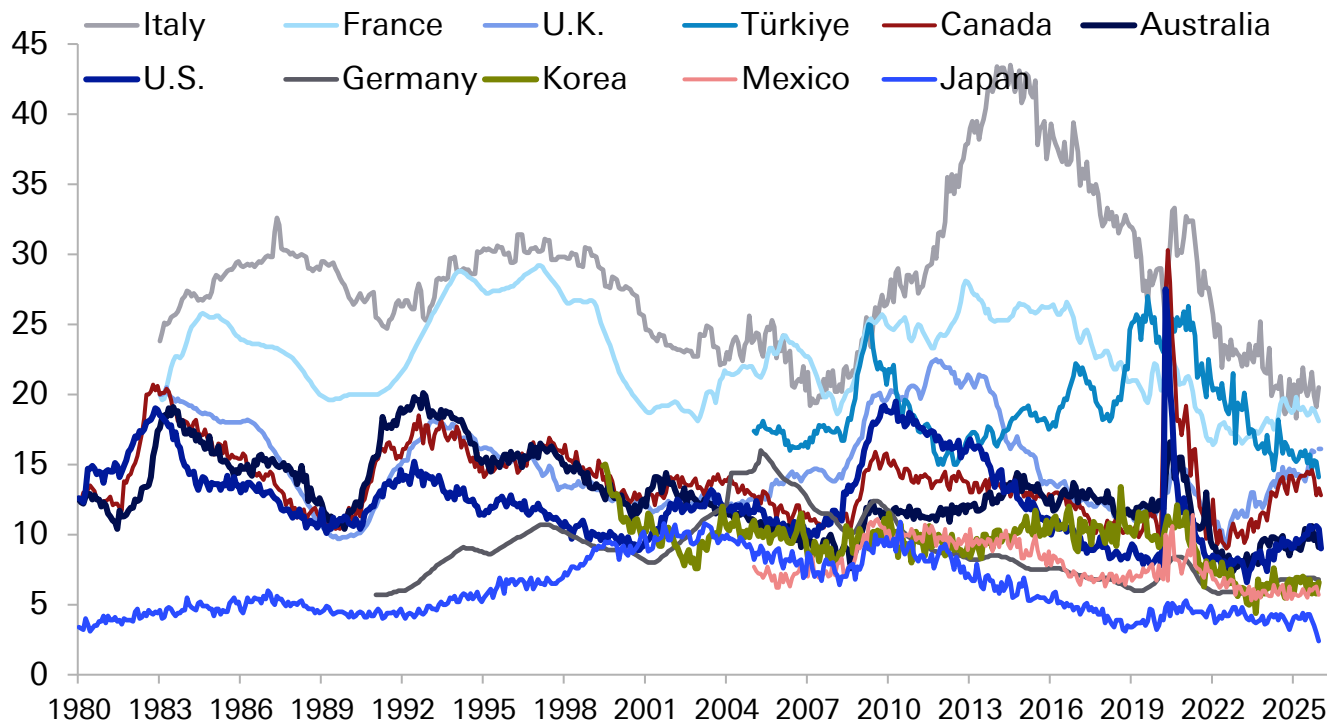


Source : Microsoft AI Economy Institute, OECD, Deutsche Bank, Note: AI Diffusion indices represent data for H2 2025

Although there are some countries seeing a pick-up in youth unemployment, perhaps due to AI, these increases are surely inconsequential versus structural or cyclical shocks like those seen in the last few decades. Surely if AI brings higher economic growth but some disruption, that will still be better for the young (and old) than weaker activity.

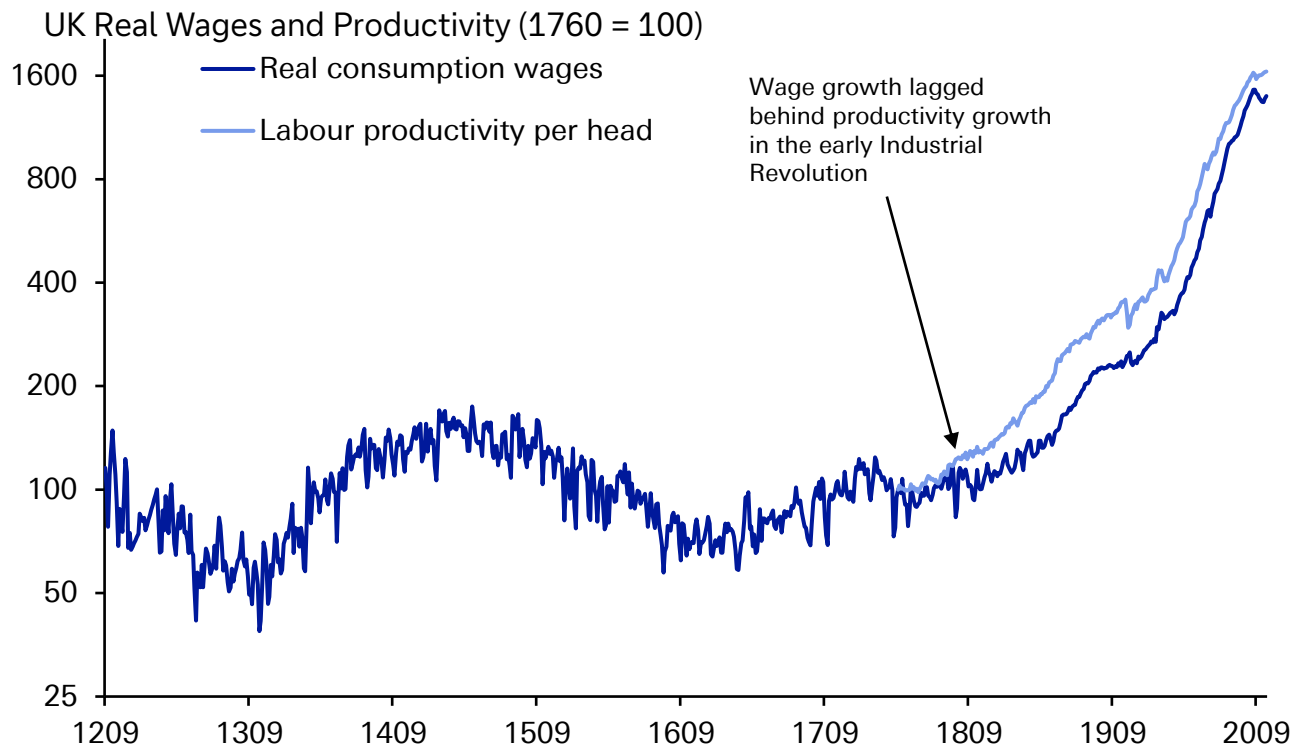


Harmonised youth unemployment rates (15-24yr (%)) - Legend ordered from current high to low.



Source: OECD, Haver, Deutsche Bank

Another issue is that productivity growth might exceed wage growth in an AI world. Indeed, this happened at the time of the first industrial revolution in the UK... the period from 1790-1840 has been dubbed Engels' pause, when wages stagnated despite the technological progress being made...

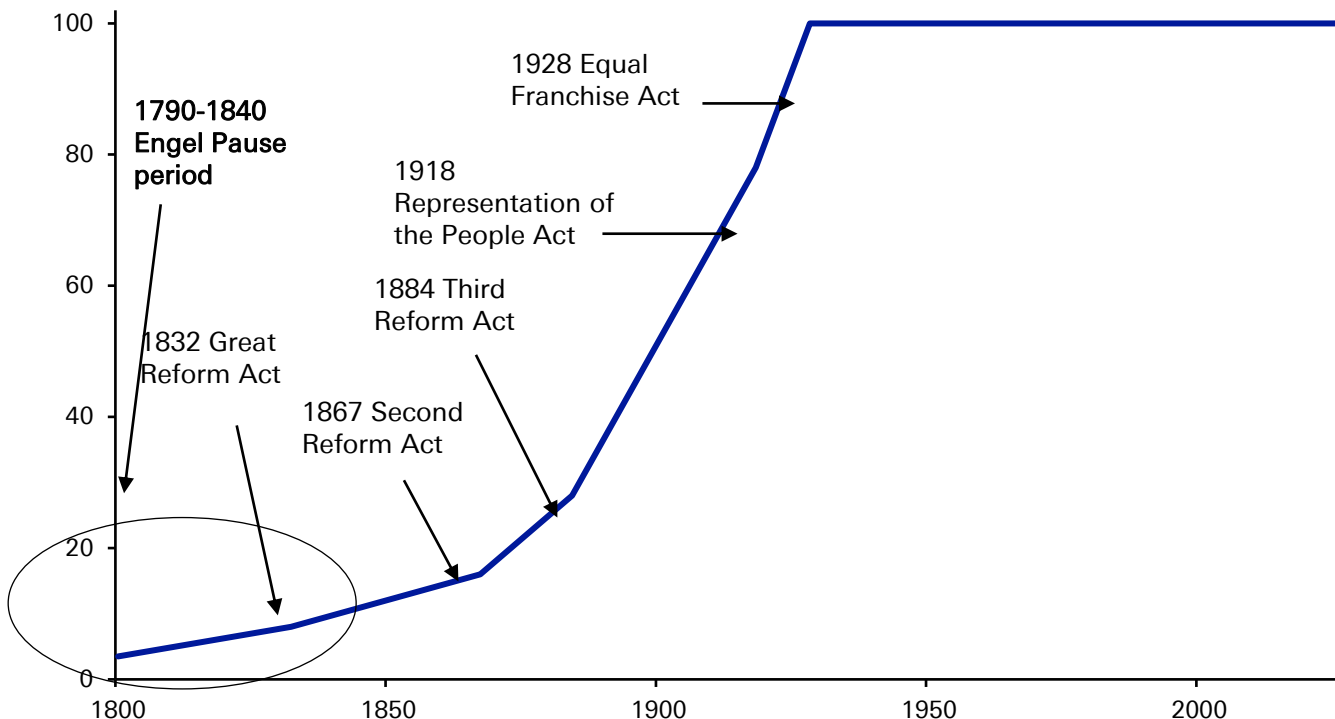


Source: Bank of England: Millennium of Macroeconomic Data, Deutsche Bank

However, a key difference back then was very few adults had the vote, so there weren't the same democratic pressures against rapid technological upheaval... Any major AI disruption to employment today will bring a huge democratic backlash....



### Percentage of UK adults with the right to vote



Source: House of Commons Library, UK Parliament, Deutsche Bank

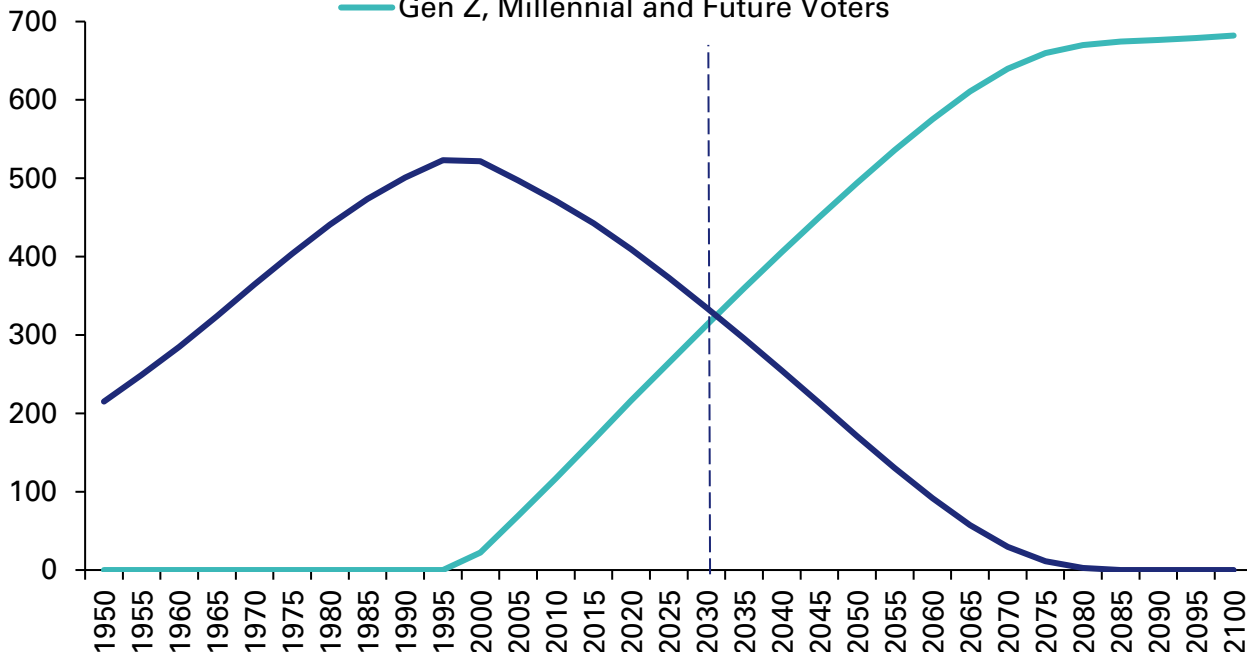


Today, a key difference is that all adults have a vote, including the young. At a G7 level, Millennial and younger votes are quickly catching up with those older. If they continue to get left behind, due to AI or other reasons (e.g. lack of home ownership), the likelihood is high they will elect politicians who protect their position through financial support, AI regulation or redistributive taxation.



## Voting Populations\* by Generations across the G7

Millions



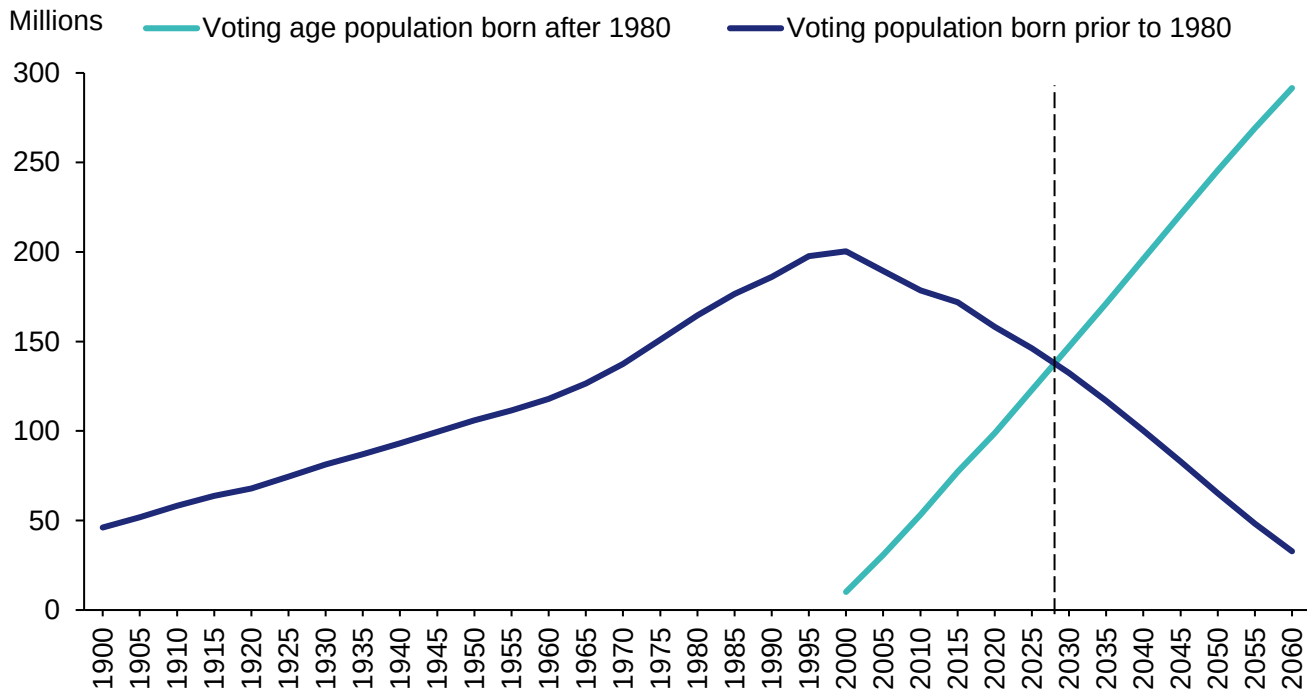
\*Voting population estimated from 17.5 years and older

Source : United Nations, Haver, Deutsche Bank

In the US, this Millennial and younger voter power shift occurs as soon as the 2028 election where they will be in the majority. Some of the extreme forecasts on AI forgets that the most-impacted countries are democracies where populism has either surged in recent years or is bubbling under the surface... It wouldn't take much for a significant political change and response...



## Voting Populations by Generations in United States

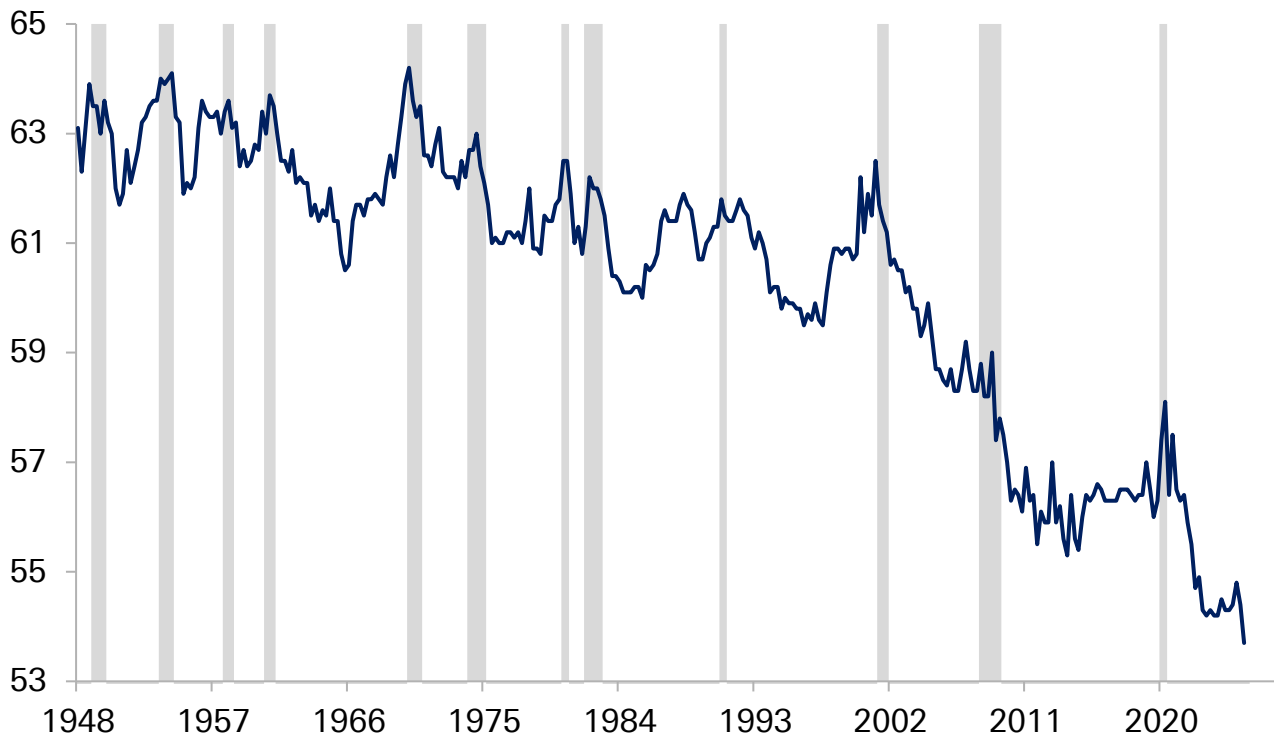


Source: US Census Bureau, Deutsche Bank

In the US, the labour share of income is already at all-time lows, so again it wouldn't take much to create a significant political backlash if AI further impacted labour...



### Total Economy: Labor share (%)

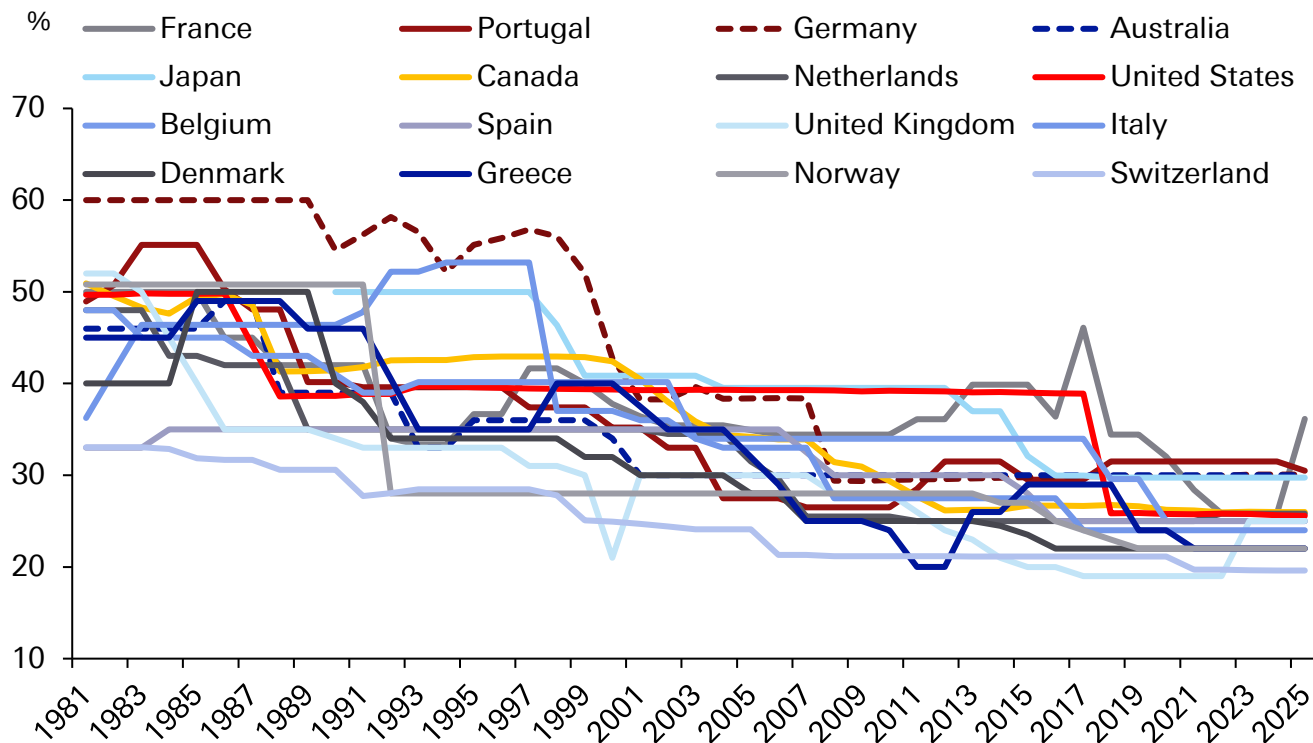


Source: BEA, Haver, Deutsche Bank

If AI did destroy jobs and boost profits, there is plenty of scope for corporate tax redistribution versus historical rates...



## Statutory corporate income tax rates (%)



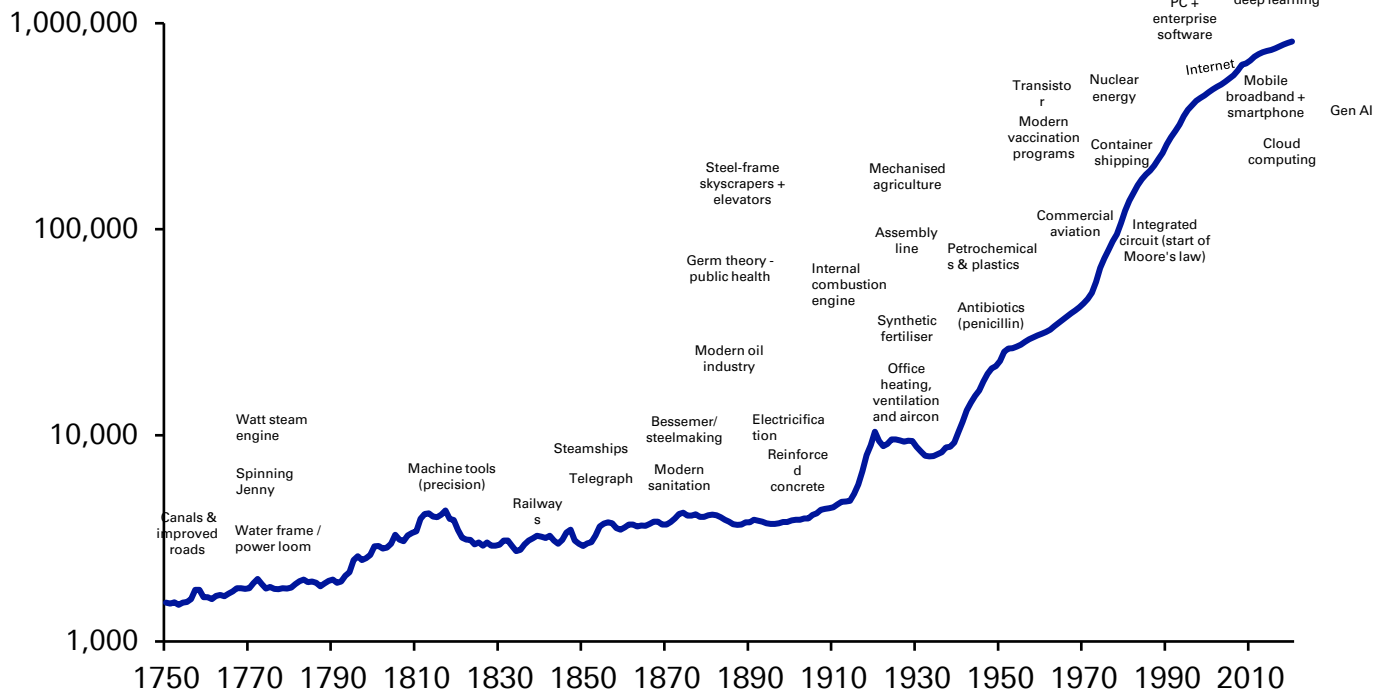
Source : OECD, Deutsche Bank, Countries are ranked from high to low tax rate in 2025

There are lots of fears that AI will be deflationary. Again, since the Industrial Revolution, humans have been remarkably innovative and yet inflation has surged. This is because “inflation is always and everywhere a monetary phenomenon” as Milton Friedman put it...



## Global Inflation since 1750

— Global Median Inflation Series

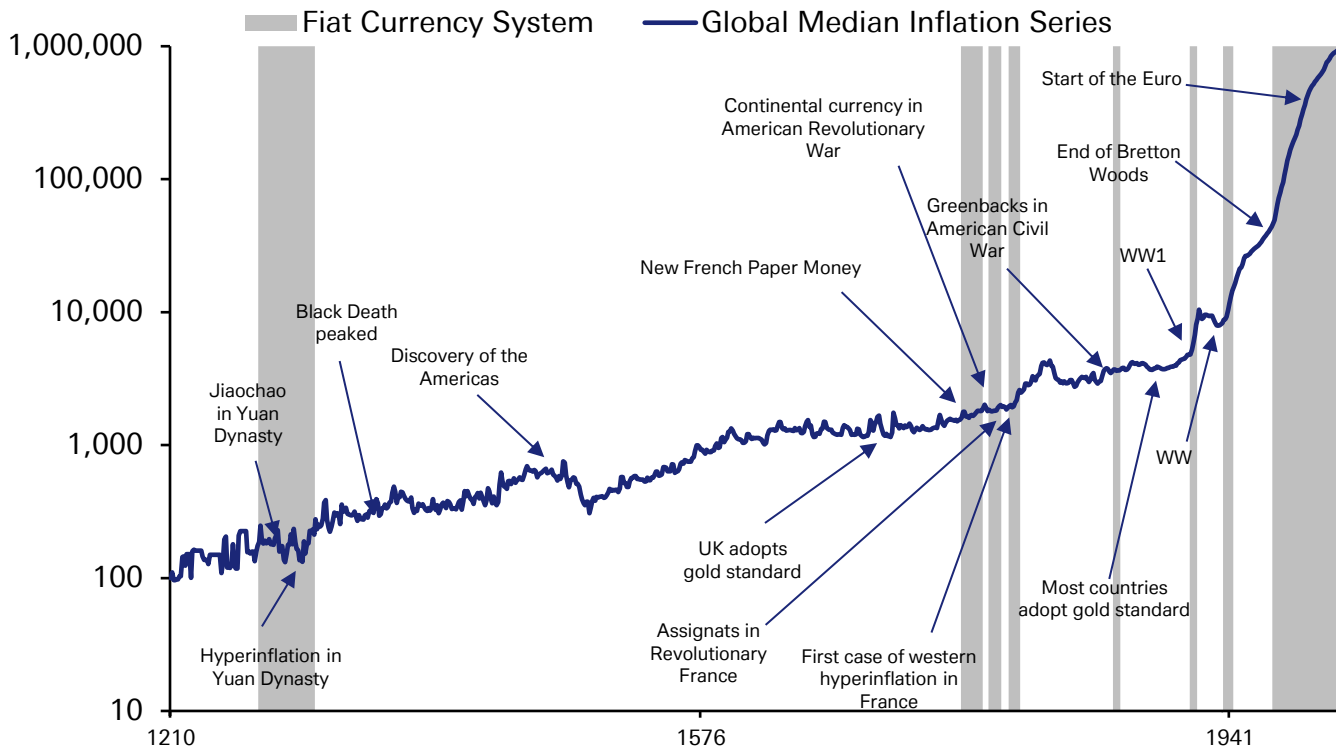


Source: Finaeon, Deutsche Bank

Fiat money regimes determine big picture inflation and not innovation. Even if AI causes large deflationary forces and unemployment, expect a huge monetary and fiscal response.



## Inflation through the history of Fiat Money since 1210



Source: Finaeon, Deutsche Bank

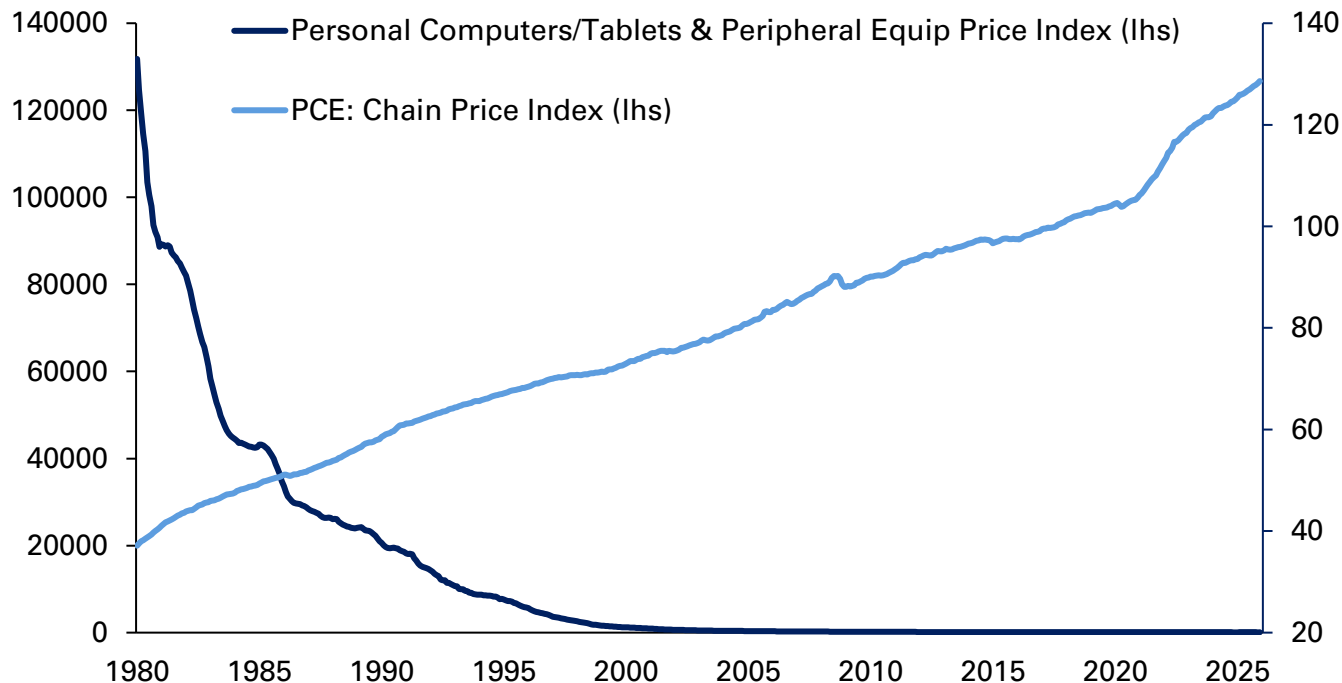
PC's have experienced enormous deflation over the last several decades... Yet actual inflation has increased substantially. Something to bear in mind when we hear how deflationary AI will be... In my view policy will be the main determinant of future inflation...



## Price Indices for Personal Computers/Tablets & Peripheral Equipment vs. General Inflation Measures (US PCE)

Index 2017=100

Index 2017=100



Source : BEA, BLS, Haver, Deutsche Bank.



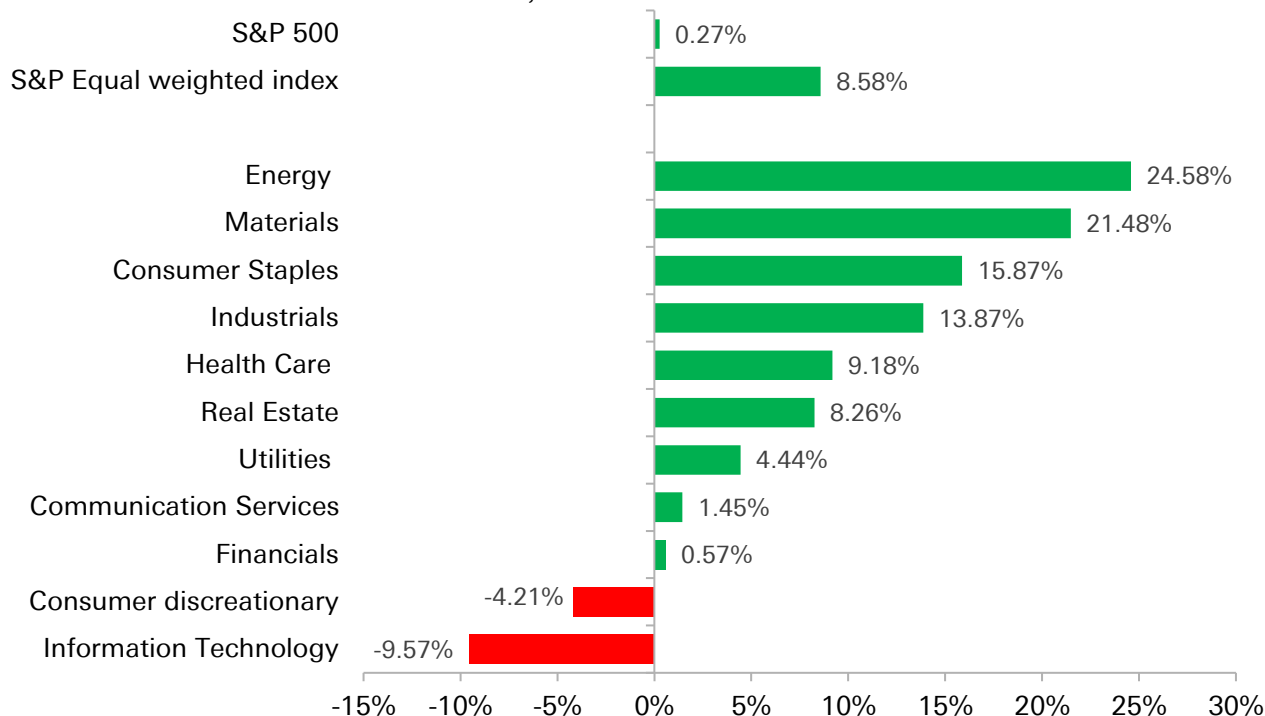
2026 so far: Rotation, dispersion and value



There's been a decent sized rotation in the US since the end of October. The S&P 500 has been broadly flat but nine out of eleven top level sectors are higher. However, the weight of tech has held back the market. Many other global markets are up strongly over this period. And the equal weight S&P 500 has performed well...



## S&P Sectors return since October 29, 2025

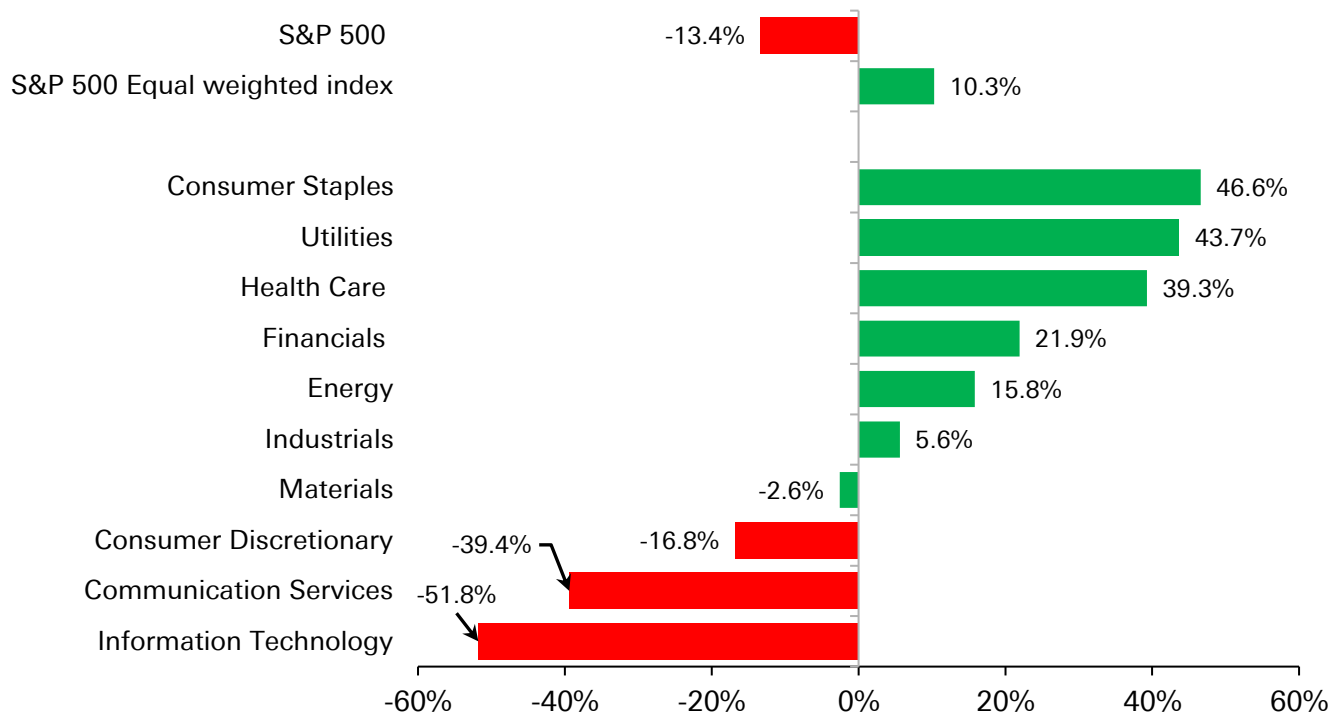


Source : Bloomberg Finance LP, Deutsche Bank., Last update : Feb 26

From the point the tech bubble burst in March 2000 until the end of 2000, we saw a huge rotation. The S&P 500 was fairly flat until September but then slipped into year-end under the weight of a continued tech sell-off even as defensives continued to do well...



## S&P Sectors return since March 27, 2000 to Dec 29, 2000

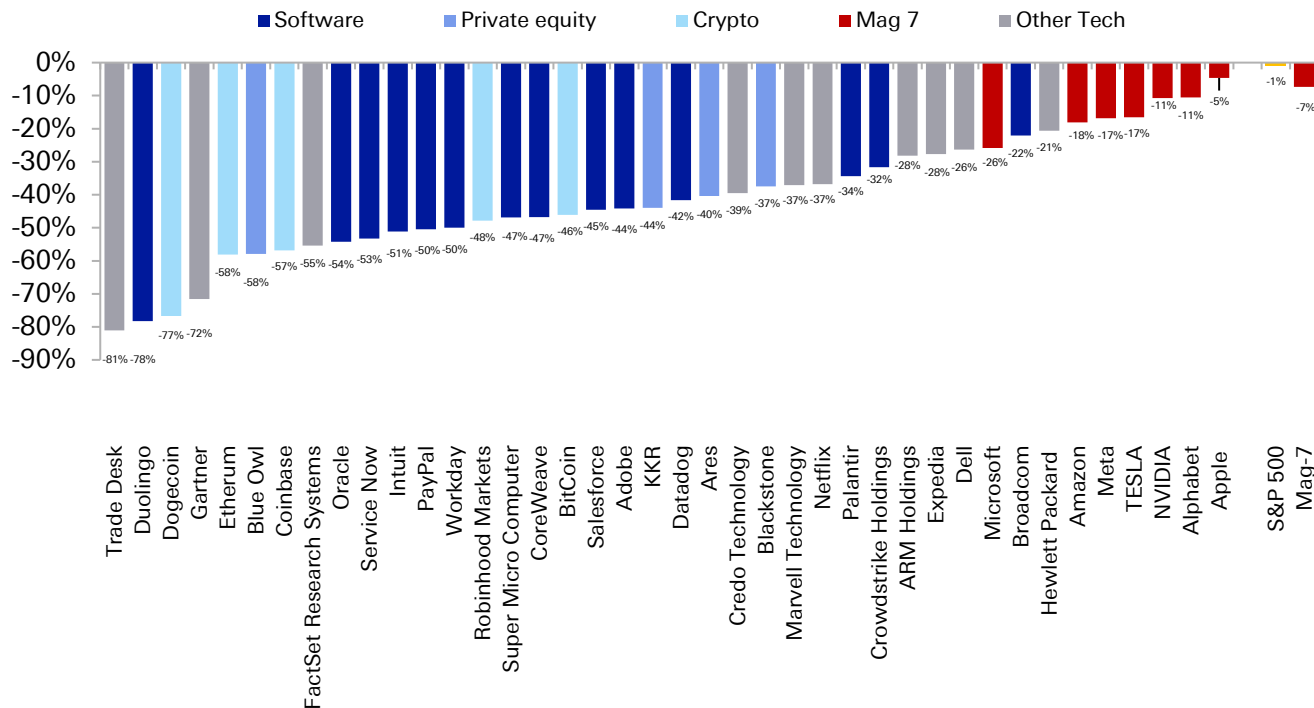


Source : Bloomberg Finance LP, Deutsche Bank

The 2026 drawdown in a lot of tech/AI stocks and related assets (e.g. crypto) has been pretty severe. It's been impressive given some of their weights in the index that the S&P 500 is only just below its all-time-highs.



## How Far Are a Selection of AI & Tech Stocks From Their highs? Drawdowns From 52 Week Highs for Selected US Tech and Related Stocks

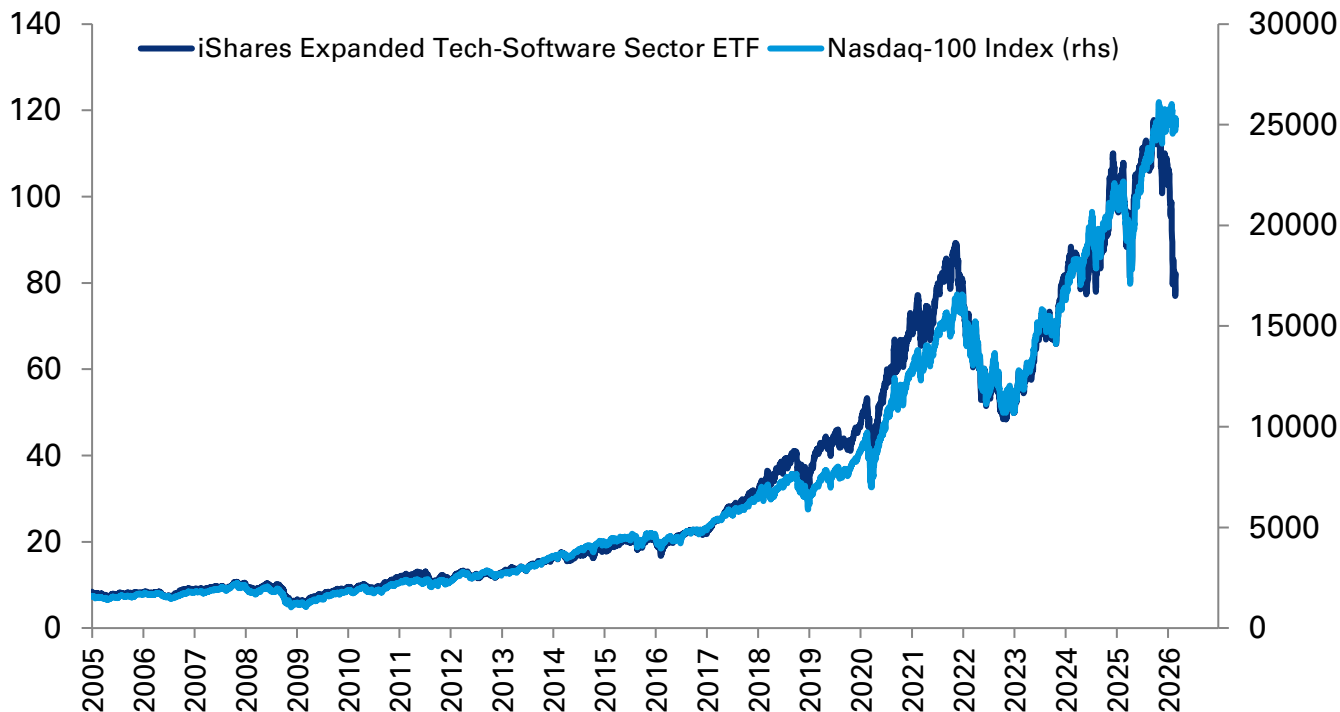


Source: Bloomberg Finance LP, Deutsche Bank, Last update: Feb 26

For two decades, tech and software have gone hand in hand. We now have winners and losers within tech and software has been a big loser in recent weeks for the first time...



## iShares Expanded Tech-Software Sector ETF vs. Nasdaq 100



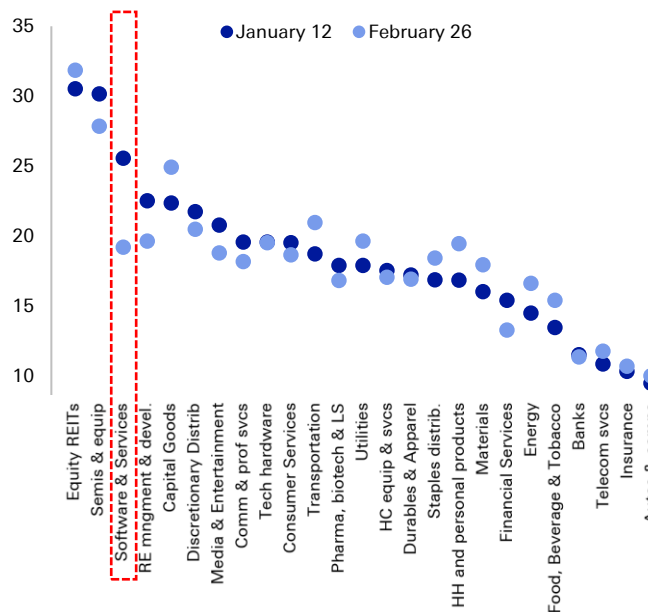
Source: Bloomberg Finance LP, Deutsche Bank

PE multiples: US software & services was the 3rd most expensive industry group in January, it's now 10th (3rd to 12th in Europe), down by -6.3x (4.8x). No other industry group has seen a similar multiple re-rating.

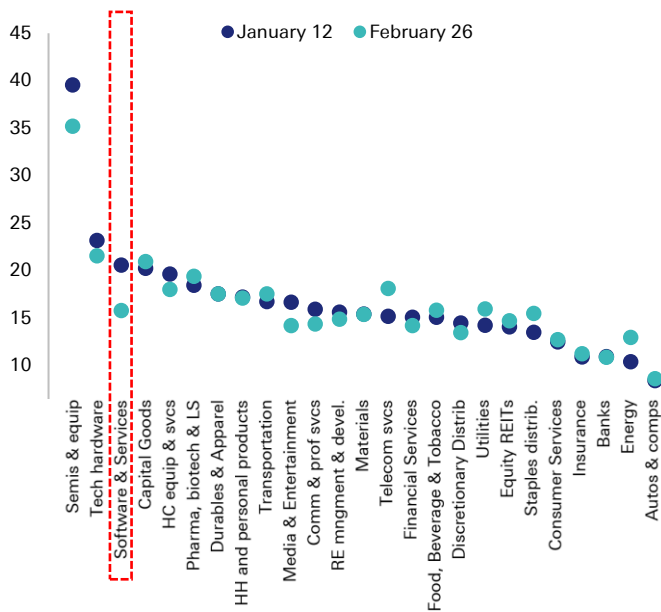


## Median blended-forward P/E multiple by industry group

Median forward P/E multiple by industry group, Russell 1000



Median blended-forward P/E multiple by industry group, Stoxx 600

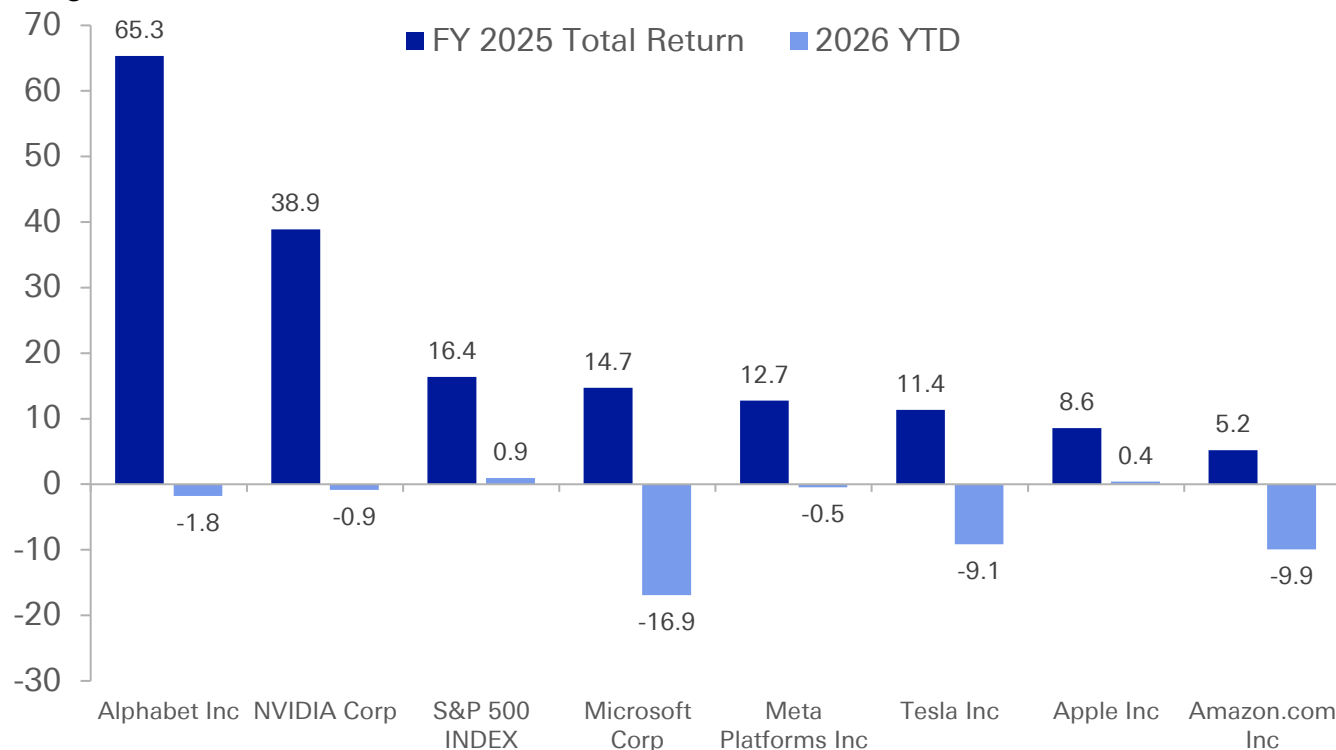


Source: Bloomberg Finance LP, Deutsche Bank

All of the Mag-7 have underperformed the S&P 500 in 2026 so far, albeit with four only just below.



## Mag 7 Total Returns

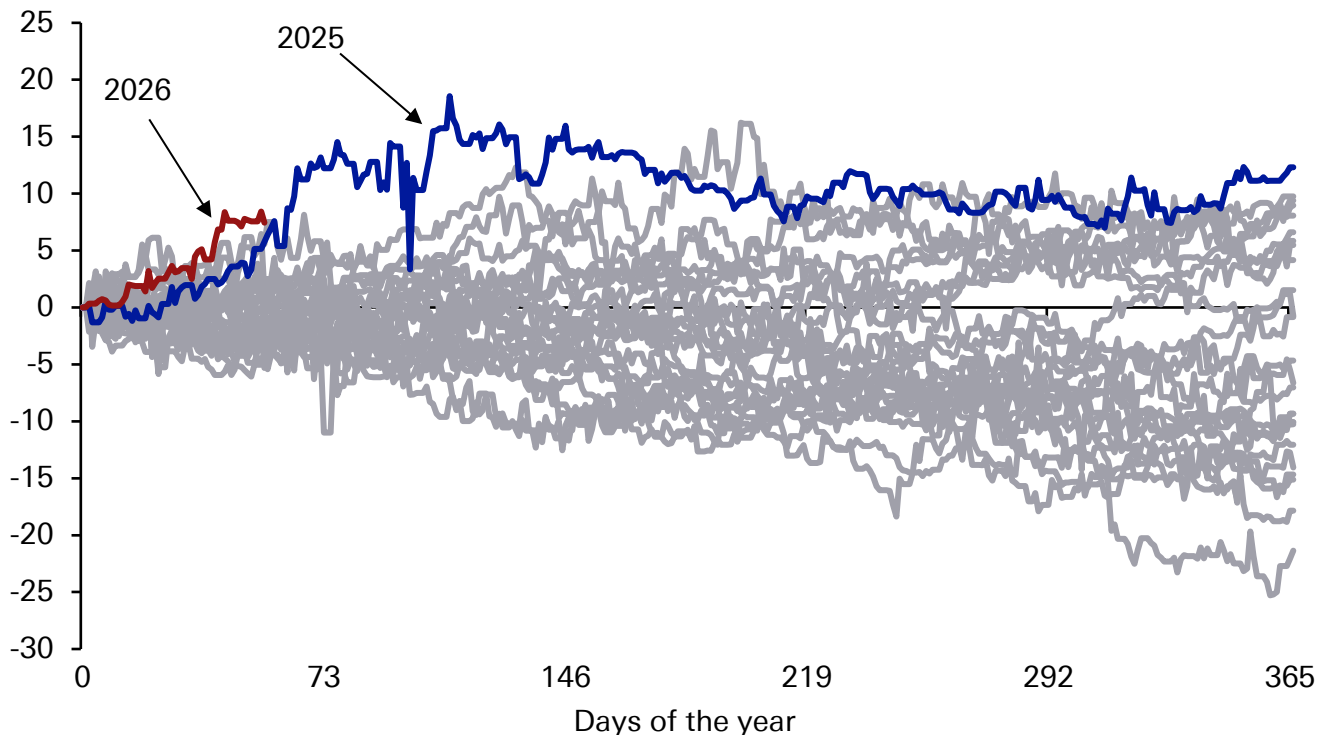


Source : Bloomberg Finance LP, Deutsche Bank., Last update Feb 26

After years of US exceptionalism in equity performance, US equities have seen a clear underperformance both in 2025, and again in 2026...



Outperformance (percentage points) between the MSCI World ex US index and the MSCI US Index since 2000

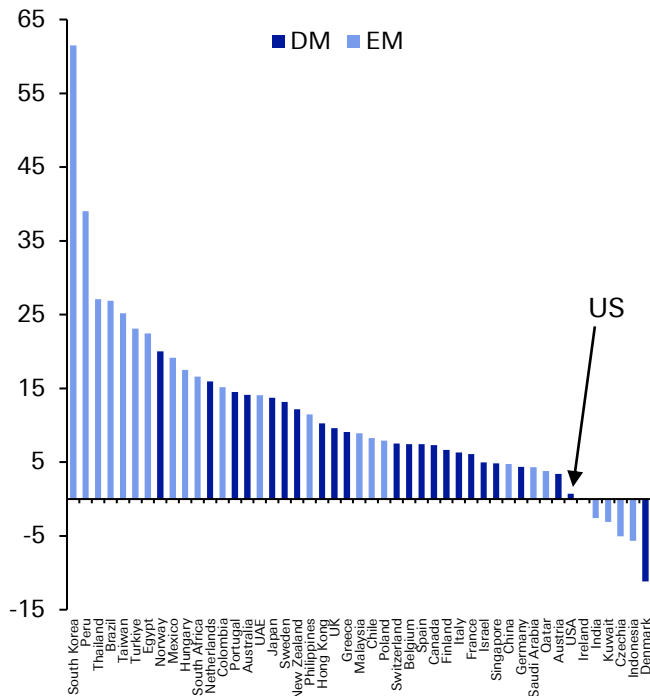
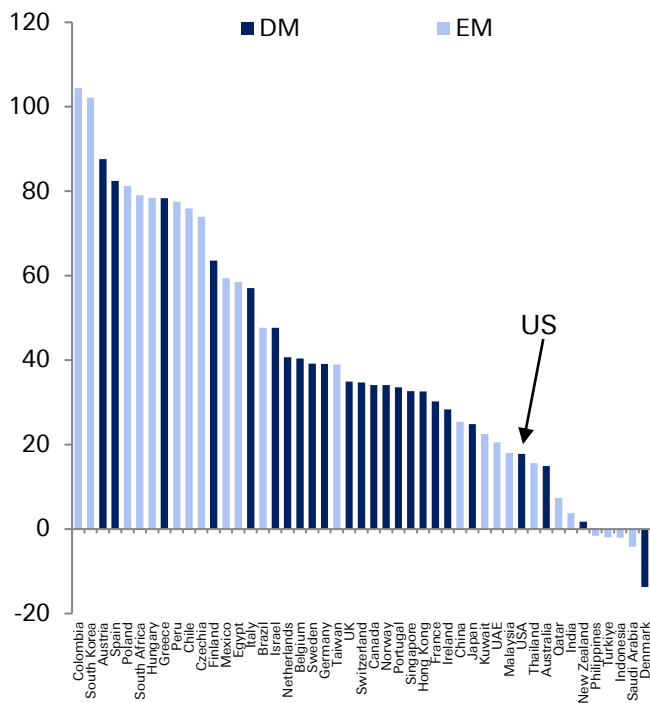


Source: Bloomberg Finance LP, Deutsche Bank

2025 was a good year for global equities, especially EM, but with the US around the bottom quartile. So far in 2026 the pattern seems to be repeating itself...



Bloomberg large/mid-cap country stock market indices: 2025 total return in USD (%) (left) and 2026 YTD (right)



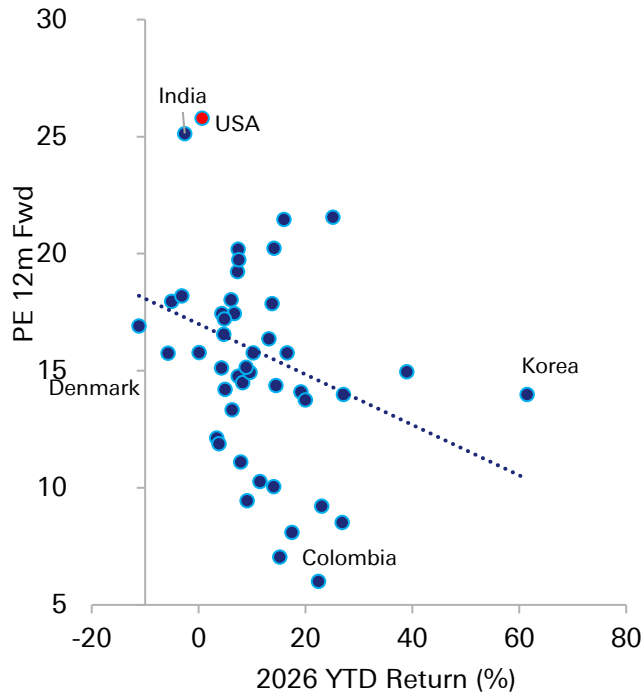
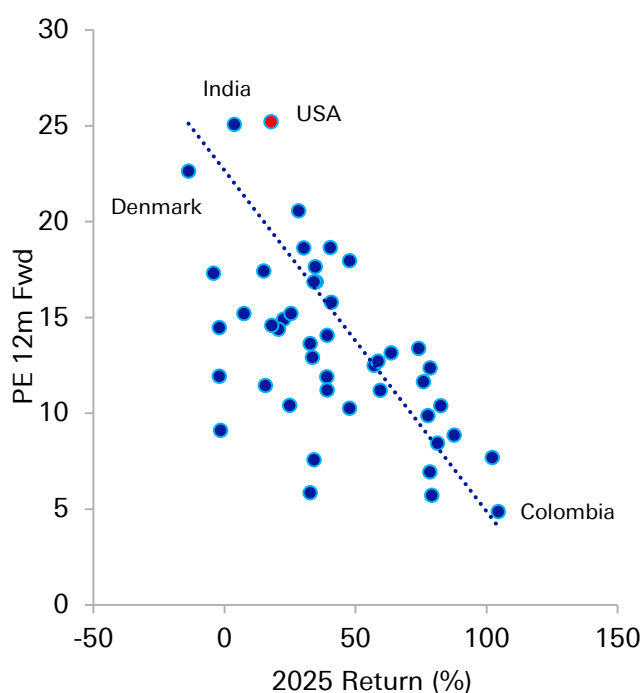
Source: Bloomberg Finance LP, Deutsche Bank, Note: 2026 YTD is updated through Feb 26



2025 Global Equity returns saw a strong correlation to starting valuations. It's early days but value seems to have been a good guide to global equity returns in January 2026 as well...



Bloomberg large/mid-cap country stock market indices, 2025 total return in USD vs. 12m fwd PE estimates at the end of 2024 (left) & Bloomberg large/mid-cap country stock market indices, 2026 YTD return in USD vs. 12m fwd PE estimates at the end of 2025 (right)

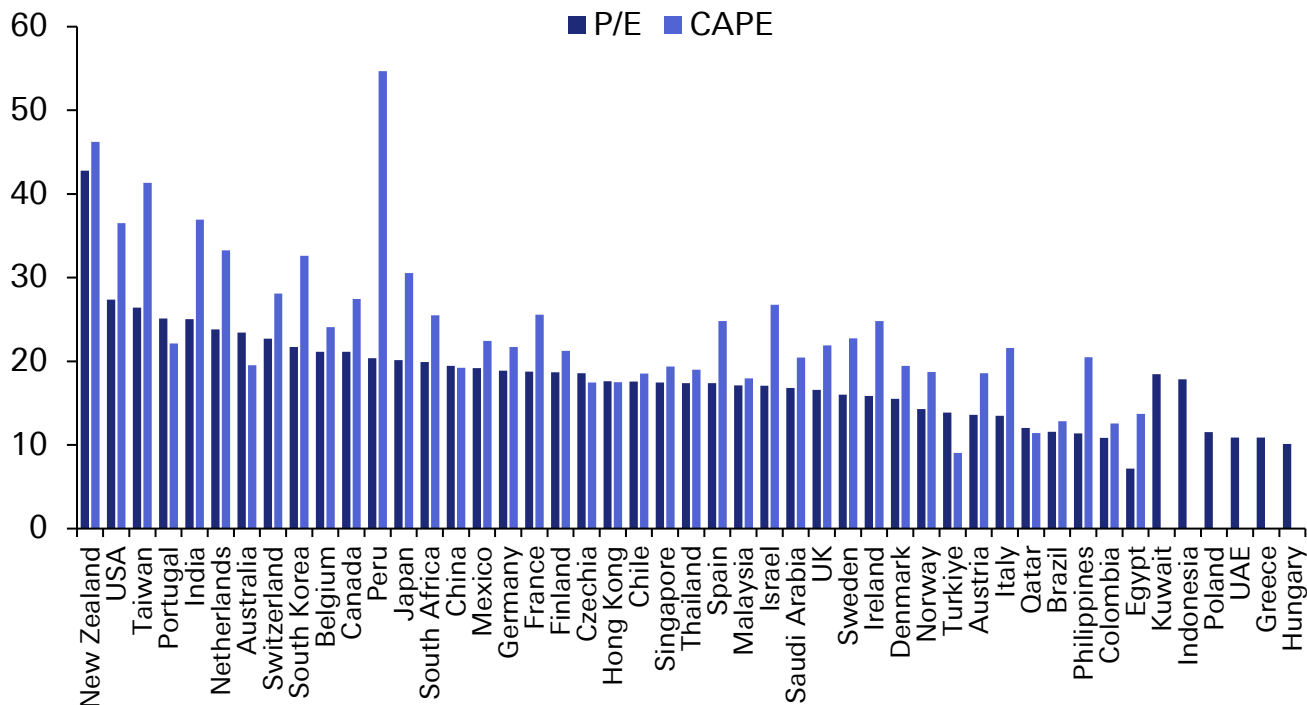


Source: Bloomberg Finance LP, Deutsche Bank Note: 2026 YTD is updated through Feb 26

If valuation is back as the key equity driver this is the high to low of Global PE ratios



## Latest global valuation ratios



Source: Bloomberg Finance LP, Deutsche Bank

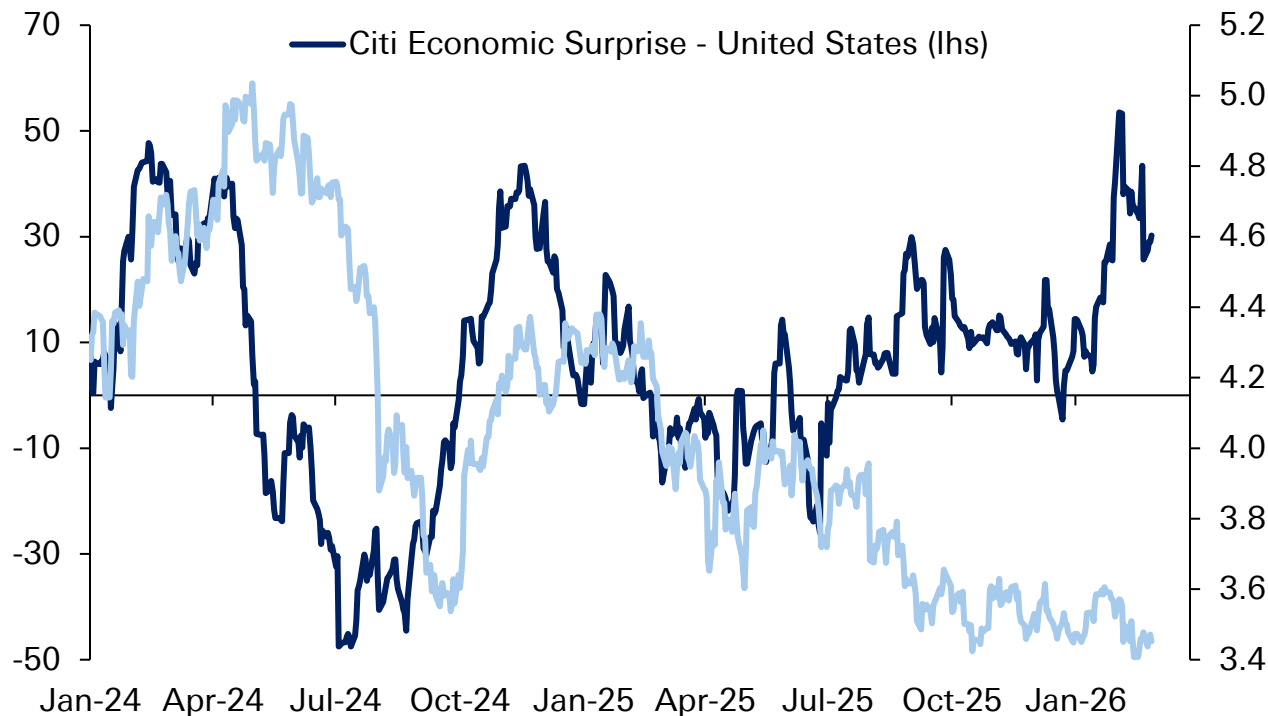


Does the US really need a rate cut... unless  
tech melts

US data surprises have been at the upper end of their 2-year range in 2026 so far. It doesn't suggest lower Fed Funds or lower yields at face value...



## Economic Surprises vs. 2yr US bond

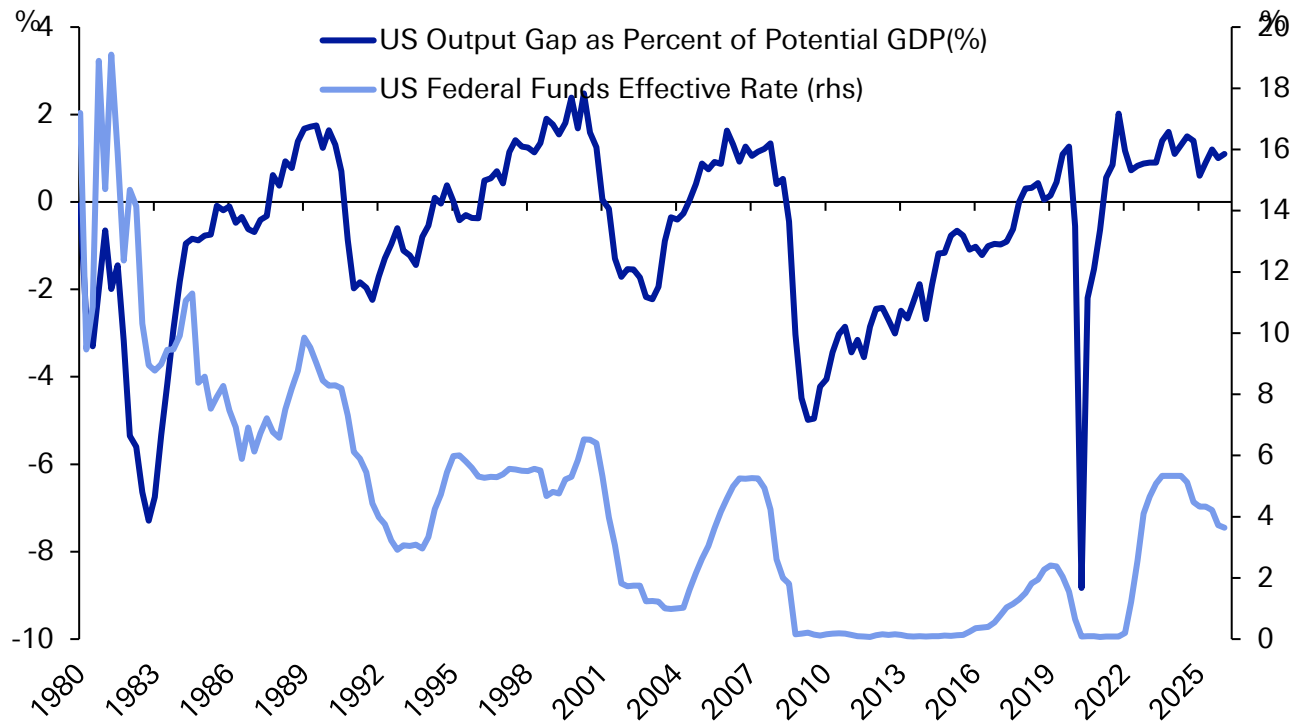


Source: Bloomberg Finance LP, Deutsche Bank

The US output gap remains firmly positive, again suggesting the economy is more running hot than cold.....



## US Output Gap and Fed funds rate

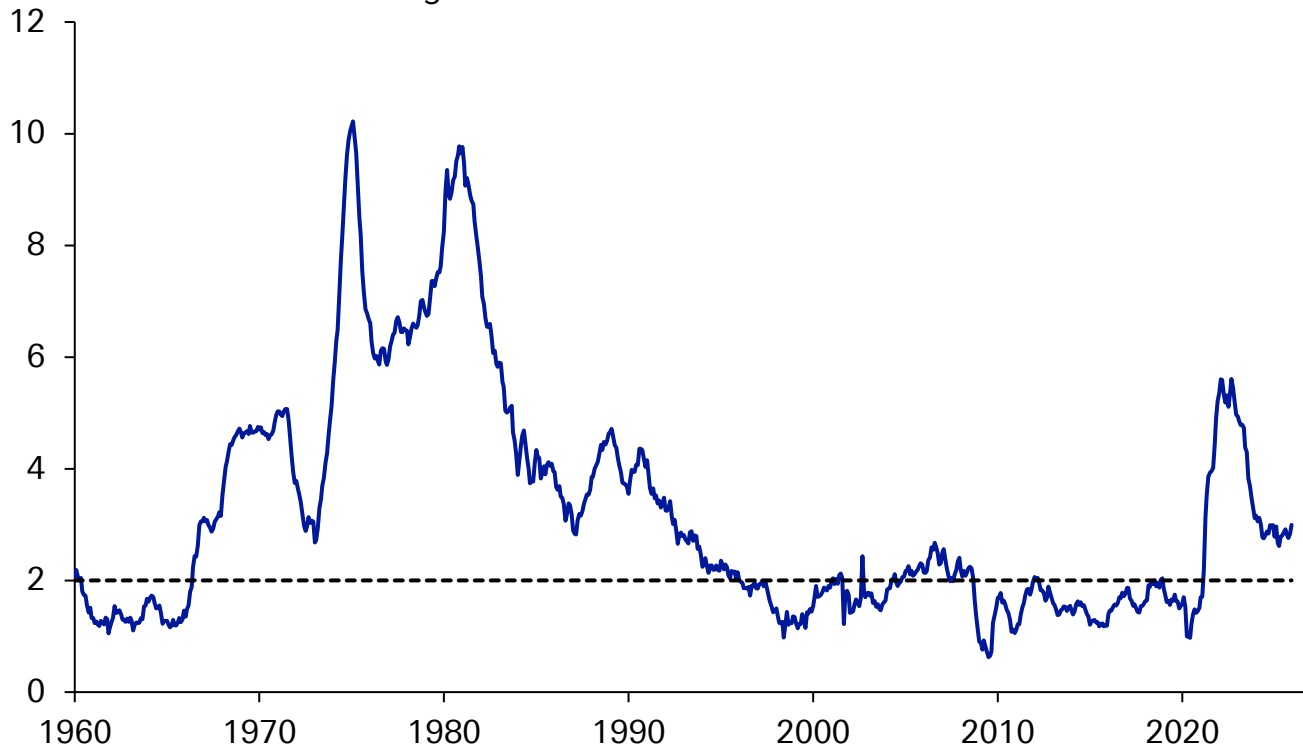


Source: CBO, FRB, Bloomberg Finance LP, Haver, Deutsche Bank

Core PCE is back up to 3.0%, and in March it'll have been above 2% for 5 full years (above 2.6% in fact)... the last time it went above 2% for 5 years, it stayed there for almost 30 in total...



## Core PCE : 12-month %Change



Source: BLS, Haver, Deutsche Bank.

Moreover, commodities have seen a huge upswing in recent months, and consensus forecasters expect US inflation to linger above target into 2027...



## Bloomberg Commodity Spot Index



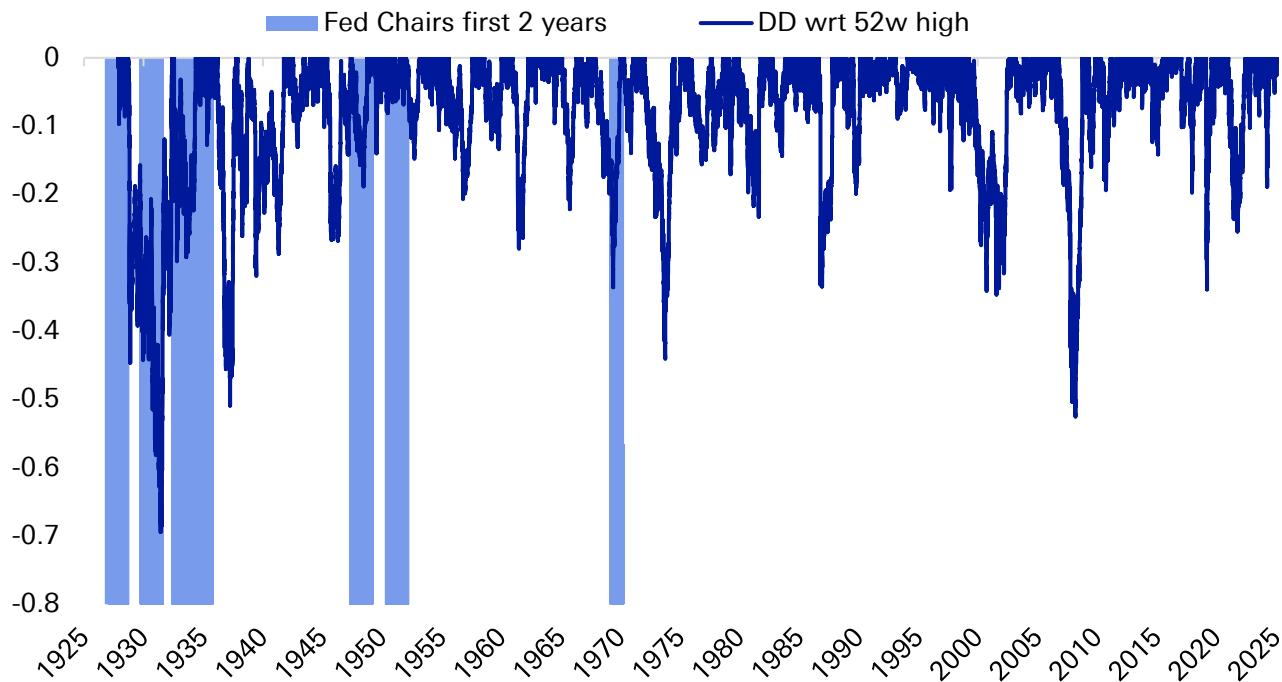
Source: Bloomberg Finance LP, Deutsche Bank

There is a perception in markets that new Fed Chairs tend to be tested soon into their tenure. However the evidence is mixed versus all history. The 1987 crash occurred very early in the Greenspan era, the GFC themes started early in the Bernanke era, but the Lehman default was 2.5 years after he started. Powell saw a very challenging late-2018 in his first year, but then Covid happened around 2 years into his term and was an outside event.



S&P 500 drawdowns from 52-week high. Blue shaded areas the first two years of a new Fed Chair's term

S&P 500 price drawdowns wrt 52w high



Source : Finaeon, Deutsche Bank





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Deutsche Bank  
Research

# The AI sell-off: Over or underdone?

March 2026

Luke Templeman | Galina Pozdnyakova

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank  
Research Institute



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# This sell-off is not the first ... but the frequency of new AI tools has accelerated in recent months



## Previous years ...

- 1) Mid-late 2022: Smaller AI firms find it difficult to attract capital during a broader risk-off move.
- 2) 2023: The 'first wave' of AI disruption fears. These typically included labour-based businesses providing simple, repetitive labour-based services – such as call centres and other offshored business process outsourcing businesses
- 3) 2024: Various small sell-offs which combined technology valuation issues with broader market concerns
- 4) Jan 2025: The release of a low-cost AI model by China's DeepSeek rattles US markets
- 5) Sept 2025: Fears grow in the software industry and reverberate around private capital managers with heavy exposure.
- 6) Oct 2025: Concerns grow for companies in the broader tech space including those that use platforms that investors fear might be replicated or disrupted by AI.

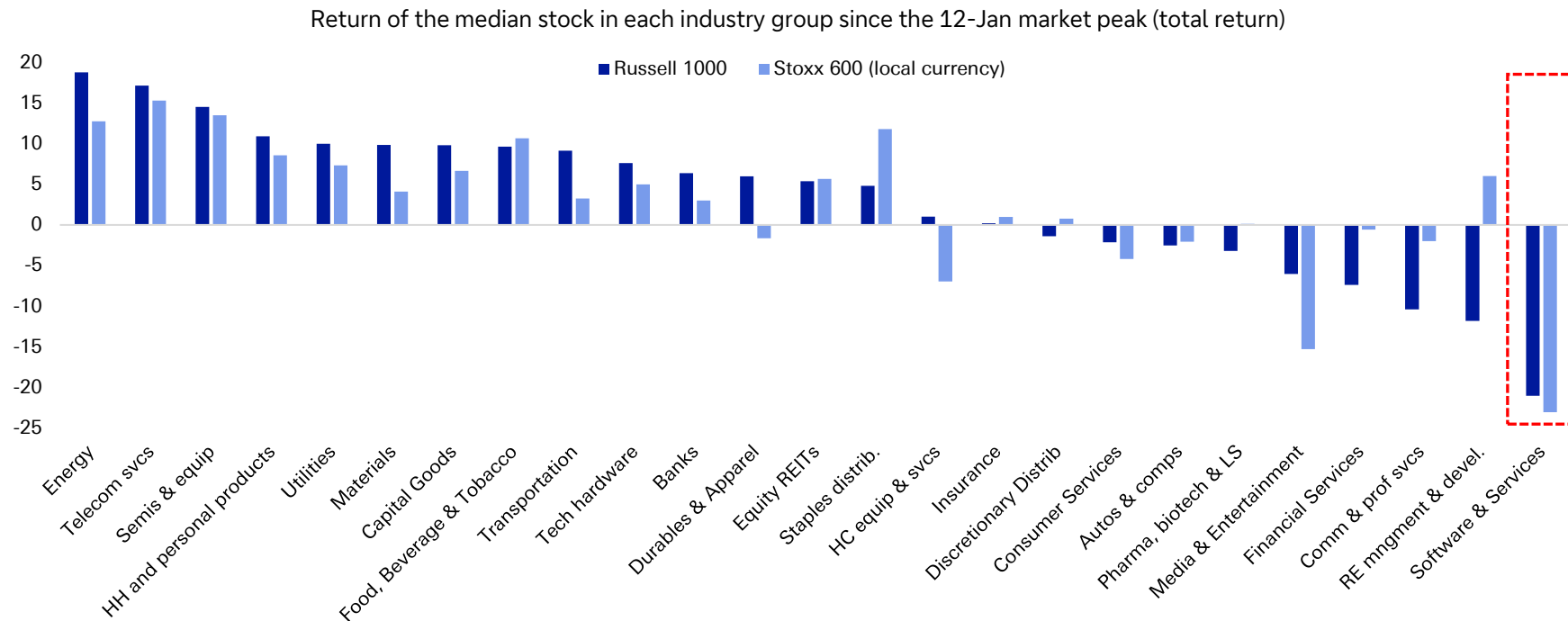
## 2026 (a small selection)

- 1) Jan: Google announces new tools for various products
- 2) Feb: Anthropic announces a variety of new AI tools
- 3) Feb: Algorhythm announced a new tool that helps companies scale their freight shipments
- 4) Feb: OpenAI announces 'Frontier', an enterprise platform for AI agents
- 5) Feb: Microsoft announces Agentic AI to automate many retail functions

### Points to note:

- The most recent sell-off is quite distinct and includes a much broader view about how AI will impact businesses and their markets
- The sell-off reflects a view that current economic structures will change meaningfully as a result of the latest AI products being released.
- Many affected firms are network-based companies whose balance sheet assets are largely intangible, and whose services are largely intangible
- There has been a rotation towards companies with real assets and goods (eg defence, commodities, banks, utilities, capital goods sectors).

## Software & services has been the worst performing industry group in both the US and Europe since the middle of January

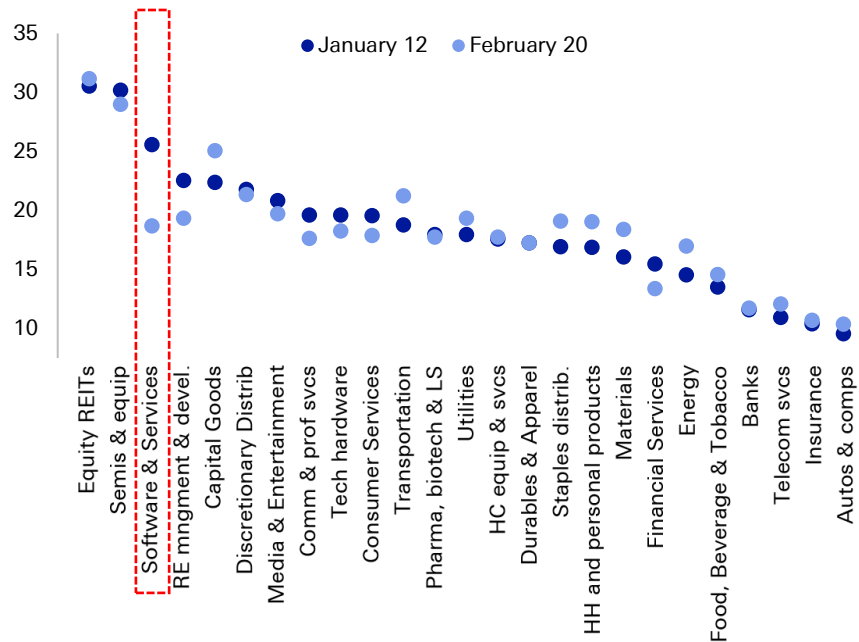


Source: Bloomberg Finance LP, Deutsche Bank.

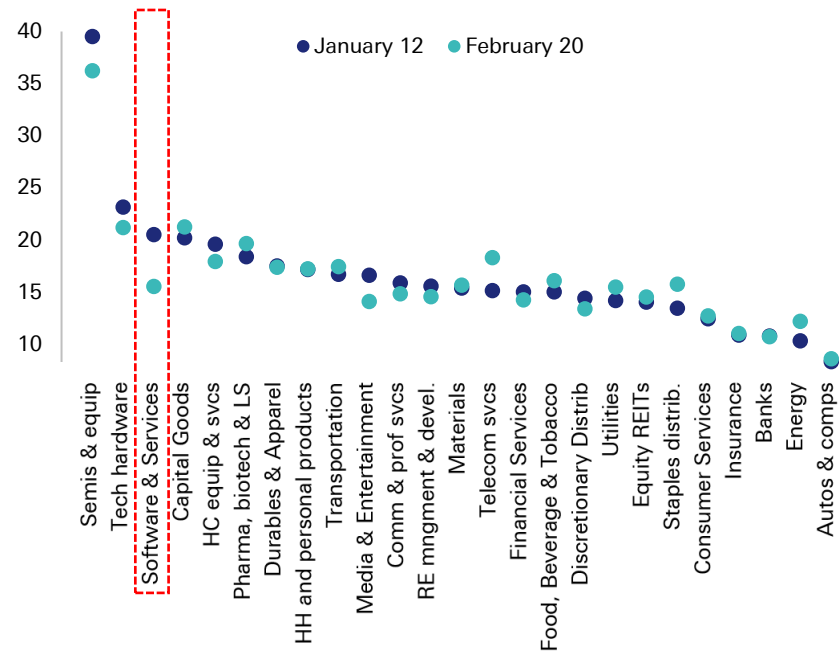
PE multiples: software & services used to be the 3<sup>rd</sup> most expensive industry group, it's now 9<sup>th</sup> (3<sup>rd</sup> to 13<sup>th</sup> in Europe), down by 5.8x (5x). No other industry group has seen a similar multiple re-rating.



Median blended-forward P/E multiple by industry group, Russell 1000



Median blended-forward P/E multiple by industry group, Stoxx 600

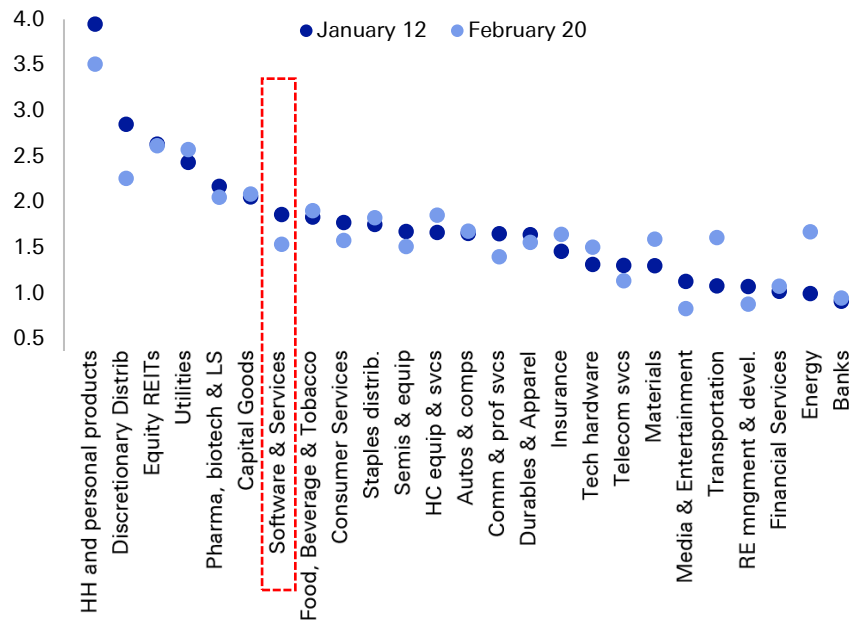


Source: Bloomberg Finance LP, Deutsche Bank.

It's a similar story for the PEG ratios. Although data is less available, software & services in the US slid from 7<sup>th</sup> to 17<sup>th</sup> most expensive industry group (3<sup>rd</sup> to 15<sup>th</sup> in Europe)

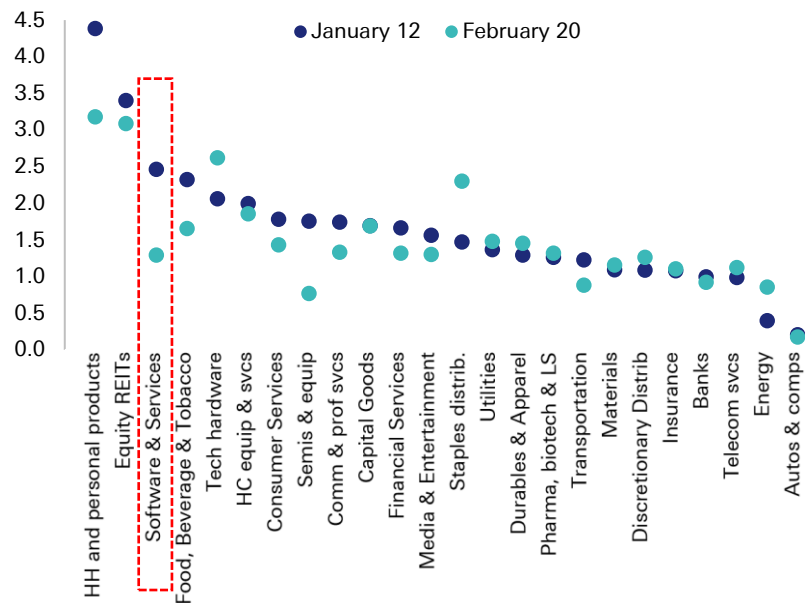


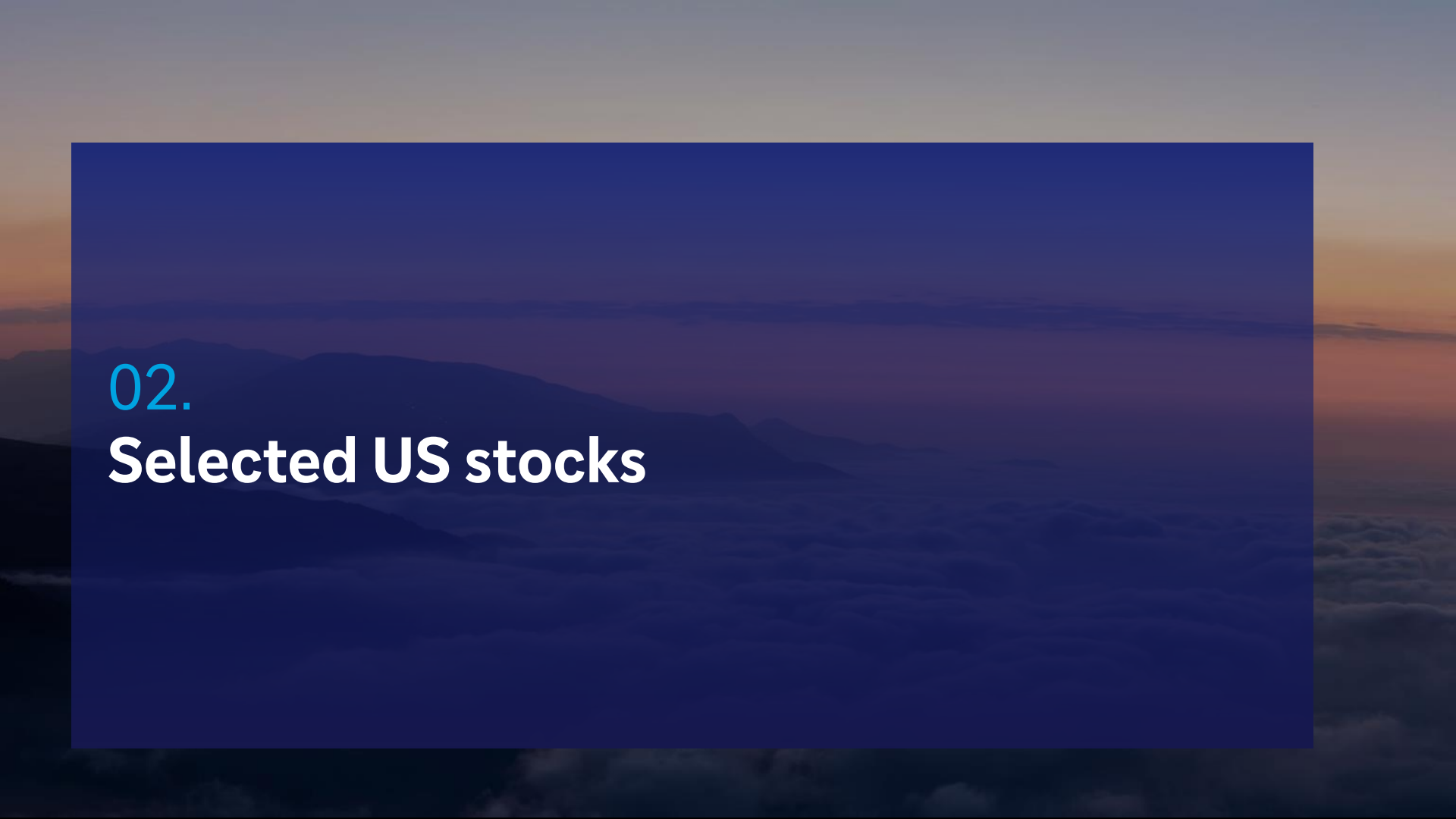
Median blended-forward PEG multiple by industry group, Russell 1000



Source: Bloomberg Finance LP, Deutsche Bank.

Median blended-forward PEG multiple by industry group, Stoxx 600





02.

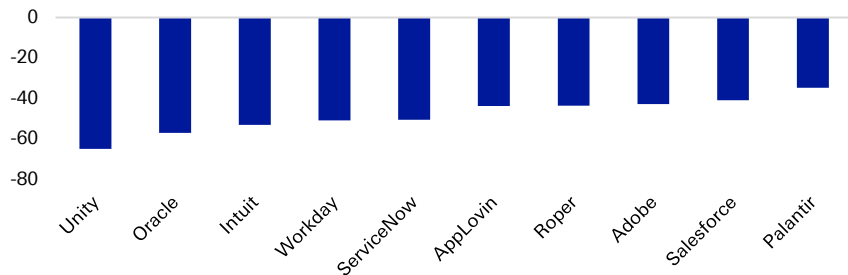
## **Selected US stocks**



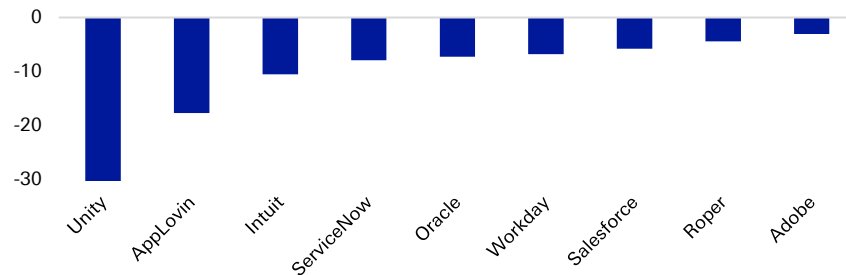
Sector focus: Software – This group had by far the largest number of battered firms, with the ten below falling by at least 30% despite being at least \$20bn market cap firms at the start of the rout. All have delivered at least 10% p.a. sales growth over the last 3 years and now trade at least 25% below analysts' target price.



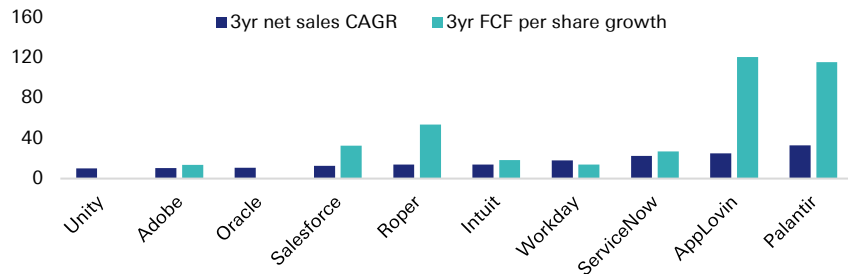
Software: price change from 52 week high



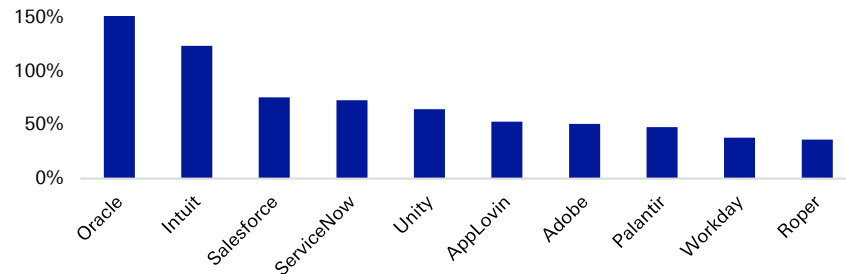
Software: change in P/E (since the 12-Jan market peak)



Software: fundamentals growth (%)



Software: upside (downside) to analyst target price\*



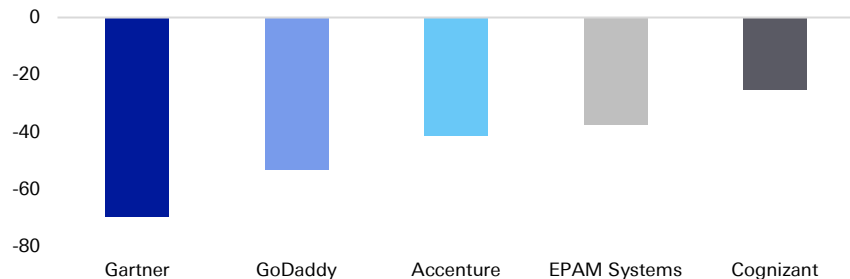
Source: Bloomberg Finance LP, Deutsche Bank.

\* Upside is relative to Deutsche Bank Research's price target if available. If not, Bloomberg consensus expectations are used.

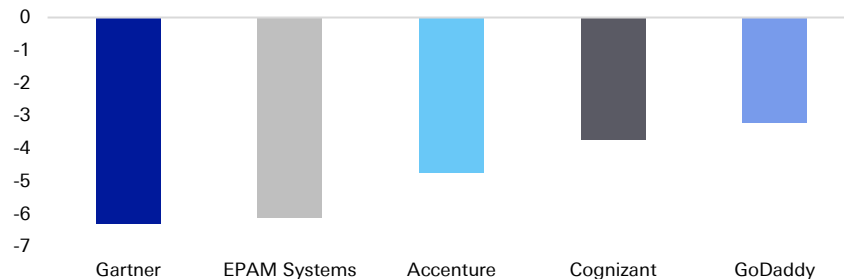
Sector focus: IT Services – Smaller and more specialised firms have also fallen but less indiscriminately and upsides to analyst targets are smaller. They have also delivered decent sales and free cash flow growth recently.



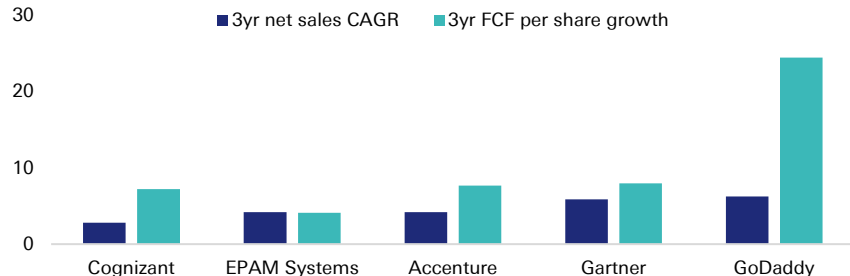
IT services: price change from 52 week high



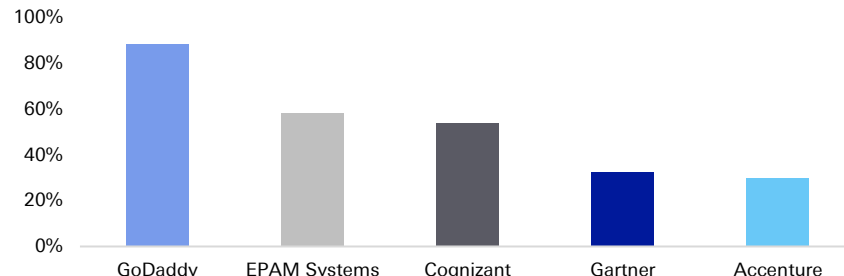
IT services: change in P/E (since the 12-Jan market peak)



IT services: fundamentals growth (%)



IT services: upside (downside) to analyst target price\*



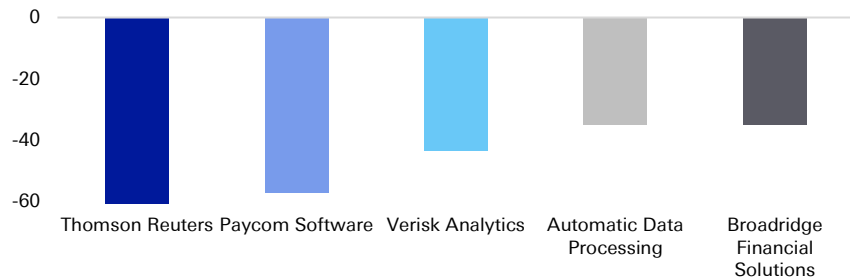
Source: Bloomberg Finance LP, Deutsche Bank.

\* Upside is relative to Deutsche Bank Research's price target if available. If not, Bloomberg consensus expectations are used.

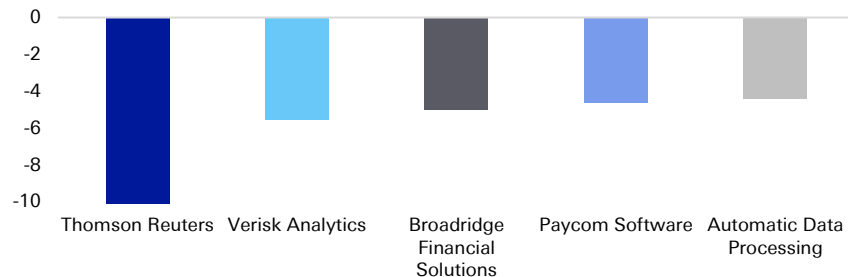
Sector focus: Professional services – It's been a similar picture for this industry group, but with weaker upside potential based on analyst targets. So, despite the spillover to other service companies, this implies that the software industry has been disproportionately hit.



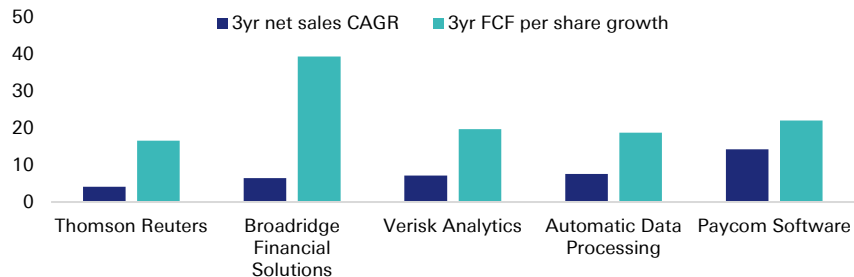
Professional services: price change from 52 week high



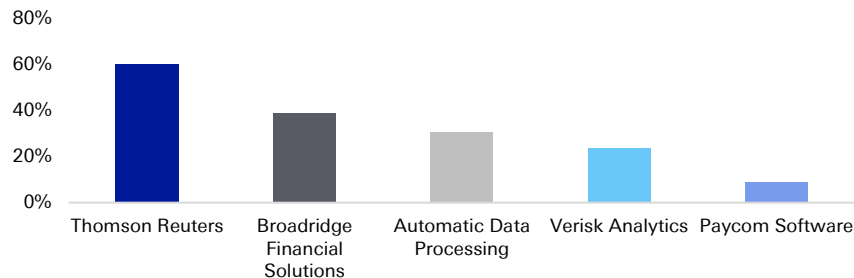
Professional services: change in P/E (since the 12-Jan market peak)



Professional services: fundamentals growth (%)



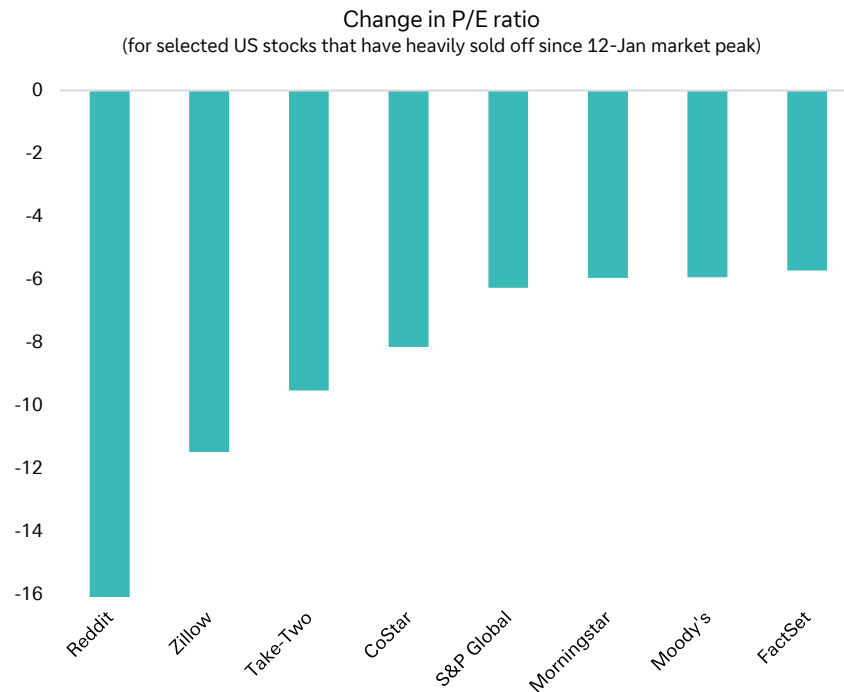
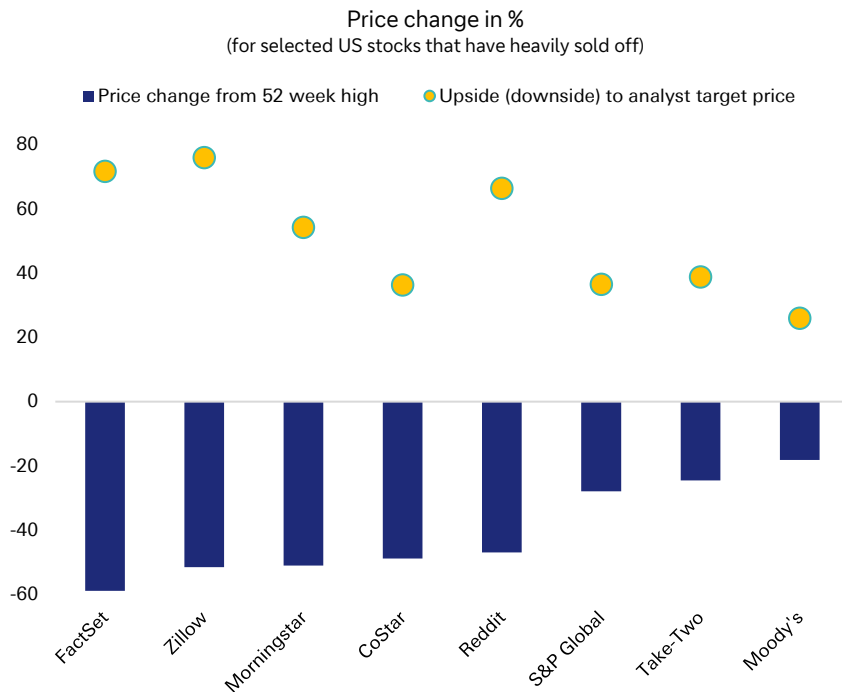
Professional services: upside (downside) to analyst target price\*



Source: Bloomberg Finance LP, Deutsche Bank. Note: Thomson Reuters is a Canadian company and is not part of the Russell 1000 index but is included for relevance

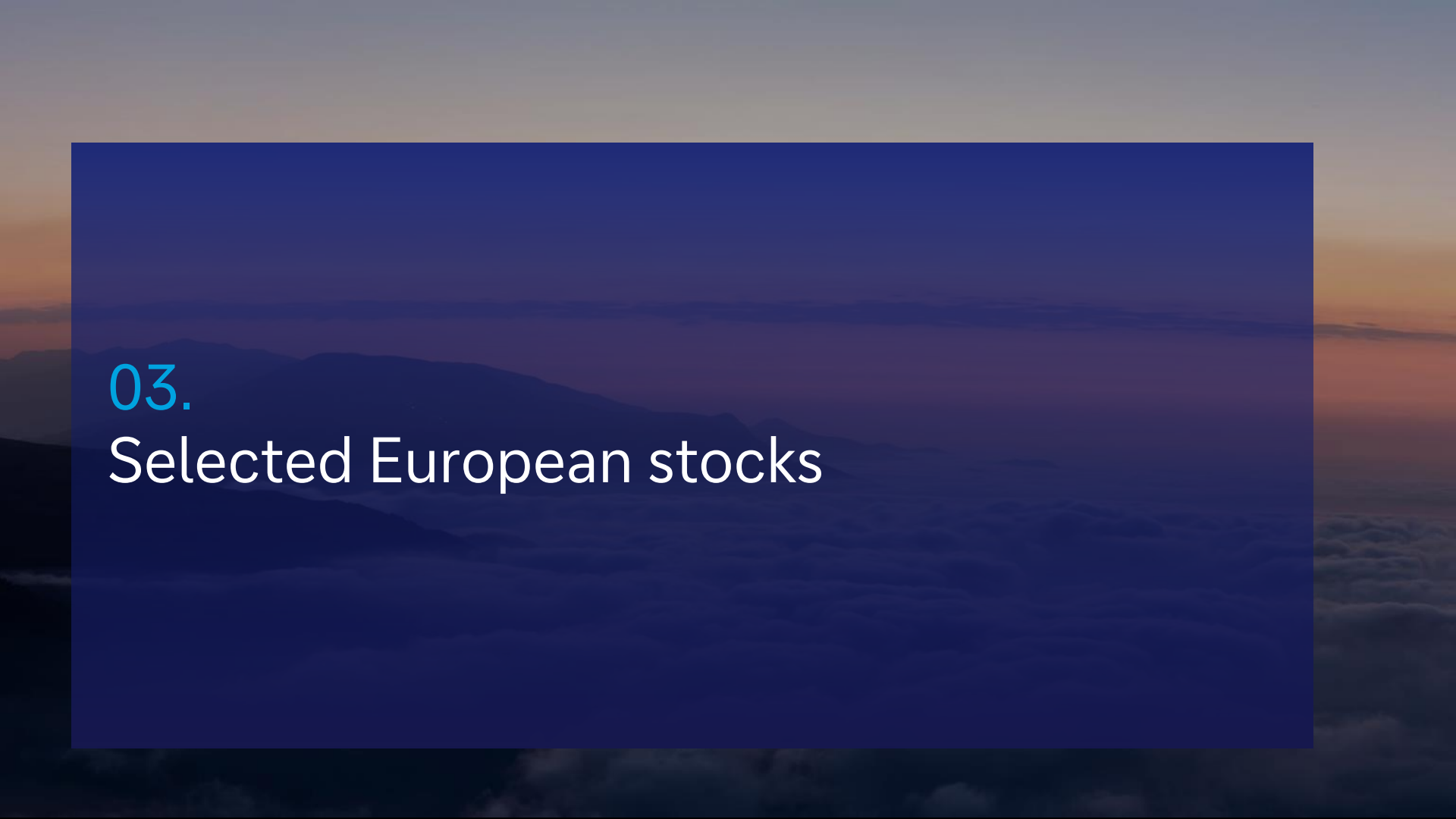
\* Upside is relative to Deutsche Bank Research's price target if available. If not, Bloomberg consensus expectations are used.

## Other notable decliners in the US featured companies involved in real estate, gaming and financial research, showing the depth and breadth of markets' AI disruption fears



Source: Bloomberg Finance LP, Deutsche Bank.

\* Upside is relative to Deutsche Bank Research's price target if available. If not, Bloomberg consensus expectations are used.

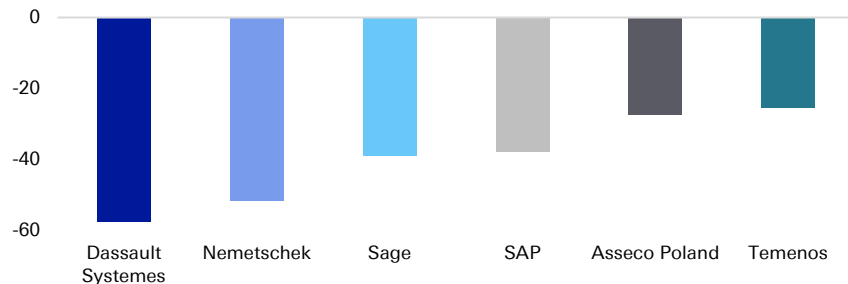


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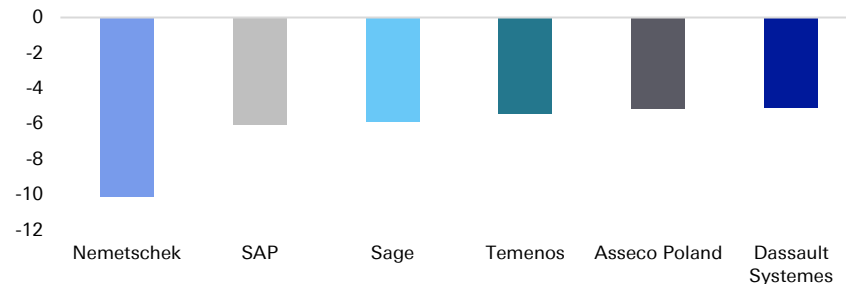
## Selected European stocks



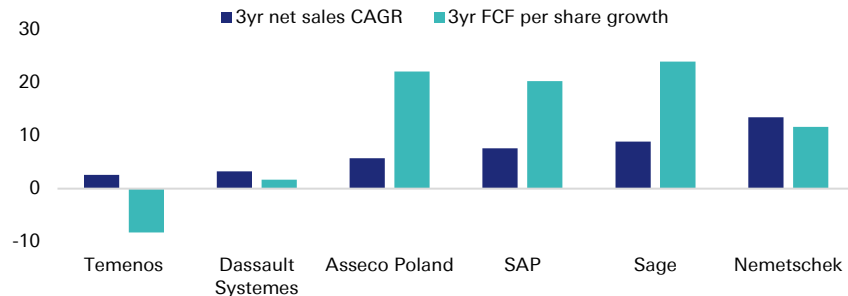
Price change from 52 week high



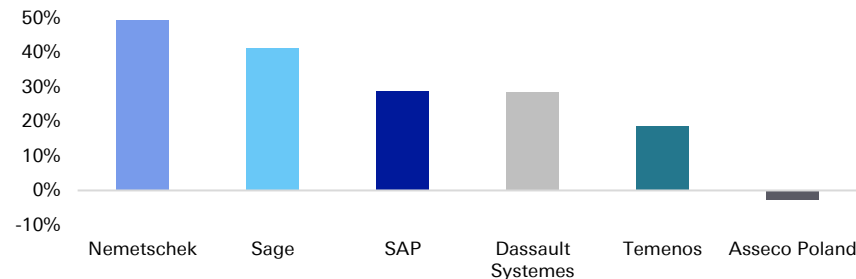
Change in P/E (from 12-Jan)



Fundamentals growth (%)



Upside (downside) to analyst target price\*

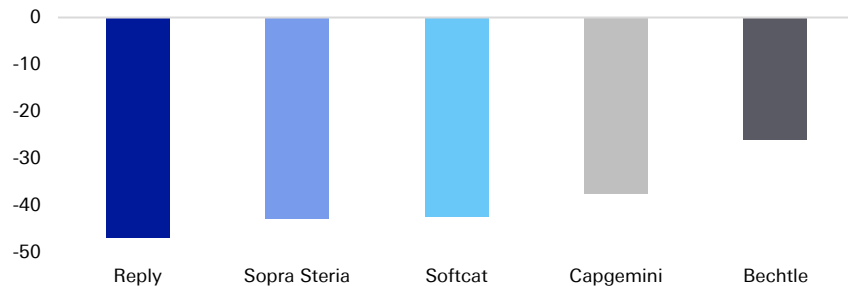


Source: Deutsche Bank.

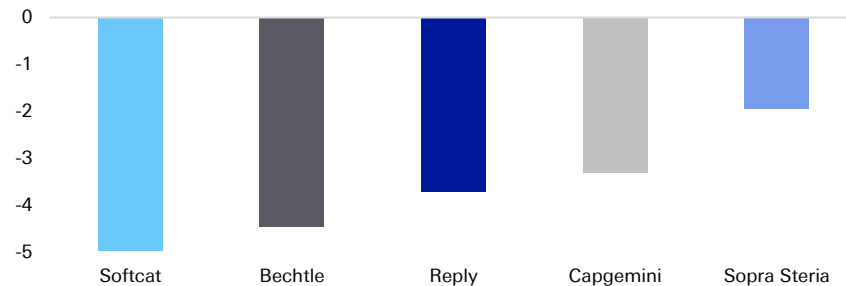
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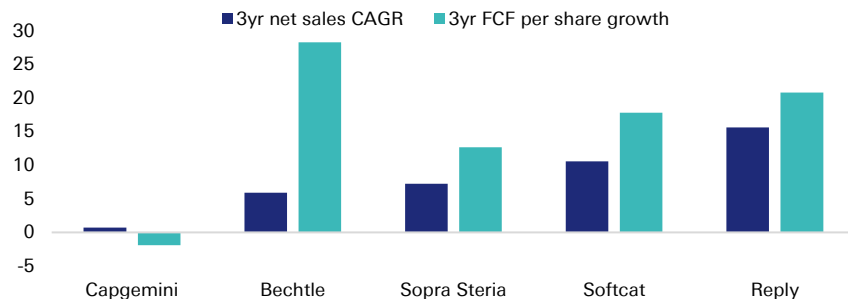
Price change from 52 week high



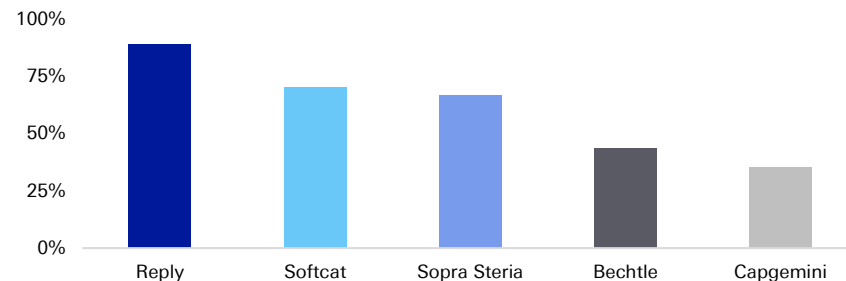
Change in P/E (from 12-Jan)



Fundamentals growth (%)



Upside (downside) to analyst target price\*

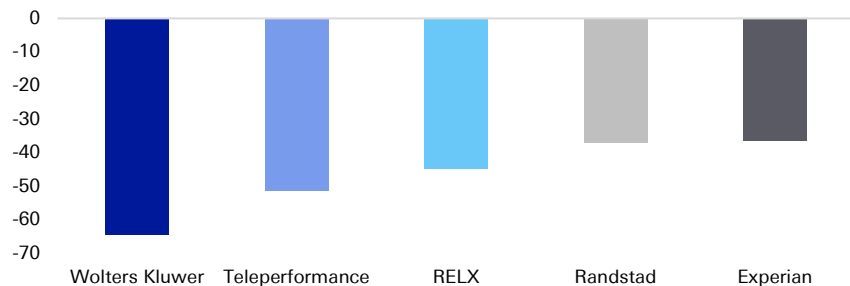


Source: Deutsche Bank.

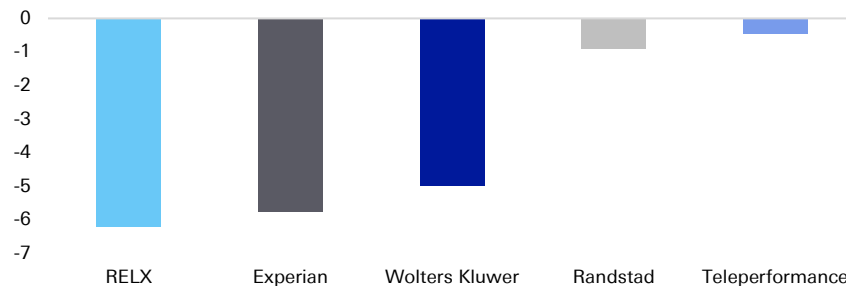
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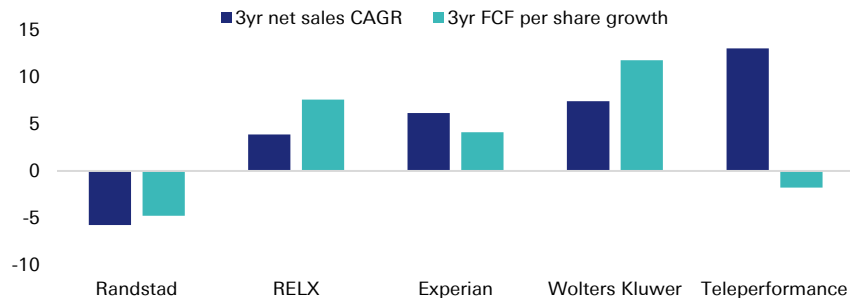
Price change from 52 week high



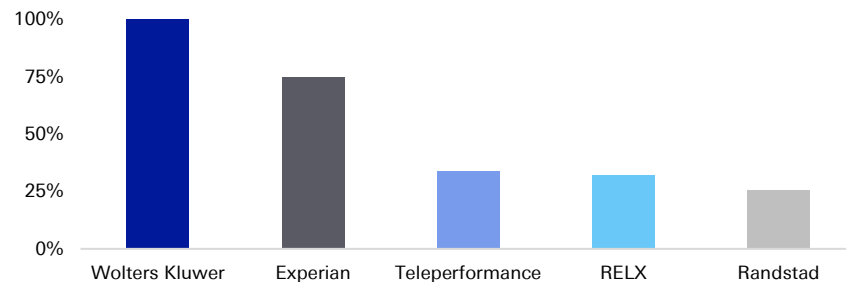
Change in P/E (from 12-Jan)



Fundamentals growth (%)



Upside (downside) to analyst target price



Source: Deutsche Bank.

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# The Brilliant World of FX: A deep dive into restricted markets

4 March 2026

In this report, we discuss "restricted markets," defined as emerging markets with current or capital account restrictions, or both. We examine their market structure and challenges of trading. Recent regulatory and technological changes offer new opportunities for market participants involved in these markets.

FX trading is increasingly electronified, particularly in the spot market. However, the electronification of FX remains fragmented in restricted markets. Foreign access to local markets is difficult due to regulations and fixed trading hours, often requiring manual FX processes. A common rule requires FX trades to be linked to underlying local investments. Consequently, much of foreign exchange spot execution is handled passively, **often via custodian standing instructions**. Many of these currencies also trade offshore via non-deliverable forwards (NDFs). Despite these complexities, local market regulations are constantly changing, fostering a more competitive FX execution environment. South Korea, for example, is liberalizing its onshore markets to increase FX flows, extending trading hours, and allowing foreign investors access through offshore banks during offshore hours. Thailand's Non-Resident Qualified Company (NRQC) scheme provides economic benefits to non-resident companies through access to onshore Thai Baht (THB) liquidity and cheaper hedging rates.

Adapting to changes in local regulations and leveraging technology, the execution space is evolving to provide foreign asset managers and companies with alternative routes for executing FX in restricted markets. Where possible, foreign asset managers may execute trades directly with the global custodian or a third-party FX bank i.e. not via standing instructions. They are responsible for executing the trades and providing documentation. While **direct dealing** offers greater control, it can be complex and may not provide access to a diverse pool of liquidity.

**Outsourcing FX solutions** is emerging as a third approach. Specialised FX providers, especially those with an onshore presence, offer expertise in navigating complex regulations and local market practices. **They provide services like efficient order execution, access to diverse liquidity pools, and efficient risk management.** Their offerings can potentially help foreign asset managers and companies achieve better pricing through wider access, and better risk management, for example, via a closer synchronization between the funding and hedging trades. Given that each market is different, there is no one-size-fits-all solution for execution. Technology continues to adapt to this dynamic environment, offering foreign asset managers and companies alternative routes for executing FX..

## Authors

Rohini Grover, Ph.D.  
Strategist



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## Introduction

FX trading has become significantly electrified especially in the spot market, as discussed extensively in [our research](#). Currencies of open economies with open current accounts, such as the G10 pairs, are traded through dozens of venues, and overall electrification has compressed spreads across these markets. However, the electrification of FX markets is more fragmented among emerging market currencies that face restrictions on their current account, capital account, or both. These subsets of emerging market currencies constitute restricted markets.

Many of these currencies have a strong presence outside their local markets, meaning they are heavily traded in offshore markets via non-deliverable forwards (NDFs). Over the past decade, BIS surveys reveal that an average of 70% of forwards in these restricted markets have been traded via NDFs. Furthermore, Asian currencies account for approximately two-thirds of global offshore currency forward (NDF) trading. NDFs on these currencies trade in venues similarly to our developed FX market execution landscape. Overall, the NDF market accounts for 4% of the \$10 trillion FX market.

Trading NDFs can also present unique regulatory challenges. For example, NDFs are subject to margin requirements such as the Uncleared Margin Rules (UMR). In our note last year '[FX clearing today](#)' we highlighted that UMR has indirectly increased the cost of non-cleared FX derivatives.

Foreign access to markets locally or **onshore markets**, however, is challenging due to regulatory demands and fixed trading hours, often requiring manual steps in the FX process. A common regulation mandates that FX trades be linked to underlying local investments. Our [latest EM handbook](#) discusses in detail the regional regulations and highlights that outside of FX dealers (market-makers) a majority of trading is supported by asset managers and corporates with an underlying local investment.

This paper examines the evolution of FX trading in restricted markets. It highlights the role of non-deliverable forwards (NDFs) in facilitating currency trading outside local jurisdictions and analyzes their impact on forward pricing in both onshore and offshore markets. The discussion then moves to the efforts made by various markets to support the onshoring of FX flows. Finally, the paper explores the electrification and execution of FX, the inherent risks associated with trading in these markets, and their future outlook.

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## Role of non-deliverable forwards in restricted markets

NDFs (Non-deliverable forwards) offer a way to gain exposure to a currency's performance without the need for funding in the underlying currencies, unlike traditional deliverable forwards. These contracts settle the difference between a pre-determined exchange rate and the spot rate at maturity (either an official domestic rate or a market-determined rate) through a single US dollar payment.

Trading predominantly offshore, NDFs enable investors to circumvent restrictions within a currency's home jurisdiction and bypass limitations on offshore delivery. Beyond hedging currency risk for direct and portfolio investors, NDFs also attract speculators (Ma et al, 2004). The presence of banks and firms with onshore and offshore operations helps to reduce discrepancies in forward rates.

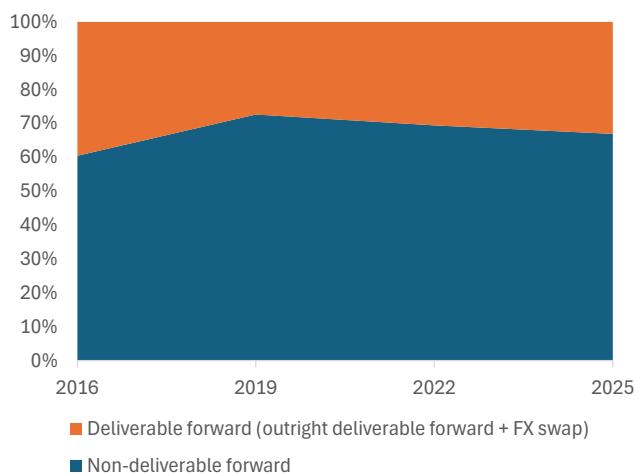


In restricted markets, onshore forward trading typically concentrates on outright deliverable forwards and deliverable FX swaps, while offshore trading specializes in NDFs. The deliverable forward market in these regions also tends to be smaller than the non-deliverable currency forward market, largely due to limited access and shorter trading hours within local currency markets.

Over the past decade, BIS surveys reveal that around 70% of total forwards (deliverable outright forwards + FX Swaps + NDFs) in these restricted markets have been traded via NDFs. Furthermore, Asian currencies account for approximately two-thirds of global offshore currency forward (NDF) trading. Consequently, foreign institutional investors and corporations hedging exposure in such markets have historically relied on NDFs. However, this reliance introduces a potential risk, also known as basis risk: the hedger is unlikely to execute the underlying spot transaction at the same rate as the fixing rate used to settle the NDF. **Hedging with NDFs retains exposure to basis risk between the fixing rate and the rate on the underlying spot transaction. This risk tends to be even higher if the two trades are not synchronized as closely as possible.**

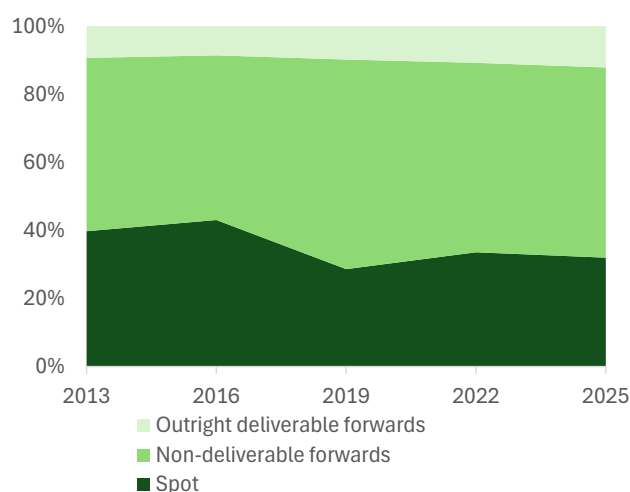
The NDF market's significant size relative to the underlying spot market highlights a strong desire among offshore participants to hedge and speculate on and engage with the currency dynamics of these restricted economies. However, the continuing existence of the NDF market alongside deliverable forwards exacts a cost in terms of lower liquidity from the division of the forward markets.

Figure 1: Forward trading dominated by non-deliverable forwards



Source : Deutsche Bank, BIS. Restricted markets: USD against INR, KRW, TWD, BRL

Figure 2: Asia: NDF volumes have outstripped underlying spot volumes



Source : Deutsche Bank, BIS. Restricted markets: USD against INR, KRW, TWD. Outright deliverable forwards = Outright forwards - NDF

## The evolution of NDF markets and impact on pricing

The evolution of NDF markets has not been uniform, and this has had varying impacts on the pricing relationship between onshore and offshore forwards across different countries. In cases where countries have cautiously maintained restrictions, NDF markets have grown significantly. However, the exact impact on the pricing gap between deliverable ('onshore') and non-deliverable ('offshore') forwards has varied by jurisdiction. For instance, in Korea, local banks' ability to

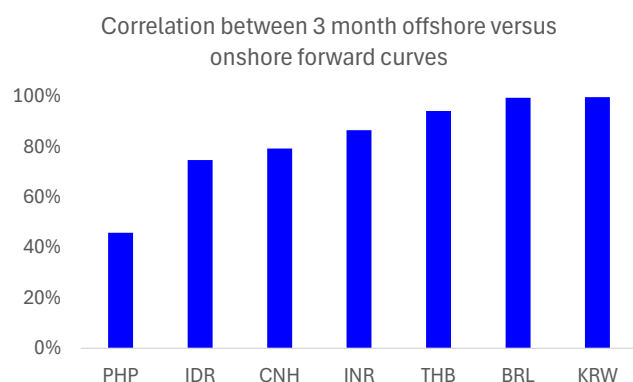


trade both the onshore forward and NDF markets has kept this gap narrow.

Differences between deliverable forward and NDF rates generally reflect the effectiveness of capital controls. A [BIS study](#) found that deviations between onshore and offshore curves were largest for the Renminbi, Indian Rupee, Indonesian Rupiah, and Philippine Peso. Previous studies on the Korean Won, Indian Rupee, and Indonesian Rupiah also demonstrate a two-way influence between deliverable forwards ('onshore') and NDFs ('offshore') during normal times. However, during more volatile periods, the NDF market tends to drive the domestic market (Misra and Behera (2006), Kim and Song (2010), Behera (2011), Cadarajat and Lubis (2012), and Goyal et al (2013)). More recently, Schmittmann and Teng (2020) noted that during the COVID-19 pandemic, there is some evidence of NDFs leading onshore markets for a few currencies.

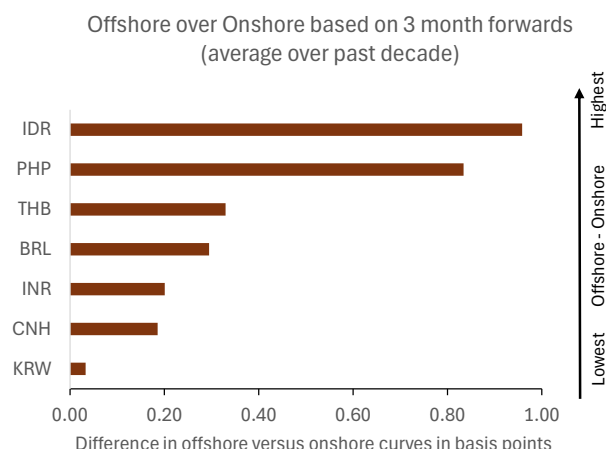
Our analysis also shows lower correlation between onshore and offshore rates for PHP, CNH, IDR, and INR compared to KRW and BRL. <sup>1</sup>Over the past decade, onshore curves have, on average, **traded 40 basis points** below their offshore counterparts in the three-month tenor across currencies (Figure 3 and Figure 4). **This suggests that non-residents investing in restricted markets can potentially achieve lower currency execution costs when accessing onshore markets.** The potential savings can be even higher as these deviations tend to widen during periods of stress, with the BIS study highlighting larger pricing differentials during the financial crisis and the May-August 2013 selloff in emerging market assets.

Figure 3: Deviations between onshore and offshore curves highest for PHP, IDR, CNH and INR



Source : Deutsche Bank. Asian pairs compared at 3 pm tokyo time for spot, onshore and offshore 3M forward. USDBRL taken at 6 pm London time.

Figure 4: Onshore curve, on average, 40 basis points below offshore curve



Source : Deutsche Bank.  $\text{Offshore(Onshore)} = (F-S)/S$ . Both of F and S taken as USD/CCY. Asian pairs compared at 3 pm tokyo time for spot, onshore and offshore 3M forward. USDBRL taken at 6 pm London time.

Beyond local market restrictions, global factors also influence both deliverable forward ('onshore') and non-deliverable forward ('offshore') rates. For instance, [previous studies](#) indicate that both onshore and offshore forward rates increase (i.e., currencies depreciate) when the VIX rises. Notably, global factors appear to have a more pronounced impact on NDF rates for the Renminbi, Indian Rupee, and

<sup>1</sup> For Asian currency pairs, we compare the three-month onshore versus offshore curves at 3 PM Tokyo time, and for BRL, at 6 PM London time.



Indonesian Rupiah.

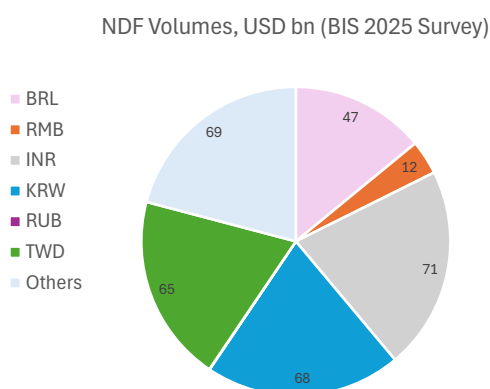
**A special note on the Renminbi:** its internationalization follows an idiosyncratic route. Chinese authorities engineered a system allowing a substantial pool of Renminbi to accumulate and be freely traded and delivered offshore, all while maintaining capital controls. This strategy has resulted in a tripartite structure for the Renminbi forward market: an onshore market (dating to 2006), an offshore NDF market (established in the 1990s), and an offshore deliverable, or CNH, market (emerging in 2010).

Lastly, when non-residents gain domestic access to forward trading, effectively allowing them to lend and borrow the local currency, the NDF market could become largely unnecessary. Its disappearance may not be immediate, however, as illustrated by the Australian dollar NDF, which persisted for several years after deliverable forwards became available in 1983.

## Onshoring of FX Flows: leaders and laggards

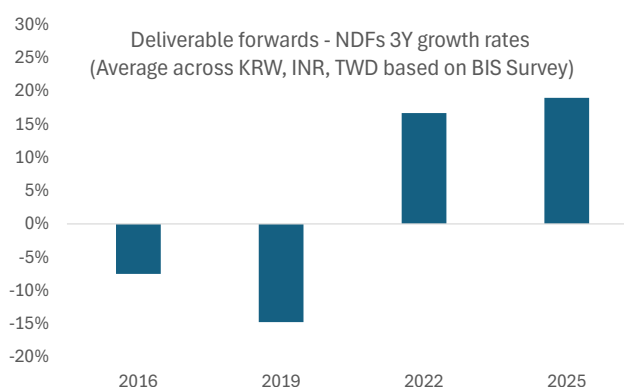
The costs associated with a larger offshore market, including reduced onshore competitiveness, amplified spillover effects during periods of offshore volatility, and a lack of transparency and oversight for domestic regulators, have prompted a re-evaluation of strategies by regulators. As our colleagues Mallika Sachdeva and Bryant Xu have previously noted in their study [Asia Thematic Strategy Notes: Onshoring FX flows: regulation is turning the market tables](#), various Asian markets have adopted distinct approaches to encourage FX onshoring. This should help non-residents gain easier access to the more liquid onshore FX markets and potentially reduce the cost of executing FX trades on the back of their investment in restricted capital markets.

Figure 5: Breakdown of NDF trading across markets based on latest BIS survey



Source : Deutsche Bank, BIS

Figure 6: The past decade has seen Asia outright deliverable forward flows outpace non-deliverable forwards



Source : Deutsche Bank, BIS, USD vs KRW, INR, TWD

Historically (2013-2019), offshore markets generally outpaced onshore growth. However, the last two BIS surveys (2019-2025) reveal a significant shift: onshore





forward trading, particularly in outright forwards in KRW and INR, is now expanding considerably faster than its offshore counterparts. This development potentially suggests the effectiveness of various market initiatives within these regions. In contrast, outside of Asia, some Latin American countries and, more recently, frontier markets are still in the nascent stages of addressing similar market development issues.

Below, we summarize the distinct approaches taken by some restricted markets to encourage FX onshoring, examining the evolution of both their onshore (within) and offshore (outside) markets.

**Korea** announced [significant reforms](#) to align its FX market with global standards. These reforms include opening the onshore interbank FX market to **foreign financial institutions (RFIs)**, allowing non-residents to trade Onshore KRW FX (Spot, Forwards and Swaps). Onshore FX trading hours have been extended until 2:00 AM local time, enabling asset managers to trade KRW without pre-funding transactions. The [scope of RFI transactions](#) will expand to cover all transactions, including FX for current transactions like import/export payments and salary remittances, and Third-Party FX (TPFX) will be promoted. Improvements in FX brokerage infrastructure, including aggregators, will allow corporate end-users to select the most favorable prices. Deutsche Bank was [the first bank to execute a trade](#) under the Registered Financial Institutions (RFI) programme. According to [Foreign Exchange Trading Trends of Foreign Exchange Banks in 2025](#) the daily average FX trading volume of foreign exchange banks reached \$80.71 billion, a 17.0% increase from 2024, marking the highest level since 2008. As per the document, this growth is attributed to extended trading hours and increased overseas securities investment by residents and domestic securities investment by foreigners.

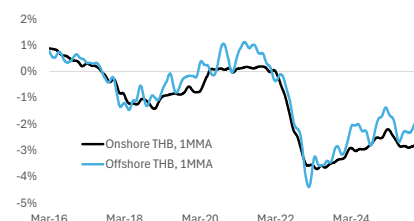
#### **Registered Financial Institutions (RFIs).**

*RFIs are Foreign FIs that are not based in Korea (Headquarters of foreign FIs, overseas branches of foreign FIs, and overseas branches of local FIs) that are allowed to directly participate in the onshore interbank FX market, if they obtain registration from the Korean authorities.*

**India** has focused on liberalizing [onshore risk management rules](#), providing market makers with more flexibility for hedging products, and allowing NDF market participation from GIFT City. This strategy has successfully redirected NDF flows onshore, enhancing RBI oversight and aligning forward curves. **Thailand** introduced the Non-Resident Qualified Company ([NROC](#)) scheme to integrate its onshore and offshore THB forward curves, allowing non-resident corporations to access onshore THB liquidity and hedge freely. The [NROC](#) scheme provides economic benefits, as the onshore THB curve has historically traded **35 basis points** below the offshore curve, and the onshore market offers greater depth in longer-dated forwards. **China** is actively internationalizing the RMB by promoting its use in cross-border trade and investments. The PBOC has focused on improving cross-border RMB settlement through international cooperation, enhanced infrastructure like the [CIPS payment platform](#) and extended onshore FX trading hours, and corporate empowerment initiatives such as cash pooling programs and increased flexibility in Panda bond financing. **Indonesia** [introduced the IDR settled Domestic Non-Deliverable Forward \(DNDF\)](#) in 2018, traded onshore and auctioned daily, which maintains hedging market access, aids Bank Indonesia intervention, and influences offshore NDFs.

**Brazil** operates extensive onshore derivatives exchanges for FX, interest rate, stock, and commodity instruments, with Brazilian FX futures and options being non-deliverable and settling in domestic currency. The latest [BIS study](#) notes a relatively lower proportion of BRL FX derivatives trading offshore (compared to INR, KRW and COP), with a significant volume occurring via futures on the Sao Paulo exchange, partly due to limited restrictions on non-resident participation. The

**Figure 7: The onshore THB curve generally cheaper and stable**



Source : Deutsche Bank, Bloomberg Finance LP





Brazilian legal and regulatory framework significantly influences this derivatives market. Constraints on over-the-counter (OTC) trading and a tax system favoring net profit/loss calculations on exchanges encourage trading migration to these platforms. Last year, Brazil [announced substantial regulatory changes](#) affecting financial and capital markets taxation – the IOF tax – which increased the levy on financial transactions.

**Argentina** has [removed the minimum](#) holding requirement for non-resident investors in local debt markets and eliminated cross-restrictions between official and parallel markets. The 48-hour advance notice for large FX purchases has also been removed. Additionally, the BCRA has issued a new 3-year FX-denominated bond (BOPREALs) to address legacy commercial debt and dividends accumulated during capital control years. These measures are expected to catalyze further foreign investments.

**Colombia** has a liquid, competitive, and efficient spot FX market by international standards, though its hedging market is predominantly characterized by NDFs. This prevalence of NDFs is primarily due to historical capital control measures and regulatory constraints inhibiting non-residents' COP settlement, hindering the development of deliverable forwards and swaps. [Between 2016 and 2022](#), offshore trading averaged around USD 7 billion, significantly surpassing combined onshore and cross-border trading of approximately USD 4 billion, while a financial transaction tax further renders deliverable derivatives more expensive than non-deliverable ones.

**Frontier markets**, as defined by the International Finance Corporation, are smaller markets with foreign investor access limits but offer promising investment opportunities, characterized by early [stages of capital market maturity](#) and often lacking financial depth and a shallow domestic investor base. The [MSCI Frontier Markets Index](#) includes large and mid-cap companies across 28 such countries, with top constituents like Vietnam, Romania, Morocco, Kazakhstan, and Slovenia. This lack of financial depth results in challenging local market liquidity conditions where direct local spot settlement can be illiquid. Consequently, investors frequently utilize NDFs to gain exposure without needing local accounts. Accessing these markets, especially for African currencies, often necessitates specialised institutions and local correspondent networks due to these characteristics.

For a detailed discussion, please see the Appendix.

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## Electronification of FX in restricted markets

Our discussion so far has focused on how regulations have shaped the FX trading landscape, historically leading to fragmentation of the forward market. Now, we turn our attention to the electronification of FX markets within this complex environment. While technology has revolutionized foreign exchange trading, boosting efficiency and transparency in G10 markets, the electronification of FX in emerging markets remains fragmented due to restrictions on capital, current accounts, or both.

[Restricted markets](#), such as India and South Korea, have developed localized electronic FX trading platforms. While these platforms remain closed to foreign participants, they are as liquid as some of the G10 FX markets. These platforms



facilitate electronic pricing via domestic data and APIs but foreign access is challenging due to regulatory demands and fixed trading hours, often requiring manual steps in the FX process. These challenges also limit the number of FX counterparties providing comprehensive coverage.

A common regulation, for instance, mandates that FX trades be linked to underlying local investments. Consequently, a significant portion of foreign exchange execution in restricted markets is handled through passive products, such as custodian standing instructions.

Despite these complexities, local market regulations are constantly changing, and the FX execution space is evolving, fostering a more competitive environment that could potentially lead to lower costs. For example, as discussed earlier, South Korea has been liberalizing its onshore markets to increase FX flows by extended trading hours and enabling foreign investors to trade Korean assets more freely.

As restricted markets continue to evolve, comprehensive FX outsourced solutions are becoming more common in the most prevalent restricted market currencies. This trend sees more [market participants](#) exploring third-party FX execution solutions with restricted currency offerings. We now discuss the different avenues of execution in detail.

## 1. Custodians

Custodians handle the settlement, safekeeping, and reporting of client assets, which includes managing cash balances and executing FX transactions. They act as facilitators for financial transactions, providing institutional investors with centralized access to their global investments. For cross-border securities transactions, global custodians often provide FX services, including executing foreign exchange and processing tax reclaims.

A key method custodians use is "**custodian standing instructions**" or "**passive products**." These are pre-agreed arrangements from clients that authorize the custodian to automatically execute necessary FX transactions, for instance, when purchasing securities. Custodians have historically been the main FX execution model. In recent years, there has been a strong push from asset managers for more transparency around transaction cost analysis and execution rates and time. Some custodians have begun to provide detailed analysis while [others have lagged](#). In some cases, custodians have also partnered with specialised providers to navigate complex restricted markets.

## 2. Direct dealing

Direct dealing refers to asset managers executing currency trades directly with the FX execution desk of global custodians or third-party FX banks. Third-party bank refers to an FX bank that isn't their global custodian. In this scenario, the asset manager is directly responsible for executing the FX transactions. This differs from methods like 'custodian standing instructions,' where the custodian executes FX trades on behalf of the asset managers.

Managers may use this method for the enhanced control and clarity it offers over their currency trade as they directly execute the trade. However, direct dealing can become more complicated especially with multiple global custodians. Managing several custodians increases operational burden. Each custodian often has its own FX bank network, with varying requirements and potentially different FX rates. A major challenge is meeting the disparate demands for security details (like ISINs,



account numbers) from each FX bank or custodian. It gets even more challenging when a manager trades directly with an FX bank that isn't their global custodian.

Direct dealing means the asset manager must take full responsibility for routing any security and account details, and sometimes for managing tax issues. This also involves bearing the operational and execution risks. Although direct dealing has become more transparent and flexible over the last decade, it still faces significant challenges. Additionally, asset managers may not have access to a diverse liquidity pool that could potentially lead to better rates. This situation has encouraged the development of other execution methods in these markets, as managers look for more reliable ways to control their workflow.

### 3. Outsourced FX

Outsourced FX solutions involve delegating foreign exchange transaction management to a specialised external provider. These solutions aim to make FX trading more efficient, transparent, and compliant, particularly in complex markets. Technology-enabled solutions are designed to manage many stages within an FX trade lifecycle that are often manual and repetitive. These solutions can also be tailored and provide a degree of control to the user.

While many global custodians offer their own outsourced FX services, specialised third-party providers also exist. Some solutions are "custodian agnostic," supporting accounts held with various custodians.

For restricted markets, outsourced solutions provide a compelling alternative. These external providers often possess expertise in specific market regulatory requirements and have established trading relationships with multiple onshore counterparties in restricted markets. This makes outsourced FX solutions particularly beneficial for asset managers seeking to streamline operations, reduce costs, manage risk, and improve transparency in trade execution. These programs help achieve execution quality and consistency across multiple providers. The outsourced provider manages specific market regulatory requirements and can flexibly navigate across different providers and currencies.

Managers can also benefit from a customized execution model tailored to their operational workflow and needs, effectively outsourcing the complexities of market regulatory requirements and the settlement process. **Such fully outsourced FX programs are gaining significant traction in restricted markets due to their ability to address challenges and deliver benefits like cost savings, improved benchmarking, enhanced risk management, and greater transparency in trade execution.**

Hedging via onshore markets can result in lower costs, with onshore forward curves typically trading below the offshore curves. In other words, the onshore and offshore markets may not always be aligned and the ability to switch between them may be another area of cost efficiency. Conversely, for hedging via offshore markets, higher efficiency gains can be achieved with greater synchronization of prices between the onshore spot trade and the corresponding offshore hedging trade. **Specialised FX providers could potentially help managers optimize their FX process on both fronts.** Overall, the automated workflow should allow managers to focus more on their core strategies, and demand for full outsourcing is expected to grow as FX preferences and approaches evolve.

Given that each market is different, there is no one-size-fits-all solution for FX



execution. Some asset managers seeking more control might prefer direct dealing, while others may look for a more automated approach through custodians or outsourced solutions. Ultimately, the decision to choose one over the other will depend on an in-depth cost-benefit analysis.

### Non-deliverable instruments in offshore markets

We now briefly discuss the execution space for trading of non-deliverable instruments in offshore markets. These instruments, especially NDFs, tend to account for 70% of the forward trading in restricted markets.

While spot remains the most electrified segment, electronic trading in forwards, particularly NDFs, has been catching up at an [increased speed](#). Like spot, trading in non-deliverable instruments utilizes similar execution venues as discussed in [our earlier paper](#), including single-dealer or multi-dealer platforms, as well as primary and secondary, or alternative venue.

Trading NDFs can present unique regulatory challenges. For example, NDFs are subject to margin requirements such as the Uncleared Margin Rules (UMR). In our note last year '[FX clearing today](#)' we highlighted that Uncleared Margin Rules (UMR) have indirectly increased the cost of non-cleared FX derivatives, incentivizing a shift toward central clearing. It affects buy-side participants, by increasing the margin requirements for uncleared derivatives that mainly constitute NDFs and options; UMR might require a firm trading NDFs to post an initial margin of Y% of the notional value. This added cost is prompting some market participants to explore central clearing solutions even for FX derivatives, despite the lack of a mandate.

[Clearing houses](#) like LCH ForexClear clear approximately 25% of the non-deliverable market. Among non-deliverable currencies, the growth in NDF clearing appears to be driven by USD/INR followed by USD/TWD and USD/KRW.

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## Risks surrounding FX trading in restricted markets

In case of trading in open markets, we [outlined](#) several risks surrounding FX trading. Here, we dig deeper into risks that can be potentially different or elevated in these markets.

[Non-synchronized custodian](#) cut-offs are a known complexity in the settlement process. Each custodian operates with its own internal procedures, technology, and staffing, which directly influence their capacity to process instructions and meet market deadlines. Clients must therefore be acutely aware of their custodian's specific deadlines to avoid potential issues such as failed trades, pre-funding requirements, or the need for bilateral settlements, all of which **introduce additional risk and cost**. This challenge is particularly acute and impactful in restricted markets due to their inherent characteristics, including higher volatility, stricter regulations, limited operational windows, and more frequent pre-funding requirements.

Another important risk to consider is **regulatory risk**, which arises from changes in regulations or their interpretation. As highlighted earlier, local market regulations are constantly changing. For example, India changed its [equity settlement](#) from T+2 to T+1. In March 2024, SEBI announced it would introduce a beta version of a T+0 rolling settlement cycle on an optional basis, which would run in parallel with



T+1. This may in the future mean INR becomes fully pre-funded.

Looking closely at NDF markets, they are advantageous for hedgers and investors because they involve major international banks as counterparties, enabling them to avoid direct credit exposure to emerging market entities and thus **lowering credit risk**. However, NDFs typically exhibit wider and more volatile spreads—especially at longer tenors—compared to deliverable forwards of major currencies, which can indicate higher liquidity risk. For instance, onshore curves have, on average, traded 40 basis points below their offshore counterparts, underscoring this difference.

Hedging with NDFs also entails exposure to **basis risk**, which is the difference between the NDF's fixing rate and the rate of the underlying spot transaction. This basis risk can be further exacerbated in restricted markets by operational challenges. A prime example is the critical need for precise synchronization between a funding trade, such as a spot transaction, and its corresponding NDF hedge. This synchronization is particularly crucial given the fixed trading hours and complex regulatory frameworks often found in these markets. Even a minor delay can lead to significant losses, especially during periods of market volatility or illiquidity. Therefore, accurate timing is essential to prevent unexpected price changes, minimize operational costs, ensure adherence to local regulations, and effectively manage limited trading options.

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## Technology and digitalization of currencies

Restricted markets, especially in Asia, have been at the forefront of digital currency adoption. This proactive stance is driven by the potential of digital currencies to enable faster, more cost-effective, and secure cross-border transactions, leading these markets to explore various avenues like stablecoins, Central Bank Digital Currencies (CBDCs), and tokenized deposits.

Singapore is at the forefront, with a robust regulatory framework for single-currency stablecoins and the widespread use of XSGD, its Singapore dollar-pegged stablecoin. Hong Kong is also making strides, establishing a stablecoin licensing framework to become a global cryptocurrency hub and encouraging institutional adoption, with projects like HKDR (Hong Kong dollar-pegged stablecoin) in development. Indonesia has seen a surge in stablecoin adoption for cross-border transactions, with local stablecoins like IDRT and XIDR gaining traction and Bank Indonesia exploring a national stablecoin.

Central Bank Digital Currencies (CBDCs) are also being actively developed in Asia and other emerging markets to modernize financial systems and improve cross-border payments. China is a leader in [CBDC development](#) with its digital yuan (e-CNY). India's e-rupee is another CBDC pilot. Brazil has adopted the CBDC Drex although recent announcements suggest the coin is not going to utilise blockchain technology. Nigeria's eNaira was Africa's first central bank digital currency (CBDC).

Tokenized deposit pilots are flourishing in key Asian financial hubs like Hong Kong and Singapore. These initiatives coexist with stablecoins and CBDCs to enhance cross-border settlement, boost financial efficiency, and build more inclusive digital finance infrastructures while maintaining strong ties to traditional banks. Hong Kong's Project Ensemble, launched by the Hong Kong Monetary Authority (HKMA), and Singapore's Project Guardian, initiated by the Monetary Authority of Singapore (MAS), are prominent examples in this space.



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## The future of FX trading in restricted markets

Our work on developed markets has shown that 45% of the FX spot market actively seeks profit, while the remaining 55% is dedicated to business functions, hedging, or facilitating international investments rather than generating alpha. For those whose FX is predominantly operational, the same electronification paths now offer part of the solution by providing a framework for increased automation. This is more crucial than ever given the headwinds of rising regulation coupled with management fee compression.

FX trading in restricted markets is largely operational, with foreign investors often executing FX trades onshore as part of their capital market activities. For instance, a common regulation mandates that FX trades be linked to underlying local investments. Consequently, a significant portion of foreign exchange execution in restricted markets is handled through passive products, such as custodian standing instructions.

This does not necessarily mean that FX in restricted markets is not electronified. It simply means that the FX workflow is not fully automated and often requires manual steps. However, as regulations in restricted markets continue to evolve, the FX execution landscape adapts accordingly. For example, comprehensive FX outsourced solutions are gaining traction among restricted market currencies.

Hedging via onshore markets can result in lower costs, with onshore forward curves typically trading below the offshore curves. In other words, the onshore and offshore markets may not always be aligned and the ability to switch between them may be another area of cost efficiency. Conversely, for hedging via offshore markets, higher efficiency gains can be achieved with greater synchronization of prices between the onshore spot trade and the corresponding offshore hedging trade. Specialised FX providers could potentially help managers optimize their FX process on both fronts.

While buy-side and sell-side participants continue to collaborate to improve the FX execution space in these markets, local regulations are also evolving to become more supportive of onshoring of FX flows. For example, recent pushes by regulators, especially in South Korea, have improved accessibility to the onshore FX market. Asset managers in offshore locations can now access these markets through offshore banks during offshore hours. This should result in more competitive onshore FX markets and improved overall efficiency for domestic and foreign investors. **The question remains whether other restricted markets will follow the Korean model.**

Lastly, technology will continue to play a key role in restricted markets. For example, Asia, within these restricted markets, has been at the forefront of digital currency adoption. This proactive stance is driven by the potential of digital currencies to enable faster, more cost-effective, and secure cross-border transactions, leading Asia to explore various avenues like stablecoins, CBDCs, and tokenized deposits.



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## Appendix: A detailed overview of restricted markets

### Korea: Non-residents access onshore markets via offshore banks

Korean regulators are introducing measures to open up the market for foreign investors. On the foreign exchange (FX) side, the Korean Ministry of Economy and Finance acknowledged that their FX market's restrictive structure had held back its growth, hindered stability, and made KRW-denominated assets less attractive. To address these issues, Korea announced [significant reforms](#) to align its FX market with global standards, with additional reforms ( 3 and 4) more recently. These include:

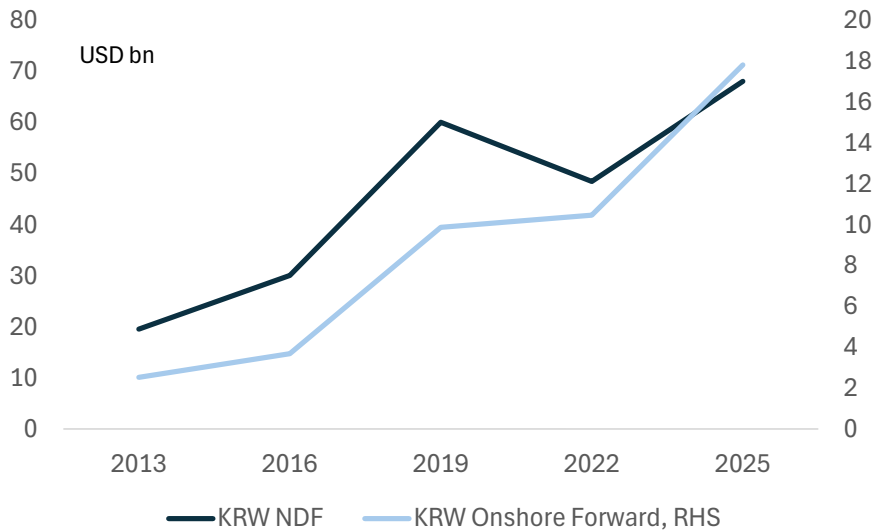
1. **Opened up the onshore interbank FX market to foreign financial institutions:** Non-residents can now trade Onshore KRW FX (Spot, Forwards and Swaps) via **Registered Financial Institutions (RFIs)**. *RFIs are Foreign FIs that are not based in Korea (Headquarters of foreign FIs, overseas branches of foreign FIs, and overseas branches of local FIs) that are allowed to directly participate in the onshore interbank FX market, if they obtain registration from the Korean authorities.*
2. **Extended onshore FX trading hours:** Local market trading hours have been extended till 2:00am local time (earlier close was 3:30pm). This allows asset managers to trade KRW without pre-funding transactions, reducing the costs and increasing the appeal of KRW assets.
3. **Facilitating RFI Transactions:** This will involve [expanding the scope of RFIs](#) to all transactions, including FX for current transactions like import/export payments. This expansion will enable RFIs to facilitate currency exchanges for multinational companies with headquarters or branches in Korea or abroad, and will also allow for salary remittance for their employees. Notably, [individual fund-level identifiers](#) are not required for trading, reporting, and monitoring for any FX, Bond, or Equity related transactions. Furthermore, Third-Party FX (TPFX) will be promoted by extending the settlement deadline for sending KRW funds to local custodian banks from 10:00 AM to 11:00 AM, and by broadening the range of institutions eligible for KRW Overdraft to include all institutions involved in securities transactions.
4. **Improving FX Brokerage Infrastructure:** Aggregators will be introduced to allow corporate end-users to select the most favorable prices through brokerage platforms.

According to "[Foreign Exchange Trading Trends of Foreign Exchange Banks in 2025](#)" the daily average foreign exchange trading volume (spot exchange and foreign exchange derivatives transactions) of foreign exchange banks last year totaled \$80.71 billion. This represents a 17.0% increase from 2024 (\$68.96 billion) and marks the highest level since the statistical reorganization in 2008. The Bank of Korea document further states that this is due to the continued impact of extended foreign exchange market trading hours in 2024, along with significant increases in transactions related to residents' overseas securities investment and foreigners' domestic securities investment.





Figure 8: Recent reforms have encouraged a shift from NDF into onshore forwards

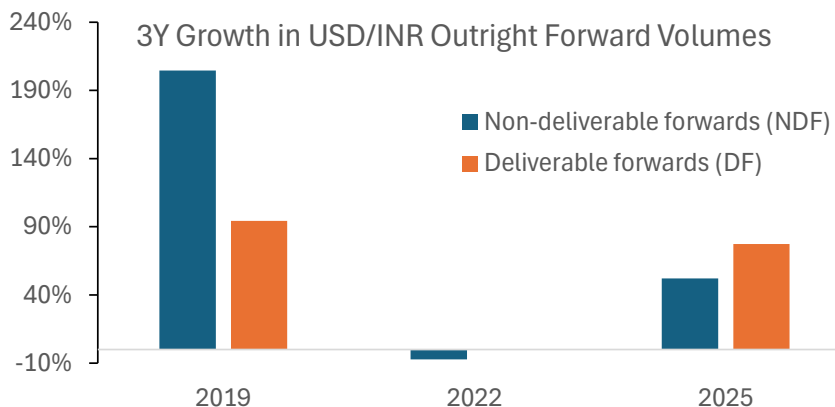


Source : Deutsche Bank, BIS

## India

A 2019 [Task Force on Offshore Rupee Markets](#) recommended increasing onshore market participation and simplifying access without altering capital controls. In response, the Reserve Bank of India (RBI) made two key changes in 2020. First, **liberalized onshore risk management rules**. Moved to a [principles-based approach](#), giving market makers more flexibility for hedging products (e.g., hedging anticipated exposures, equal treatment for residents/non-residents, expanded derivative access). **Second, allowed NDF market participation from GIFT City**: Indian and foreign bank units can now trade FX-settled NDFs with non-residents from GIFT City, an international financial services center operating outside India's main capital controls. This strategy has successfully redirected NDF flows onshore, enhancing RBI oversight and aligning deliverable and non-deliverable forward curves. The latest [RBI draft](#) requires reporting of all OTC foreign exchange derivative contracts involving INR, undertaken globally by the related parties of the Authorised Dealer Category-I (AD Cat-I) banks.

Figure 9: RBI reforms have potentially led to more flows onshore



Source : Deutsche Bank, BIS





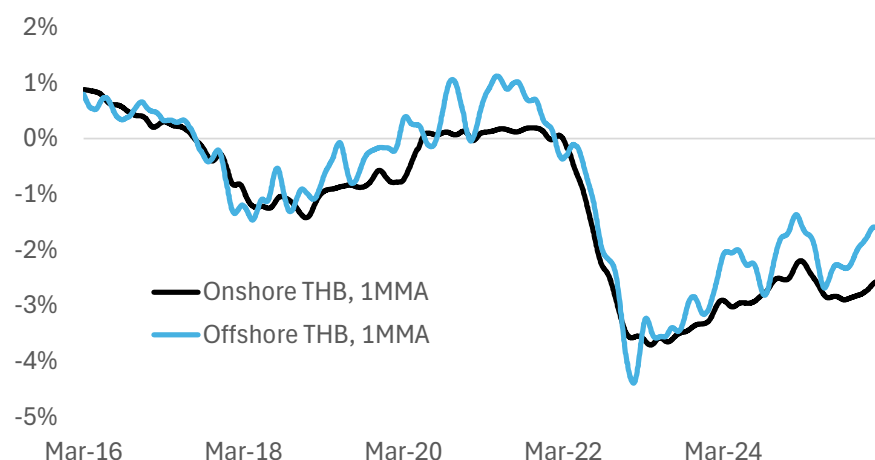
## Thailand

Thailand's Baht (THB) has a unique currency control system, leading to a split between onshore and offshore THB forward curves due to limitations on non-resident THB balances and documentation for onshore trading.

In 2021, Thailand launched the Non-Resident Qualified Company ([NRQC](#)) scheme to integrate these markets. This scheme allows non-resident corporations with Thai trade and investments to access onshore THB liquidity and hedge freely. Further relaxations were announced, effective December 1, 2025, allowing non-residents to conduct transactions without underlying documents and maintain unlimited THB deposits, making the [NRQC scheme](#) generally applicable without case-by-case approval.

The NRQC scheme provides economic benefits, as the onshore THB curve has historically traded **35 basis points** below the offshore curve, and the onshore market offers greater depth in longer-dated forwards. This incentivizes corporates to shift hedging onshore, resulting in cost savings and increased efficiency, as also highlighted by our colleagues in [Asia Thematic Strategy Notes: Onshoring FX flows: regulation is turning the market tables](#).

Figure 10: The onshore THB curve generally cheaper and stable



Source : Deutsche Bank, Bloomberg Finance LP

## China

We highlighted earlier China's internationalization follows an idiosyncratic route. Chinese authorities engineered a system allowing a substantial pool of Renminbi to accumulate and be freely traded and delivered offshore, all while maintaining capital controls. This strategy has resulted in a tripartite structure for the Renminbi forward market: an onshore market (dating to 2006), an offshore NDF market (established in the 1990s, CNY NDF), and an offshore deliverable, or CNH, market (emerging in 2010, CNH DF).

China is actively [internationalizing the RMB](#) by promoting its use in cross-border trade and investments. The PBOC has focused on improving cross-border RMB settlement through:

- **International cooperation:** Expanding bilateral local currency-RMB swap lines with over 40 countries and establishing an [RMB liquidity agreement](#)

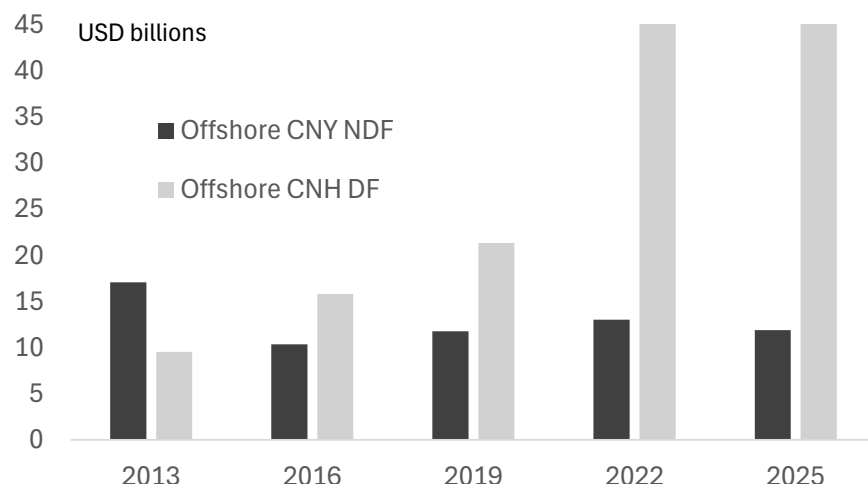


for central banks.

- **Enhanced infrastructure:** Continuously building out the [CIPS payment platform](#) (with 193 Direct and 1573 Indirect Participants across 124 countries as of December 2025, up from 19 Direct Participants and 176 Indirect Participants from 50 countries in 2015). Extending onshore FX market trading hours to 3 AM Beijing Time (effective January 3, 2023) to cover global time zones.
- **Corporate empowerment:** Initiatives like cash pooling programs for MNCs and increased flexibility in using Panda bond financing proceeds.

According to our [latest EM handbook](#), China, with its rapidly growing bond market (now the world's second-largest), is progressively opening up to foreign investors. This involves initiatives like promoting Dim Sum bonds, encouraging local-currency settlement, and testing the digital RMB, ultimately granting corporations increased flexibility in their funding and hedging strategies.

Figure 11: CNH has largely surpassed CNY NDFs



Source : Deutsche Bank, BIS

## Other Asian markets

Malaysia shifted currency flows from offshore to onshore in 2016 by prohibiting licensed banks from facilitating NDF trades, following USD/MYR volatility. This policy reduced NDF liquidity and speculative activity, forcing hedging onshore. The BNM supported this with a [Dynamic Hedging Framework](#), liberalizing onshore hedging for non-residents and significantly boosting onshore forward liquidity. Further foreign exchange policy liberalizations in [April 2021](#) granted exporters greater flexibility in managing foreign currency proceeds, eased FX hedging restrictions for residents, and enhanced capital market efficiency by simplifying onshore foreign exchange activities.

[Indonesia](#) introduced the IDR-settled Domestic Non-Deliverable Forward (DNDF) in 2018, traded onshore and auctioned daily by Bank Indonesia. While not primarily designed to shift offshore hedging onshore due to documentation requirements, the DNDF serves a critical function: it maintains hedging market access, provides an additional tool for BI intervention, and allows the onshore DNDF curve to influence offshore NDFs, rather than being solely driven by offshore



volatility during stress. The DNDF has effectively contained offshore NDF points during global turmoil, enabling non-residents to continue hedging economically and preventing forced asset sales.

[Taiwan](#) allowed partial integration between onshore and NDF markets, as Taiwanese banks are permitted to participate in the NDF market up to 20% of their net open FX positions. They are restricted to trading NDFs solely with locally registered banks and their offshore sister branches. Onshore banks are prohibited from booking NDFs with corporations. Taiwan CBC also imposes limits on nonresident investment in local currency bonds and mandates onshore currency transaction reporting requirements.

### Brazil

Brazil operates extensive and developed onshore derivatives exchanges for an emerging economy, handling FX, interest rate, stock, and commodity instruments. Brazilian FX futures and options are non-deliverable, settling in domestic currency.

Latest [BIS study](#) shows distinctive nature of the BRL FX market, noting a comparatively low proportion of its FX derivatives trading offshore. The BRL is unique as a significant volume of its FX derivatives trading occurs via futures on the Sao Paulo exchange, a characteristic less prominent for other currencies. Crucially, the limited restrictions on non-resident participation in BRL futures contribute to this onshore liquidity concentration.

[This market structure](#), characterized by predominant exchange-traded, non-deliverable transactions, evolved due to a combination of factors including tendency to economic instability that generated demand for hedging. Historical high and volatile inflation, followed by increased foreign currency borrowing post-Real Plan, consistently drove the need for risk management tools in both interest rate and FX markets.

The Brazilian legal and regulatory framework significantly influences this derivatives market. Constraints on over-the-counter (OTC) trading and a tax system favoring net profit/loss calculations on exchanges encourage trading migration to these platforms. Additionally, strict limitations on foreign currency use and trading, such as restricted FX spot market access and the Brazilian Real's non-convertibility outside the country, necessitate non-deliverable, local currency-settled derivatives as substitutes for cash transactions.

Last year, Brazil [announced substantial tax changes](#) affecting financial and capital markets taxation – the IOF tax – which increased the levy on financial transactions.

### Colombia

The Colombian FX market has a liquid, competitive, and efficient spot FX market by international standards, contrasted with a hedging market predominantly characterized by Non-Deliverable Forwards (NDFs). As per the [IMF report](#), the spot FX market has relatively high trading volumes, with daily average inter-dealer turnover around USD 750 million and dealer-to-customer volume around USD 600 million, largely driven by nonfinancial customers, domestic pension funds, and foreign institutional investors. The interbank FX market is competitive, with tight bid-ask spreads.

[Colombia's FX turnover](#) has increased, but a significant portion of this surge is due to offshore NDF trading, keeping its overall turnover lower than the average across



emerging markets. Between 2016 and 2022, offshore trading averaged around USD 7 billion, significantly surpassing combined onshore and cross-border trading of approximately USD 4 billion. This rapid offshore growth contrasts with the slower increase in onshore and cross-border activity. The prevalence of NDFs as the primary hedging instrument stems from historical capital controls and regulatory constraints inhibiting non-residents' COP settlement, which in turn hinders the development of deliverable forwards and swaps. Furthermore, a financial transaction tax renders deliverable derivatives more expensive than non-deliverable ones.

### Argentina

According to the [IMF report](#), non-resident investors are no longer subject to a minimum holding requirement in order to participate in local debt markets, as well as to the cross-restriction between the official and parallel markets. Additionally, the requirement of a 48-hour advance notice for large FX purchases has been removed. The BCRA has also issued a new 3-year FX-denominated bond (BOPREALs) targeted at firms with legacy commercial debt and dividends accumulated during the capital control years, providing greater predictability about the resolution of these legacy external obligations. These measures are expected to catalyze further foreign investments.

### Frontier markets

The term "frontier markets," coined by the International Finance Corporation in 1992, refers to markets that are relatively smaller with foreign investor access limits but offer promising investment opportunities. The [MSCI Frontier Markets Index](#) includes large and mid-cap companies across 28 such countries, with top constituents like Vietnam, Romania, Morocco, Kazakhstan, and Slovenia. Other notable frontier countries include Nigeria, Ghana, Kenya, Uganda, Zambia, Uzbekistan, Uruguay, Costa Rica and Pakistan.

[Frontier Markets](#) are characterized by their early stages of capital market maturity, often lacking financial depth and a shallow domestic investor base. Many of these markets fall below the emerging market median in terms of overall financial development and the depth of their local financial markets. This lack of financial depth also translates into challenging local market liquidity conditions, where bid-offer spreads and the price impact of trades are significantly larger compared to other emerging markets.

Direct local spot settlement in these markets can be illiquid, leading investors to use Non-Deliverable Forwards (NDFs) for exposure without local accounts. Accessing these markets, particularly for African currencies, often requires specialised institutions and local correspondent networks.

For a more detailed discussion of the regulatory framework please see our [latest EM handbook](#).



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## Appendix 1

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Deutsche Bank  
Research

# Iran: Its politics, economy, and impact

March 2026

Luke Templeman | Camilla Siazon | Peter Sidorov | Galina Pozdnyakova

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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- 1. Population:** 91.5mn, 17<sup>th</sup> largest population in the world and home to the world's most populous Shia Muslim country.
- 2. Strait of Hormuz:** c25% of global seaborne oil trade and c20% global LNG trade go through the waterway.
- 3. Oil reserves:** Iran has the world's third-largest proved reserves of crude oil, behind Venezuela and Saudi Arabia
- 4. Poverty:** About 1/3 of Iran's population live in poverty (<\$8.30 per day, World Bank)
- 5. Employment:** Only 38% of Iranians are employed (and just 12% of women). Human capital has been eroded by decades of 'brain drain' to Western countries. Meanwhile, Iran is one of the world's biggest hosts of refugees, particularly those from Afghanistan
- 6. Inflation:** In the first five months of the 25/26 fiscal year was 40% yoy, mainly driven by food and housing costs
- 7. Banks:** Over 70% of Iranian banks fail to meet capital adequacy ratio requirements and nearly half have a negative CAR
- 8. Trade concentration:** Iran's top three trading partners in non-oil exports and imports account for 60% and 70% of the total
- 9. GDP:** Before the recent US strikes on Iran, GDP was expected to contract 2.3% pa for this year and next. Impacted by UN sanctions and falling Chinese demand for Iranian oil.
- 10. Chinese demand:** 11% of China's crude oil and condensate imports come from Iran

Source: World Bank, Bloomberg Finance LP, Deutsche Bank.

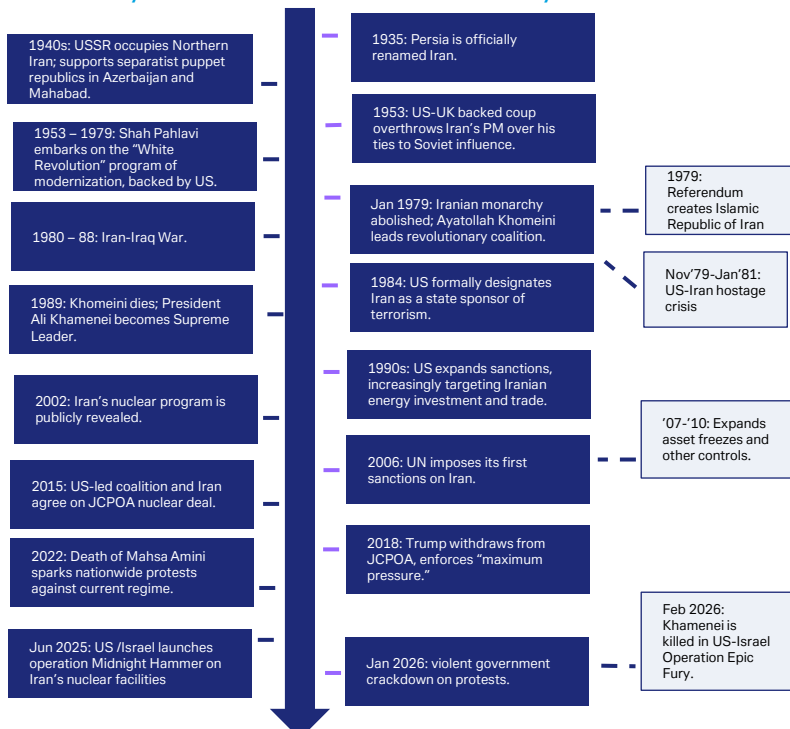


# To put current events in context, the following illustrates Iran's rise from a US-backed state into an Islamic Republic

Initially a battleground for Cold War influence, Iran has transitioned to present itself as a revolutionary, anti-Western state



## Key events in Iran's recent history



Source: dbLumina, Bloomberg Finance LP, Deutsche Bank.

## Iran continues to make overtures east

**China:** Beijing was a source of weapons to Iran until it officially stopped in 2005. More recently, China and Iran have maintained a deep economic relationship, with Beijing being its largest trade partner. In 2021, both sides signed a 25-year strategic partnership worth \$400bn.

**Russia:** Relations between the two have grown closer since the war in Ukraine. In March 2021, Iran and Moscow signed a 20-year strategic framework that deepens cooperation across defence, energy, technology, and sanctions-evasion mechanisms.

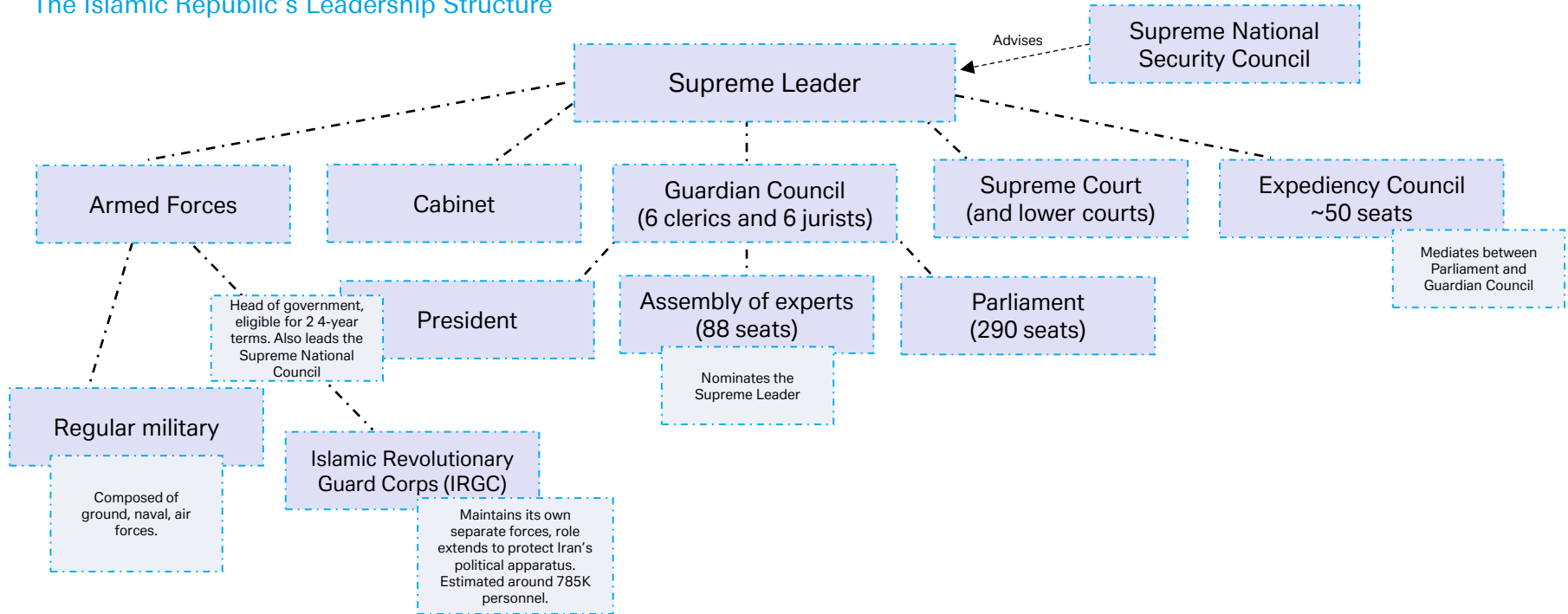
**Lebanon & Yemen:** Iran's relationship with its neighbours are indirect and run mainly through powerful non-state allies (i.e. Hezbollah and the Houthis). Through them, they are able to wield some influence and leverage resulting in the "Axis of Resistance" (a network of groups cooperating against the US, Israel and the Gulf).

Iran governs as an Islamic theocracy. The Ayatollah oversees national policies and commands the armed forces.

With Khamanei gone, an interim leadership council governs Iran while the Assembly of Experts seeks to pick a new leader



## The Islamic Republic's Leadership Structure



Source: Council on Foreign Relations

## Iran's position in the Middle East and its relationship with its neighbours

Iran is at a very weak point following the collapse of the Assad regime in Syria and Israel's significant destruction of Hamas and Hezbollah over the last two years. Still, Iran has the military capabilities to inflict serious harm to its neighbours.



Countries that have experienced strikes from Iran

- Bahrain
- Cyprus
- Iraq
- Israel
- Jordan
- Kuwait
- Oman
- Qatar
- Saudi Arabia
- UAE



Countries where Iran-backed groups have a presence

- West Bank and Gaza Strip (Hamas)
- Lebanon (Hezbollah)
- Iraq (Various groups)
- Yemen (Houthis)

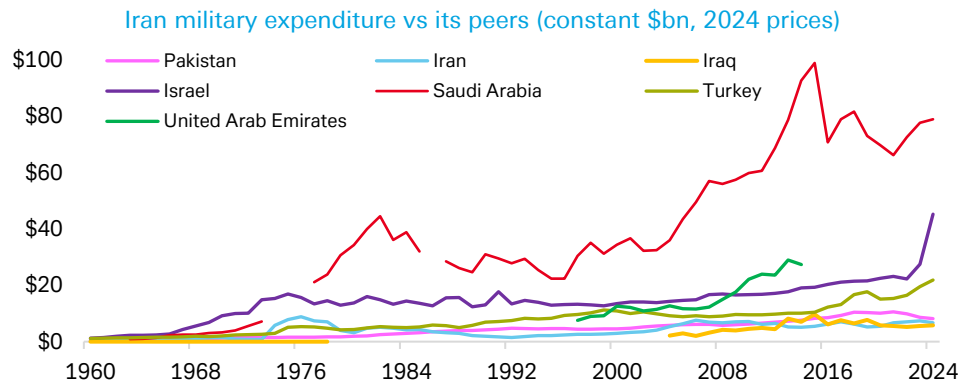
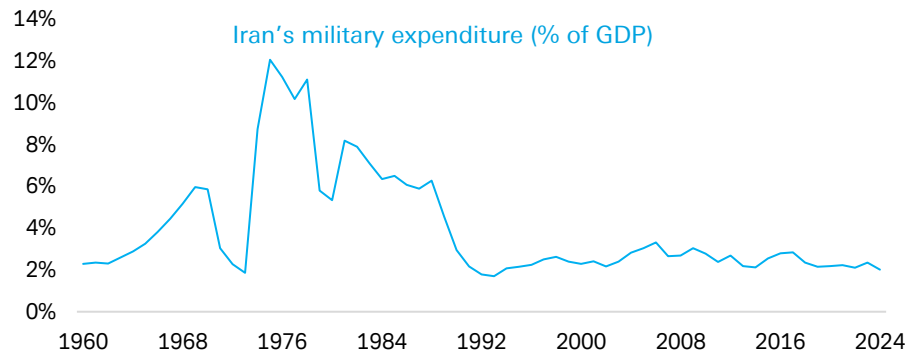
Source: Media reports, Deutsche Bank

# The state of Iran's military

Iran's military budget is modest compared to its regional peers, despite its sensitive geographical position



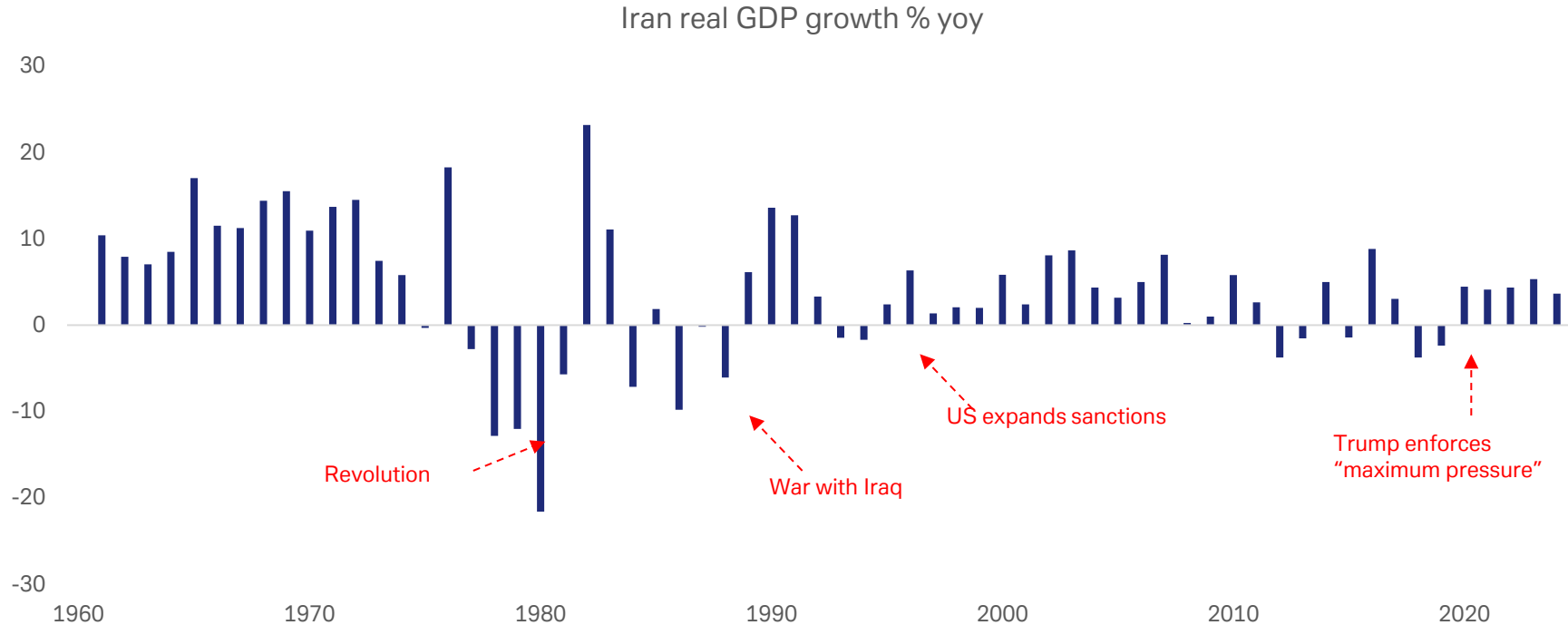
- ❑ Iran maintains the largest stockpile of domestically produced missile and unmanned aerial vehicle systems (US ATA 2025). However, estimating the approximate size is difficult given the lack of public information.
- ❑ Previous official US estimates place the number of ballistic missiles to be over 3,000, although Tehran's capacity weakened after the US/Israel strikes in June 2025.
- ❑ Iran holds at least 14 Iranian ballistic missile variants, according to the US National Air and Space Intelligence Center (Ballistic missiles travel faster than drones and cruise missiles, making them harder to intercept).
- ❑ Personnel in Iran's conventional military totals around 420k, whilst the IRGC is reportedly around 785k, according to IISS.
- ❑ The Iranian Parliament recently approved an increase to \$9.23bn in military expenditure for 2026-2027.
  - ❑ Official numbers are not available, but Iranian media [reports](#) rough breakdowns of \$4.62bn allocated to the Ministry of Defence and Armed Forces Logistics and \$1.88bn to the IRGC.



Source: SIPRI calculations and estimates. Data for certain years unavailable. Note that different sources calculate figures in different ways.

# Iran's economy

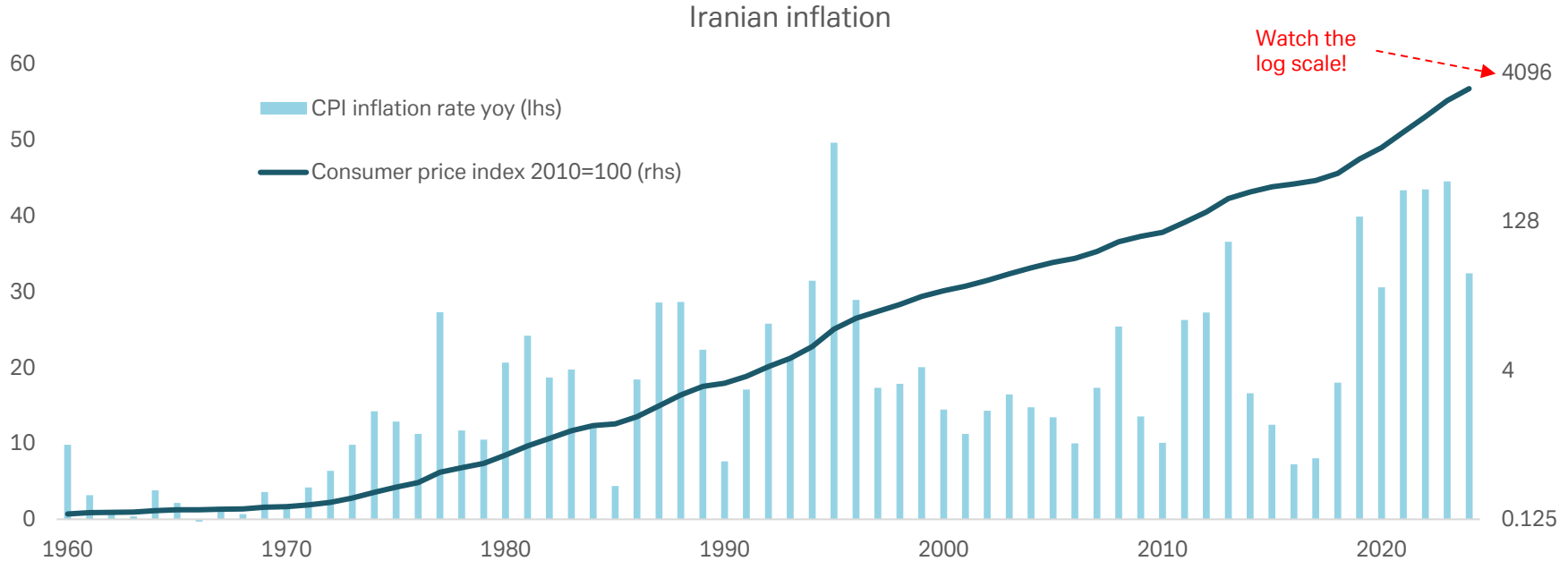
The export of hydrocarbons has been critical to Iran's ability to produce semi-consistent growth over the last three decades



Source: World Bank, Deutsche Bank

## Iran's economy

Spiralling food and housing prices have been a tragic result of crippling inflation, particularly for the 1/3 of people in the country who live below the poverty line.



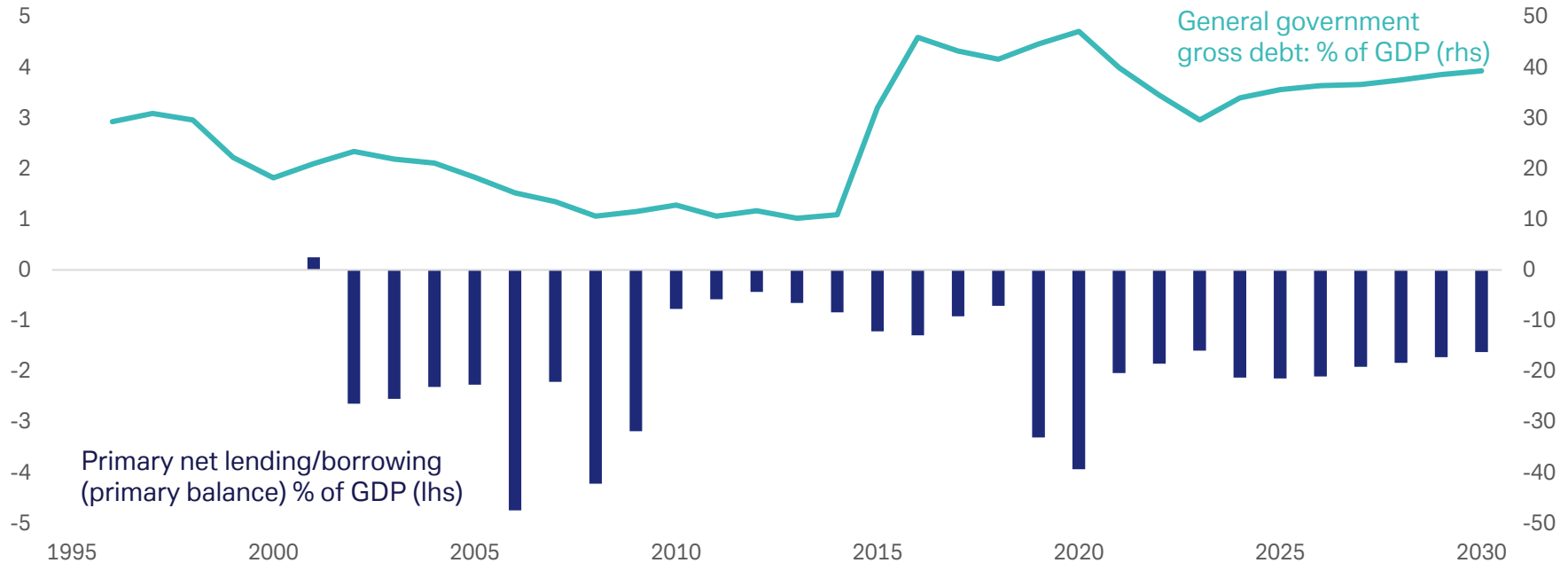
Source: World Bank, Deutsche Bank

## Iran's economy

Some figures are subject to estimates by the IMF, however, they give an indication of the trends in Iran's economy and show how the government is able to maintain its finances despite presiding over an economy that produces below its potential



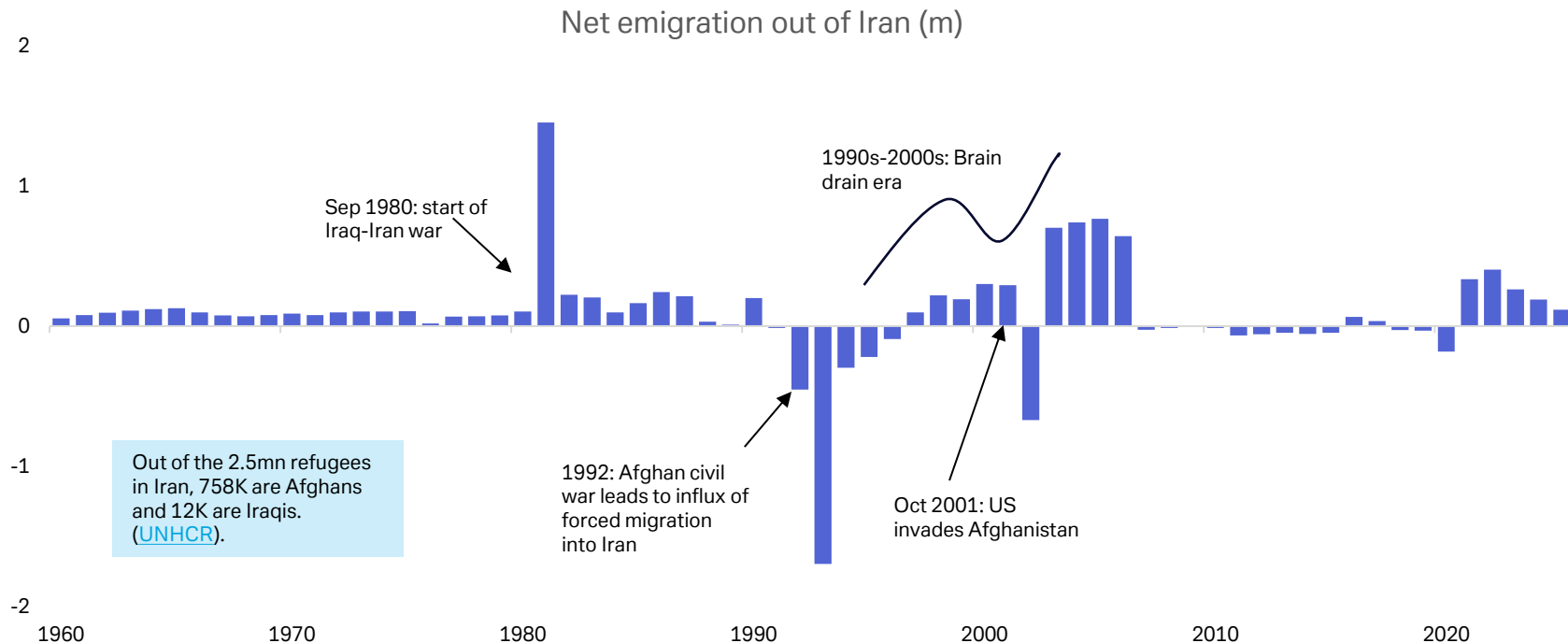
Iran budget and debt position



Source: IMF, Deutsche Bank

## Iran's migration picture is caught between absorbing a large proportion of Afghan refugees, whilst steadily losing a disproportionate share of its own educated, urban middle class

Key drivers of the “brain drain” are high inflation and currency collapse, sanctions-driven stagnation, corruption, and repression of dissent



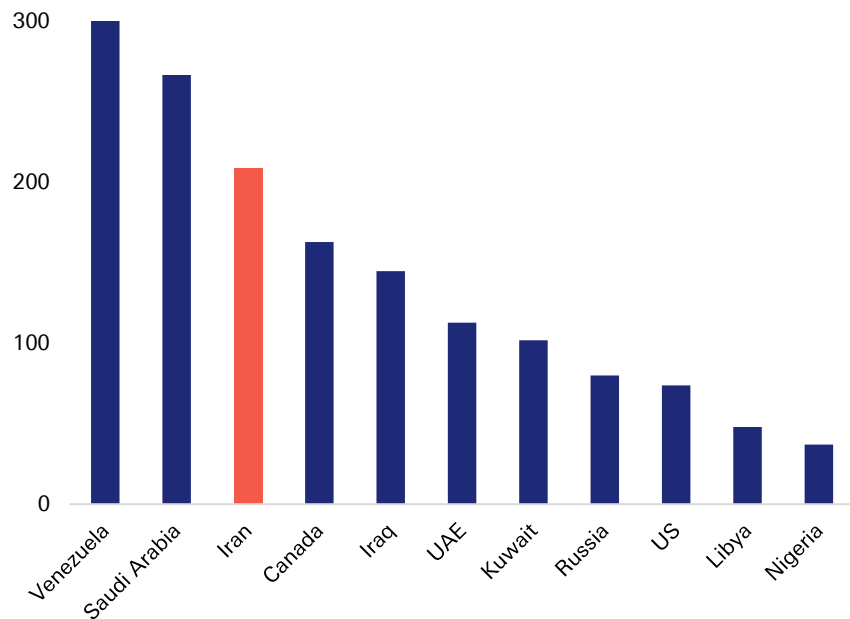
Source: World Bank, Deutsche Bank Research.



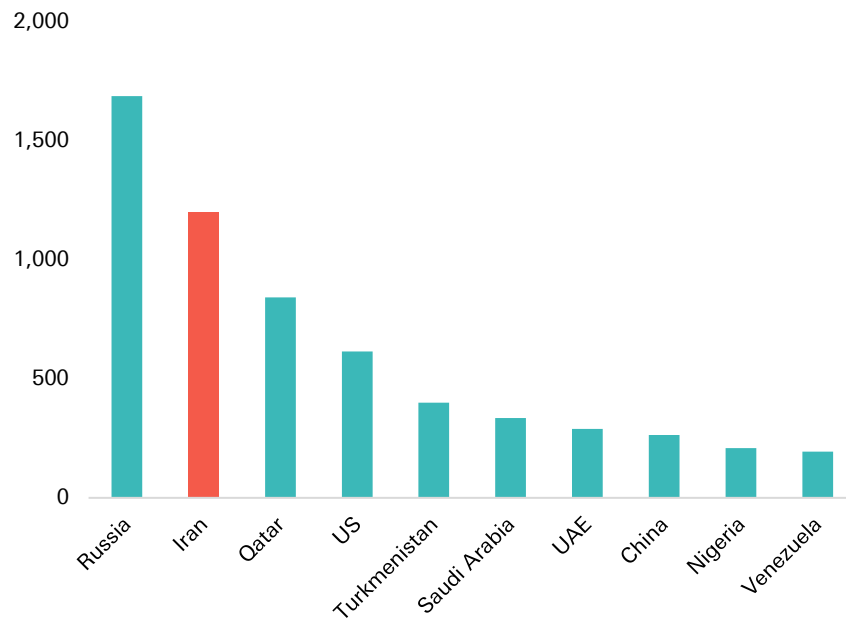
Iran is a top-3 global holder of crude and natural gas. That means the impact on commodity markets will reverberate beyond the immediate disruptions to the oil trade.



Largest proved reserve holders of crude oil  
2024 oil reserves (bn barrels)



Largest proved reserve holders of natural gas  
2024 gas reserves (tn cubic feet)



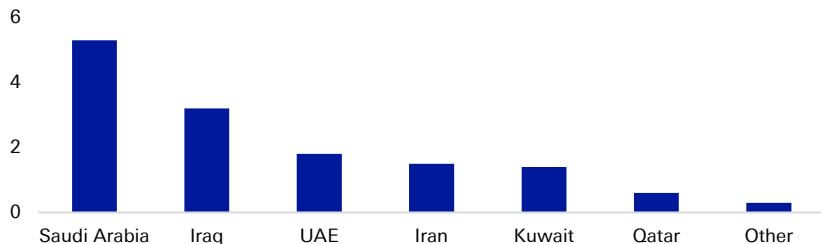
Source: EIA, Deutsche Bank.

How important is the Strait of Hormuz? More than 25% of global seaborne oil trade and c 20% global LNG trade go through the strait. Asian economies are the key consumer.

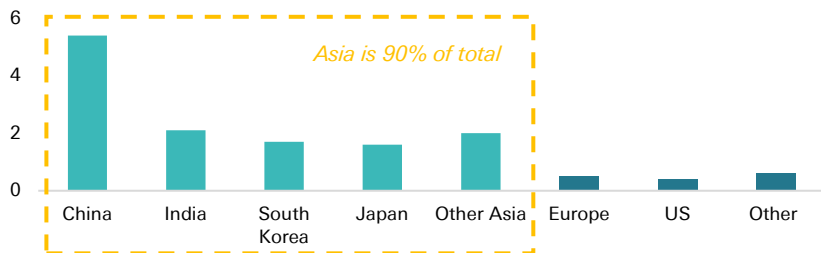


## Volume of crude oil and condensate transported through the Strait of Hormuz...

... by origin (mbpd)

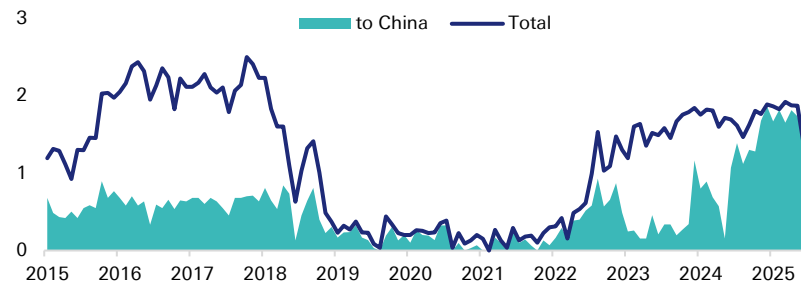


... by destination (mbpd)

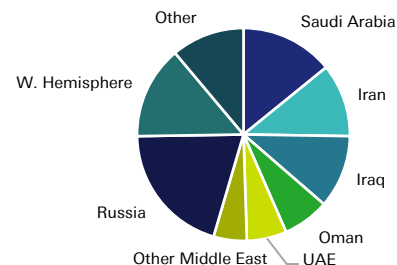


Source: EIA, Bloomberg Finance LP, Deutsche Bank.

Iran crude oil exports, mbpd



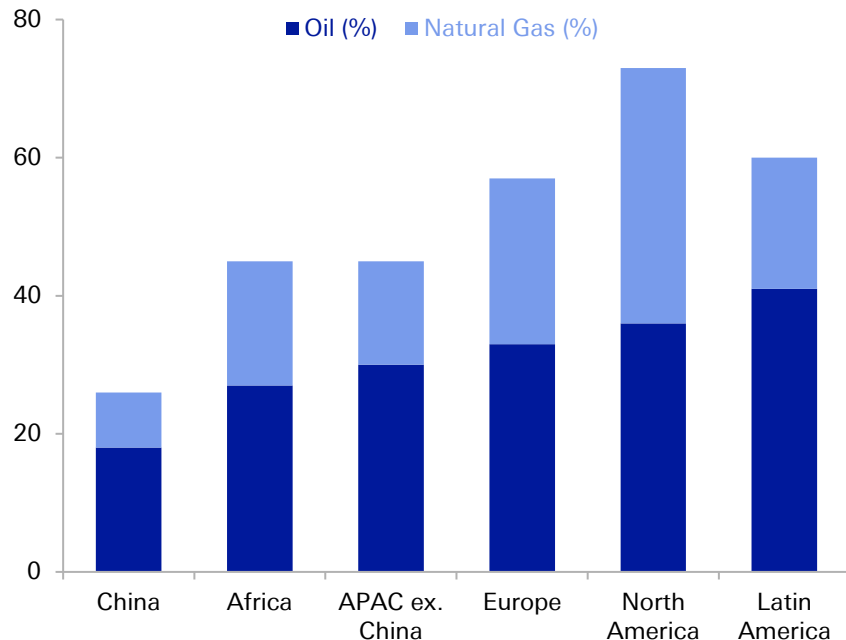
China's crude oil and condensate imports by source, 2024



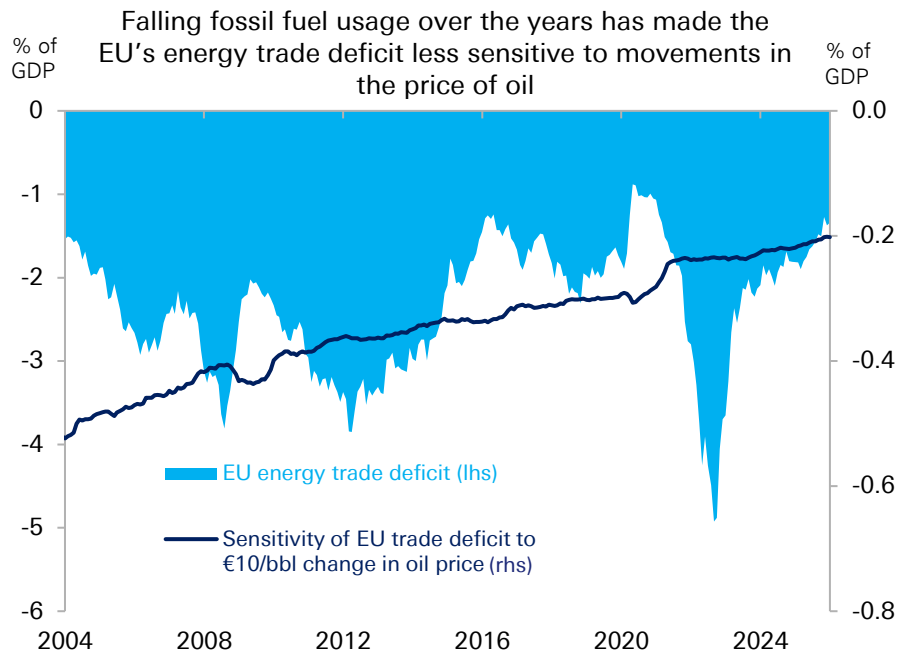
The potential energy shock: China and the EU are the world's two largest energy importers, however, the fossil fuel intensity of their economies is declining, potentially limiting the impact



Share of oil and gas in total energy supply



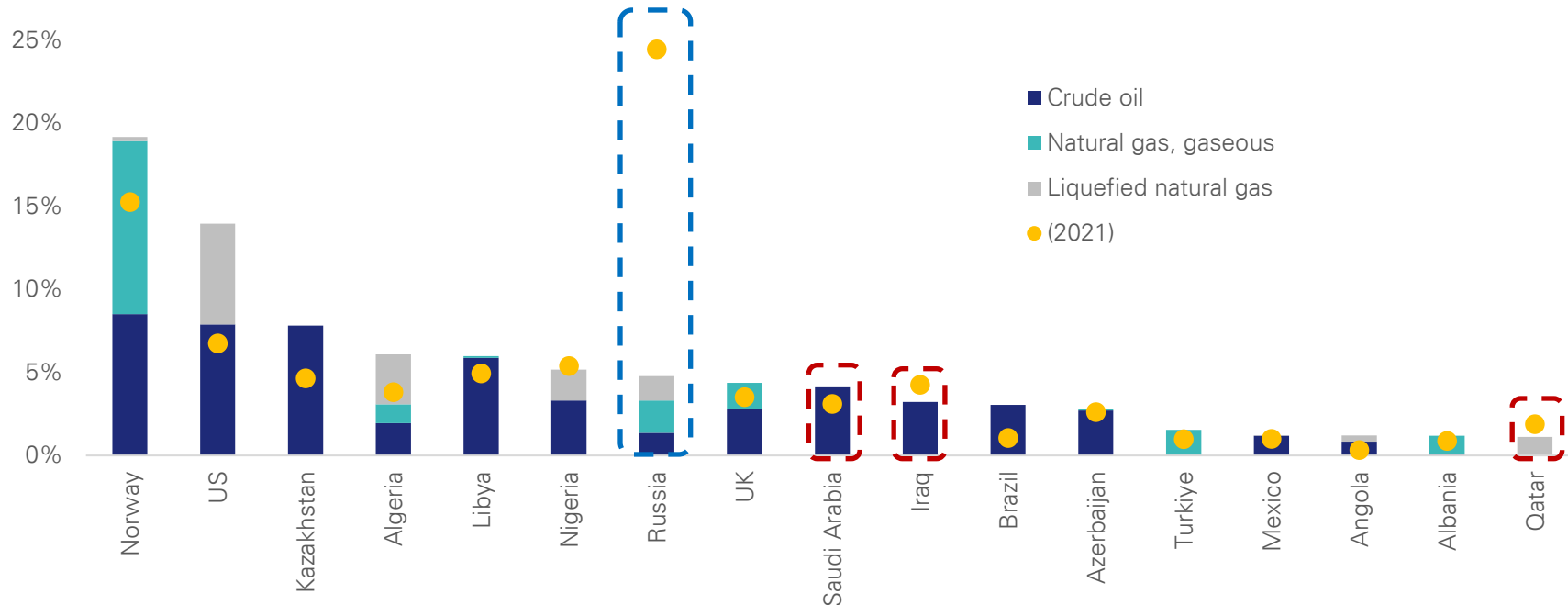
Source: Deutsche Bank, IEA, Eurostat



Europe has become more diversified in its energy imports and has a low direct exposure to the core Middle East countries (unlike its substantial exposure to Russia in 2022)



European Union - Top import sources of crude oil and natural gas (2025)

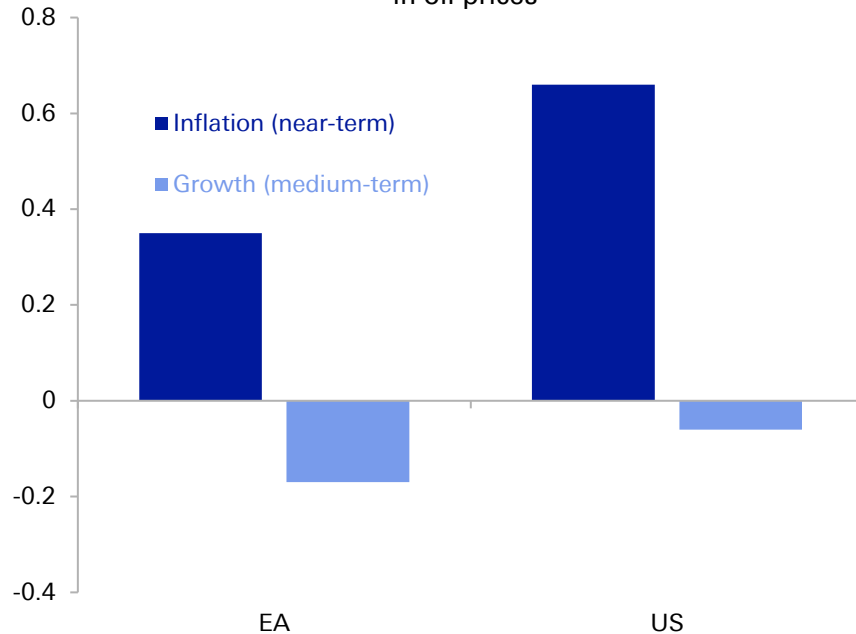


Source: Deutsche Bank, Eurostat

When it comes to oil shocks, the US is more sensitive on inflation while Europe is more sensitive on growth. However, the sharp rise in European natural gas prices does increase its exposure.

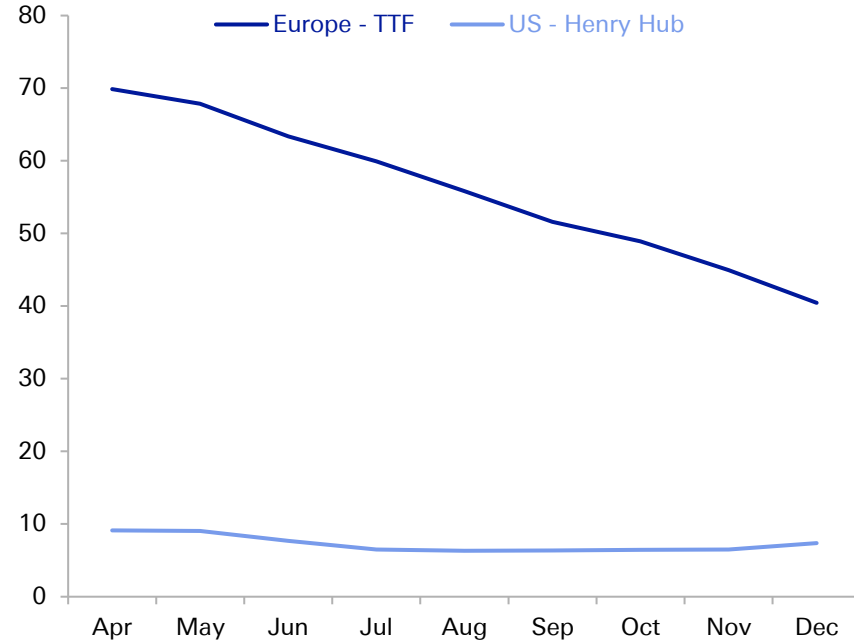


The inflation and growth impact of a \$20/bbl increase in oil prices

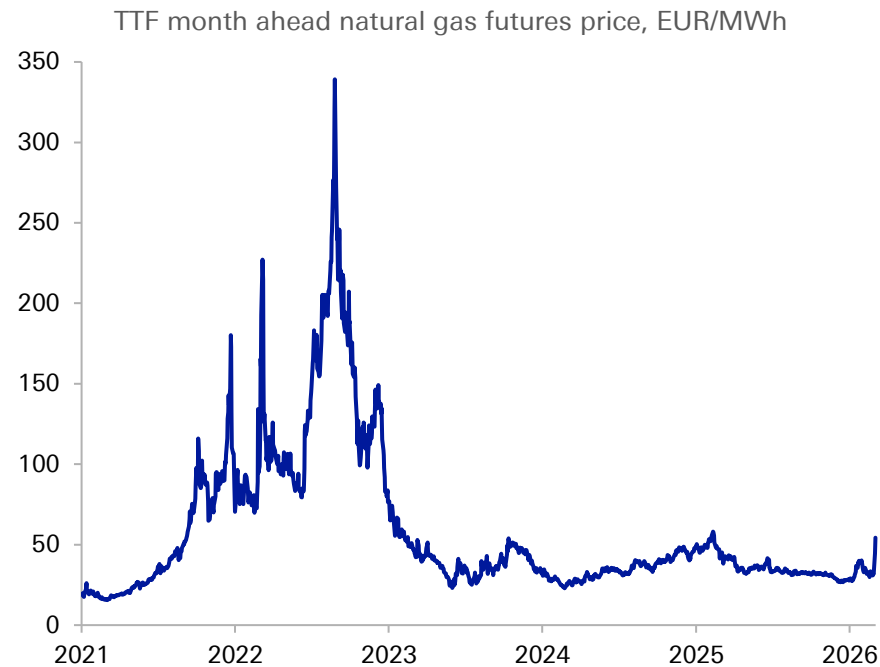
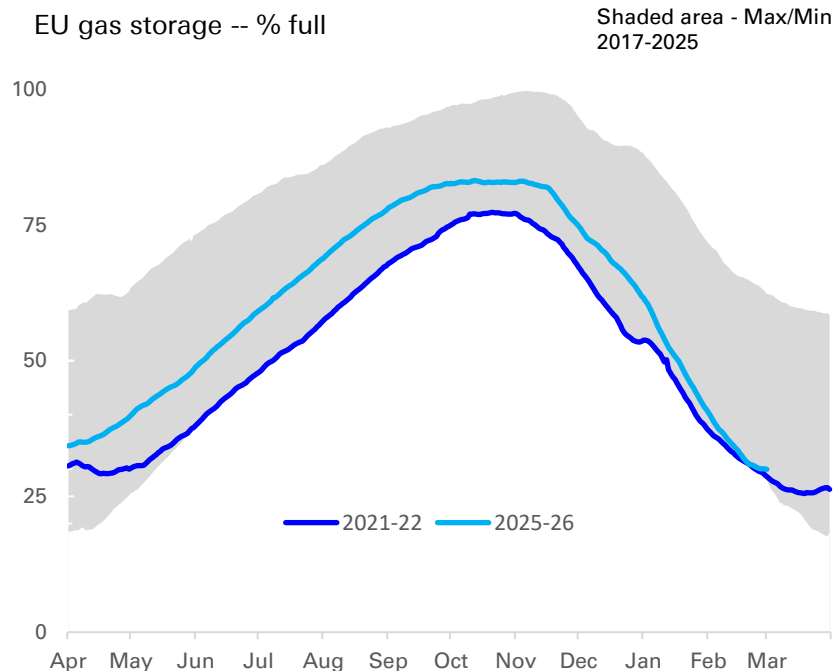


Source: Deutsche Bank, ECB, FRBUS, Bloomberg Finance LP

% increase in natural gas futures (Feb 27 - Mar 3)

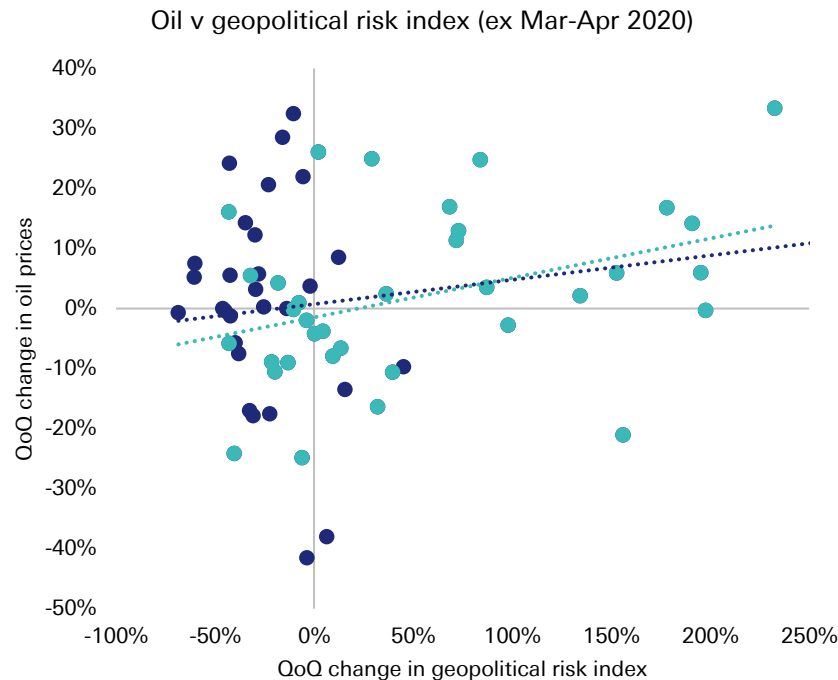
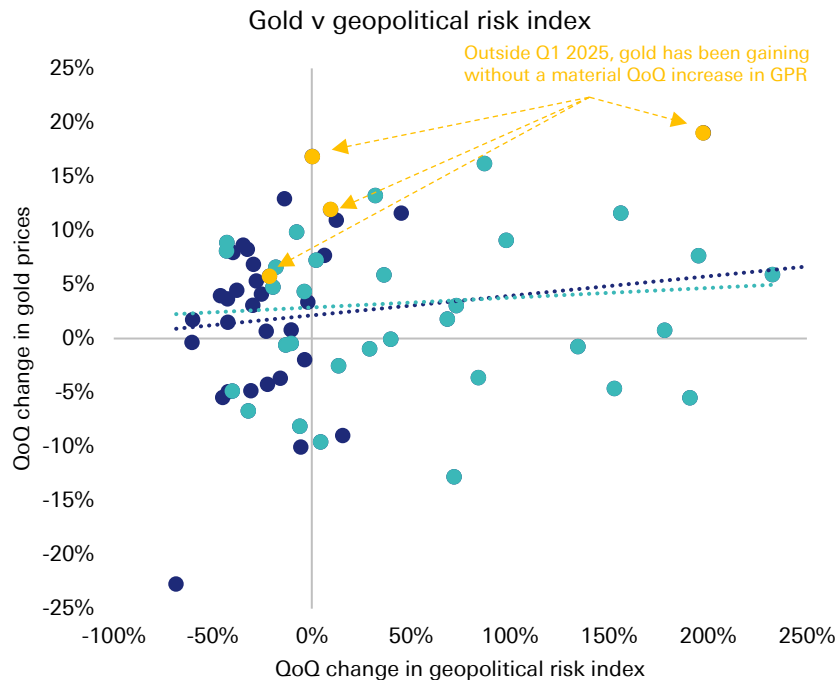


Europe's low storage levels does provide some concern about a LNG price shock, but the size of the gas price move is an order of magnitude smaller than it was in 2022.



Source: Gas Infrastructure Europe, Bloomberg Finance LP, Deutsche Bank

Among commodities, oil is a better proxy for geopolitical risk in markets (outside covid) than gold even with the US less reliant on oil than in the 1970s.



Source: Bloomberg Finance LP, Caldara, Dario and Matteo Iacoviello (2022), "Measuring Geopolitical Risk," American Economic Review, Deutsche Bank.



# Appendix 1

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# Iran Shock: Key transmission channels

3 March 2026

2026 is barely two months old and has already demonstrated how shock prone the world has become: the Greenland threat, euro FX appreciation, AI fallout and now war in the Middle East.

In light of the conflict in the Middle East, we use this note to discuss some the ramifications for the European macro outlook. In each we benchmark events in 2026 against the geopolitical and energy shock suffered by Europe in 2022.

First, the rise of geopolitical uncertainty. There is inevitably a focus on energy costs and the risk of higher inflation. However, economic uncertainty is the enemy of growth and there tends to be a stronger correlation between geopolitical uncertainty and economic uncertainty when there is a nexus with energy, trade or financing. Using daily data, we estimate the increase in geopolitical uncertainty so far would cost the euro area about 0.25% of GDP if it persists, though this could be lower if the resilience seen during the 2025 tariff-driven uncertainty shock holds up. In comparison, the uncertainty shock in 2022 was estimated at more than 1% of GDP. That said, fiscal policy leaned against the shock in 2022. It's less obvious that fiscal policy has the space to lean against the shock in 2026.

Second, the impact of rising oil and gas prices on inflation. There is good news, so far at least. Energy curves are pricing a mostly transitory shock, albeit enough to lift 2026 HICP inflation by 0.2bp. This is only a fraction of the energy inflation surge we saw in 2022. If a big energy boost is avoided, non-linear dynamics – triggered by more frequent price changes – become less likely too. With trade between Europe and Asia already avoiding the Suez canal before the latest events, and with Europe's energy imports more diversified, the risk of an escalating regional conflict amplifying an inflation shock via supply chains is also lower.

Third, the role that stable inflation expectations will play in steadying the ECB's hand. A central bank can afford to look through an energy supply shock as long as inflation expectations remain well anchored. We look at recent data and how consistent they are with the definition of well-anchored expectations. In 2022, the de-anchoring occurred when HICP was rising rapidly above 2%. In 2026, the increase in energy costs so far is more consistent with removing the risk of undershooting of the 2% inflation target than with clear overshooting.

This is a situation with considerable uncertainty. We don't know if we are at the end of the beginning or the beginning of the end of the military operations in the Middle East. The market had been worrying lately about a larger HICP undershoot and whether this could compel the ECB to ease further. Rising energy prices muddy the waters but broadly speaking don't yet challenge the ECB baseline of inflation on target in the medium-term. The ECB's data-dependent, meeting-by-meeting, no pre-commitment approach to policy feels wholly appropriate. It can afford to assess how the situation in the Middle East evolves before determining whether and how to respond.

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## Introduction

The Euro Area has recently demonstrated remarkable resilience, with its economy consistently exceeding market expectations over the past few quarters. The robust labor market, the stimulus from Next Generation EU (NGEU), and the lingering positive effects of earlier monetary policy easing have collectively enabled the economy to navigate previous shocks, including recent tariff disputes and the potential spread of French political tensions. However, the frequency of these shocks is notably high and increasing. The escalating situation in the Middle East will once again test the Euro Area's resilience.

Against this backdrop, we present our initial thoughts on how the recent Iranian shock may translate into economic impacts. In our view, three key shock propagation channels warrant close monitoring:

**1. Uncertainty:** Geopolitical uncertainty is highly likely to rise. The daily US Political Uncertainty Indicator already registered a significant jump over the weekend, a magnitude not massively dissimilar to that observed immediately after Liberation Day. Geopolitical uncertainty correlates well with our Economy-wide Uncertainty Indicator. Our empirical estimates suggest that if the current increase in political uncertainty were to persist for a full year, it could reduce Euro Area GDP growth by 0.25 pp. Risks are probably tilted towards a smaller impact – economy-wide policy uncertainty appears to have become less sensitive to direct policy uncertainty in recent times, with the resilience of domestic demand likely providing some buffer against such shocks.

The negative news, however, is that we are likely only at the beginning of this unfolding shock with the possibility that the uncertainty indicator could rise further. The Iranian shock, particularly given its nexus to energy, bears comparison to the Ukraine shock. For reference, the 2022 Russian invasion of Ukraine, which generated a sharp rise in energy costs, resulted in a 3-standard-deviation shock to economic uncertainty. Our models suggested such a shock could cut GDP growth by over 1 pp, plunging the economy into recession. Yet, it did not. Fiscal intervention definitively offset the effect, with Member States implementing various measures to reduce energy bills for households and firms.

This time, however, the situation may differ. Defence spending cannot create the same fiscal buffer, and public finances, already stretched in many European countries, may not support additional stimulus, nor do all countries have the same fiscal maneuvering room.

**2. Energy Prices and Inflation:** The second shock propagation factor involves oil and how changes in energy prices will transmit to inflation. Crude oil recently surged by ~10% to nearly \$80 per barrel (bbl), while natural gas prices jumped 40% to EUR 46 per MWh following Qatar's suspension of LNG production. Our updated model indicates this will elevate the 2026 Euro Area HICP profile by c.0.2 percentage points, keeping inflation above the 2% target from March onwards. If this shock persists for 12 months, 2026 HICP would be 0.3 pp higher (2.1%). Although this necessitates close monitoring, the current energy shock seems far less severe than the 2022 energy shock: even if oil and gas were to double from their 2025 levels, energy inflation would reach approximately 18% compared to 37% in 2022.

**3. Inflation Expectations:** A third critical element to closely watch is inflation



expectations, which are key for the ECB and any potential policy actions the Bank may decide to undertake. The ECB will closely monitor longer-term inflation expectations, as their anchoring is crucial for maintaining a steady monetary policy, allowing them to look through short-term disruptions. Past instances of de-anchoring expectations, such as the 2022 energy shock, have prompted ECB policy responses. During that crisis, not only short-term but also longer-term professional and market expectations rose significantly.

While current inflation expectations appear well-anchored, several factors warrant close monitoring: household expectations, being particularly sensitive to food and petrol prices, are the most vulnerable to de-anchoring; the ECB's concern often lies with shifts in the middle of the expectation distribution, not just the tails; and longer-term expectations are more critical than short-term spikes, which can be rational responses to data changes.

**4. Supply chain risks:** Relatively good news is that the risk of an energy price shock being amplified by broader supply-side disruptions to shipping and energy trade appears contained. First, the physical trade of non-energy goods has already adapted, with shipping between Europe and Asia largely avoiding the Suez Canal before the latest events. This suggests a lower risk of an additional inflation shock from disruptions to non-energy supply chains. Second, Europe's direct energy availability is also buffered by its current import mix, which is towards suppliers like Norway and the US, with a relatively low share of imports coming directly from Persian Gulf states.

**Overall, we see risks as clearly lower than in 2022 though some areas of vulnerability have increased.** When comparing the current shock to the most "comparable" past event—the energy crisis following the Ukraine/Russia war—there are both positive and negative elements to consider.

On the positive side, even assuming a protracted increase in oil prices, we are still far from the magnitude of increase recorded during the 2022 energy shock, when Brent crude moved above \$100/bbl for nearly six months. This should reduce the risk of inflation expectations de-anchoring. The EU's energy supply has also become more geographically diversified and the fossil fuel intensity of its economy has declined, so the current shock is likely to be a price shock rather than raising risks of physical shortages as we saw with natural gas supplies in 2022. Finally, the Iran conflict has less in the way of other direct ramifications for Europe, such as via trade disruption.

On the negative side, however, the Euro Area probably has less fiscal buffer now than in 2022. The European Commission can reactivate its General Escape Clause if needed, but it would apply to public finances that are already stretched in several European countries. And while the 2022 shock came at a time of a strong post-Covid rebound, the current shock comes when the Euro Area economy is suffering from various external headwinds. First, the lingering energy competitiveness shock – and with the current risks coming on top of already low gas storage at the end of the winter season, this is unlikely to fade any time soon. Second, ongoing trade uncertainty stemming from the US. Third, the Second China Shock which has undermined Europe's (and especially Germany's) manufacturing competitiveness.

We are likely only at the beginning of this unfolding shock, with several factors still unknown. If things worsen, policymakers may need to step in again, with probably more focus on the ECB, as monetary policy typically acts fast and faster than fiscal

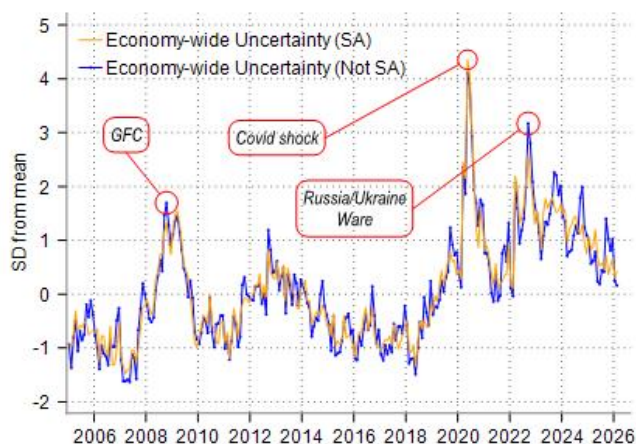


policy. The ECB's data-dependent, meeting-by-meeting, no-pre-commitment approach to policy feels wholly appropriate, allowing it to assess how the situation in the Middle East evolves before determining whether and how to respond

## # 1 Channel: Uncertainty

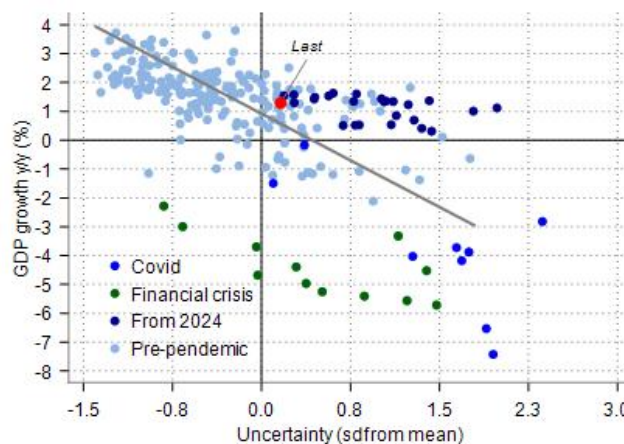
Uncertainty serves as a critical channel for shock propagation within the economy. As our regular readers know, we have developed an [Economy-wide Uncertainty](#) indicator that measures uncertainty by focusing on survey data, specifically how panelists respond to questions regarding expectations. The more concentrated their answers are towards a single outcome, the lower the perceived uncertainty. Our economy-wide uncertainty indicator demonstrates a strong correlation with GDP growth, and our analysis suggests that a one-standard-deviation increase in this indicator would typically reduce GDP growth by 0.4 pp.

Figure 1: DB Economy-wide Policy Uncertainty



Source : Deutsche Bank, EC, Macrobond

Figure 2: DB Economy Wide Policy Uncertainty and GDP growth



Source : Deutsche Bank, EC, Macrobond

**Assessing Recent Events:** Our economy-wide uncertainty indicator relies on European Commission surveys, which are released monthly. Therefore, we will need to await the end of the month to obtain the first comprehensive assessment of how European business and household confidence has evolved as a result of the latest events in the Middle East, and consequently, to infer changes in overall uncertainty. In the interim, two analyses can be conducted to assess how uncertainty might evolve and what might be the impact on the economy:

- **#1. Relying on Daily Indicators:** The Economic Policy Uncertainty website provides daily data on US policy uncertainty. This indicator, when analyzed on a monthly basis, exhibits a good correlation with the European Policy Uncertainty Indicator. Historically, a one-standard-deviation (1 sd) increase in US policy uncertainty has translated into a 0.6 sd increase in the European policy uncertainty indicator. Our research indicates that a persistent shock (lasting 12 months) of one standard deviation to policy uncertainty elevates economic uncertainty by 0.5 sd, which would subsequently reduce GDP growth by 0.2 pp.

While the level of the daily US policy uncertainty indicator has been



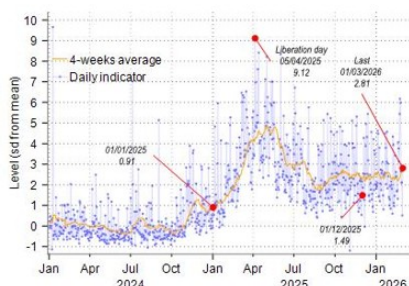


relatively low in recent weeks, it jumped on March 1<sup>st</sup> after the events in the Middle East. The indicator increased of 2.3 sd, a surge not dissimilar to after Liberation Day, suggesting a notable initial jump.

Considering the established relationships between US policy uncertainty, European policy uncertainty, and overall Euro Area economy-wide policy uncertainty, a sustained shock of this magnitude, protracted over a full year, could potentially cause a drag of 0.27 pp on Euro Area GDP growth.

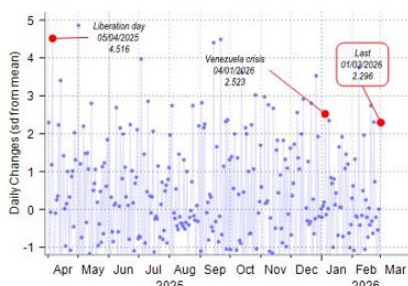
On the positive side, economy-wide policy uncertainty appears to have become less sensitive to changes in policy uncertainty in recent times. The strength of its domestic demand can create some buffers. The negative news, however, is that we are likely only at the beginning of this new shock. With several factors still unknown, there is a distinct possibility that the uncertainty indicator could rise further.

Figure 3: US daily policy uncertainty... its levels



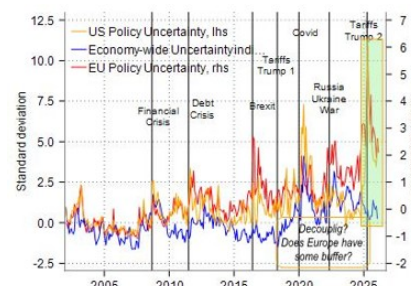
Source : Deutsche Bank, [Economic Policy Uncertainty](#), Macrobond

Figure 4: ..and its daily changes



Source : Deutsche Bank, [Economic Policy Uncertainty](#), Macrobond

Figure 5: US and European monthly uncertainty



Source : Deutsche Bank, [Economic Policy Uncertainty](#), EC, Macrobond

- **# 2. Drawing Lessons from Past Similar Events:** To contextualize the current situation, it is useful to look at similar events in the past. The closest comparable example is the onset of the Ukraine war in February 2022. On that occasion, our economy-wide uncertainty indicator rose by 3 standard deviations. If uncertainty were to rise by such an amount, our model would suggest that the economy would enter a recession, similar to what occurred after the Global Financial Crisis (when the Uncertainty Indicator rose by 2 sd) and during the COVID-19 pandemic (a rise of 4 sd).

However, following the Ukraine/Russia war, the economy did not enter a recession. A key factor explaining this divergence was the vigorous fiscal reaction. The European Commission had activated the General Escape Clause following the COVID-19 shock. Given the gravity of the pandemic, the relaxation of fiscal policy was insufficient to prevent a Euro Area recession. Nevertheless, the Commission extended the activation of the General Escape Clause to address the energy shock prompted by the war. To put this in context, [in Autumn 2021](#) the European Commission projected a stimulative fiscal impulse for 2022 worth less than 1 pp of GDP. One year later, in [Autumn 2022](#), the estimated fiscal stance for 2022 was revised to 2.25 pp of GDP, representing an additional fiscal boost of over 1 pp.

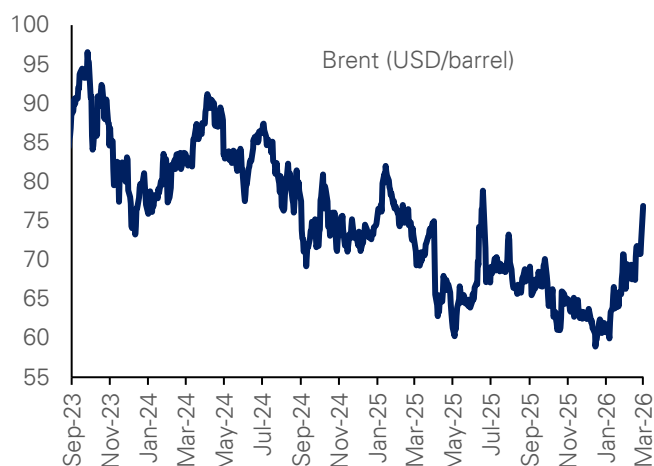


This significant fiscal boost helped prevent the uncertainty shock from translating into another recession, effectively creating an economic buffer. Today, however, the situation might be different. The Commission has activated the National Escape Clause (NEC), but solely for defence-related expenditures. These expenditures typically have a [low fiscal multiplier](#), are politically challenging to implement, and not all Member States possess the same [fiscal maneuvering room](#). Notably, France, Italy, and Spain have not requested the activation of the NEC. While circumstances might change and the EU could potentially call for the activation of the General Escape Clause again, the starting point today is one in which fiscal policy has less room for manoeuvre. With the ECB also running off its balance sheet today vs reinvesting in 2022, monetary policy is less accommodating of fiscal easing. In other words, there might be more downside risk in the event of a more persistent energy shock.

## # 2 Channel: Inflation

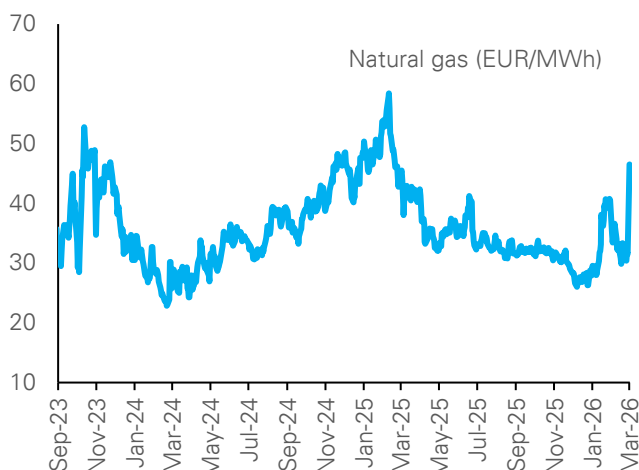
Energy prices operate as the main transmission channel of geopolitical risk to inflation. Since Friday, crude oil has surged by c.10% to nearly USD 80 bbl. The move in natural gas has been more pronounced, with prices climbing nearly 50% to EUR 46 MWh after Qatar announced the suspension of natural gas production at some of its facilities.

Figure 6: Oil up by 10%...



Source : Deutsche Bank Research, Bloomberg Finance LP

Figure 7: ...gas by almost 50%



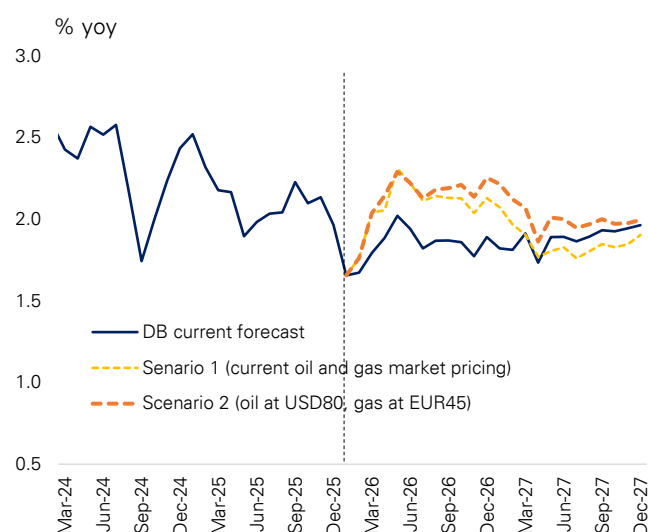
Source : Deutsche Bank Research, Bloomberg Finance LP

To illustrate the direct impact of higher energy prices to consumers, we have updated our inflation model with the latest market pricing data. Relative to our baseline scenario, the 2026 headline HICP profile shifts up by about 0.2pp, with inflation staying above the 2% target from March onwards. As the energy impulse fades, headline inflation would fall back to 1.8-1.9% in 2027. However, if the shock turned out to be more persistent, and both oil and gas stayed at their current levels over the next 12 months, headline HICP would be 0.3pp higher than currently assumed in 2026 (2.1%) and 0.2pp higher in 2027 (2.1%).



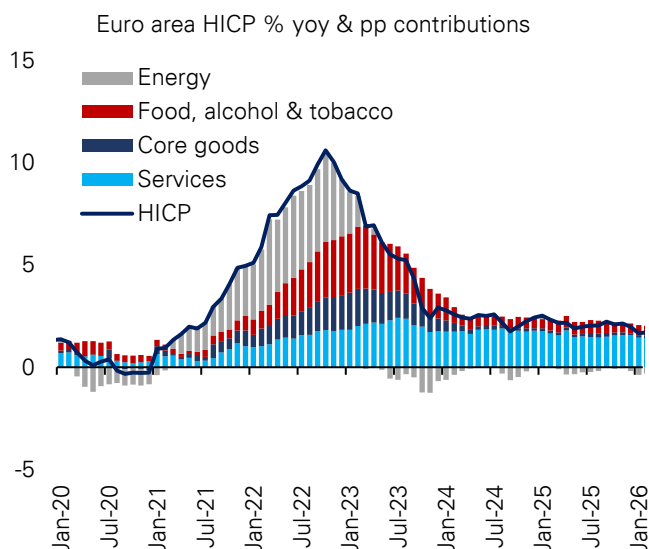
While this would imply no undershooting of the inflation target and warrant a more cautious approach from the ECB, it is worth highlighting that we are still far from replaying the inflation surge experienced during the 2022 energy shock. For context, energy inflation reached 37% in 2022. If oil and gas were to double from their 2025 levels (to USD 136 bbl and EUR 73 MWh, respectively), our model shows energy inflation would rise to ~18%. Our commodity strategists [expect](#) oil prices to remain around USD 80 bbl for the time being, with potential for a USD 80-100 bbl range if shipping traffic in the Strait of Hormuz remains threatened and USD 200 bbl is the Strait is fully closed for an extended period of time.

Figure 8: Surging energy prices could push headline inflation back above target



Source : Deutsche Bank Research, Eurostat, Haver Analytics

Figure 9: Food prices amplified inflationary pressures in 2022



Source : Deutsche Bank Research, Eurostat, Haver Analytics

Beyond the direct boost to inflation, renewed energy price increases risk triggering indirect and second-round effects. Food prices for instance, which also played a significant role in the 2022 inflation surge, could add to the inflationary pressures. [ECB research](#) has shown that higher energy costs were the main contributor to the rise in food inflation over that period. While we see risks to food inflation currently tilted to the downside, this could shift rapidly.

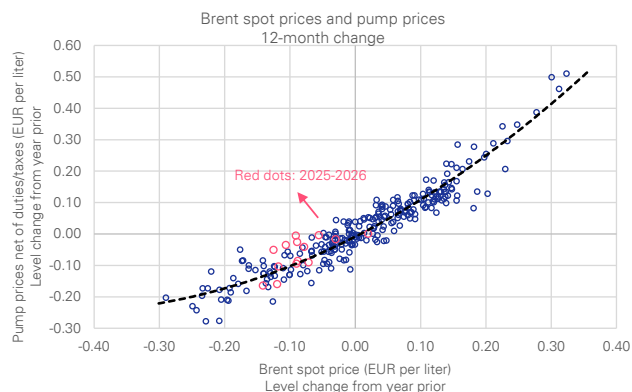
The model outputs described above represent our central estimate, but we identify both amplifying and mitigating factors that create a balanced risk profile around this baseline. On the upside for inflation, there is a risk of an asymmetric price response; empirically, end-consumer prices have historically tended to react more strongly to oil price shocks on the upside than on the downside ([Figure 10](#)). Conversely, a mitigating factor is the EU's strategic energy reserves. Member states are required to hold emergency oil stocks equivalent to at least 90 days of net imports, a rule with which compliance appears to be high ([Figure 11](#)). However, it is worth noting that as of late 2025, the stock levels of larger member states appear a little lower. The reserves could be released, if not to keep prices low, then to prevent any major shortages from materializing.

Finally, note that these developments will leave the upcoming ECB staff projections



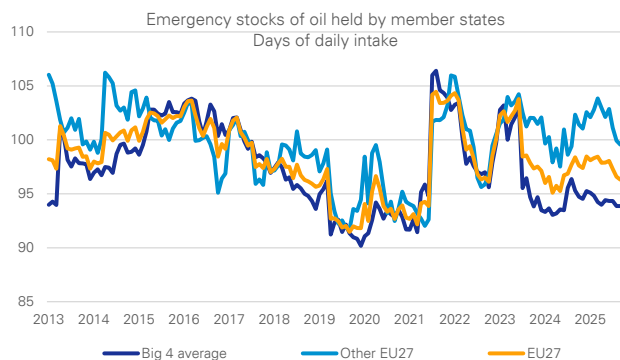
already outdated. The new forecasts, set for release on March 19, will be based on technical assumptions set between February 11-24. Over that window, oil prices were around USD 67 bbl while gas was around EUR 31 MWh.

Figure 10: Asymmetric pass-through from crude to pump prices



Source : Deutsche Bank Research, Eurostat

Figure 11: EU countries strategic oil stocks provide a buffer



Source : Deutsche Bank Research, Eurostat

### #3 Channel: Inflation expectations

One of the data signals the ECB will be watching closely during this period is longer-term inflation expectations. The ECB's monetary policy strategy is focused on inflation in the medium term, and if inflation expectations remain well anchored, it will give the ECB confidence to look through any short-term disruptions from current events and keep policy on hold throughout this year. In the past risks of inflation expectations de-anchoring have coincided with policy responses from the ECB, both to the upside (the pandemic and energy shock from Russia's invasion of Ukraine) as well as the downside (the period of unconventional policy following the euro debt crisis)

The energy shock resulting from Russia's invasion of Ukraine in 2022 is likely to provide the closest analogy to the current events given what we know so far. Although inflation expectations had been rising because of the pandemic (and the broader economic recovery before that), both professional (SPF) and market-implied inflation expectations rose significantly following Russia's invasion in February 2022. Importantly, it was not just short-term expectations that rose, but longer-term expectations also rose and remained elevated.





Figure 12: Expectations from the Survey of Professional Forecasters rose across all horizons (%)



Source : Deutsche Bank Research, Haver Analytics LP

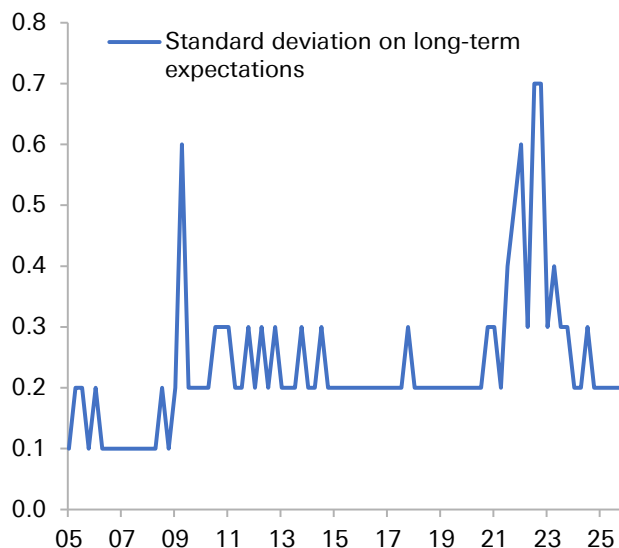
Figure 13: Market-implied expectations also rose and remained high for a while (%)



Source : Deutsche Bank Research, Bloomberg Finance LP

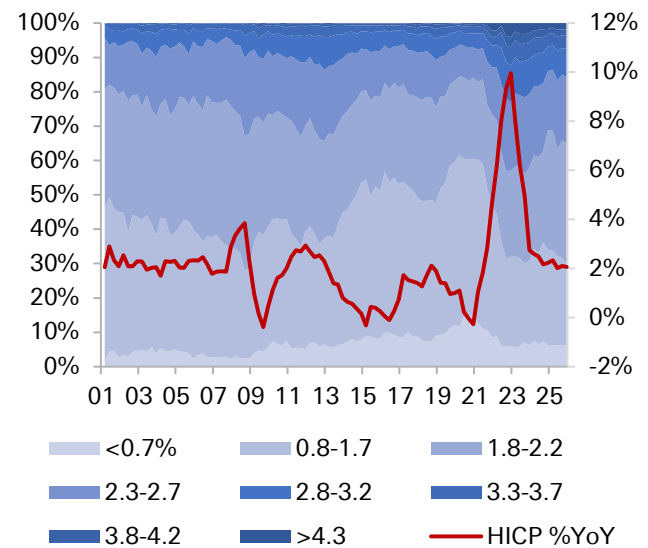
The increase in long-term professional expectations (SPF) also saw the dispersion of expectations increase, with the standard deviation of the distribution rising to a record high. The increase was driven by the mid-to-upper tail of the distribution. The share of respondents with expectations above 2.3% rose from an average of 21% in 2021 to an average of 38% in 2022, and 42.4% in 2023 (its peak). It was this broad-based increase that gave the ECB concern that inflation expectations risked de-anchoring, and contributed to their decision to respond to the risk of second-round effects by hiking policy rates.

Figure 14: The rise in long-term SPF expectations was caused by a change in the shape of the distribution



Source : Deutsche Bank Research, Haver Analytics LP

Figure 15: The distribution of long-term SPF expectations shifted when headline inflation was high



Source : Deutsche Bank Research, Haver Analytics LP



In a [recent report we examined](#) how well various measures of anchored inflation expectations are in the latest data. Our findings suggest that heading into the current crisis, inflation expectations appear to be well anchored. While that does not guarantee that expectations will remain anchored in response to current events, it does provide a solid foundation that could allow expectations to remain anchored. That analysis as well as past inflation expectations episode suggest a few factors warrant close monitoring at this time:

First, our analysis showed household inflation expectations displayed the weakest evidence in support of well anchored inflation expectations. On top of this, the academic literature has shown that households' inflation expectations are sensitive to food and petrol prices – two of the prices they most commonly interact with. Therefore, higher oil prices could pose a risk to the anchoring of household inflation expectations, and we may see household expectations become the first to de-anchor.

Second, longer-term expectations matter more than shorter-term expectations. Spikes in shorter-term expectations are often rational, and appropriate responses to changes in data. It does not necessarily indicate second-round pressures (such as wage-price spirals). The ECB's recent press statements have specifically focused on "longer-term" inflation expectations.

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#### #4. Shipping and Energy Trade: More limited impact through supply chain chokepoints

While the potential closure of the Suez Canal due to the US-Iran conflict raises concerns about supply chain disruptions and additional price pressures for the European economy, our European shipping analysts highlight two fundamental factors cushioning the impact on European value chains and costs, leaving aside the movement of commodity prices, addressed in the previous section.

- First, the container shipping market has built significant capacity. The global fleet has grown from 30.9m TEUs at the end of 2024 to 33.1m TEUs as of early February, with forecasts for further capacity growth of 4.4% in 2026. This, combined with a sizable order book equivalent to 35% of the current fleet (the highest in over 15 years), creates a significant supply-demand imbalance. This slack has kept freight rate movements modest and far from the dynamics seen during the COVID crisis. While spot rates from Shanghai have jumped on the recent news, this is from relatively low levels seen before the Chinese New Year.
- Second, the "delta" or marginal impact of any Suez blockage is less severe than one might assume because traffic has already been minimal since 2024. Current daily container transits through the canal were averaging only ~23k TEUs year-to-date, a fraction of the 150-200k TEUs per day seen prior to 2024.

As such, our shipping specialists do not currently expect meaningful value chain disruptions outside of energy trade. Based on these market fundamentals, they anticipate a more limited increase in freight rates than seen during previous crisis episodes (see their latest [leading indicator dashboard](#) for charts and details).

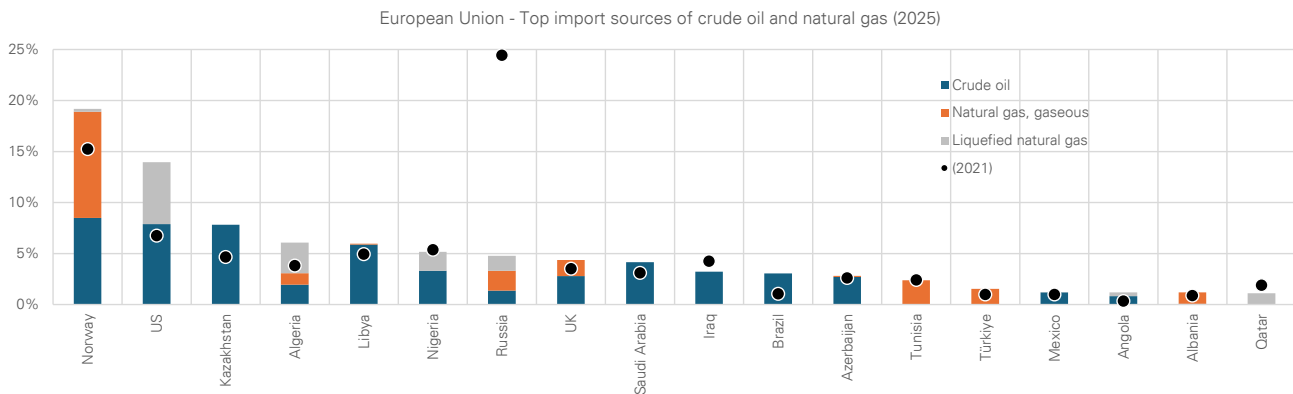
**Diversification.** When it comes to energy trade, a notable recomposition of the



EU's energy suppliers has occurred since Russia's 2022 invasion of Ukraine. [Figure 16](#) shows a shift in import shares across oil and natural gas between 2021 and 2025, where Russia's share of EU energy imports fell from 25% to 5%. The import shares of other key suppliers increased, notably Norway, the US, Kazakhstan and Algeria, whose shares collectively rose by 16pp over the same period. This revised sourcing structure informs the EU's exposure to current geopolitical risks.

Moreover, Europe exhibits a relatively low direct import reliance on the Persian Gulf region – only Saudi Arabia and Qatar appear in the list of major import sources, based on the data we collected from Eurostat. The primary transmission of the current crisis to the European economy is hence more likely to be via global energy prices, rather than through a direct disruption to the physical supply of energy.

Figure 16: EU energy imports: A post-Russia recomposition



Source : Deutsche Bank Research, Eurostat (datasets [nrg\\_ti\\_gas](#) and [nrg\\_ti\\_oil](#)). Notes: 2025 trade share covers January to September. Energy categories are aggregated based on their calorific value. These specific Eurostat energy datasets, compiled from national statistical bodies, are used in place of standard trade data to ensure the accurate capture of natural gas flows, which are often missed by customs-based reporting, as some [research](#) shows. Data for Türkiye, Albania and Tunisia are likely reflect pipeline transit volumes from Azerbaijan, Russia and Libya.



## Appendix 1

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## Chapter 2: Critical minerals

3 March 2026

- Rare earths are vital components for a state's defense and technology complex
- Without rare earths – and critical minerals more broadly – the three other chokepoints of chips, cables, and cloud cannot function.
- The key bottlenecks are in processing (separation, refining, alloying, magnets), are predominantly located in China.
- Using the historical analogy of Fordism in the 20th century, we look at how critical minerals have evolved from industrial outputs into strategic “chokepoints” across key technology and defense needs.

Our new series, “The Great Rebalancing,” focuses on five critical “chokepoints” that are redefining global power: Currencies, Critical Minerals, Chips, Cables, and Cloud.

Each chapter of our series will explore how and why these chokepoints define a new matrix of global power. Currencies allocate risk, critical minerals anchor the physical economy, chips generate compute power, cables move data and finance, and cloud platforms organize markets. Influence over this layered system shapes geopolitical standing – determining which nations can scale their economies, exert control, and apply strategic pressure.

This series argues that a “great rebalancing” is underway, shifting the global order from liberal institutionalism, where interdependence fosters cooperation, toward what we call “infrastructure realism”. In this new paradigm, control over key systems and inputs shapes outcomes. Under “infrastructure realism”, material and digital infrastructures become decisive instruments of state power. The US-China trade tensions of 2025 starkly illustrated this shift. The financial system — from bond markets to payment rails to digital money — acts as the meta-infrastructure that coordinates and prices all others.

Thus, this series addresses the wider question: which major powers are best positioned for influence and what are their strategic advantages? The 2025 US National Intelligence Annual Threat Assessment identified China as the primary actor challenging US global interests, citing its use of “asymmetric and conventional hard power tactics” and its drive to establish “alternative systems” to compete with the US. While the US-China power competition is longstanding, 2025 highlighted the asymmetric economic dependencies and capabilities shaping the relationship between the two powers.

As these five chapters will demonstrate, power now rests on the control of critical infrastructure. Both the US and China are in a race to reduce strategic dependencies on each other, and the outcome will define the next era.

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## Introduction: From inputs to chokepoints, the competition for industrial control

In this chapter, we examine how the breadth and scale of the 2025 US-China critical minerals crisis supports our central thesis that the global economic order is moving towards infrastructure-realism.<sup>1</sup> In Chapter 1 of this series, we analyzed this through the lens of currency and payments architecture (see [here](#)). In Chapter 2, we demonstrate how control over the industrial processes for rare earths (and critical minerals more broadly) is the next layer in this system of power. This control spans every stage, from mining and processing to setting technical standards and securing offtake agreements. As the world undergoes a rebalancing towards an infrastructure-determined order, it is likely that conflict will accelerate for control over these strategic chokepoints.

Rare earths, often described as the “vitamins” of high-tech manufacturing, are integral components vital for defense systems, electric vehicles, power grids and data centres. Concerns that China could use its rare earth dominance as political leverage are not new; in 2010, for example, Japan faced shortages and price spikes after a halt in shipments from China led many to link the event to a preceding diplomatic clash.<sup>2</sup> In addition, critical mineral self-sufficiency has been many countries’ longstanding priority for decades (in particular Europe’s), especially as part of their goal towards a net-zero energy transition.

Still, 2025 revealed the extent to which rare earths have become an explicitly binding constraint in US-China relations. Why? The answer brings us to our main argument: rare earths and critical minerals have grown to become indispensable to a country’s national security; a hallmark of an infrastructure-realist world.<sup>3</sup> In October 2025, US Treasury Secretary Scott Bessent warned that Beijing’s controls pointed “a bazooka at the supply chains and the industrial base of the entire free world.”<sup>4</sup>

The financial implications of this are significant. The Atlantic Council, for instance, estimates an effective Chinese export ban on rare earths would lead to severe shortages across the energy, automotive and defense sectors within weeks. It also calculates that a one-year disruption of dysprosium, neodymium, and manganese would reduce US GDP by \$1.6bn, \$154mn, and \$96mn, respectively.<sup>5</sup> Addressing this vulnerability, the 2025 US National Security Strategy identifies critical minerals as a national security chokepoint, arguing that the US “must never be dependent on any outside power” for inputs essential to defense and economic resilience.<sup>6</sup>

The core of the critical minerals crisis is not geological scarcity, but industrial concentration. China’s capacity to exert geopolitical influence through its critical minerals sector in 2025 stems directly from its decades-long effort to industrialize its critical minerals supply chains. Today, only China possesses the capacity to process rare earths into high-purity oxides, metals, and magnets at scale. The dependency is stark: the US and the EU import a respective 71% and 46% of their rare earths from China, and the US

<sup>1</sup> In our series introduction, we define “infrastructure realism” as a new world order where state control over key systems and inputs shape outcomes. Under infrastructure realism, material and digital infrastructures constitute the decisive instruments of state power.

<sup>2</sup> Previously published: Marion Laboure and Cassidy Ainsworth-Grace, “US vs China: Understanding the Rare Earth Metals Tensions in 10 Charts,” Deutsche Bank Research, January 18, 2023, [https://research.db.com/research/Article?rid=9d202c96-b32e-4e3d-a0ff-9a418104a6f5\\_604&kid=RP0001&documentType=R&wt\\_cc1=IND-2758331-5233](https://research.db.com/research/Article?rid=9d202c96-b32e-4e3d-a0ff-9a418104a6f5_604&kid=RP0001&documentType=R&wt_cc1=IND-2758331-5233); and Marion Laboure and Cassidy Ainsworth-Grace, “Tangled Wires: The Geopolitical Landscape of Semiconductors & Rare Earth Metals,” Deutsche Bank Research, June 7, 2023, [https://research.db.com/research/Article?rid=e5a16334-9484-490f-9217-7ab2649de709\\_604&kid=RP0001&documentType=R&wt\\_cc1=IND-2758331-5233](https://research.db.com/research/Article?rid=e5a16334-9484-490f-9217-7ab2649de709_604&kid=RP0001&documentType=R&wt_cc1=IND-2758331-5233).

<sup>3</sup> A tenuous truce was reached in November 2025, after both sides agreed to a one-year delay for Beijing’s rare earth export controls.

<sup>4</sup> Doug Palmer, “Bessent Says US in Talks with China to Prevent New Trade War,” Politico, October 13, 2025. <https://www.politico.com/news/2025/10/13/bessent-china-talks-trade-00606345>.

<sup>5</sup> Reed Blakemore, Alexis Harmon, and Peter Engelke, “Critical Minerals in Crisis: Stress Testing US Supply Chains against Shocks” (issue brief, Atlantic Council, October 9, 2025), <https://www.atlanticcouncil.org/in-depth-research-reports/issue-brief/critical-minerals-in-crisis-stress-testing-us-supply-chains-against-shocks/>.

<sup>6</sup> Now under the current administration, the US views mineral self-sufficiency as integral to its current goal of strengthening the country’s military industrial complex. This is because the USGS recognizes the US is at least 50% import reliant on many critical minerals essential for defense, including cobalt, graphite, gallium, and several rare earths.

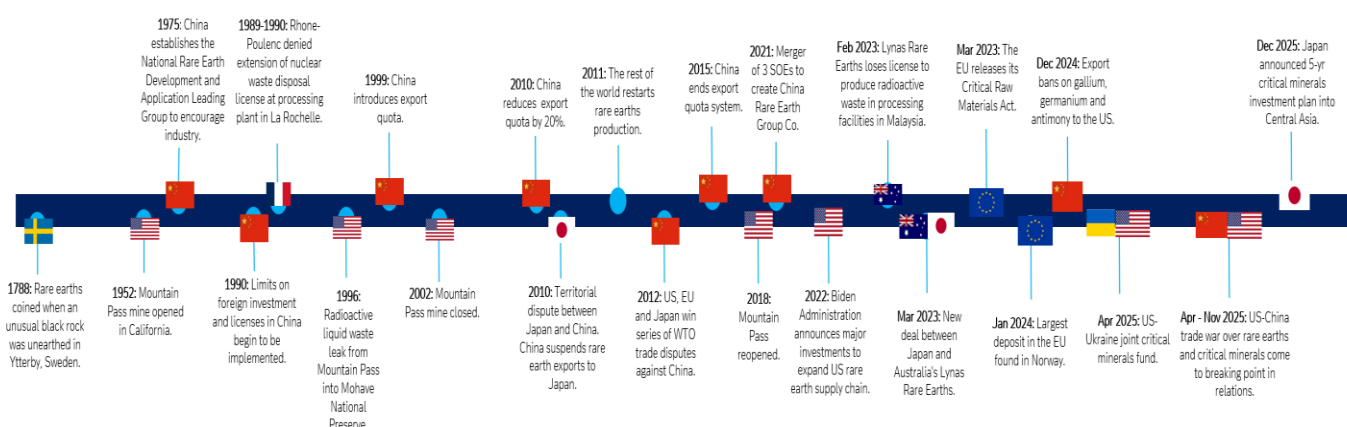


imports 100% of its heavy rare earths from China.<sup>7</sup> Thus, the scale and impact of Beijing's dominance over critical minerals is a clear assertion of infrastructure realism. By establishing and scaling its control over this foundational layer of the physical economy, China has gained a powerful tool to shape outcomes and advance its economic priorities. China's latest strategic move amidst tensions over Taiwan exemplify this, with Beijing restricting "dual-use" rare earth magnets and critical minerals to 20 Japanese companies in February.<sup>8</sup>

This ascension of power via industrialization calls to mind a useful historical analogy: just as US industrial strategy during the period of Fordism in the 20th century drove its leadership in sectors like automobiles and defense, China has strategically focused on industrializing its critical minerals supply chains in the 21st century.

While we anticipate competition in this sphere will become more crowded over the next decade, our analysis suggests that China is well-positioned to remain the dominant actor for the foreseeable future. In response, we expect the US to employ increasingly assertive and protectionist policies as it seeks to reindustrialize its own mineral supply chains. Under this tense push-and-pull, minerals used as leverage will be the foremost feature of the global critical minerals system in the next decade.

**Figure 1: Timeline of the global critical minerals industry**



Source: Deutsche Bank Research.

## 1. Critical minerals and rare earths: Laying out the industrial question

In 1992, Deng Xiaoping famously declared, "the Middle East has oil, China has rare earths." However, unlike oil, where geopolitics are largely shaped by supply volumes and demand, the landscape of rare earths and the broader critical minerals sector is far more complex.

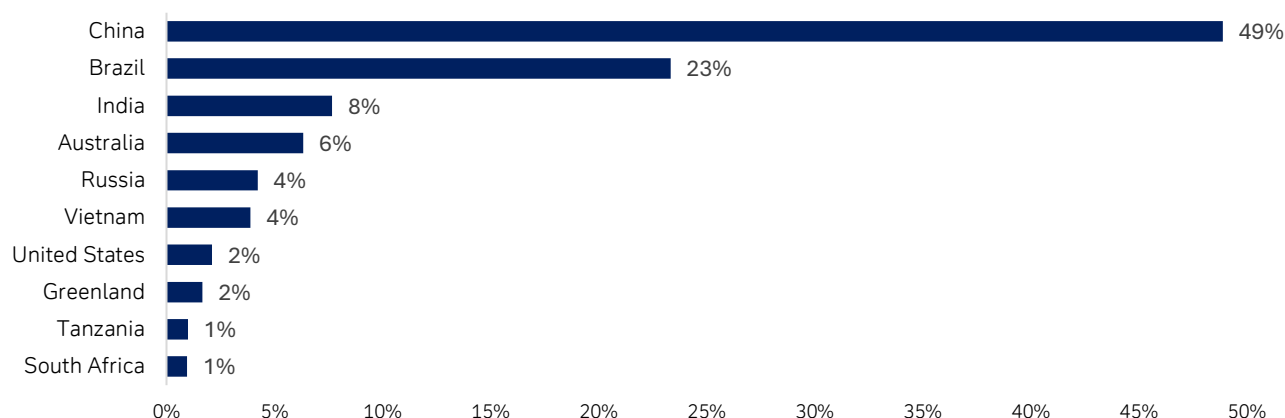
For one, while rare earths and most other critical minerals are not geologically scarce, their upstream supply is highly concentrated. Approximately five countries hold nearly 90% of known rare earth reserves, creating significant vulnerabilities in the global supply chain.

<sup>7</sup> Heavy rare earth metals are a subgroup of rare earths, characterized by higher atomic weights, smaller ionic radii, and greater scarcity compared to light rare earths. They are essential for high temperature magnets in electric vehicles, defense systems, and green technology. "MINERAL COMMODITY." 2026. <https://pubs.usgs.gov/periodicals/mcs2026/mcs2026.pdf>. For Europe, Eurostat. 2025. "Imports of Rare Earth Elements Saw 30% Drop in 2024." @EU\_Eurostat. Eurostat. April 9, 2025. <https://ec.europa.eu/eurostat/en/web/products-eurostat-news/w/ddn-20250409-1>.

<sup>8</sup> McMorow, Ryan, Harry Dempsey, Joe Leahy, and Leo Lewis. 2026. "China Slams Dozens of Japanese Companies with Export Curbs." @FinancialTimes. Financial Times. February 24, 2026. <https://www.ft.com/content/afa45318-1c56-435a-a3c8-8b82a93714a0?shareType=nongift>.



**Figure 2: Mineable rare earths reserve as a % of total global minable reserves**



Source: Deutsche Bank, USGS.

Even more pronounced is the concentration of midstream processing capacity. This is because the core challenge with critical minerals lies not in their extraction, but in their transformation into usable, high-value industrial inputs. The capital, infrastructure, and skilled labour necessary to extract, separate, and refine critical minerals are considerable and slow to scale. As we will demonstrate in the next section, it is Beijing's dominant position in these industrial processing stages that has enabled it to convert its technical dominance into geopolitical leverage.

### 1.1 The material base of modern power: Why critical minerals matter...

A mineral or metal is deemed “critical” if it meets two criteria: (i) it is of high economic or strategic importance, particularly for the energy transition, defense, technology, and industrial production; and (ii) its supply chain is vulnerable to disruption, either due to foreign dependence, geographic concentration, limited substitutability, or geopolitical risks. Criticality, therefore, is a measure of strategic vulnerability, not of geological scarcity.

Over time, the list of materials falling under this definition has expanded to include lithium, cobalt, nickel, copper, graphite, gallium, germanium, and tungsten. While many of these metals have been in use for centuries, their strategic importance grew significantly during the 20th century with the rise of industrial warfare and mass electrification. In the 21st century, critical minerals have become foundational to national security, evolving from mere inputs for consumer goods to core components of military, technological and industrial infrastructure.<sup>9</sup> Below are some examples of how critical minerals have come to represent a strategic chokepoint, and how they are essential for the three other layers we discuss in this series (chips, cables and cloud).<sup>10</sup>

- **Defense:** According to defense analytics firm Govini, 80,006 components across 1,908 U.S. weapons platforms rely on minerals subject to Chinese export controls.<sup>11</sup> Breaking this down across weapon systems, this dependency affects 92% of US Navy systems, 85% of the Air Force, 70% of the Army, and 62% of the Marine Corps.
- **Advanced technology:** Critical minerals enable digital performance, energy efficiency, data storage, and high-speed connectivity. In semiconductors, these

<sup>9</sup> While early lists were concise, the European Union's list of critical raw materials, first published in 2011 with 14 materials, expanded to 34 by 2023. Similarly, the United States' formalization of its critical minerals list has seen significant changes even in recent years. The 2025 list of critical minerals, has now risen to 60. Included the addition of 10 mineral commodities that were not in the 2022 edition—boron, copper, lead, metallurgical coal, phosphate rock, potash, rhenium, silicon, silver, and uranium.

<sup>10</sup> Advanced semiconductors rely on gallium, germanium, tungsten, and rare earths. Cables: Rare earth magnets, specialty alloys, and energetic materials underpin missiles, radars, submarines, and aircraft. Cloud: Data centres' energy and storage infrastructure depend on copper, rare earths, and battery minerals.

<sup>11</sup> Govini, “From Rock to Rocket: Critical Minerals and the Trade War for National Security,” 2025,

<https://www.govini.com/insights/from-rock-to-rocket-critical-minerals-and-the-trade-war-for-national-security>.



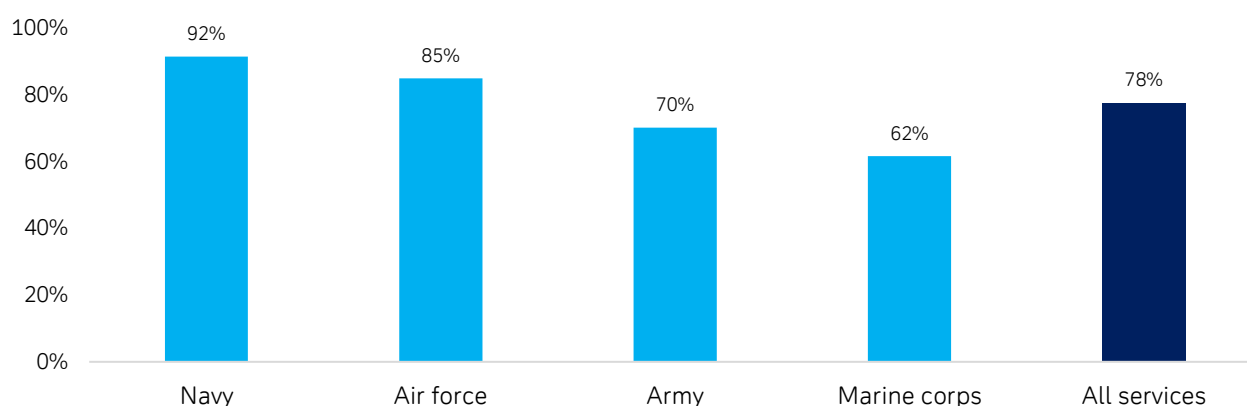


metals are essential for processing, storing, and transmitting information. For example, gallium – for which China accounts for >99% of global supply – is a crucial material for powering chips in smartphones, satellite systems, 5G infrastructure, power electronics, and AI hardware.<sup>12</sup>

- Clean tech and net zero transition: The transition to electric vehicles, renewable energy generation and electrified infrastructure is heavily dependent on a stable supply of critical minerals. Minerals such as lithium, nickel and cobalt are crucial to battery performance, while rare earth magnets are indispensable for wind turbines and EV motors.

Ultimately, critical minerals are a foundational layer of modern economic, technological, and military power. Demand for these materials tends to rise in tandem with geopolitical tension, as access becomes constrained. This dynamic is central to the current era, as nations like the US seek to reinforce their industrial and defense capabilities.

**Figure 3: Critical minerals in US weapons systems, 2025**



Source: Deutsche Bank Research, Govini.

### 1.2 ... and especially rare earth metals

Rare earth metals, a subset of critical minerals, is a group of 17 metallic elements that possess magnetic and conductive properties that have significantly improved the performance, efficiency, longevity, and reliability of countless modern technologies.

First discovered in Sweden in 1788, it was not until the 20<sup>th</sup> century that their commercial applications became ubiquitous, used across a wide variety of intermediate and consumer products. Today, the US Geological Survey (USGS) identifies rare earths as essential components in over 200 commercial products, ranging from consumer goods like mobile phones, computer hard drives, cameras, floodlights, and microwave filters, to industrial applications such as petroleum refining.<sup>13</sup>

The primary strategic chokepoint in the rare earth supply chain is in processing. The unique chemical characteristics of rare earths make the separation stage of the supply chain a technologically demanding and capital-intensive endeavour requiring highly specialized expertise. As Thyssen and Binnemans (2011) show in their scientific study on rare earths, these minerals have extraordinarily similar chemical and physical properties, leading to a decades-long struggle for scientists to isolate and place them on the periodic table. This similarity renders traditional chemical extraction methods ineffective.

Producing high-purity rare earth oxides – and subsequently, metals and alloys – requires complex extraction and waste handling, making the process capital-intensive,

<sup>12</sup> "Critical Minerals in Artificial Intelligence," SFA (Oxford), 2025, <https://www.sfa-oxford.com/knowledge-and-insights/critical-minerals-in-low-carbon-and-future-technologies/critical-minerals-in-artificial-intelligence/>.

<sup>13</sup> Marion Laboure and Cassidy Ainsworth-Grace, "US vs China: Understanding the Rare Earth Metals Tensions in 10 Charts," Deutsche Bank Research, January 18, 2023.



environmentally sensitive and technologically demanding.<sup>14</sup> As a result, many countries can mine rare-earth metals, but China is currently the only nation with the capacity to separate and process them at a mass industrial scale.

## 2. China's industrialization of critical mineral supply chains: A historical perspective

No country has capitalized on the critical minerals era more effectively than China. Through a long-term, state-directed industrial strategy and significant investment, China now exercises a near-monopoly over key mineral supply chains, turning industrial scale into a source of significant leverage and a bottleneck for the rest of the world. This bottleneck is particularly challenging to overcome because it is rooted in technological capacity, is capital-intensive, and is slow to replicate. Focusing on rare earths as the primary example, the remainder of this section explains how China built, consolidated, and ultimately learned to leverage its industrial advantage in the geopolitical sphere.

### 2.1 China's midstream dominance: the key is capacity, more than reserves

Up until the late 1980s, the United States was the global leader in rare earth production. The discovery of a large, high-grade rare earth deposit at Mountain Pass, California in 1949 paved the way for mass production of commercial quantities of rare earth elements, which in turn reduced prices and spurred their wider commercial application.<sup>15</sup> France was also once at the forefront of rare-earth processing via its chemical giant Rhone-Poulenc (later acquired by Solvay).

However, western advantage was overtaken as China's leadership recognized the strategic value of its own vast rare earth reserves and launched a concerted effort to develop its domestic production capacity. To do this, the state supported programs that would work on enhancing mining techniques, invested in R&D for rare earth applications, and provided financial backing from state-controlled banks, allowing domestic rare earth companies to operate and scale even without immediate profitability. As a result, the country's domestic production grew at a rapid average of 40% annually from 1978 to 1989.<sup>16</sup>

As China's rare earth exports increased over the 1990s, and the United States began to prioritize environmental and labor concerns that made rare earth processing less viable, production steadily shifted to China as it became increasingly difficult for non-Chinese producers to compete on price. Globalization, combined with China's emergence as the world's workshop, further encouraged the US and Europe to step back from domestic production as supply chains migrated to China. With cheaper domestic supplies at their disposal, Chinese manufacturers also had the opportunity to scale the production of key end-products such as high-strength magnets and electronics, crowding out foreign competition in downstream manufacturing industries as well.

To consolidate its advantage in rare earth production, authorities simultaneously reduced their export quotas – by 54% between 2005–2010 – to secure resources for domestic demand and regain control over operations.<sup>17</sup> This caused international prices for rare earths to increase dramatically, only for Beijing to later release significant volumes of subsidized output to the markets. This self-reinforcing cycle created an oversupply of unprocessed materials, drove down prices, and solidified China's position as the dominant buyer and processor.

<sup>14</sup> Pieter Thyssen and Koen Binnemans, "Accommodation of the Rare Earths in the Periodic Table: A Historical Analysis," in *Handbook on the Physics and Chemistry of Rare Earths*, vol. 41 (Elsevier, 2011) 1-93, [Why Rare Earths Aren't Really "Rare"-and Why China Still Controls the Choke Point](#) and [Accommodation of the Rare Earths in the Periodic Table: A Historical Analysis](#). As Thyssen and Binnemans also document, many early rare earths OR elements were in fact complex mixtures: metals such as "ceria" and "yttria" took over a century to fully deconstruct, while "didymium" was later separated into Praseodymium and Neodymium.

<sup>15</sup> "Rare-Earth Metal Prices in the USA Ca. 1960 to 1994," USGS, 2026, <https://www.usgs.gov/publications/rare-earth-metal-prices-usa-ca-1960-1994-0>.

<sup>16</sup> "China's Rare Earths Industry and Its Role in the International Market," U.S.-China Economic and Security Review Commission, November 2010, <https://www.uscc.gov/sites/default/files/Research/RareEarthsBackgrounderFINAL.pdf>.

<sup>17</sup> Ibid.

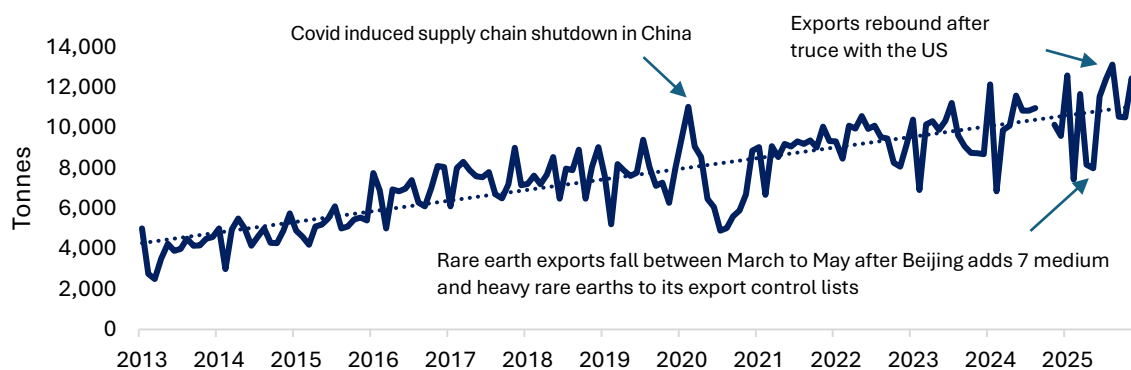


China's control over global prices is reinforced by a lack of external visibility into true rare earth and critical mineral pricing. This is because China's two domestic rare earth exchanges operate with limited transparency, while the Shanghai Metals Market publishes indicative price ranges rather than real-time spot prices. Outside of China, the OTC market for these metals is highly opaque, and trading volumes on the London Metal Exchange are virtually non-existent. The cessation of public reporting on China's rare earth mining and refining quotas further contributed to price uncertainty and corporate volatility as foreign companies struggled to navigate the less transparent environment.

As a result of these factors, China has dominated rare earth processing conversion capacity since the 1990s, with its share of global production peaking at 97.5% in 2009. Today, China maintains dominant positions in rare earth mining (~60–70%), rare earth processing (~90%), and magnet manufacturing (~93%). However, while the country overwhelmingly dominates the midstream and downstream segments, it still relies on large volumes of raw metal imports to refine and process. Thus, China is also one of the largest consumers of raw element imports across critical mineral supply chains. To close this gap, Beijing has expanded its strategic stockpiles and increased overseas investments in mines (e.g. Indonesia).<sup>18</sup> The record \$213.5bn investment in the Belt and Road initiative in 2025—including \$32.6bn directed to metals and mining to secure processing capacity and offtake rights – is a recent example of this strategic focus. Through these initiatives, China is also strengthening its position in the upstream segment, moving towards even greater influence across the entire critical mineral value chain.

Finally, over the years, Beijing has consolidated state-owned enterprises in the rare earths industry, enabling vertical integration to control the full value chain. In December 2021, Beijing announced the creation of China Rare Earth GroupCo, merging three major state-owned enterprises and two research institutions. This new entity now controls roughly 70% of China's domestic rare earth production.

**Figure 4: China rare earth exports**



Source: Deutsche Bank Research, China General customs. Note: Data for Sep and Oct 2024 not available.

## 2.2 Critical minerals as “Mineral Fordism”

To understand the relationship between China's dominance in critical minerals infrastructure and its role in the international order, it is useful to draw an analogy with the global era of American industrialization in the 20th century, a system often referred to as “Fordism”.

Fordism was a system of mass production and consumption in the 1920s that relied on standardized parts and moving assembly lines. Its defining feature was vertically integrated production at scale (e.g., the assembly line). Using Stefan Link's definition in *Forging Global Fordism*,<sup>19</sup> Fordism also functioned as a tool of state-led modernization,

<sup>18</sup> Marion Laboure and Camilla Siazon, “The Top 10 Strategic Commodities to Watch,” Deutsche Bank Research Institute, October 22, 2025. <https://www.dbresearch.com/PROD/RIPROD/PDFVIEWER.caliass?pdfViewerPdfUrl=PROD0000000000606588>.

<sup>19</sup> Stefan J. Link, “Forging Global Fordism: Nazi Germany, Soviet Russia, and the Contest over the Industrial Order,” (Princeton: Princeton University Press, 2020).



enabling countries to accelerate their industrial development and rearmament in the lead-up to World War II.

This dynamic was evident in how both Germany and the Soviet Union actively sought American technology and production methods to rapidly modernize their economies and spread their own state-led ideologies. As Link demonstrates, the open patent system and permissive technology transfer between the US and the USSR and Germany during this time allowed engineers from these countries to study, copy and replicate the methods from the Fordist production systems in Detroit to ultimately build their own industrial and military capabilities.<sup>20</sup> Fordism's mass industrialization, combined with the US's post-war soft power, consumer capitalism and embedded liberal institutionalism, helped cement US hegemony in the 20th century.<sup>21</sup>

Just as Fordism underpinned the US's economic and military rise, China has managed to pioneer and master a new production system for critical minerals that now forms the basis of its geopolitical influence. Through the vertical integration of rare earth companies and dominance as the world's largest downstream off-taker, China has built the world's most complete processing ecosystem for critical minerals thus far. And just as Ford standardized physical parts by achieving massive economies of scale and driving down costs, China has effectively set global prices and production standards. This is most evident in its pricing power over many critical minerals, which has made it the most rational choice for Western companies to mine raw materials globally and ship them to China for processing.

Today, the US is seeking to re-industrialize its own rare earth supply chains but faces what Link describes as the constraints of the "late developer." Unlike the tech transfers that took place between the US and Germany and the Soviet Union during Fordism, in today's infrastructure-realist world, there appears to be less scope for such cooperation between strategic competitors. The West cannot simply replicate China's industrial model without also absorbing the immense economic, social, and environmental costs that China accepted during its own decades-long supply-chain transformation.

### **2.3 China's critical minerals leverage as a tool of infrastructure realism: a rare earths case study**

China's rare earths strategy represents a modern form of "mineral Fordism." In this section, we examine how its near-total dominance over rare earths and critical minerals has enabled the country to leverage this industrial advantage for geopolitical purposes.

While China has long recognized the advantage of its rare earth dominance and has periodically used export quotas and restrictions since the 1990s, we argue that its initial aim for these controls was to strengthen its own domestic capabilities rather than to exert any geopolitical influence.

The example many cite as the first time China used its market dominance for political leverage was in September 2010, when shipments of rare earths from China to Japan were halted for two months. Many observers concluded that this was a retaliatory measure, since the disruption occurred soon after a hostile clash between Chinese and Japanese boats in the long-disputed waters near the Senkaku Islands (called the Diaoyu Islands by China). Contrary to popular narrative, however, there is little evidence that China publicly announced a targeted export ban on Japan in 2010. While China threatened "strong countermeasures" against Japan, it never formally confirmed exports cuts of rare earths to Japan,<sup>22</sup> leaving analysts to infer intent from trade data. China's premier at the time also

<sup>20</sup> Ibid. According to Link, Germany ultimately internalized these methods more effectively than the Soviet Union.

<sup>21</sup> Ibid.

<sup>22</sup> A study by Simon Evenett and Johannes Fritz found that while China certainly applied export restrictions on rare earths during the 2010s, there was no evidence of a targeted ban on Japan. In fact, it was Australia, not Japan, that witnessed sharp falls in monthly REM exports from China during this period. Simon Evenett and Johannes Fritz, "Revisiting the China–Japan Rare Earths Dispute of 2010," *CEPR*, July 19, 2023. <https://cepr.org/voxeu/columns/revisiting-china-japan-rare-earths-dispute-2010>.



said that “China is not using rare earths as a bargaining chip.”<sup>23</sup> Nevertheless, the market impact was clear: rare earth metal prices surged nearly tenfold following the disruption. Afterwards, Japan (in conjunction with the US and EU) filed a successful WTO trade complaint alleging China’s use of export quotas and other restrictions on rare earths, leading China to dissolve any restrictions by 2015.

The more systematic use of rare earths as leverage began as US-China competition intensified during the first Trump administration. Then, Beijing’s export restrictions on its rare earths and critical minerals became more systematic and concentrated. Indications that the government was beginning to view its rare earth dominance as a strategic tool for national security emerged in 2019, when tensions over the US-China trade war reached a new peak. During a high-profile visit to one of China’s major rare earths mining and processing facilities in Ganzhou, President Xi Jinping made remarks to state media that rare earths could be a “new Long March” battleground.<sup>24</sup> This episode can be interpreted as the first informal signal that Beijing had started to consider its near-monopoly over the industry as a point of leverage in its trade dispute with the US. These concerns were amplified one year later when China’s National People’s Congress passed a new export control law, creating a legal framework to restrict exports of sensitive materials on national security grounds. While rare earths were not named outright, the new law allowed Beijing to legally halt its exports to specific countries if it chose to do so.

By the 2020s, as global competition evolved into a tech race, China’s actions became more explicit. In 2023, it banned the export of specialized rare earth processing equipment and imposed export restrictions on gallium, germanium and certain types of graphite.<sup>25</sup> In 2024, these restrictions were tightened into a full export ban of gallium and germanium to the US. By 2025, China had developed a range of policy tools related to critical minerals, effectively transforming its industrial dominance into a bargaining tool.

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<sup>23</sup> Su Qiang, “Rare Earth Will Not Be Used as Bargaining Chip,” *China Daily*, October 8, 2010.

<sup>24</sup> The “Long March” comment is seen as a reference to the massive, year-long tactical retreat by the CCP in 1934, as the Red Army escaped the KMT forces during the Civil War. It cemented Mao Zedong’s leadership, allowed the weakened party to survive, and is now celebrated as a foundational symbol of perseverance, sacrifice, and strength. Xi’s visit occurred soon after the US had backlisted Huawei and came amidst fears that Beijing would restrict or embargo rare earth exports to the US. Zhou Xin, Wendy Wu, and Kinling Lo, “Xi Jinping Visits Rare-Earth Minerals Facility amid Talk of Use in US Trade War,” *South China Morning Post*, May 20, 2019. <https://www.scmp.com/economy/china-economy/article/3010977/xi-jinping-visits-rare-earth-minerals-facility-amid-talk-use>.

<sup>25</sup> Marion Laboure and Cassidy Ainsworth-Grace, “Rare Earth Metals: What to Expect in 2024 in 10 Slides,” Deutsche Bank Research, March 26, 2024, [https://research.db.com/research/Article?rid=eec2bc46-c2ac-4f07-bcbe-2c75d836a3ee-604&id=RP0001&documentType=R&wt\\_cc1=IND-3146993-0000](https://research.db.com/research/Article?rid=eec2bc46-c2ac-4f07-bcbe-2c75d836a3ee-604&id=RP0001&documentType=R&wt_cc1=IND-3146993-0000).



**Figure 5: How China controls the whole value chain**

| Mineral                        | Use cases                                                                                                                                                                          | Upstream dominance (mining & reserves)                                                                                                                                                 | Midstream dominance (refining & processing)                                                                                                                                    | Key dynamics & strategic actions                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Nickel</b>                  | EV batteries, defence alloys, aerospace materials.                                                                                                                                 | Indonesia holds the largest reserves and leads mined production (67%), but China controls the value chain.                                                                             | China controlled 65% of global refined nickel output in 2024, primarily through ownership of Indonesian refining capacity.                                                     | China, the largest nickel consumer, is increasing stockpiles due to escalating US trade tensions.                                                                                                                                                                                                                                                                                                                               |
| <b>Lithium</b>                 | Lithium-ion batteries for smartphones and laptops, power tools and grid-scale energy storage systems.                                                                              | Australia, Chile, and China produced ~72% in 2025. China controls 21% of upstream supply.                                                                                              | China controls ~70% of lithium chemical refining. The top three refiners (China, Argentina, Chile) account for ~95% of global capacity.                                        | China's control over the lithium value chain is growing. The US imported over 90% of its lithium-ion battery cells from China in 2024. US response: investment in Canadian mining (e.g., Thacker Pass Mine). In the UK, the country's first commercial lithium plant opened in Feb 2026.                                                                                                                                        |
| <b>Cobalt</b>                  | EV and defence batteries, jet engine alloys, industrial gas turbines, electroic warfare systems.                                                                                   | ~74% of mined cobalt comes from the DRC, but the IEA projects Indonesia to bring its market position down to 50% by 2040. Chinese companies owned 80% of DRC production in 2024 (EIA). | China refined ~78% of global cobalt market in 2024; primarily for lithium-ion batteries.                                                                                       | China remains the top buyer, driven by its battery industry. The US was 79% net import-reliant for cobalt in 2025. US response: the US is diversifying supply chains with new mining and processing projects, facing challenges from weak prices (e.g., US stake in Trilogy Metals in Alaska).                                                                                                                                  |
| <b>Copper</b>                  | Aircraft, ground vehicles, ships, submarines, missiles, AI data centers.                                                                                                           | Chile has the world's largest reserves and leads global mining production (23%).                                                                                                       | China dominates downstream supply chains, controlling 48% of global refining capacity.                                                                                         | US response: the Trump administration has designed a copper policy to onshore mining and refining production in the US.                                                                                                                                                                                                                                                                                                         |
| <b>Gallium &amp; Germanium</b> | Essential semiconductor materials used in fiber optics, infrared optics, solar technologies, advanced military and biomedical applications                                         | China controls 99% of gallium production and nearly all of germanium production.                                                                                                       | Control over high-purity refined output used for chips and defence is more balanced, with Europe, the US, Japan and the Canada overseeing some parts of the midstream segment. | No US domestic gallium production has existed since 1987, while some limited germanium production occurs. The US also stockpiles germanium in its National Defense stockpile, but no gallium. In 2025, the DoE announced ~\$6mn in funding for gallium.                                                                                                                                                                         |
| <b>Tungsten</b>                | Steels, aerospace, and military projectiles.                                                                                                                                       | China controls 82% of Tungsten production; also has the world's largest Tungsten reserves.                                                                                             | China refines over 80% of the world's tungsten concentrate into intermediate products.                                                                                         | Tungsten concentrate production ex- China = 21% of total world production. China remains the top consumer of Tungsten. In 2025, China placed several restrictions on critical minerals including Tungsten to the US.                                                                                                                                                                                                            |
| <b>Rare earths</b>             | 17 minerals critical for defence and tech – F-35 fighter jets, submarines, Tomahawk missiles, radar systems, unmanned aerial vehicles etc), and advanced semiconductor technology. | China dominates the REM supply chain, controlling ~69% of rare earth mining in 2025.                                                                                                   | China controls >90% of refining capacity.                                                                                                                                      | IEA projects China will still dominate refining (76%) and mining (51%) by 2030, implying rare earths should remain the frontline in the US-China resource war.                                                                                                                                                                                                                                                                  |
| <b>Graphite</b>                | Batteries, brake linings, lubricants, powdered metals, refractory applications, and steelmaking.                                                                                   | China controls 78% of graphite production. The US has had no domestic natural graphite production since the 1950s.                                                                     | China maintains dominance across mining, processing, and anode manufacturing.                                                                                                  | The US and EU import around 46% (2021-2025) and 38% (2024) of their graphite from China, respectively. China requires export permits for high-grade graphite (Dec 2023), while the US imposed 93.5% anti-dumping duties on Chinese graphite and anode materials (July 2025). US response: the US is developing its upstream supply chains; the USGS currently tracks five active exploration or development projects in the US. |

Sources: Deutsche Bank Research, [USGS](#), [IEA](#), EIA.

### 3. The new mineral map: Bloc formation and infrastructure partnerships

In response to supply disruptions and heightened geopolitical tensions surrounding critical minerals in 2025, governments and companies have stepped up their efforts to restructure their critical minerals supply chains. What is emerging is not diversification in the classical sense, but a re-mapping of supply chains along strategic, security-aligned blocs. Thus far, the following themes have emerged, characterized by uneven progress and significant constraints.

#### 3.1 State consolidation

Similar to China's approach to vertical integration, Washington is seeking partial state ownership in domestic critical mineral companies. Since January 2025, the Trump administration has announced multiple partnerships with domestic and international minerals companies as it seeks to build private sector willingness to invest. Recent examples include a 10% stake in Trilogy Metals in Alaska to target cobalt, copper and related minerals; a \$150mn stake in a Louisiana-based gallium producer;<sup>26</sup> a 5% stake in the Canadian company Lithium Americas; and a 5% stake in its Thacker Pass mine in Nevada, scheduled to open in 2028. Notably, in July 2025, the Pentagon also became the

<sup>26</sup> Steff Chávez and Camilla Hodgson, "Pentagon Invests \$150mn in US Gallium Company to Secure Strategic Supplies," *Financial Times*, January 12, 2026, <https://www.ft.com/content/00b2ba44-9f20-4d41-9b01-3df9dde5dae7>.





largest shareholder of MP Materials. In response to these policy efforts, shares of MP Materials, Lynas, and USA Rare Earths soared in 2025.

The US is not the only country pursuing this strategy; other countries such as Australia and Canada are implementing similar measures to encourage private sector investment. Just as the Chinese state supported its nascent rare earths industry during the 1980s, so too are other governments today. This intervention aims to yield notable dividends by countering the historical private sector reluctance to build or maintain capacity, a hesitation often influenced by significant price fluctuations, at times exacerbated by supply dynamics originating from China, which has often led to facility closures and threatened the viability of Western domestic mining, processing and downstream manufacturing capacity.

### 3.2 Stockpiling

In response to supply chain risks, the US and the EU, among others, have announced or expanded critical minerals stockpiles. However, this strategy has yielded limited gains. Many speciality critical minerals cannot be effectively stockpiled in their final, usable forms because they are alloyed into highly specific forms and combinations for diverse end-uses. As such, existing public and private stockpiles of key processed materials, like neodymium or dysprosium, would be depleted within weeks or months under an effective export ban of critical minerals.<sup>27</sup> We saw this play out in April 2025, when automakers raised alarms over rare earth shortages following China's implementation of export controls, with Ford's CEO describing US dependency on Chinese rare earths as a "hand-to-mouth situation."<sup>28</sup>

Ultimately, stockpiling raw ore does not solve the core problem. The binding constraint remains midstream processing capacity, not geological resources: without maintaining the full value chain, stockpiling offers limited protection. The Atlantic Council notes that even "sufficient" raw inputs stockpiles may be ineffective if midstream bottlenecks persist. It also argues that stockpiles are more effective at derisking new domestic minerals projects and countering price manipulation than filling market gaps during a major supply crisis.<sup>29</sup>

### 3.3 Mineral deals as the new security treaties

The third emerging theme we have detected is the evolution of mineral alliances beyond traditional multilateralism, moving towards arrangements with strategic security implications. This strategic reorientation is largely led by the US, as evident in its latest National Security Strategy, which states "America's economic partners should no longer expect to earn income from the United States through overcapacity and structural imbalances but instead pursue growth through managed cooperation tied to strategic alignment and by receiving long-term U.S. investment."<sup>30</sup>

For the US, these allied mineral deals serve multiple objectives:<sup>31</sup> gaining access to partner stockpiles (e.g. in Japan and South Korea), forming buyers' consortia to increase purchasing power, and using trade agreements to insulate domestic firms from price volatility. This dynamic has been visible in recent US engagement with Ukraine and Greenland.

<sup>27</sup> Reed Blakemore, Alexis Harmon and, Peter Engelke, "Critical Minerals in Crisis: Stress Testing US Supply Chains against Shocks," (issue brief, Atlantic Council, October 9, 2025), <https://www.atlanticcouncil.org/in-depth-research-reports/issue-brief/critical-minerals-in-crisis-stress-testing-us-supply-chains-against-shocks/>.

<sup>28</sup> "Hand-To-Mouth": Ford CEO Confirms Factory Shutdown amid China's Rare Earth Chokehold," *Rare Earth Exchanges*, June 14, 2025, <https://rareearthexchanges.com/news/hand-to-mouth-ford-ceo-confirms-factory-shutdown-amid-chinas-rare-earth-chokehold/>.

<sup>29</sup> Reed Blakemore, Alexis Harmon and, Peter Engelke, "Critical Minerals in Crisis: Stress Testing US Supply Chains against Shocks," (issue brief, Atlantic Council, October 9, 2025), <https://www.atlanticcouncil.org/in-depth-research-reports/issue-brief/critical-minerals-in-crisis-stress-testing-us-supply-chains-against-shocks/>.

<sup>30</sup> "National Security Strategy of the United States of America," The White House, 2025, <https://www.whitehouse.gov/wp-content/uploads/2025/12/2025-National-Security-Strategy.pdf>.

<sup>31</sup> Reed Blakemore, Alexis Harmon and, Peter Engelke, "Critical Minerals in Crisis: Stress Testing US Supply Chains against Shocks," (issue brief, Atlantic Council, October 9, 2025), <https://www.atlanticcouncil.org/in-depth-research-reports/issue-brief/critical-minerals-in-crisis-stress-testing-us-supply-chains-against-shocks/>.



However, in an environment characterized by increased national resource focus and infrastructure realism, such multilateral partnerships have clear limitations during a crisis. Potential partner nations may be constrained by their own domestic needs and unable to act as a supplier for the US. Also, many potential allies remain indirectly dependent on Chinese flows in one way or another, which could limit their willingness to participate fully.

### 3.4 Resource nationalism and export controls

Other countries with significant stakes in critical mineral mining are increasingly imposing export restrictions (e.g., licensing requirements, taxes, quotas, and outright bans) on their upstream raw materials, with the aim of protecting and developing their domestic downstream industries.<sup>32</sup> This trend is significant, as it could complicate China's own strategy of securing upstream raw mineral inputs from abroad.

A notable example is Indonesia, which holds the world's largest nickel reserves. It banned the export of raw nickel ore in 2014 to promote domestic industrialization. However, many resource-rich countries are still unable to replicate this strategy as they lack the processing self-sufficiency. Below, we survey other key players in the critical minerals race and assess their progress and constraints thus far.

### 3.5 United States: Mineral diplomacy meets a renewed Monroe Doctrine<sup>33</sup>

The US has sought to diversify its critical mineral supply chains away from China since the 2000s. For example, the Biden administration took proactive measures such as funding commercial-scale rare-earth oxide production and supporting MP Materials in 2022. However, unlike the Biden administration, which made a clean energy transition a key goal, the second Trump administration views critical mineral sufficiency as integral to its goal of strengthening the US military industrial base. This is again a characteristic of the infrastructure realism paradigm discussed earlier, where the struggle for critical minerals is increasingly perceived as a contest for global hegemony. Examples that demonstrate how critical minerals are becoming increasingly entwined with US defense include the "One Big Beautiful Bill Act" (July 2025), which allocated \$9.5bn to domestic critical mineral capabilities, including \$7.5bn for Defense and \$2bn for energy dominance and Defense Production Act purposes. Later in October 2025, the Pentagon also announced plans to procure up to \$1bn in critical minerals.

This convergence of critical mineral sufficiency and US military security perhaps explains why amongst its many domestic supply chain solutions, the current administration has also embarked on assertive geopolitical expansion to strengthen its minerals industrial policy. This is a marked departure from the multilateralism of economic security guarantees that defined the previous order of "Liberal Internationalism," and a clear step toward infrastructure-realism. The first clear example of this was in February 2025, when Washington said it would not continue supporting Ukraine's war efforts against Russia, unless compensated with Ukraine's vast natural resources.<sup>34</sup> This position reportedly triggered strong reactions from Ukraine, with President Zelensky stating that would be akin to "selling a country." The situation escalated dramatically when the US temporarily suspended all military aid and intelligence sharing to Ukraine in March. But by April, both sides had come to an agreement and a joint infrastructure fund was established.<sup>35</sup> Under

<sup>32</sup> David Guberman, Samantha Schreiber, and Anna Perry, "Office of Industry and Competitiveness Analysis Export Restrictions on Minerals and Metals: Indonesia's Export Ban of Nickel," (working paper, Office of Industry and Competitiveness Analysis, U.S. International Trade Commission, 2024)

[https://www.usitc.gov/publications/332/working\\_papers/ermm\\_indonesia\\_export\\_ban\\_of\\_nickel.pdf](https://www.usitc.gov/publications/332/working_papers/ermm_indonesia_export_ban_of_nickel.pdf)

<sup>33</sup> The Monroe Doctrine was a foundational policy during the early 1800s that prioritized US sovereignty over the Western Hemisphere from European influence; in the 21<sup>st</sup> century.

<sup>34</sup> Seb Starcevic, "Trump Demands \$500B in Rare Earths from Ukraine for Continued Support," *POLITICO Pro*, February 2025. <https://subscriber.politicopro.com/article/2025/02/trump-demands-500b-in-rare-earths-from-ukraine-for-continued-support-00203529>.

<sup>35</sup> Marion Laboure, Peter Sidorov, and Camilla Siazon, "Thematic Research: US-Ukraine Deal: Strategic Win and Security Tradeoff?," Deutsche Bank Research, May 1, 2025, [https://research.db.com/research/Article?rid=d4e3ea2a-dd1a-482b-83e3-e627c5b9d033-604&kid=RP0001&documentType=R&wt\\_cc1=IND-3146993-0000](https://research.db.com/research/Article?rid=d4e3ea2a-dd1a-482b-83e3-e627c5b9d033-604&kid=RP0001&documentType=R&wt_cc1=IND-3146993-0000).





the agreement, Ukraine committed to direct 50% of its royalties, rent payments, and license fees from its critical minerals' assets to the US.

This episode represents a shift in US foreign policy from traditional aid to economic statecraft, a hallmark of infrastructure realism. In 2026, the Trump administration accelerated its pursuit of critical mineral security by renewing its call to purchase Greenland, which holds the world's eighth-largest rare earth deposit and other critical minerals. President Trump has consistently framed Greenland as vital to US national security, first proposing its purchase in 2019. And while the Trump administration has not formally designated Venezuela as a strategic mineral priority after US forces entered the country and deposed its president in January 2026, US officials have signalled interest in Venezuela's mineral endowment, citing Venezuela's steel and critical minerals as potential areas of US-Venezuela collaboration.<sup>36</sup> These incidents underscore a US dual strategy of securing critical minerals, as well as exploring opportunities for expanded influence or access to strategically important regions.

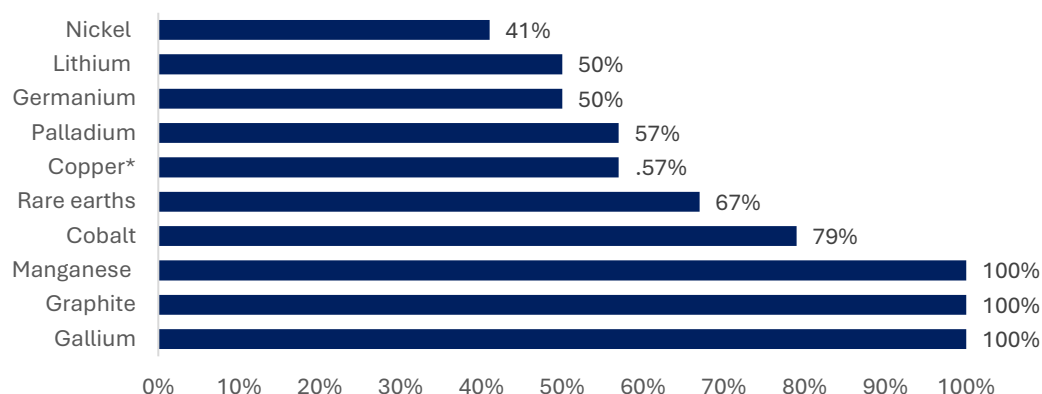
### 3.6 How effective has the US response effort been?

In October 2025, Secretary Scott Bessent suggested US dependence on China's rare earths could be reduced within 18–24 months.<sup>37</sup> In practice, however, a typical rare earth project requires around 9 years, from exploration to construction.

Thus, US progress is still years away from building sufficient critical mineral chains, with its midstream and downstream capacity continuing to be the primary bottlenecks. Its biggest rare earth company investments, Lynas and MP Materials, still lag on processing capabilities, and according to CSIS, MP Materials' output of NdFeB magnets of 1,000 tons by end-2025 represents less than 1% of China's 2024 output.<sup>38</sup>

In January 2026, the Trump administration pivoted away from tariffs toward international price coordination, directing USTR and Commerce to pursue price floors with partners.<sup>39</sup> This underscores the US's recognition that domestic capabilities alone are insufficient. In his Executive Order, President Trump said that if price floor agreements have not progressed by 13 July 2026, he may implement additional measures including import restrictions and minimum import prices for certain types of critical minerals.

**Figure 6: US net import reliance as a % of consumption in 2025, select critical minerals**



Source: Deutsche Bank Research, USGS. Note: Copper refined.

<sup>36</sup> "Trump wants Venezuela's oil. What about minerals?," Politico Pro, January 2026, <https://subscriber.politicopro.com/article/eenews/2026/01/06/trump-wants-venezuelas-oil-what-about-minerals-00712231>. One day after US forces captured Venezuelan President Nicolas Maduro, Commerce Secretary Howard Lutnick said: "you have steel, you have minerals, all the critical minerals" that Trump would "fix and bring it back – for the Venezuelans."

<sup>37</sup> Demetri Sevastopulo, "China 'Made a Real Mistake' by 'Firing Shots' on Rare Earths, Says Scott Bessent," *Financial Times*, October 31, 2025, <https://www.ft.com/content/399ad440-ecd3-4fb0-b97d-a0d8e0a8a22c>.

<sup>38</sup> Gracelin Baskaran and Meredith Schwartz, "The Consequences of China's New Rare Earths Export Restrictions," CSIS, April 14, 2025, <https://www.csis.org/analysis/consequences-chinas-new-rare-earths-export-restrictions>.

<sup>39</sup> Ernest Scheyder and Kanishka Singh, "Trump Says No Critical Minerals Tariffs for Now, Will Seek Overseas Supplies," *Reuters*, January 14, 2026, <https://www.reuters.com/world/china/trump-says-no-critical-minerals-tariffs-now-will-look-overseas-supplies-2026-01-14/>.



### 3.7 European Union: The challenge of re-industrialization

Once a leader in the critical minerals supply chain, the EU's position has been significantly affected by the intensifying competition between the US and China, with Industry Commissioner Stephane Sejourne describing the bloc as “collateral damage”.<sup>40</sup> According to a recent report from the European Court of Auditors, the bloc is import-reliant on China for several critical minerals, including Gallium (71%), Magnesium (76%) and Germanium (45%).<sup>41</sup>

As the critical minerals crisis intensified in 2025, Europe accelerated its earlier moves for self-sufficiency but has been slower than the US in assembling the necessary toolkit. Unlike the US, the EU continues to place significant emphasis on recycling initiatives and government funding. The most recent measures include the Critical Raw Materials Act (2023) and ReSourceEU (2025), a ~€3bn initiative that builds on the CRMA's goals<sup>42</sup> to target critical raw material extraction (>10%), processing (40%), and recycling (15%) by 2030.<sup>43</sup> Under CRMA, the EU strives to ensure it limits its import reliance from a single third country below 65%.

ReSourceEU will help with its focus on preventing scrap aluminium from leaving the bloc and recycling magnets used in car batteries, supported by a €2bn annual fund backed by the European Investment Bank. While recycling could meet 20% of the EU's current permanent magnet demand of 20,000 tonnes, the EU has indicated it would legally compel companies to diversify their supply sources if voluntary efforts prove insufficient. Despite CRMA and ReSourceEU, however, many of Europe's strategic projects will struggle to meet their targets by 2030 (see figure 7), in particular recycling (12% vs 25% target), and processing (24% vs 40% target).<sup>44</sup>

To be clear, Europe does maintain some reserves, including the Fen Complex in Norway, which remains Europe's largest known rare-earth deposit since it was discovered in 2024. And company-level initiatives are underway to restart rare-earth production, such as UK manufacturer Less Common Metals Ltd's expansion in France, Solvay in France, and LKAB in Sweden.

Europe has also undertaken several strategic partnerships with non-EU countries (e.g. Chile, Mexico, New Zealand) to boost its minerals resiliency, in a hallmark of traditional multilateralism. But as the European Court of Auditors has found, these partnerships have so far improved contribution but generated little effect on the secure supply of these raw materials.

The most pressing issue appears to be in the EU's processing abilities due to a lack of technology and a decline of facilities needed in the bloc. The ECA report notes that at present, the EU's processing capacity is shutting down, and the auditors remain uncertain whether it will actually be able to recover.<sup>45</sup> Our European analysts anticipate that Europe may continue to face challenges in asserting its full influence in global agenda-setting throughout 2026 as long as it remains reliant on both US and Chinese supply chains.

<sup>40</sup> Alice Hancock and Andy Bounds, “EU Plans Minerals Stockpile Centre to Stop US Snapping up Supplies.” @FinancialTimes. Financial Times. November 19, 2025. <https://www.ft.com/content/70964f99-54e9-494c-861d-139259d33893>.

<sup>41</sup> “Special Report 04/2026: Critical Raw Materials for the Energy Transition.” 2026. European Court of Auditors. 2026. <https://www.eca.europa.eu/en/publications/SR-2026-04>.

<sup>42</sup> “New Measures to Secure Raw Materials and Strengthen the EU's Economic Security,” European Commission, December 3, 2025, [https://commission.europa.eu/news-and-media/news/new-measures-secure-raw-materials-and-strengthen-eus-economic-security-2025-12-03\\_en](https://commission.europa.eu/news-and-media/news/new-measures-secure-raw-materials-and-strengthen-eus-economic-security-2025-12-03_en).

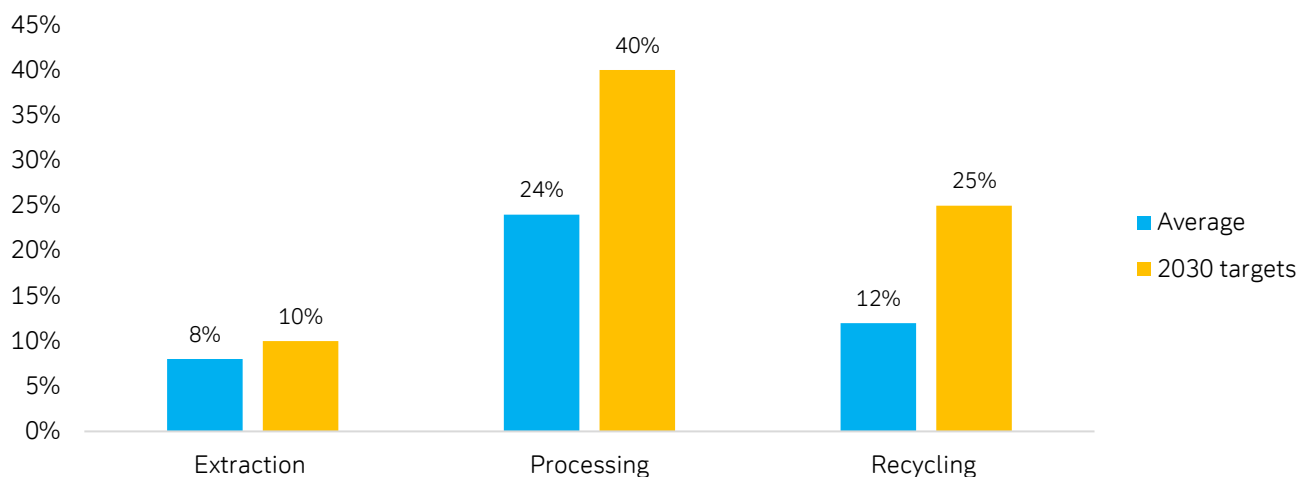
<sup>43</sup> Ibid.

<sup>44</sup> “Special Report 04/2026: Critical Raw Materials for the Energy Transition.” 2026. European Court of Auditors. 2026. <https://www.eca.europa.eu/en/publications/SR-2026-04>.

<sup>45</sup> “Special Report 04/2026: Critical Raw Materials for the Energy Transition.” 2026. European Court of Auditors. 2026. <https://www.eca.europa.eu/en/publications/SR-2026-04>.



**Figure 7: Average EU production capacity of specific critical minerals vs its 2030 targets**



Note: Average of EU production taken from 2016 – 2020. Production capacity is the typical amount of a specific raw material that the EU can produce annually, on average, using its existing industrial facilities and resources. Source: European Court of Auditors, European Commission, Deutsche Bank Research.

### 3.8 Japan: A constrained success

Since facing China's export embargo in 2010, Japan has significantly reduced its reliance on Chinese rare earths from ~90% in 2010 to 60-70% today. The 2010 crisis compelled Tokyo to aggressively diversify its supply chains via long-term offtake agreements and overseas investments, notably in Australia's Lynas Rare Earths. Japan also began deep-sea rare earth mining tests in the Pacific in January 2026.<sup>46</sup>

However, Japan remains the world's largest rare-earth importer. In addition, new Chinese export controls in January and February 2026 — reportedly linked to Taiwan-related signalling<sup>47</sup> — also show that Japan is still vulnerable to disruptions from China's rare earth export controls.<sup>48</sup>

Japan's experience illustrates both the feasibility and the inherent limits of diversification within the framework of infrastructure realism.

<sup>46</sup> "Japan to Conduct Deep-Sea Rare Earth Mining Test in Pacific - Japan Today," *Japan Today*, January 8, 2026, <https://japantoday.com/category/national/japan-to-conduct-deep-sea-rare-earth-mining-test-in-pacific>.

<sup>47</sup> "China Bans Dual-Use Goods Exports for Japan Military over Taiwan Remarks," *CNBC*, January 7, 2026, <https://www.cnbc.com/2026/01/07/china-bans-dual-use-goods-exports-for-japan-military-over-taiwan-remarks.html>.

<sup>48</sup> These latest restrictions apply to a wide range of dual-use products that could contribute to Japan's military capabilities (China's definition of dual use encompasses critical chokepoints like rare earths, gallium, germanium and magnets). Unlike in 2010, China's 2026 move is explicitly intentional.



**Figure 8: Select examples of the Trump Administration's critical minerals' policy**

| Date                     | Partner                                                               | Key details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>April 30, 2025</b>    | Ukraine                                                               | Joint fund that will receive 50% of royalties and license fees from natural resource projects in Ukraine. The deal is more favourable to Ukraine than earlier versions negotiated between Washington and Kyiv. The US International Development Finance Corporation oversees the US side of the deal, and the US secures "first in line" preferential rights for offtake.                                                                                                                                                                         |
| <b>July 1, 2025</b>      | India, Japan, Australia                                               | Announced the Quad Critical Minerals Initiative, which works with private sector partners to facilitate increased investments.                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>October 20 2025</b>   | Australia                                                             | Both governments pledged to deploy \$1bn each within six months to jointly advance strategic projects. Resulting offtake will be directed to buyers in the US. Includes provisions to collaborate on supply-side interventions, including price support, streamlined permitting, geological mapping, recycling initiatives and complementary strategic stockpiling. Also includes joint mechanisms to curb the Chinese acquisition of new mining assets through tools like domestic investment screening.                                         |
| <b>October 2025</b>      | Japan, Malaysia, Thailand                                             | The US signed MOUs with Thailand and Malaysia and a critical minerals framework with Japan. MOUs are focused on enhancing cooperation between the US and other countries over its critical minerals sectors, no funding obligation is included as of yet.                                                                                                                                                                                                                                                                                         |
| <b>November 6, 2025</b>  | Central Asia (C5+1)                                                   | Several government-to-government and company agreements on critical minerals.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>November 18, 2025</b> | Saudi Arabia                                                          | Established a Strategic Framework for Cooperation on securing uranium, metals, permanent magnets, and critical mineral supply chains. The DoD also announced it will finance a 49% equity stake in a new rare earths refinery in SA; MP Materials will provide the technical expertise. Both countries also announced a joint civil nuclear energy cooperation. Important given Russia and China control over half of global uranium enrichment capacity, which poses serious risk to US energy security (US imports 30% of uranium from Russia). |
| <b>December 4, 2025</b>  | DRC                                                                   | Secures preferential access for US companies to vast cobalt, copper, and lithium reserves. The partnership was brokered alongside a peace deal between the DRC and Rwanda.                                                                                                                                                                                                                                                                                                                                                                        |
| <b>December 2025</b>     | Australia, Greece, Israel, Japan, Qatar, Korea, Singapore, UAE and UK | "Pax Silica" initiative launched by the US, which aims to build silicon and critical mineral supply chains integral to AI technologies. Focus is on private-public partnership with companies like ASML, Rio Tinto and Samsung.                                                                                                                                                                                                                                                                                                                   |

Source: Deutsche Bank Research, USGS.



## 4. Scenarios for the critical minerals system (2026-2035)

The global critical minerals system is undergoing a structural shift. The central challenge is no longer geological availability, but rather how access, processing, and control are organized. In this new paradigm, critical minerals are moving from traded commodities toward strategic infrastructure, where power is derived from control over refining, standards, and long-term offtake rather than spot markets.

Against this backdrop, we outline four potential scenarios for 2026–2035, defined along two axes: whether supply chains converge or fragment, and whether allocation is driven primarily by state strategy or by market mechanisms (Figure 9). From 2025 onward, we find that the decades-long trend of critical minerals increasingly treated as security assets has accelerated. Over the next decade, we believe states will be more willing to use stronger policy tools—including conditional security guarantees—to secure supply and processing access. Thus, a state-driven supply chain ecosystem where minerals are used as strategic leverage a likely scenario over the next decade.

Who might emerge as a potential challenger to China? A recent CSIS comparative analysis suggests the United States is emerging as the leading global rare earths processing hub, supported by its existing infrastructure, strong R&D capacity, and bipartisan policy measures that have mobilized private investment.<sup>49</sup> Australia, Saudi Arabia, and Canada also show potential in this new landscape.<sup>50</sup>

**Figure 9: Four possible scenarios for the global critical minerals system (2026 – 2035)**

|            | Converging supply chains                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Fragmenting supply chains                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | Club-based resilience ●●●○                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Minerals used as leverage ●●●●                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| State-led  | <ul style="list-style-type: none"> <li>- Mineral alliances (e.g. U.S.–EU–Japan–Korea–Australia) formalize coordination on stockpiles, refining capacity, and joint procurement.</li> <li>- China-Global South deepens ties with DRC/Indonesia/Africa/Latin America) via infrastructure and finance in exchange for resource access and processing rights.</li> <li>- Public investment, strategic offtakes, and critical mineral clubs emerge.</li> <li>- Shared ESG and tech standards reduce duplication.</li> </ul> | <ul style="list-style-type: none"> <li>- Export controls, local content mandates, and strategic embargoes.</li> <li>- China maintains dominance across refining; West struggles to catch up.</li> <li>- Producer states extract rents, tie access to security deals.</li> <li>- Trade becomes volatile, framed as national security.</li> <li>- Prices higher and more volatile; trade disputes/ WTO cases neutralised by “national security” exceptions, more financing via state-backed structures (guarantees, strategic funds, tied offtakes).</li> </ul> |
|            | Open market scaling ●●○○                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Chaotic competition ●○○○                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Market-led | <ul style="list-style-type: none"> <li>- Markets respond to price signals: private investment in recycling, innovation, and substitution.</li> <li>- Vertical integration and long-term contracts stabilize flows.</li> <li>- Moderate state support (e.g., tax credits) but minimal politicization.</li> </ul>                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>- Companies scramble for supply, face rising geopolitical and ESG risk.</li> <li>- M&amp;A and hoarding intensify; transparency declines.</li> <li>- Regulatory divergence and subsidies lead to trade frictions.</li> </ul>                                                                                                                                                                                                                                                                                           |

Source: Deutsche Bank Research. Note: Dots indicate relative likelihood over the 2026–2035 horizon.

<sup>49</sup> Gracelin Baskaran and Meredith Schwartz, “Developing Rare Earth Processing Hubs: An Analytical Approach,” CSIS, 2025, <https://www.csis.org/analysis/developing-rare-earth-processing-hubs-analytical-approach#h2-results>.

<sup>50</sup> Ibid.



## 5. Conclusion: Critical minerals become infrastructure

This chapter has demonstrated how critical minerals have evolved from mere industrial inputs into strategic chokepoints across key technology and defense sectors. As their use-cases have evolved, countries have recognized they can no longer solely rely on market economics or corporate decision making to steer critical minerals but must designate their need as a national security priority. Furthermore, as the global system rebalances towards an “infrastructure realism” paradigm, China’s control over this foundational layer creates binding bottlenecks across downstream chokepoints such as semiconductors and cloud computing. In our next chapter on Chips, we will go on to discuss the intertwined relationship between critical minerals and semiconductors, and how this has led to a dynamic of reciprocal actions and reactions, particularly between China and the US.

Still, we believe the status quo between a leading China and the rest of the world in critical minerals will unfold only gradually. The IEA projects China will continue to dominate mineral refining (76%) and mining (51%) through 2030.<sup>51</sup> Japan required more than a decade to reduce its direct reliance on Chinese rare earths from 90% to 60% — illustrating the time, cost, and persistence required to unwind dependence once industrial ecosystems are entrenched.

Managed interdependence, therefore, will likely define the rare earths and critical minerals landscape for the next decade, as the US and other countries attempt to replicate the industrial edge Beijing began building in the 1990s. Returning to the “Mineral Fordism” analogy, historian Stefan Link observed that “all late developers must turn, for capital and technology, to those they seek to emulate” —a path that is structurally more difficult in today’s geopolitically fragmented system.

As this slow transformation unfolds, the use of minerals for geopolitical leverage is likely to become an established feature of the global system, with China selectively deploying its mineral sector capabilities to advance trade, security, and foreign policy goals. In an infrastructure-realist world, control over processing, standards, and access—not ownership alone—will increasingly define power. In light of this, over the next five years, we expect geopolitical friction related to critical minerals to manifest in several areas: (i) ongoing competition between China and the US over critical minerals; and (ii) competition between China and the rest of the world over critical minerals; and (iii) competition between the US and the rest of the world, as it continues to pursue its dual goals of gaining access to strategically important regions and ensuring critical mineral security.

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<sup>51</sup> “Rare earth elements – Analysis”, IEA, May 21, 2025, <https://www.iea.org/reports/rare-earth-elements-2>.



## Appendix 1

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